

**ACADEMIC REGULATIONS
AND
SYLLABUS
CHOICE BASED CREDIT SYSTEM
MLR18**

**COMPUTER SCIENCE &
ENGINEERING**

for

Bachelor of Technology (B.Tech)

For the batches admitted from 2018-19

B. Tech. - Regular Four Year Degree Program
(For batches admitted from the academic year 2018 - 19)

&

B. Tech. - Lateral Entry Scheme
(For batches admitted from the academic year 2019 - 20)

(2)



MLRInstitute of Technology

(Autonomous)

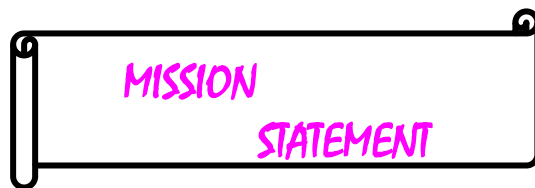
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VISION OF THE DEPARTMENT

Promote Innovation-centric education to produce globally competent graduates in Computer Science and Engineering education & research capable of building a strong and developed nation.



MISSION OF THE DEPARTMENT

- M1:** Provide rigorous course work and State-of-the-Art laboratories for the students to make them globally competent.
- M2:** Strengthen the department interaction with Multi National Companies to enhance graduate technological advancement skills and research capabilities.
- M3:** Impart human values and ethics to the graduates for serving the society with highest regard to the mother land.

Program Educational Objectives (PEOs):

- PEO1:** Graduates of the program will have a globally competent professional career in software Industry.
- PEO2:** Graduates of the program will pursue higher education and research.
- PEO3:** Graduates of the program will have entrepreneur skills to solve societal problems.

FOREWORD

The autonomy is conferred on MLR Institute of Technology by UGC, based on its performance as well as future commitment and competency to impart quality education. It is a mark of its ability to function independently in accordance with the set norms of the monitoring bodies like UGC and AICTE. It reflects the confidence of the UGC in the autonomous institution to uphold and maintain standards it expects to deliver on its own behalf and thus awards degrees on behalf of the college. Thus, an autonomous institution is given the freedom to have its own **curriculum, examination system and monitoring mechanism**, independent of the affiliating University but under its observance.

MLR Institute of Technology is proud to win the credence of all the above bodies monitoring the quality in education and has gladly accepted the responsibility of sustaining, if not improving upon the standards and ethics for which it has been striving for more than a decade in reaching its present standing in the arena of contemporary technical education. As a follow up, statutory bodies like Academic Council and Boards of Studies are constituted with the guidance of the Governing Body of the College and recommendations of the JNTU Hyderabad to frame the regulations, course structure and syllabi under autonomous status.

The autonomous regulations, course structure and syllabi have been prepared after prolonged and detailed interaction with several expertise solicited from academics, industry and research, in accordance with the vision and mission of the college in order to produce quality engineering graduates to the society.

All the faculty, parents and students are requested to go through all the rules and regulations carefully. Any clarifications, if needed, are to be sought, at appropriate time with principal of the college, without presumptions, to avoid unwanted subsequent inconveniences and embarrassments. The Cooperation of all the stake holders is sought for the successful implementation of the autonomous system in the larger interests of the college and brighter prospects of engineering graduates.

PRINCIPAL

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B. Tech. - Regular Four Year Degree Programme
(For batches admitted from the academic year 2018 - 19)
&
B. Tech. - Lateral Entry Scheme
(For batches admitted from the academic year 2019 - 20)

For pursuing four year under graduate Bachelor Degree Programme of study in Engineering (B.Tech) offered by MLR Institute of Technology under Autonomous status is herein referred to as MLRIT (Autonomous):

All the rules specified herein approved by the Academic Council will be in force and applicable to students admitted from the Academic Year 2018-19 onwards. Any reference to "Institute" or "College" in these rules and regulations shall stand for M L R Institute of Technology (Autonomous).

All the rules and regulations, specified hereafter shall be read as a whole for the purpose of interpretation as and when a doubt arises, the interpretation of the Chairman, Academic Council is final. As per the requirements of statutory bodies, the Principal, M L R Institute of Technology shall be the chairman Academic Council.

1. ADMISSION

1.1. Admission into first year of four year B. Tech. degree Programmes of study in Engineering

1.1.1. Eligibility:

A candidate seeking admission into the first year of four year B. Tech. degree Programmes should have:

- (i)** Passed either Intermediate Public Examination (I.P.E) conducted by the Board of Intermediate Education, Telangana, with Mathematics, Physics and Chemistry as optional subjects or any equivalent examination recognized by Board of Intermediate Education, Telangana or a Diploma in Engineering in the relevant branch conducted by the Board of Technical Education, Telangana or equivalent Diploma recognized by Board of Technical Education for admission as per guidelines defined by the Regulatory bodies of Telangana State Council for Higher Education (TSCHE) and AICTE.
- (ii)** Secured a rank in the EAMCET examination conducted by TSCHE for allotment of a seat by the Convener, EAMCET, for admission.

1.1.2. Admission Procedure:

Admissions are made into the first year of four year B. Tech. Degree Programmes as per the stipulations of the TSCHE.

- (a) Category A seats are filled by the Convener, TSEAMCET.
- (b) Category B seats are filled by the Management.

1.2. Admission into the second year of four year B. Tech. degree Program in Engineering

1.2.1 Eligibility:

A candidate seeking admission under lateral entry into the II year I Semester B. Tech. degree Programmes should have passed the qualifying exam (B.Sc. Mathematics or Diploma in concerned course) and based on the rank secured by the candidate at Engineering Common Entrance Test ECET (FDH) in accordance with the instructions received from the Convener, ECET and Government of Telangana.

1.2.2 Admission Procedure:

Admissions are made into the II year of four year B. Tech. degree Programmes through Convener, ECET (FDH) against the sanctioned strength in each Programmes of study as lateral entry students.

2. PROGRAMMES OFFERED

MLR Institute of Technology, an autonomous college affiliated to JNTUH, offers the following B. Tech. Programmes of study leading to the award of B. Tech. degree under the autonomous scheme.

- 1) B.Tech. - Aeronautical Engineering
- 2) B.Tech. - Computer Science and Engineering
- 3) B.Tech. - Electronics and Communication Engineering
- 4) B.Tech. - Electrical & Electronics Engineering
- 5) B.Tech. - Information Technology
- 6) B.Tech. - Mechanical Engineering

3. DURATION OF THE PROGRAMMES

3.1 Normal Duration

- 3.1.1 B. Tech. degree programme extends over a period of four academic years leading to the Degree of Bachelor of Technology (B.Tech.) of the Jawaharlal Nehru Technological University Hyderabad.
- 3.1.2 For students admitted under lateral entry scheme, B. Tech. degree programme extends over a period of three academic years leading to the Degree of Bachelor of Technology (B. Tech.) of the Jawaharlal Nehru Technological University Hyderabad.

3.2 Maximum Duration

- 3.2.1 The maximum period within which a student must complete a full-time academic programme is 8 years for B. Tech. If a student fails to complete the academic programme within the maximum duration as specified above, he shall forfeit the seat in B.Tech and his admission shall stand cancelled.
- 3.2.2 For students admitted under lateral entry scheme in B. Tech. degree programme, the maximum period within which a student must complete a full-time academic programme is 6 years. If a student fails to complete the academic programme within the maximum duration as specified above, he shall forfeit the seat in B.Tech and his admission shall stand cancelled.
- 3.2.3 The period is reckoned from the academic year in which the student is admitted first time into the degree Programme.

4. AWARD OF B.Tech. DEGREE

A student will be declared eligible for the award of the B.Tech. degree if he/she fulfils the following academic regulations:

- 4.1 The candidate shall pursue a course of study for not less than four academic years and not more than eight years.
- 4.2 The candidate shall register for 160 credits and secure 160 credits.
- 4.3 The degree will be conferred and awarded by Jawaharlal Nehru Technological University Hyderabad on the recommendations of the Chairman, Academic Council.

5. PROGRAMME STRUCTURE

- 5.1 UGC/AICTE specified Definitions/ Descriptions are adopted appropriately for various terms and abbreviations used in these Academic Regulations/ Norms, which are listed below.

Semester Scheme:

Each UGP is of 4 Academic Years (8 Semesters), each year divided into two Semesters of 22 weeks (≥ 90 working days), each Semester having - 'Continuous Internal Evaluation (CIE)' and 'Semester End Examination (SEE)'. Choice Based Credit System (CBCS) and Credit Based Semester System (CBSS) as denoted by UGC, and Curriculum/Course Structure as suggested by AICTE are followed.

- 5.1.2 The B.Tech. Programme of MLR Institute of Technology are of Semester pattern, with 8 Semesters constituting 4 Academic Years, each Academic Year having TWO Semesters (First/Odd and Second/Even Semesters). Each Semester shall be of 22 Weeks duration (inclusive of Examinations), with a minimum of 90 Instructional Days per Semester.

- 5.1.3 Credit Courses:

a) All Courses are to be registered by a student in a Semester to earn Credits. Credits shall be assigned to each Subject/ Course in a L: T: P: C (Lecture Periods: Tutorial Periods: Practical Periods : Credits) Structure, based on the following general pattern ..

- One Credit - for One hour/Week/Semester for Theory/Lecture(L)/Tutorial(T) Courses; and
- One Credit - for Two hours/Week/Semester for Laboratory/Practical (P) Courses, Mini Project...
- Mandatory Courses will not carry any Credits.

- 5.1.4 **Course Classification:**

All Courses offered for the UGP are broadly classified as:

- Basic Science Courses (BSC) include Mathematics, Physics, Chemistry, Biology etc
- **Engineering Science Courses (ESC)** courses include Materials, Workshop, Basics of Electrical/Electronics/ Mechanical/Computer Science & Engineering, Engineering Graphics, Instrumentation, Engineering Mechanics, Instrumentation etc.
- **Humanities and Social Science including Management Courses (HSMC)** courses include English, Communication skills, Management etc.
- Professional Core Courses(PCC) are core courses relevant to the chosen specialization/branch.
- **Professional Elective Courses (PEC)** are courses relevant to the chosen specialization/ branch offered as electives.
- **Open Elective Courses (OEC)** courses from other technical and/or emerging subject areas offered in the College by the Departments of Engineering, Science and Humanities.
- **Mandatory Course:** Course work on peripheral subjects in a programme, wherein familiarity considered mandatory. To be included as non-Credit, Mandatory Courses, with only a pass in each required to qualify for the award of degree from the concerned institution.
- **Project Work** and/or internship in industry or elsewhere, seminar.

5.1.5 Course Nomenclature:

The Curriculum Nomenclature or Course-Structure Grouping for the each of the UGP E&T (B.Tech. Degree Programme), is as listed below (along with AICTE specified Range of Total Credits).

S. No.	Broad Course Classification	Course Group/ Category	Course Description	Range of Credits
1)	BSC,ESC & HSMC	BSC – Basic Sciences Courses	Includes - Mathematics, Physics and Chemistry Subjects	25*
2)		ESC - Engineering Sciences Courses	Includes fundamental engineering subjects.	24*
3)		HSMC – Humanities and Social Sciences including Management	Includes subjects related to Humanities, Social Sciences and Management.	12*
4)	PCC	PCC – Professional Core Courses	Includes core subjects related to the Parent Discipline/ Department/ Branch of Engg.	48*
5)	PEC	PEC– Professional Elective Courses	Includes Elective subjects related to the Parent Discipline / Department / Branch of Engg.	18*
6)	OEC	OEC – Open Elective Courses	Elective subjects which include inter-disciplinary subjects or subjects in an area outside the Parent Discipline/ Department / Branch of Engg.	18*
7)	PWC	Project Work	Major Project.	15*
8)		Industrial Training/ Mini-Project	Industrial Training/ Internship/ Mini-Project.	
9)		Seminar	Seminar / Colloquium based on core contents related to Parent Discipline/ Department/ Branch of Engg.	
10)	MC	Mandatory Courses	Mandatory Courses (non-credit)	-
Total Credits for UGP (B. Tech.)Programme				160 (100%)

- Minor variations as per AICTE guidelines

6. COURSE REGISTRATION

- 6.1 A 'Faculty Advisor or Counsellor' shall be assigned to each student, who advises him/her about the UGP, its Course Structure and Curriculum, Choice/Option for Subjects/Courses, based on his/her competence, progress, pre-requisites and interest.
- 6.2 Academic Section of the College invites 'Registration Forms' from students prior (before the beginning of the Semester), ensuring 'DATE and TIME Stamping'. The Registration Requests for any 'CURRENT SEMESTER' shall be completed BEFORE the commencement of SEEs (Semester End Examinations) of the 'PRECEDING SEMESTER'
- 6.3 A Student can apply for Registration, which includes approval from his faculty advisor, and then should be submitted to the College Academic Section through the Head of Department

(a copy of the same being retained with Head of Department, Faculty Advisor and the Student).

- 6.4 A student may be permitted to register for his/her course of CHOICE with a Total of prescribed credits per Semester (permitted deviation being $\pm 12\%$), based on his PROGRESS and SGPA/CGPA, and completion of the 'PRE-REQUISITES' as indicated for various courses in the Department Course Structure and Syllabus contents.
- 6.5 Choice for 'additional Courses' must be clearly indicated, which needs the specific approval and signature of the Faculty Advisor/Counsellor.
- 6.6 If the Student submits ambiguous choices or multiple options or erroneous (incorrect) entries during Registration for the Course(s) under a given/specified Course Group/ Category as listed in the Course Structure, only the first mentioned Course in that Category will be taken into consideration.
- 6.7 Dropping of Courses or changing of options may be permitted, ONLY AFTER obtaining prior approval from the Faculty Advisor, 'within 15 Days of Time' from the commencement of that Semester. Course Options exercised through Registration are final and CAN NOT be changed, and CAN NOT be inter-changed; further, alternate choices will also not be considered. However, if the Course that has already been listed for Registration (by the Head of Department) in a Semester could not be offered due to any unforeseen or unexpected reasons, then the Student shall be allowed to have alternate choice - either for a new Subject (subject to offering of such a Subject), or for another existing Subject (subject to availability of seats), which may be considered. Such alternate arrangements will be made by Head of the Department, with due notification and time-framed schedule, within the FIRST WEEK from the commencement of Class-work for that Semester.

7. COURSES TO BE OFFERED

- 7.1 A typical section (or class) strength for each Semester shall be 60.
- 7.2 courses may be offered to the Students, only if minimum of 20 students ($1/3^{\text{rd}}$ of the section strength) opt for it.
- 7.2 More than ONE TEACHER may offer the SAME SUBJECT (Lab/Practical's may be included with the corresponding Theory Subject in the same Semester) in any Semester. However, selection choice for students will be based on - 'CGPA Basis Criterion' (i.e., the first focus shall be on early Registration in that Semester, and the second focus, if needed, will be on CGPA of the student).
- 7.3 If more entries for Registration of a Subject come into picture, then the concerned Head of the Department shall take necessary decision, whether to offer such a Subject/Course for TWO (or multiple) SECTIONS or NOT.
- 7.4 OPEN ELECTIVES will be offered by a department to the students of other departments.

8. ATTENDANCE REQUIREMENTS

- a. A student will be eligible to appear for the End Semester Examinations, if he acquires a minimum of 75% of attendance in aggregate of all the Subjects/Courses (excluding Mandatory or Non-Credit Courses) for that Semester.
- b. Condoning of shortage of attendance in aggregate up to 10% (65% and above, and below 75%) in each Semester may be granted by the College Academic Committee on genuine and valid grounds, based on the student's representation with supporting evidence by following the govt. rules in vogue.
- c. A stipulated fee shall be payable towards condoning of shortage of attendance.
- d. Shortage of Attendance below 65% in aggregate shall in No case be condoned.
- e. A student shall not be promoted to the next Semester unless he/she satisfies the attendance requirements of the current Semester. The student may seek readmission for the Semester when offered next. He / She shall not be allowed to register for the subjects of the Semester while he/she is in detention. A student detained due to shortage of attendance, will have to repeat that Semester when offered next.

- f. Students whose attendance is less than 75% are not entitled to get the scholarship / fee reimbursement in any case as per the TS Govt. Rules in force.

9. ACADEMIC REQUIREMENTS FOR PROMOTION / COMPLETION OF REGULAR B.TECH PROGRAMME COURSE STUDY.

- 9.1 A student shall be deemed to have satisfied the Academic Requirements and earned the Credits allotted to each Course, if he secures not less than 35% marks in the End Semester Examination, and a minimum of 40% of marks in the sum Total of the CIE (Continuous Internal Evaluation) and SEE (Semester End Examination) taken together; in terms of Letter Grades, this implies securing P Grade or above in that Course.
- 9.2 A student shall be deemed to have satisfied the Academic Requirements and earned the Credits allotted to – Internship, if he/she secures not less than 40% of the Total marks to be awarded for each. The student would be treated as failed, if he/she - (i) does not submit a report on his/her Internship, or does not make a presentation of the same before the Evaluation Committee as per the schedule, or (ii) does not present the Seminar as required in the IV year I/II Semester, or (iii) secures less than 40% of marks in Internship evaluations. He/She may reappear once for each of the above evaluations, when they are scheduled again; if he/she fails in such 'one reappearance' evaluation also, he/she has to reappear for the same in the next subsequent Semester, as per the schedule.
- 9.3 A Student will not be promoted from I Year to II Year, unless he/she fulfils the Attendance requirements
- 9.4 A Student will not be promoted from II Year to III Year, unless he/she fulfils the Attendance and Academic Requirements and (i) secures a Total 50% of Credits up to II Year II Semester from all the relevant regular and supplementary examinations.
- 9.5 A Student will not be promoted from III Year to IV Year, unless he/she fulfils the Attendance and Academic Requirements and (i) secures a Total 50% of Credits up to III Year II Semester, from all the regular and supplementary examinations.
- 9.6 After securing the necessary 160 Credits as specified for the successful completion of the entire UGP, resulting in 160 Credits for UGP performance evaluation, i.e., the performance of the Student in these 160 Credits shall alone be taken into account for the calculation of the final CGPA.
- If a Student registers for some more 'extra courses' (in the parent Department or other Departments/Branches of Engg.) other than those listed courses Totalling to 160 Credits as specified in the Course Structure of his/her Department, the performances in those 'extra courses' (although evaluated and graded using the same procedure as that of the required 160 Credits) will not be taken into account while calculating the SGPA and CGPA. For such 'extra courses' registered, % marks and Letter Grade alone will be indicated in the Grade Card, as a performance measure, subject to completion of the Attendance and Academic Requirements as stated in items 8 and 9.1-9.6.
- 9.7 Students who fail to earn minimum of 160 Credits as per the Course Structure, and as indicated above, within 8 Academic Years from the Date of Commencement of their I Year shall forfeit their seats in B.Tech Programme and their admissions shall stand cancelled.
- When a Student is detained due to shortage of attendance/lack of credits in any Semester, he may be re-admitted into that Semester, as and when offered. However the regulations at the time of admissions hold good.

10. EVALUATION - DISTRIBUTION AND WEIGHTAGE OF MARKS

- 10.1 The performance of a student in each Semester shall be evaluated Course-wise (irrespective of Credits assigned) with a maximum of 100 marks for Theory. The B.Tech Project Work (Major Project) will be evaluated for 100 marks in Phase-I and 200 Marks in Phase-II.
- 10.2 For all Theory Courses as mentioned above, the distribution shall be 30 marks for CIE, and 70 marks for the SEE.

- 10.3 a) For Theory Subjects (inclusive of Minor Courses), during the Semester, there shall be 2 mid-term examinations for 30 marks each. Each mid-term examination consists of one subjective paper for 25 marks, and assignment for 5 marks for each subject.

Question paper contains 2 Parts (Part-A and Part-B.) The distribution of marks for PART-A and PART-B will be 10 marks & 15 marks respectively for UG programme.

Pattern of the question paper is as follows:

PART-A

Consists of **one compulsory question** with five sub questions each carrying two mark. For the I-Mid examinations the sub question would be from first 2 ½ units and for the II-Mid examination the sub question would be from the remaining 2 ½ units.

PART-B

Consists of five questions (out of which students have to answer three questions) carrying five marks each. Each question there will be an “either” “or” choice (that means there will be two questions from each unit and the student should answer any one question). The questions can consist of sub questions also.

- b) The first mid-term examination shall be conducted for the first 50% of the syllabus, and the second mid-term examination shall be conducted for the remaining 50% of the syllabus.
- c) First Assignment should be submitted before the commencement of the first mid-term examinations, and the Second Assignment should be submitted before the commencement of the second mid-term examinations. The assignments shall be specified/given by the concerned subject teacher.
- d) If any candidate is absent for the MID term examination or those who want to improve their internal marks in any subject can opt for Computer Based Test (CBT) as and when offered. The CBT is a 45 minutes duration ONLINE exam consisting of 30 objective questions from the entire syllabus of the subject. Best marks is consider as a final marks from the average of two mid examinations or CBT examination marks. The CBT can be taken after the payment of prescribed fee. There is no Internal Improvement for the courses Machine Drawing, Production Drawing, Engineering Drawing, Engineering Graphics and practical courses.
- 10.4 For Practical Courses, there shall be a Continuous Internal Evaluation (CIE) during the Semester for 30 internal marks, and 70 marks are assigned for Lab/Practical End Semester Examination (SEE). Out of the 30 marks for internals, day-to-day work in the laboratory shall be evaluated for 15 marks; and for the remaining 15 marks - two internal practical tests (each of 15 marks) shall be conducted by the concerned laboratory teacher and the average of the two tests is taken into account. The SEE for Practical's shall be conducted at the end of the Semester by Two Examiners appointed by the Chief Controller of Examinations in consultation with the Head of the Department.
- 10.5 For the Subjects having Design and/or Drawing, (such as Engineering Graphics, Engineering Drawing, Machine Drawing, Production Drawing Practice, and Estimation), the distribution shall be 30 marks for CIE (10 marks for day-to-day work and 20 marks for internal tests) and 70 marks for SEE. There shall be two internal tests in a semester and the average of the two shall be considered for the award of marks for internal tests.
- 10.6 **Open Elective Course:** Students can choose One Open Elective Course (OEC-I) during V Semester, one (OEC-II) during VI Semester, one (OEC-III) in VII Semester, and one (OEC-IV) in VIII Semester from the list of Open Elective Courses given. However, Students cannot opt for an Open Elective Courses offered by their own (parent) Department, if it is already listed under any category of the Subjects offered by parent Department in any Semester.
- 10.7 There shall be an internship to be taken up during the vacation after II year II Semester (IV Semester) examination. However, the internship and its report shall be evaluated in III Year I Semester (VI Semester) SEE. The internship shall be submitted in a report form and

- presented before the committee. It shall be evaluated for 100 marks. The committee consists of head of the department, and a senior faculty member(s)(minimum Two) of the department. There shall be no internal marks for internship.
- 10.8 There shall be a Comprehensive Viva in VI Semester and will be conducted SEE by through a test or a committee consisting of One External Examiner, Head of the Department and two Senior faculty members of the Department. The comprehensive viva is intended to assess the student's understanding of the subjects he/she studied during the B.Tech course of study. The Comprehensive Viva is evaluated for 100 marks by the test or committee. There shall be no CIE for Comprehensive Viva.
- 10.9 Each Student shall start the Project Work Phase-I during the IV B.Tech I Semester(VII Semester), as per the instructions of the Project Guide/Project Supervisor assigned by the Head of Department. Total 100 marks allotted for the Project Work Stage-I. 40% of marks shall be evaluated Project Guide/Project supervisor CIE (Continuous Internal Evaluation) based on the reports submitted and conduct presentations. Remaining 60% of marks shall be evaluated by committee comprising of the Head of the Department, project supervisor and senior faculty member from concerned department based on Viva/Seminar Presentation. He/She must secure the 40% of the marks from CIE. For Project work Phase-II in IV Year II Sem. There is an internal marks of 50, the evaluation should be done by the supervisor. The There is an external marks of 150 and the same evaluated by the external examiner appointed by the Chief Controller of Examinations and he secures a minimum of 35% of marks in the Semester End Examination and a minimum aggregate of 40% of the total marks in the Semester End Examination and Continuous Internal Evaluation taken together.
- 10.10. **End Semester examination:**
- Question paper contains 2 Parts (Part-A and Part-B) having the questions distributed equally among all units.
 - The distribution of marks for PART-A and PART-B will be 20 marks & 50 marks respectively for UG programme. Pattern of the question paper is as follows:
- PART-A**
- Consists of one question which are compulsory. The question consists of ten sub-questions one from each unit and carry 2 marks each.
- PART-B**
- Consists of 5 questions carrying 10 marks each. Each of these questions is from one unit and may contain sub questions. Each question there will be an "either" "or" choice (that means there will be two questions from each unit and the student should answer any one question).
- 10.11 For Mandatory Non-Credit Courses offered in a Semester, after securing $\geq 65\%$ attendance and has secured not less than 35% marks in the SEE, and a minimum of 40% of marks in the sum Total of the CIE and SEE taken together in such a course, then the student is **PASS** and will be qualified for the award of the degree. No marks or Letter Grade shall be allotted for these courses/activities. However, for non credit courses '**Satisfactory**' or '**Unsatisfactory**' shall be indicated instead of the letter grade and this will not be counted for the computation of SGPA/CGPA.
- 10.12. For social innovation and Engineering Exploration courses, there shall be a Continuous Internal Evaluation (CIE) during the Semester for 30 internal marks, and 70 marks are assigned End Semester Examination (SEE). Internal evaluations done through presentations in CIE-1 & CIE-2. The SEE for shall be conducted at the end of the Semester by Two Examiners appointed by the Chief Controller of Examinations in consultation with the Head of the Department.

11. AWARD OF DEGREE

After a student has satisfied the requirement prescribed for the completion of the Programme and is eligible for the award of B. Tech. Degree he shall be placed in one of the following four classes Shown in Table.

Table: **Declaration of Class based on CGPA (Cumulative Grade Point Average)**

Class Awarded	Grade to be Secured
First Class with Distinction	CGPA $\geq$ 8.00
First Class	$\geq$ 6.50 to $<$ 8.00 CGPA
Second Class	$\geq$ 5.50 to $<$ 6.50 CGPA
Pass Class	$\geq$ 5.00 to $<$ 5.50 CGPA
FAIL	CGPA $<$ 5

a) Improvement of Grades and Completion of the Course

- Candidates who have passed in a theory course in a Semester are allowed to appear for improvement of Grade in the next immediate supplementary examination for a maximum of three subjects only. Candidates will not be allowed to improve grade in the Laboratory, Seminars, Internships and Project Work.
- Improved grade will not be counted for the award of prizes/medals and Rank. However the previous grade will be considered for the award of prizes/medals and rank in case of toppers.
- If the candidate does not show improvement in the grade, his/her previous grade will be taken into consideration.

12. LETTER GRADE AND GRADE POINT

- 12.1 Marks will be awarded to indicate the performance of each student in each Theory Subject, or Lab/Practical's, or Seminar, or Project, or Internship*/Mini-Project, Minor Course etc., based on the %marks obtained in CIE+SEE (Continuous Internal Evaluation + Semester End Examination, both taken together), and a corresponding Letter Grade shall be given. * For internship only SEE will be taken into consideration. There is no CIE for Internship.

- 12.2 As a measure of the student's performance, a 10-point Absolute Grading System using the following Letter Grades (UGC Guidelines) and corresponding percentage of marks shall be followed...

% of Marks Secured (Class Intervals)	Letter Grade (UGC Guidelines)	Grade Points
90% and above ($\geq$ 90% , $\leq$ 100%)	O (Outstanding)	10
Below 90% but not less than 80% ($\geq$ 80% , $<$ 90%)	A ⁺ (Excellent)	9
Below 80% but not less than 70% ($\geq$ 70% , $<$ 80%)	A (Very Good)	8
Below 70% but not less than 60% ($\geq$ 60% , $<$ 70%)	B ⁺ (Good)	7
Below 60% but not less than 50% ($\geq$ 50% , $<$ 60%)	B (above Average)	6
Below 50% but not less than 40% ($\geq$ 40% , $<$ 50%)	C (Average)	5
Below 40% ($<$ 40%)	F (FAIL)	0
Absent	AB	0

- 12.3 A student obtaining F Grade in any Subject shall be considered 'failed' and will be required to reappear as 'Supplementary Candidate' in the End Semester Examination (SEE), as and when offered. In such cases, his Internal Marks (CIE Marks) in those Subject(s) will remain same as those he obtained earlier.

- 12.4 A Letter Grade does not imply any specific % of Marks.
- 12.5 In general, a student shall not be permitted to repeat any Subject/Course (s) only for the sake of 'Grade Improvement' or 'SGPA/CGPA Improvement'. However, he has to repeat all the Subjects/Courses pertaining to that Semester, when he is detained.
- 12.6 A student earns Grade Point (GP) in each Subject/Course, on the basis of the Letter Grade obtained by him in that Subject/Course (excluding Mandatory non-credit Courses). Then the corresponding 'Credit Points' (CP) are computed by multiplying the Grade Point with Credits for that particular Subject/Course.

Credit Points (CP) = Grade Point (GP) x Credits For a Course

- 12.7 The Student passes the Subject/Course only when he gets $GP \geq 4$ (P Grade or above).
- 12.8 The Semester Grade Point Average (SGPA) is calculated by dividing the Sum of Credit Points (ΣCP) secured from ALL Subjects/Courses registered in a Semester, by the Total Number of Credits registered during that Semester. SGPA is rounded off to TWO Decimal Places. SGPA is thus computed as

$SGPA = \{\sum_{i=1}^N C_i G_i\} / \{\sum_{i=1}^N C_i\}$ For each Semester,

where 'i' is the Subject indicator index (takes into account all Subjects in a Semester), 'N' is the no. of Subjects 'REGISTERED' for the Semester (as specifically required and listed under the Course Structure of the parent Department), C_i is the no. of Credits allotted to that ix Subject, and G_i represents the Grade Points (GP) corresponding to the Letter Grade awarded for that i Subject.

Illustration of Computation of SGPA Computation

Course	Credit	Grade Letter	Grade Point	Credit Point (Credit x Grade)
Course1	3	A	8	$3 \times 8 = 24$
Course2	4	B+	7	$4 \times 7 = 28$
Course3	3	B	6	$3 \times 6 = 18$
Course4	3	O	10	$3 \times 10 = 30$
Course5	3	C	5	$3 \times 5 = 15$
Course6	4	B	6	$4 \times 6 = 24$

Thus, **SGPA =139/20 =6.95**

- 12.9 The Cumulative Grade Point Average (CGPA) is a measure of the overall cumulative performance of a student over all Semesters considered for registration. The CGPA is the ratio of the Total Credit Points secured by a student in ALL registered Courses in ALL Semesters, and the Total Number of Credits registered in ALL the Semesters. CGPA is rounded off to TWO Decimal Places. CGPA is thus computed from the I Year Second Semester onwards, at the end of each Semester, as per the formula

**$CGPA = \{\sum_{j=1}^M C_j G_j\} / \{\sum_{j=1}^M C_j\}$... for all S Semesters registered
(i.e., up to and inclusive of S Semesters, $S \geq 2$),**

where 'M' is the TOTAL no. of Subjects (as specifically required and listed under the Course Structure of the parent Department) the Student has 'REGISTERED' from the 1st Semester onwards up to and inclusive of the Semester S (obviously $M > N$), 'j' is the Subject indicator index (takes into account all Subjects from 1 to S Semesters), C_j is the no. of Credits allotted to the jth Subject, and G_j represents the Grade Points (GP) corresponding to the Letter Grade awarded for that jth Subject. After registration and completion of I Year I Semester however, the SGPA of that Semester itself may be taken as the CGPA, as there are no cumulative effects.

For CGPA Computation

Semester 1	Semester 2	Semester 3	Semester 4	Semester 5	Semester 6
Credits : 20 SGPA : 6.9	Credits : 22 SGPA : 7.8	Credits : 25 SGPA : 5.6	Credits : 26 SGPA : 6.0	Credits : 26 SGPA : 6.3	Credits : 25 SGPA : 8.0

Thus, **CGPA =**
$$\frac{20 \times 6.9 + 22 \times 7.8 + 25 \times 5.6 + 26 \times 6.0 + 26 \times 6.3 + 25 \times 8.0}{144} = 6.73$$

- 12.10 For Merit Ranking or Comparison Purposes or any other listing, ONLY the 'ROUNDED OFF' values of the CGPAs will be used.
- 12.11 For Calculations listed in Item 12.6–12.10, performance in failed Subjects/Courses (securing F Grade) will also be taken into account, and the Credits of such Subjects/Courses will also be included in the multiplications and summations. However, Mandatory Courses will not be taken into consideration.
- 12.12 Conversion formula for the conversion of GPA into indicative percentage is
[CGPA Earned-0.50]x10= % of marks scored

13. DECLARATION OF RESULTS

Computation of SGPA and CGPA are done using the procedure listed in 12.6– 2.10.

No SGPA/CGPA is declared, if a candidate is failed in any one of the courses of a given Semester.

14. WITH HOLDING OF RESULTS

If the student has not paid fees to College at any stage, or has pending dues against his name due to any reason what so ever, or if any case of indiscipline is pending against him, the result of the student may be withheld, and he will not be allowed to go into the next higher Semester. The Award or issue of the Degree may also be withheld in such cases.

15. REVALUATION

Students shall be permitted for revaluation after the declaration of end Semester examination results within due dates by paying prescribed fee. After revaluation if there is any betterment in the grade, then improved grade will be considered. Otherwise old grade shall be retained.

16. SUPPLEMENTARY EXAMINATIONS

- 16.1 Supplementary examinations for the odd Semester shall be conducted with the regular examinations of even Semester and vice versa, for those who appeared and failed or absent in regular examinations. Such candidates writing supplementary examinations may have to write more than one examination per day.

16.2 ADVANCED SUPPLEMENTARY EXAMINATION

Advanced supplementary examinations will be conducted for IV year II Semester after announcement of regular results.

17. TRANSCRIPTS

After successful completion of prerequisite credits for the award of degree a Transcript containing performance of all academic years will be issued as a final record. Duplicate transcripts will also be issued if required after the payment of requisite fee and also as per norms in vogue.

18. RULES OF DISCIPLINE

- 18.1 Any attempt by any student to influence the teachers, Examiners, faculty and staff of controller of Examination for undue favours in the exams, and bribing them either for marks or attendance will be treated as malpractice cases and the student can be debarred from the college.
- 18.2 When the student absents himself, he is treated as to have appeared and obtained zero marks in that course(s) and grading is done accordingly.
- 18.3 When the performance of the student in any subject(s) is cancelled as a punishment for indiscipline, he is awarded zero marks in that subject(s).
- 18.4 When the student's answer book is confiscated for any kind of attempted or suspected malpractice the decision of the Examiner is final.

19. MALPRACTICE PREVENTION COMMITTEE

A malpractice prevention committee shall be constituted to examine and punish the students who involve in malpractice / indiscipline in examinations. The committee shall consist of:

- a) Controller of examinations - Chairman
- b) Addl. Controller of examinations.- Member Convenor
- c) Subject expert - member
- d) Head of the department of which the student belongs to. - Member
- e) The invigilator concerned - member

The committee shall conduct the meeting after taking explanation of the student and punishment will be awarded by following the malpractice rules meticulously.

Any action on the part of candidate at the examination like trying to get undue advantage in the performance at examinations or trying to help another, or derive the same through unfair means is punishable according to the provisions contained hereunder. The involvement of the Staff who are in charge of conducting examinations, valuing examination papers and preparing / keeping records of documents relating to the examinations, in such acts (inclusive of providing incorrect or misleading information) that infringe upon the course of natural justice to one and all concerned at the examination shall be viewed seriously and will be recommended for appropriate punishment after thorough enquiry.

20. TRANSITORY REGULATIONS

Student who has discontinued for any reason, or has been detained for want of attendance or lack of required credits as specified, or who has failed after having undergone the Degree Programme, may be considered eligible for readmission to the same Subjects/Courses (or equivalent Subjects/Courses, as the case may be), and same Professional Electives/Open Electives (or from set/category of Electives or equivalents suggested, as the case may be) as and when they are offered (within the time-frame of 8 years from the Date of Commencement of his I Year I Semester).

21. AMENDMENTS TO REGULATIONS

The Academic Council of MLR Institute of Technology reserves the right to revise, amend, or change the regulations, scheme of examinations, and / or syllabi or any other policy relevant to the needs of the society or industrial requirements etc., without prior notice.

22. STUDENT TRANSFERS

There shall be no Branch transfers after the completion of Admission Process. Transfer of students from other colleges or universities are permitted subjected to the rules and regulations of TSCHE (TE Department) and JNTUH in vogue.

23. GRADUATION DAY

The College shall have its own Annual Graduation Day for the award of Degrees issued by the College/University.

24. AWARD OF MEDALS

Institute will award Medals to the outstanding students who complete the entire course in the first attempt within the stipulated time.

25. SCOPE

- i) Where the words “he”, “him”, “his”, occur in the write-up of regulations, they include “she”, “her”.
- ii) Where the words “Subject” or “Subjects”, occur in these regulations, they also imply “Course” or “Courses”.
- iii) The Academic Regulations should be read as a whole, for the purpose of any interpretation.
- iv) In case of any doubt or ambiguity in the interpretation of the above rules, the decision of the Chairman of the Academic Council is final.

Academic Regulations for B. Tech. (Lateral Entry Scheme)

(Effective for the students getting admitted into II year from the Academic Year 2018-2019 onwards)

1. The Students have to acquire 120 credits from II to IV year of B.Tech Programme (Regular) for the award of the degree.
2. Students, who fail to fulfil the requirement for the award of the degree in 6 consecutive academic years from the year of admission, shall forfeit their seat.
3. The same attendance regulations are to be adopted as that of B. Tech. (Regular)
4. **Promotion Rule:**
5. A Student will not be promoted from III Year to IV Year, unless he/she fulfils the Attendance and Academic Requirements and (i) secures a Total of 50% Credits up to III Year II Semester, from all the regular and supplementary examinations.

6. **Award of Class:**

After the student has satisfied the requirements prescribed for the completion of the programme and is eligible for the award of B. Tech. Degree, he/she shall be placed in one of the following four classes: The marks obtained for 120 credits will be considered for the calculation of CGPA and award of class shall be shown separately.

Table: **Declaration of Class based on CGPA (Cumulative Grade Point Average)**

Class Awarded	Grade to be Secured
First Class with Distinction	CGPA $\geq$ 8.00
First Class	$\geq$ 6.50 to $<$ 8.00 CGPA
Second Class	$\geq$ 5.50 to $<$ 6.50 CGPA
Pass Class	$\geq$ 5.00 to $<$ 5.50 CGPA
FAIL	CGPA $<$ 5

7. All other regulations as applicable for B. Tech. Four-year degree programme (Regular) will hold good for B.Tech (Lateral Entry Scheme).

MALPRACTICES RULES - DISCIPLINARY ACTION FOR /IMPROPER CONDUCT IN EXAMINATIONS

S. No	Nature of Malpractices / Improper Conduct	Punishment
1 (a)	Possesses or keeps accessible in examination hall, any paper, note book, programmable calculators, Cell phones, pager, palm computers or any other form of material concerned with or related to the subject of the examination (theory or practical) in which he is appearing but has not made use of (material shall include any marks on the body of the candidate which can be used as an aid in the subject of the examination)	Expulsion from the examination hall and cancellation of the performance in that subject only.
(b)	Gives assistance or guidance or receives it from any other candidate orally or by any other body language methods or communicates through cell phones with any candidate or persons in or outside the exam hall in respect of any matter.	Expulsion from the examination hall and cancellation of the performance in that subject only of all the candidates involved. In case of an outsider, he will be handed over to the police and a case is registered against him.
2	Has copied in the examination hall from any paper, book, programmable calculators, palm computers or any other form of material relevant to the subject of the examination (theory or practical) in which the candidate is appearing.	Expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the candidate has already appeared including practical examinations and project work and shall not be permitted to appear for the remaining examinations of the subjects of that Semester/year. The Hall Ticket of the candidate is to be cancelled and sent to the Principal.
3	Impersonates any other candidate in connection with the examination.	The candidate who has impersonated shall be expelled from examination hall. The candidate is also debarred and forfeits the seat. The performance of the original candidate who has been impersonated, shall be cancelled in all the subjects of the examination (including practical's and project work) already appeared and shall not be allowed to appear for examinations of the remaining subjects of that Semester/year. The candidate is also debarred for two consecutive Semesters from class work and all examinations. The continuation of the course by the candidate is subject to the academic regulations in connection with forfeiture of seat. If the imposter is an outsider, he will be handed over to the police and a case is registered against him.

4	Smuggles in the Answer book or additional sheet or takes out or arranges to send out the question paper during the examination or answer book or additional sheet, during or after the examination.	Expulsion from the examination hall and cancellation of performance in that subject and all the other subjects the candidate has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that Semester/year. The candidate is also debarred for two consecutive Semesters from class work and all examinations. The continuation of the course by the candidate is subject to the academic regulations in connection with forfeiture of seat.
5	Uses objectionable, abusive or offensive language in the answer paper or in letters to the examiners or writes to the examiner requesting him to award pass marks.	Cancellation of the performance in that subject.
6	Refuses to obey the orders of the Addl. Controller of examinations / any officer on duty or misbehaves or creates disturbance of any kind in and around the examination hall or organizes a walk out or instigates others to walk out, or threatens the addl. Controller of examinations or any person on duty in or outside the examination hall of any injury to his person or to any of his relations whether by words, either spoken or written or by signs or by visible representation, assaults the addl. Controller of examinations, or any person on duty in or outside the examination hall or any of his relations, or indulges in any other act of misconduct or mischief which result in damage to or destruction of property in the examination hall or any part of the College campus or engages in any other act which in the opinion of the officer on duty amounts to use of unfair means or misconduct or has the tendency to disrupt the orderly conduct of the examination.	In case of students of the college, they shall be expelled from examination halls and cancellation of their performance in that subject and all other subjects the candidate(s) has (have) already appeared and shall not be permitted to appear for the remaining examinations of the subjects of that Semester/year. The candidates also are debarred and forfeit their seats. In case of outsiders, they will be handed over to the police and a police case is registered against them.
7	Leaves the exam hall taking away answer script or intentionally tears of the script or any part thereof inside or outside the examination hall.	Expulsion from the examination hall and cancellation of performance in that subject and all the other subjects the candidate has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that Semester/year. The candidate is also debarred for two consecutive Semesters from class work and all examinations. The continuation of the course by the candidate is subject to the academic regulations in connection with forfeiture of seat.

8	Possess any lethal weapon or firearm in the examination hall.	Expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the candidate has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that Semester/year. The candidate is also debarred and forfeits the seat.
9	If student of the college, who is not a candidate for the particular examination or any person not connected with the college indulges in any malpractice or improper conduct mentioned in clause 6 to 8.	Student of the colleges expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the candidate has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that Semester/year. The candidate is also debarred and forfeits the seat. Person(s) who do not belong to the College will be handed over to police and, a police case will be registered against them.
10	Comes in a drunken condition to the examination hall.	Expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the candidate has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that Semester/year.
11	Copying detected on the basis of internal evidence, such as, during valuation or during special scrutiny.	Cancellation of the performance in that subject and all other subjects the candidate has appeared including practical examinations and project work of that Semester/year examinations.
12	If any malpractice is detected which is not covered in the above clauses 1 to 11 shall be reported to the principal for further action to award suitable punishment.	

COURSE STRUCTURE

I B.TECH.- I-SEMESTER									
Course Code	Course Title	Course Area	Hours per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		Internal (CIE)	External (SEE)	Total
A4BS02	Linear Algebra and Calculus	BSC	3	1	-	4	30	70	100
A4BS08	Applied Physics	BSC	3	1	-	4	30	70	100
A4EE01	Basic Electrical Engineering	ESC	3	1	-	4	30	70	100
A4ES02	Engineering Graphics and Design	ESC	1	-	4	3	30	70	100
A4BS10	Applied Physics Lab	BSC	-	-	3	1.5	30	70	100
A4EE02	Basic Electrical Engineering Lab	ESC	-	-	2	1	30	70	100
A4AE63	Workshop Practices	ESC	-	-	4	2	30	70	100
A4HS03	Social Innovation	HSMC	-	-	2	1	30	70	100
Total			10	3	15	20.5	240	560	800
Mandatory Course (Non-Credit)									
A4MC04	Technical Seminar - I	MC	-	-	2	-	30	70	100

I B.TECH. - II-SEMESTER									
Course Code	Course Title	Course Area	Hours per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		Internal (CIE)	External (SEE)	Total
A4BS04	Advanced Calculus	BSC	3	1	-	4	30	70	100
A4BS11	Chemistry	BSC	4	-	-	4	30	70	100
A4CS01	Programming for Problem Solving	ESC	3	-	-	3	30	70	100
A4HS01	English	HSMC	2	-	-	2	30	70	100
A4CS02	Programming for Problem Solving Lab	ESC	-	-	4	2	30	70	100
A4BS12	Chemistry Lab	BSC	-	-	3	1.5	30	70	100
A4HS02	English Language and Communication Skills Lab	HSMC	-	-	2	1	30	70	100
A4HS04	Engineering Exploration	HSMC	-	-	2	1	30	70	100
Total			12	1	11	18.5	240	560	800
Mandatory Course (Non-Credit)									
A4MC05	Technical Seminar-II	MC	-	-	2	-	30	70	100

II B.TECH.- I SEMESTER									
Course Code	Course Title	Course Area	Hours per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		Internal (CIE)	External (SEE)	Total
A4BS05	Probability and Statistics	BSC	2	1	-	3	30	70	100
A4CS03	Discrete Structures	PCC	3	1	-	4	30	70	100
A4CS04	Data Structures	PCC	3	1	-	4	30	70	100
A4CS05	Database Management Systems	PCC	3	-	-	3	30	70	100
A4EC71	Electronic Devices	ESC	3	-	-	3	30	70	100
A4CS06	Data Structures Lab	PCC	-	-	3	1.5	30	70	100
A4CS07	Database Management Systems Lab	PCC	-	-	3	1.5	30	70	100
Total			14	3	6	20	210	490	700
Mandatory Course (Non-Credit)									
A4MC01	Environmental Studies	MC	2	-	-	-	30	70	100

II B.Tech.- II SEMESTER									
Course Code	Course Title	Course Area	Hours per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		Internal (CIE)	External (SEE)	Total
A4EC72	Digital Electronics	ESC	3	-	-	3	30	70	100
A4CS08	Computer Organization and Architecture	PCC	3	1	-	4	30	70	100
A4CS09	Object Oriented Programming	PCC	3	1	-	4	30	70	100
A4CS10	Design & Analysis of Algorithms	PCC	3	1	-	4	30	70	100
A4HS06	Business Economics and Financial Analysis	HSMC	3	-	-	3	30	70	100
A4EC73	Electronic Devices and Digital Logic Lab	ESC	-	-	3	1.5	30	70	100
A4CS11	Object Oriented Programming Lab	PCC	-	-	3	1.5	30	70	100
Total			15	3	6	21	210	490	700
Mandatory Course (Non-Credit)									
A4MC02	Gender Sensitization	MC	-	-	2	-	30	70	100

III B.Tech.- I SEMESTER

Course Code	Course Title	Course Area	Hours per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		Internal (CIE)	External (SEE)	Total
A4IT04	Web Technologies	PCC	3	1	-	4	30	70	100
A4CS12	Formal Language and Automata Theory	PCC	3	-	-	3	30	70	100
A4CS13	Operating Systems	PCC	3	1	-	4	30	70	100
PEC	PE 1	PEC	3	-	-	3	30	70	100
OEC	Open Elective-I	OEC	3	-	-	3	30	70	100
A4IT05	Web Technologies Lab	PCC	-	-	3	1.5	30	70	100
A4HS07	Advanced English Communication Skills Lab	HSMC	-	-	3	1.5	30	70	100
A4CS55	Summer Internship	PWC	-	-	-	2	-	100	100
Total			15	2	6	22	210	590	800
Mandatory Course (Non-Credit)									
A4MC03	Constitution of India	MC	2	-	-	-	30	70	100

III B.Tech.- II SEMESTER

Course Code	Course Title	Course Area	Hours per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		Internal (CIE)	External (SEE)	Total
A4CS14	Compiler Design	PCC	3	1	-	4	30	70	100
A4CS15	Computer Networks	PCC	3	-	-	3	30	70	100
A4CS16	Linux Programming	PCC	3	-	-	3	30	70	100
PEC	PE 2	PEC	3	-	-	3	30	70	100
PEC	PE 3	PEC	3	-	-	3	30	70	100
OEC	Open Elective-II	OEC	3	-	-	3	30	70	100
A4CS17	Network Simulation Lab	PCC	-	-	3	1.5	30	70	100
A4CS18	Linux Programming Lab	PCC	-	-	3	1.5	30	70	100
A4CS56	Comprehensive Viva	PWC	-	-	-	1	-	100	100
Total			18	1	6	23	240	660	900
Mandatory Course (Non-Credit)									
A4MC06	Essence of Indian Traditional Knowledge	MC	2	-	-	-	30	70	100

IV B.Tech.- I SEMESTER

Course Code	Course Title	Course Area	Hours per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		Internal (CIE)	External (SEE)	Total
A4CS19	BigData Analytics	PCC	3	-	-	3	30	70	100
PEC	PE 4	PEC	3	-	-	3	30	70	100
PEC	PE 5	PEC	3	-	-	3	30	70	100
OEC	Open Elective-III	OEC	3	-	-	3	30	70	100
A4CS20	BigData Analytics Lab	PCC	-	-	4	2	30	70	100
A4CS57	Project Phase - 1	PWC	-	-	8	4	100	-	100
Total			12	-	12	18	250	350	600

IV B.Tech. - II SEMESTER

Course Code	Course Title	Course Area	Hours per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		Internal (CIE)	External (SEE)	Total
PEC	PE 6	PEC	3	-	-	3	30	70	100
A4HS05	Operations Research	HSMC	3	-	-	3	30	70	100
HSMC	Open Elective-IV (Humanities)	HSMC	3	-	-	3	30	70	100
A4CS58	Project Phase - II	PWC	-	-	16	8	50	150	200
Total			9	-	16	17	140	360	500

PROFESSIONAL ELECTIVES

PE-I		PE-II	
A4CS21	Artificial Intelligence	A4IT01	Python Programming
A4IT06	Principles of Programming Language	A4CS24	Multimedia Computing
A4CS22	Distributed Databases	A4CS25	Data Mining
A4CS23	Human Computer Interaction	A4CS26	Software Engineering
PE-III		PE-IV	
A4CS27	Internet of Things	A4CS31	Software Testing
A4CS28	Cloud Computing	A4CS32	Cryptography and Network Security
A4CS29	Natural Language Processing	A4CS33	Web Services and Service Oriented Architecture
A4CS30	Mobile Application Development	A4CS34	Ad-hoc and Sensor Networks
PE-V		PE-VI	
A4CS35	Distributed Systems	A4CS39	Software Project Management
A4CS36	Soft Computing	A4CS40	Machine Learning
A4CS37	Agile Software Development	A4CS41	Information Retrieval System
A4CS38	E-Commerce	A4CS42	Predictive Analytics

OPEN ELECTIVE COURSES

OPEN ELECTIVE COURSE-I			
S. No.	Course Code	Course Name	Offering Department
1	A4AE17	Fabrication Processes	Aeronautical Engineering
2	A4AE18	Fundamentals of Avionics	
3	A4AE19	Introduction to jets and rockets	
4	A4CS43	Fundamentals of Databases	Computer Science and Engineering
5	A4CS44	Introduction to C++	
6	A4CS45	Core Java Programming	
7	A4EC59	Logic Design	Electronics & Communication Engineering
8	A4EC60	Principles of Communications	
9	A4EC61	Measurements and Instruments	
10	A4EE52	Electrical Wiring and Safety Measures	Electrical & Electronics Engineering
11	A4EE53	Electrical Materials	
12	A4EE54	New trends Electrical Energy	
13	A4IT11	Fundamentals of Data Structures	Information Technology
14	A4IT12	Software Engineering Principles	
15	A4IT13	Computer Organization and Operating Systems	
16	A4ME60	Elements Of Mechanical Engineering	Mechanical Engineering
17	A4ME61	Fundamentals Of Engineering Materials	
18	A4ME62	Fundamentals Of Theory Of Machines	
19	A4HS10	Business Economics and Financial Analysis	HS
20	A4HS11	Basics of Entrepreneurship	
21	A4HS12	Human Values and Professional Ethics	

OPEN ELECTIVE COURSE-II			
S. No.	Course Code	Course Name	Offering Department
1	A4AE25	Introduction to aircraft Industry	Aeronautical Engineering
2	A4AE26	Non destructive testing Methods	
3	A4AE27	Fundamentals of Finite element method	
4	A4CS46	Introduction to Computer Networks	Computer Science and Engineering
5	A4CS47	Introduction to Linux Programming	
6	A4CS48	Fundamentals of Artificial Intelligence	
7	A4EC62	Fundamentals of Integrated Circuits	Electronics & Communication Engineering
8	A4EC63	Signal Transmission through linear systems	
9	A4EC64	Fundamentals of VLSI Design	
10	A4EE55	Power Plant Engineering	Electrical & Electronics Engineering
11	A4EE56	Analysis of Linear Systems	
12	A4EE57	Neural Networks and Fuzzy Logic	
13	A4IT14	Basics of Python Programming	Information Technology
14	A4IT15	Human Computer Interaction	
15	A4IT16	Software Testing Fundamentals	
16	A4ME63	Fundamentals of Mechatronics	Mechanical Engineering
17	A4ME64	Basics Of Thermodynamics	
18	A4ME65	Fundamentals of Smart Materials	
19	A4HS13	Advanced Entrepreneurship	HS

OPEN ELECTIVE COURSE-III			
S. No.	Course Code	Course Name	Offering Department
1	A4AE40	Guidance and control of aerospace vehicles	Aeronautical Engineering
2	A4AE41	Wind tunnel Techniques	
3	A4AE42	Introduction to Aerospace Technology	
4	A4CS49	Soft Computing	Computer Science and Engineering
5	A4CS50	Introduction to Cloud Computing	
6	A4CS51	Web Programming	
7	A4EC65	Introduction of Microprocessors and Microcontrollers	Electronics & Communication Engineering
8	A4EC66	Fundamentals of Image processing	
9	A4EC67	TV Engineering	
10	A4EE58	Electrical Engineering Materials	Electrical & Electronics Engineering
11	A4EE59	Non-Conventional Power Generation	
12	A4EE60	Solar Energy and Applications	
13	A4IT17	E- Commerce	Information Technology
14	A4IT18	Cyber Forensics	

15	A4IT19	Discrete Structures	
16	A4ME66	Fundamentals of Operation Research	Mechanical Engineering
17	A4ME67	Fundamentals of Energy Sources and Applications	
18	A4ME68	Basics of Robotics	
19	A4HS14	Indian Ethos & Business Ethics	HS

OPEN ELECTIVE-IV			
S. No.	Course Code	Course Name	Offering Department
1	A4HS15	Management Science	HS
2	A4HS16	Intellectual Property Rights	
3	A4HS17	Number Theory	
4	A4HS18	Physics and Technology of Thin films	
5	A4HS19	Polymer chemistry	
7	A4CS52	Introduction to Network Security	Computer Science and Engineering
8	A4CS53	Data Analytics	
9	A4CS54	.NET & C# Programming	
10	A4EC68	Introduction to signal Processing	Electronics & Communication Engineering
11	A4EC69	Introduction to mobile communication	
12	A4EC70	Basic Embedded systems Design	
13	A4EE61	Instrumentation and Control	Electrical & Electronics Engineering
14	A4EE62	Energy Audit and Management Systems	
15	A4EE63	Energy Storage Systems	
16	A4IT20	Introduction to Mobile Application Development	Information Technology
17	A4IT21	Distributed Databases	
18	A4IT22	Big Data	
19	A4ME69	Introduction to Material Handling	Mechanical Engineering
20	A4ME70	Non Conventional Energy Sources	
21	A4ME71	Aspects of Heat Transfer	

I B.TECH I SEMESTER SYLLABUS

LINEAR ALGEBRA AND CALCULUS

I B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4BS02	BSC	L	T	P	C	CIE	SEE	Total
		3	1	-	4	30	70	100

COURSE OBJECTIVES:

To learn

1. Usage of the appropriate test to find the convergence and divergence of the given series.
2. The concept of differential equations and solve them using appropriate methods.
3. Concept of Rank of a matrix, Consistency and solving system of linear equations.
4. The Rank and Nullity of vectors.
5. Concept of eigen values, eigen vectors and diagonalization of the matrix.

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Apply the appropriate test to find the convergence and divergence of the given series.
2. Identify the different types of differential equations and solve them using appropriate methods.
3. Solve the system of linear equations using rank of a matrix.
4. Find Rank and Nullity of given vectors.
5. Diagonalize the matrix using eigen values and eigen vectors.

UNIT-I SEQUENCES AND SERIES

Basic definitions of Sequences and series – Convergence and divergence –Comparison Test- Ratio Test –Raabe's Test - Cauchy's n^{th} root Test –Integral Test – Absolute and Conditional convergence – Power Series.

UNIT-II ORDINARY DIFFERENTIAL EQUATIONS

Introduction- Exact and reducible to Exact differential equations -Equations not of first degree: Equations solvable for p-equations, solvable for y- equations and solvable for x and Clairaut's type. Linear differential equations of second and higher order with constant coefficients - Non-Homogeneous term of the type $Q(x) = e^{ax}$, $\sin ax$, $\cos ax$, $e^{ax}v(x)$, $x^n v(x)$ - Method of variation of parameters.

UNIT-III THEORY OF MATRICES

Real matrices: Symmetric-skew-symmetric and orthogonal matrices –Complex matrices: Hermitian, Skew –Hermitian and Unitary matrices –Elementary row and column transformations –Elementary matrix- Finding rank of a matrix by reducing to Echelon form and Normal form-Finding the inverse of a matrix using elementary row/column transformations (Gauss-Jordan method)-Consistency of system of linear equations (homogeneous and non-homogeneous) using the rank of a matrix –Solving $m \times n$ and $n \times n$ linear system of equations by Gauss Elimination.

UNIT-IV VECTOR SPACES

The n-dimensional Vectors –Vector space – linear dependence of vectors –Basis and dimensions –linear transformations-range and kernel of a linear map – rank and nullity - rank and nullity theorem – inverse of a linear transformation-composition of linear map- Matrix associated with a linear map.

UNIT-V EIGEN VALUES, EIGEN VECTORS AND INNER PRODUCT SPACES

Eigen values and Eigen vectors of a matrix- Eigenbases - Diagonalization- Inner product space – Norm of a vector – Schwarz's Inequality – Normed vector space – Orthogonal and orthonormal sets – Gram Schmidt orthogonalization process.

TEXT BOOKS:

1. Ervin Kreyszig, Advanced Engineering Mathematics, 9th Edition, John Wiley & Sons, 2006.
2. B.S.Grewal, Higher Engineering Mathematics, Khanna publishers, 36th Edition, 2010.
3. V. Krishnamurthy, V.P. Mainra and J.L. Arora, An introduction to Linear Algebra, Affiliated East – West press, Reprint 2005.

REFERENCE BOOKS:

1. G.B.Thomas, calculus and analytical geometry,9th Edition, Pearson Reprint 2006.
2. N.P Bali and Manish Goyal ,A Text of Engineering Mathematics,Laxmi publications,2008.
3. E.L.Ince, Ordinary differential Equations,Dover publications,1958.

WEB REFERENCES:

1. https://www.efunda.com/math/math_home/math.cfm
2. <https://www.ocw.mit.edu/resources/#Mathematics>
3. <https://www.sosmath.com/>
4. <https://www.mathworld.wolfram.com/>

E -TEXT BOOKS:

1. <https://www.e-booksdirectory.com/details.php?ebook=10166>
2. <https://www.e-booksdirectory.com/details.php?ebook=10166>

MOOCS Course:

1. <https://swayam.gov.in/>
2. <https://onlinecourses.nptel.ac.in/>

APPLIED PHYSICS

I B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4BS08	BSC	L	T	P	C	CIE	SEE	Total
		3	1	0	4	30	70	100

COURSE OBJECTIVES:

The course should enable the students to:

1. Learn the behaviour of matter waves and applications of Schrodinger wave equations in periodic potential energy of electron.
2. Understand the formation of energy bands in solids.
3. Gain the knowledge of carrier concentration and recombination process of semiconductor materials.
4. Learn the basic principles of laser and optical fibre.
5. Understand the development of nano technology and synthesis of nano materials by using different techniques.

COURSE OUTCOMES:

The student will be able to:

1. Conclude the dual nature of material particles and able to explain how moving particles are associated with its energies
2. Analyse the energy bands in solids and accordingly classify the materials
3. Evaluate the mobility of charge carrier concentration of a given semiconductor material.
4. Justify how the graded index optical fiber is more efficient than step index optical fiber in fiber optic communication system.
5. Recommend appropriate synthesis method and explain the characterization techniques.

UNIT-I

QUANTUM MECHANICS

Introduction to quantum physics: Black body radiation, Planck's law, photoelectric effect, Compton effect, Waves and Particles, de Broglie Hypothesis, Matter Waves, Davisson and Germer's Experiment, Heisenberg's Uncertainty Principle, Schrodinger's Time Independent Wave Equation-Physical Significance of the wave Function, Particle in One Dimensional Potential Box.

UNIT-II

INTRODUCTION TO ELECTRONIC MATERIALS

Band theory - Free electron theory, Origin of Energy Band formation in Solids, Estimation of Fermi energy level, Density of states and energy band diagrams, Kronig-Penny model (to introduce origin of band gap), Energy bands in solids, E-k diagram, Direct and indirect bandgaps.

Classification of Materials: Conductors, Semiconductors & Insulators, Effective mass of an Electron. Fermi-Dirac Statistics (Qualitative treatment).

UNIT-III

SEMICONDUCTORS AND OPTOELECTRONICS

Semiconductors: Intrinsic and extrinsic semiconductors, Dependence of Fermi level on carrier-concentration and temperature (equilibrium carrier statistics). Carrier generation and recombination, Carrier transport: diffusion and drift, Hall effect and its applications.

Semiconductors design- PN junction-diode, Zener diode, fabrication and characterization techniques- Heterojunctions and associated band-diagrams.

Optoelectronic devices: properties of photo detectors, solar cells, Semiconductor laser, Four-point probe measurement for carrier density.

UNIT-IV

LASER & OPTICAL FIBER

Laser: Characteristics of Laser beams, Energy levels in atoms, Einstein's theory of matter radiation interaction and A and B coefficients; amplification of light by population inversion, different types of lasers: gas lasers -He-Ne, solid-state lasers -ruby, Nd-YAG, Applications of Lasers.

Fiber Optic Communication: Structure of Optical fibers, Basic principle of fibers, Acceptance angle and

Numerical aperture, Types of Optical Fibers-Step Index and Graded Index fibers; Modes of fibers-SMSI, MMSI, MMGI., Optical fiber Communication System with block diagram. Applications of fibers, fiber optic sensors – Basic principle, Intrinsic, Extrinsic sensors. Working of Pressure and Temperature Sensors.		
UNIT-V	INTRODUCTION TO ENGINEERED MATERIALS	
Fundamentals of nano particles, nano scale, properties, Techniques for synthesis of nano materials -Sol-gel, Chemical vapour deposition (CVD) methods. Characterization of nanomaterials: Imaging methods- SEM, TEM, Scanning Probe Microscopy: STM. Fabrication method- quantum wire, Applications of Nano materials in engineering and Biomedical fields.		
TEXT BOOKS:		
1. Engineering Physics, B.K. Pandey, S. Chaturvedi – Cengage Learning 2. Haliday and Resnick, Physics – wiley 3. R. Robinett, “Quantum Mechanics”, OUP Oxford, 2006.IIndEdn. 4. P.K Palanisamy, Engineering Physics, Sitech Publications, 2013, IVthEdn. 5. Essentials of Nano Tecnology by Jeremy Ramsden.		
REFERENCE BOOKS:		
1. E. Hecht, “Optics”, Pearson Education, 2008. 2. J. Singh, Semiconductor Optoelectronics: Physics and Technology, Mc Graw-Hill inc.(1995) 3. R. Robinett, “Quantum Mechanics”, OUP Oxford, 2006. 4. O. 1Svelto, “Principles of Lasers”, Springer Science & Business Media, 2010. 5. D. A. Neamen, “Semiconductor Physics and Devices”, Times Mirror High EducationGroup, Chicago,1997.		
WEB REFERENCES:		
1. https://www.edx.org/course?search_query=semiconductor+physics 2. https://www.edx.org/course/nanotechnology-fundamentals-purdue-nano530x 3. https://www.edx.org/course/physics-electronic-polymers-pep-purdue-nano600		
E-TEXT BOOKS:		
1. http://www.phys.sinica.edu.tw/TIGP-NANO/Course/2010_Fall/classnotes/NanoB_week14.pdf 2. https://www.scribd.com/document/70908178/Semiconductor-Devices-Basic-Principles-Jasprit-Singh 3. https://www.scribd.com/doc/105174065/Fundamentals-of-Photonics 4. ftp://nozdr.ru/biblio/kolxo3/P/PE/PEo/Thyagarajan%20K.,%20Ghatak%20A.%20Lasers..%20Fundamentals%20and%20Applications%20(2ed.,%20GTP,%20Springer,%202010)(ISBN%20144196441X)(O)(674s)_PEo_.pdf 5. https://subodhtripathi.files.wordpress.com/2012/01/optical-fiber-communications-by-gerd-keiser_2.pdf 6. http://www.hailienene.com/resources/nano-technology.pdf		
MOOC COURSE:		
1. http://nptel.ac.in/courses/115103030/(Four-point probe measurement for carrier density) 2. http://nptel.ac.in/courses/115102025/ (Fundamental concepts of semiconductors) 3. http://nptel.ac.in/courses/118104008/1 (Fundamentals of Nano technology) 4. http://nptel.ac.in/courses/118104008/13 (Nano structures, synthesis and characterization) 5. http://nptel.ac.in/courses/104104085/2(Lasers and its applications)		

BASIC ELECTRICAL ENGINEERING

I B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EE01	ESC	L	T	P	C	CIE	SEE	Total
		3	1	-	4	30	70	100

COURSE OBJECTIVES

The course should enable the students

1. To understand and analyze basic electric and magnetic circuits
2. To study the working principles of electrical machines
3. To introduce the components of low voltage electrical installations.

COURSE OUTCOMES: The students will be able to

1. To analyze and solve electrical circuits using network laws and theorems.
2. To analyze basic Electric and Magnetic circuits.
3. To study the working principles of Electrical Machines.
4. To introduce components of Low Voltage Electrical Installations

UNIT-I DC CIRCUITS

Electrical circuit elements (R, L and C), voltage and current sources, Kirchhoff current and voltage laws, analysis of simple circuits with dc excitation. Superposition, Thevenin and Norton Theorems. Time-domain analysis of first-order RL and RC circuits.

UNIT-II AC CIRCUITS

Representation of sinusoidal waveforms, peak and rms values, phasor representation, real power, reactive power, apparent power, power factor. Analysis of single-phase ac circuits consisting of R, L, C, RL, RC, RLC combinations (series and parallel), resonance. Three-phase balanced circuits, voltage and current relations in star and delta connections.

UNIT-III TRANSFORMERS & ELECTRICAL INSTALLATIONS

Magnetic materials, BH characteristics, ideal and practical transformer, equivalent circuit, losses in transformers, regulation and efficiency. Auto-transformer and three-phase transformer connections. **Components of LT Switchgear:** Switch Fuse Unit (SFU), MCB, ELCB, MCCB, Types of Wires and Cables, Earthing. Types of Batteries, Important Characteristics for Batteries

UNIT-IV DC MACHINES

Principle and operation of DC Motor, Construction of DC machine. Types of DC motor, losses and Torque equation. DC generator construction, working principle and its EMF equation. Types of dc generators, efficiency of dc generator, problems on Emf equation. Applications.

UNIT-V AC MACHINES

Generation of rotating magnetic fields, Construction and working of a three-phase induction motor, Significance of torque-slip characteristic. Loss components and efficiency, starting and speed control of induction motor. Construction, working of Single-phase induction motor. Construction and working of synchronous generators, Emf equation and problems.

TEXT BOOKS:

1. D. P. Kothari and I. J. Nagrath, "Basic Electrical Engineering", Tata McGraw Hill, 2010.
2. D. C. Kulshreshtha, "Basic Electrical Engineering", McGraw Hill, 2009.
3. Basic Electrical Engineering by M.S. Naidu and S. Kamakshiah TMH
4. Mehta V K, —Principles of Electrical Engineering II, S. Chand & Company.

REFERENCE BOOKS:

1. L. S. Bobrow, "Fundamentals of Electrical Engineering", Oxford University Press, 2011.
2. E. Hughes, "Electrical and Electronics Technology", Pearson, 2010.
3. V. D. Toro, "Electrical Engineering Fundamentals", Prentice Hall India, 1989.
4. Nagsarkar T K and Sukhija M S, —Basics of Electrical Engineering II, Oxford press. Basic concepts of Electrical Engineering, P.S. Subramanyam, BS Publications.

ENGINEERING GRAPHICS & DESIGN								
I B. TECH- I SEMESTER								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ES02	ESC	L	T	P	C	CIE	SEE	Total
		1	0	4	3	30	70	100
COURSE OBJECTIVES:								
1. Create awareness and emphasize the need for Engineering Drawing in various branches of engineering. 2. Enable the student with various concepts of dimensioning, conventions and standards related to engineering drawings. 3. Follow the basic drawing standards and conventions. 4. Develop skills in three-dimensional visualization of engineering component.								
UNIT-I	INTRODUCTION							
Introduction to Engineering Drawing covering: Principles of Engineering Graphics and their significance, usage of drawing instruments, lettering, Conic sections including the Rectangular Hyperbola (General method only); Cycloid, Epicycloids, Hypocycloid and Involute.								
UNIT-II	DRAWING OF PROJECTIONS OR VIEWS: ORTHOGRAPHIC PROJECTION IN FIRST ANGLE PROJECTION ONLY							
Principles of orthographic projections – conventions – first and third angle projections. Projections of points-Projection of lines inclined to both the planes. PROJECTIONS OF PLANES: Projections of regular planes, inclined to both planes.								
UNIT-III	INTRODUCTION TO COMPUTER AIDED DRAFTING							
INTRODUCTION TO COMPUTER AIDED DRAFTING: Generation of points, lines, curves, polygons, simple solids, dimensioning. LAYERS: Concept of layers, working with layers, creating, display, locking, unlocking, and delete commands, Coordinate system in AutoCAD, UCS, WCS, MCS PROJECTION OF SOLIDS-Solids inclined to both planes(Auxiliary plane method) DEVELOPMENT OF SURFACES OF SOLIDS: Theory of development ,development of lateral surface along with base.								
UNIT-IV	ISOMETRIC DRAWINGS							
Divisions of pictorial projection, theory of Isometric Drawing- Isometric view and Isometric projections; Drawing Isometric circles, Dimensioning Isometric Objects; Conversion of Isometric view to Orthographic views and Orthographic to isometric views.								
UNIT-V	3D MODELING							
Types of 3D models, 3D Coordinate Systems, basic commands in 3D, PEDIT command. CREATING SOLID MODELS: creating pre-defined Solid Primitives, Dynamic UCS, methods of creating solids by - Extrude, Revolve, Swept, Loft, & Presspull, in 3Dcreating solid models, Dynamic UCS. MODIFYING 3D OBJECTS: Fillet, Chamfer, Rotate, Mirror, Array, Slicing solid Models. EDITING 3D OBJECTS: SOLVIEW, SOLDRAW, SOLPROF.								
TEXT BOOKS:								
1. D. P. Kothari and I. J. Nagrath, “Basic Electrical Engineering”, Tata McGraw Hill, 2010. 2. D. C. Kulshreshtha, “Basic Electrical Engineering”, McGraw Hill, 2009. 3. Basic Electrical Engineering by M.S. Naidu and S. Kamakshaiah TMH 4. Mehta V K, —Principles of Electrical Engineeringll, S. Chand & Company								

REFERENCE BOOKS:

1. L. S. Bobrow, "Fundamentals of Electrical Engineering", Oxford University Press, 2011.
2. E. Hughes, "Electrical and Electronics Technology", Pearson, 2010.
3. V. D. Toro, "Electrical Engineering Fundamentals", Prentice Hall India, 1989.
4. Nagsarkar T K and Sukhija M S, —Basics of Electrical EngineeringII, Oxford press. Basic concepts of Electrical Engineering, P.S. Subramanyam, BS Publications.

APPLIED PHYSICS LABORATORY

I B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4BS10	BSC	L	T	P	C	CIE	SEE	Total
		-	-	3	1.5	30	70	100

COURSE OBJECTIVES:

The course should enable the students to:

1. To provide an experimental foundation for the theoretical concepts introduced in the lectures.
2. To teach how to make careful experimental observations and how to think about and draw conclusions from such data.
3. To help students understand the role of direct observation in physics and to distinguish between inferences based on theory and the outcomes of experiments.
4. To introduce the concepts and techniques which have a wide application in experimental science but have not been introduced in the standard courses.
5. To teach how to write a technical report which communicates scientific information in a clear and concise manner.

COURSE OUTCOMES:

The student will be able to:

1. To make careful experimental observations and draw conclusions from such data.
2. To distinguish between inferences based on theory and the outcomes of experiments.
3. To write a technical report which communicates scientific information in a clear and concise manner.

LIST OF EXPERIMENTS

WEEK-1 | LIGHT EMITTING DIODE (LED)

1. Analyze the V-I characteristics of GREEN LED Source by varying input voltage from zero to two volts. Analyze the graph obtained.
2. Analyze the results obtained for V-I characteristics of RED LED Source by varying input voltage with interval 0.2 volts.
3. Analyze the V-I characteristics of YELLOW LED Source for the resistance of 100Ω. What is the effect of doubling the resistance on the V-I characteristics.

WEEK-2 | SOLAR CELL

1. Study the V-I characteristics of a solar cell for the voltage interval of 0.4 V when the given electric bulb is at a distance of 10cm. Will there be any change if the distance is doubled?
2. Study the P-V characteristics of a solar cell for the voltage interval of 0.3 V for four different distances of the given electric bulb. Discuss your observations.
3. Study the P-I characteristics of a solar cell for the given electric bulb for the distances 30 cm. What is expected if the distance is doubled?

WEEK-3 | OPTICAL FIBER

To determine the numerical aperture and acceptance angle of an optical fiber.

WEEK-4 | HALL EFFECT

1. Investigate the deflection of the carriers in the conductor under the function of the magnetic field. Based on this measurement, calculate the density of the carriers and the sign of the charges in the conductor.
2. Verify the Hall effect in extrinsic semiconducting samples and determine the type of given semiconductor and density of majority charge carriers.

WEEK-5 | ENERGY GAP OF PN JUNCTION DIODE

1. Using PN junction diode, determine the energy gap of a semiconductor by applying 2V of reverse bias by increasing the temperature from 30-70 degree centigrade. Analyze the obtained results.
2. Evaluate the energy gap of Si PN junction diode by applying 1.5V of reverse bias for values of temperature 70, 60, 50, 40, 30 centigrade.
3. Evaluate the energy gap of a semiconductor by applying 0.5V of reverse bias for temperatures at 75, 65, 55, 45, 35 centigrade.

WEEK-6	THERMISTER
1. Study the variation of Resistance versus Temperature using Thermister. Plot graph between Resistance vs Temperature and analyze the results. 2. Evaluate the Temperature dependent resistance of a given material by using Thermister.	
WEEK-7	LCR CIRCUIT
1. To determine the Resonance frequency and Quality factor of a LCR Circuit 2. Study the Resonance frequency of an electrical circuit by varying values of inductance, Capacitance and resistors and analyze the results.	
WEEK-8	PIN PHOTO DIODE
To determine the V-I characteristics of PIN photo diode.	
WEEK-9	RC CIRCUIT
1. Analyze the time constant of R-C circuit by varying Resistance and Capacitance values in a electrical circuit. 2. Determine the time constant of a given RC circuit by plotting a graph between Charging current versus time.	
WEEK-10	TORSIONAL OSCILLATOR
1. Making use of a torsional oscillator of 300gms circular disc, determine the rigidity modulus of given steel wire for 65 cm & 55 cm lengths. Analyze the results obtained. 2. Determine the rigidity modulus of the given copper wire for 50, 40, 30 lengths by using torsional oscillator of 400gms circular disc. Analyze the results obtained. 3. Give your analysis of $L&T^2$ behavior of a torsional oscillator. You may pick your own values for the analysis.	
WEEK-11	LASER - DIFFRACTION GRATING
1. Using a diffraction grating element of 2500 LPI determine the wavelength of LASER source for first and second order diffraction when the distance between the screen and grating is 50cm. What is your analysis? 2. Determine the wavelength of a LASER source for first three orders of diffraction by maintaining a Distance of 30 cms between grating material and the screen. Use diffraction grating element of 15000 LPI. What differences do you observe for the three orders.	
WEEK-12	MELDE'S EXPERIMENT
1. Determine the longitudinal frequency of tuning fork by using 100cm length of the thread by varying masses of 5gms, 10gms. Discuss your findings. 2. Find out the transverse frequency of tuning fork by using 80cm length of the thread and by varying masses of 10gms and 40gms. Analyze the results obtained.	
REFERENCE BOOKS:	
1. "Semiconductor Physics and Devices: Basic Principles" by Donald A Neamen. 2. "Optics, Principles and Applications" by K K Sharma. 3. "Principles of Optics" by M Born and E Wolf. 4. "Oscillations and Waves" by Satya Prakash and Vinay Dua. 5. "Waves and Oscillations" by N Subrahmanyam and Brij Lal.	
WEB REFERENCES:	
1. http://www.arxiv.org/pdf/1510.00032 2. http://www.nptel.ac.in/courses/122103010/ 3. http://www.researchgate.net/.../276417736_Video_Presentations_in_Engineering-Ph... 4. http://www.wileyindia.com/engineering-physics-theory-and-practical.html	

Note: Students can perform any 8 experiments

BASIC ELECTRICAL ENGINEERING LAB

I B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EE02	ESC	L	T	P	C	CIE	SEE	Total
		-	-	2	1	30	70	75

COURSE OBJECTIVES:

The course should enable the students to:

1. Get an exposure to common electrical components and their ratings.
2. Make electrical connections by wires of appropriate ratings.
3. Understand the usage of common electrical measuring instruments.
4. Understand the basic characteristics of transformers and electrical machines.

COURSE OUTCOMES:

1. Get an exposure to basic electrical laws.
2. Demonstrate different types of electrical circuits to different excitations.
3. Calculate the relation between the basic electrical parameters.
4. Analyze the basic characteristics of transformers and electrical machines.

LIST OF EXPERIMENTS

WEEK-1 INTRODUCTION AND USE OF MEASURING INSTRUMENTS & SAFETY PRECAUTIONS

To study the usage of electrical instruments and the required precautions to be taken.

WEEK-2 KIRCHOFF'S LAWS(KVL & KCL)

To verify kvl and kcl

WEEK-3 SUPERPOSITION THEOREM

To verify superposition theorem

WEEK-4 THEVENIN'S AND NORTON'S THEOREM

To obtain equivalent circuit of a complex network

WEEK-5 STEADY STATE AND TRANSIENT RESPONSE OF R-L, R-C & R-L-C CIRCUITS

To find the steady and transient response of r-l, r-c & r-l-c circuits for step input.

WEEK-6 OPEN CIRCUIT, SHORT CIRCUIT & LOAD TEST ON SINGLE PHASE TRANSFORMER

To calculate the efficiency of single phase transformer.

WEEK-7 CUT OUT VIEW OF DC MACHINE

Demonstration on constructional and cut out view of dc machine

WEEK-8 CUT OUT VIEW OF INDUCTION MOTOR

Demonstration on constructional and cut out view of single phase induction motor

WEEK-9 MAGNETIZATION CHARACTERISTICS OF DC SHUNT GENERATOR

To find the magnetization characteristics of dc shunt generator

WEEK-10 BRAKE TEST ON DC SHUNT MOTOR

To find the torque-speed characteristics of dc shunt motor.

WEEK-11 THREE PHASE TRANSFORMER CONNECTIONS

To calculate the relation between phase and line voltages

WEEK-12 BRAKE TEST ON 3-PHASE INDUCTION MOTOR

To find the torque-slip characteristics of induction motor

REFERENCE BOOKS:

- | |
|--|
| <ol style="list-style-type: none">1. Department Lab Manual2. Chakrabarthi, " Circuit Theory", Dhanpat Rai Publications, 6th Edition,20063. V K Mehta, Rohit Mehta, "Principles of Electrical Machines", S Chand Publications, 1st Edition,20064. Nagrath & DP Kothari, "Electrical Machines", Mcgraw Hill Education Publications, 4th Edition, 2010. |
|--|

WEB REFERENCES:

- | |
|---|
| <ol style="list-style-type: none">1. http://www.ee.iitkgp.ac.in2. http://www.citchennai.edu.in |
|---|

WORKSHOP PRACTICES

I B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE63	ESC	L	T	P	C	CIE	SEE	Total
		-	-	4	2	30	70	100

COURSE OBJECTIVES:

Student will

1. Get the hands-on experience on various trades.
2. Perform various machining operations.
3. Capable to make useful products using one or more operations

COURSE OUTCOMES:

Student will be able to:

1. Fabricate components with their own hands
2. Get practical knowledge of the dimensional accuracies and tolerances possible with different manufacturing processes
3. Assemble different components
4. Produce small devices of their interest

LIST OF EXPERIMENTS

WEEKS	BASIC TRADES	BASIC MANUFACTURING
	Fitting	Machine Shop
WEEK 1	Filing Four Sides of Work piece	Facing & Step Turning on Lathe
WEEK 2	L- Fit	Milling and Drilling
	Carpentry	Black Smithy
WEEK 3	Half Lap Joint	Convert round rod to S-hook
WEEK 4	Dove Tail Joint	Convert round rod to Chisel
	Tin Smithy	Casting
WEEK 5	Tin Smithy- Prepare a Rectangular Tray	Preparation of Mould Cavity for Multi Piece Pattern
WEEK 6	Prepare A Square Tin	Casting of Simple pattern
	Electrical	Welding Shop
WEEK 7	House Wiring Parallel and Series Connection	Lap/Butt joint Using Arc Welding
WEEK 8	House Wiring Two Way Switch	Lap/Butt joint Using Gas Welding
	Electronics	Plastic Moulding & Glass Cutting
WEEK 9	Soldering Parallel Connection	Injection moulding of Simple Components
WEEK 10	Soldering Series Connection	Glass Cutting
WEEK 11	Revision/Practice	Revision/Practice

TEXT BOOKS:

1. Workshop Manual by P. Kannaiah and K. L. Narayana.
2. Rao P.N., "Manufacturing Technology", Tata McGraw Hill House, Vol. I and Vol. II.

REFERENCE BOOKS:

1. Hajra Choudhury S.K., Hajra Choudhury A.K. and Nirjhar Roy S.K., "Elements of Workshop Technology", Media promoters and publishers private limited, Mumbai, Vol. I 2008 and Vol. II 2010.
2. H. S. Bawa, "Workshop Practice", Tata McGraw-Hill Publishing Company Limited, New Delhi, (2007).
3. Kalpakjian S. and Steven S. Schmid, "Manufacturing Engineering and Technology", Pearson Education India Edition, 4th edition, 2002.

E-TEXT BOOKS:

1. <https://blogpuneet.files.wordpress.com/2013/07/introduction-to-basic-manufacturing-processes-and-workshop-technology.pdf>
2. <https://soaneemrana.org/onewebmedia/Manufacturing%20Processes%20By%20H.N.%20Gupta.pdf>

MOOC COURSE:

1. <https://www.class-central.com/course/edx-fundamentals-of-manufacturing-processes-7224>

SOCIAL INNOVATION

I B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4HS03	HSMC	L	T	P	C	CIE	SEE	Total
		-	-	2	1	30	70	100

COURSE OBJECTIVES:

The course should enable the students to:

1. Understand social innovation concepts and approaches.
2. Understand the community problems, social and economical change.
3. Identify new and unaddressed social needs.
4. Analysis of social innovation disclosures in different sectors.
5. Design innovative solutions with Social impact through application of new models of leadership, collective intelligence and creativity techniques.

UNIT-I INTRODUCTION TO SOCIAL INNOVATION

Core definitions, core elements and common features of social innovation, a topology of social innovations, history of social innovation, social and economic change, Swachh Bharat, Unnat Bharat Abhiyan, National Service Scheme (NSS).

UNIT-II INTERACTION AND ENGAGEMENT WITH SOCIETY

Engage with community, interact with them to understand the community problems, Understanding social and economical change – individuals, organizations and movements.

UNIT-III PROCESS OF SOCIAL INNOVATION

Understanding the pain/need, description and problem definition, social and economic constraints for affordable and appropriate technology.

UNIT-IV SOCIAL INNOVATION ACROSS FOUR SECTORS IN INDIA AND GLOBAL SCENARIO

The four sectors – the non-profit sector, public sector, the private sector, the informal sector, links between and cross sectors.

UNIT-V SOCIAL INNOVATION – CASE STUDIES

Designing and implementing social innovations, report writing and documentation, presentation of the case studies with a focus on impact and vision on society.

TEXT BOOKS:

1. The Power of Social Innovation: How Civic Entrepreneurs Ignite Community Networks for Good 1st Edition by Stephen Goldsmith, Michael R. Bloomberg, Gigi Georges, Tim Glynn Burke.
2. The Open Book of Social Innovation: Ways to Design, Develop and Grow Social Innovation Paperback – March, 2010 by Robin Murray, Julia Caulier-Grice, Geoff Mulgan.

REFERENCE BOOKS:

1. Social innovator series: ways to design, develop and grow social innovation, the open book of social innovation by robin murray julie caulier-grice geoff mulgan.
2. The International Handbook on Social Innovation: Collective Action, Social Learning and Transdisciplinary Research Paperback by Frank Moulaert, Diana MacCallum.
3. Guide to Social Innovation by Johannes HAHN and Laszlo ANDOR

WEB REFERENCES:

1. http://s3platform.jrc.ec.europa.eu/documents/20182/84453/Guide_to_Social_Innovation.pdf
2. <https://www.si-drive.eu/wp-content/uploads/2016/12/SI-DRIVE-CA-short-2016-11-30-Druckversion.pdf>

E-TEXT BOOKS:

1. <https://epdf.tips/the-power-of-social-innovation-how-civic-entrepreneurs-ignite-community-networks.html>
2. <https://youngfoundation.org/wp-content/uploads/2012/10/The-Open-Book-of-Social-Innovation.pdf>
3. <http://www.idmais.org/desislabs/wp-content/media/social.pdf>

MOOC COURSE:

1. <https://iversity.org/en/courses/social-innovation-101-en>
2. <https://www.class-central.com/tag/social%20innovation>
3. <https://www.mooc-list.com/tags/social-innovation>

I B.TECH II SEMESTER SYLLABUS

ADVANCED CALCULUS

II Semester: CSE/IT

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4BS04	BSC	L	T	P	C	CIA	SEE	Total
		3	1	-	4	30	70	100
Contact Classes: 44	Tutorial Classes: 08	Practical Classes: Nil			Total Classes: 52			

COURSE OBJECTIVES:

To learn

1. Evaluation of length, areas & volumes of different curves of revolution.
2. The partial derivatives of several variable functions.
3. Concept and application of Laplace transforms.
4. Fourier series for periodic functions.
5. Numerical techniques.

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Evaluate the improper integrals using beta and gamma functions.
2. Find the Maxima and Minima of several variable functions.
3. Solve the differential equations using Laplace transform techniques.
4. Find the Fourier series of the periodic functions.
5. Apply various numerical techniques to solve differential equations.

UNIT-I	BETA GAMMA FUNCTIONS AND APPLICATIONS OF DEFINITE INTEGRALS	
Beta- Gamma Functions and their Properties-Relation between them- Evaluation of improper integrals using Gamma and Beta functions. Application of definite integrals: Lengths, evaluate surface areas and volumes of revolution of curves (only in Cartesian co-ordinates).		
UNIT-II	CALCULUS OF SEVERAL VARIABLES	
Limit, Continuity - Partial derivative- Partial derivatives of higher order -Total derivative - Chain rule, Jacobians -functional dependence & independence. Applications: Maxima and Minima of functions of two variables without constraints and Lagrange's method (with constraints)		
UNIT-III	LAPLACE TRANSFORMS	
Laplace transforms of elementary functions- First shifting theorem - Change of scale property - Multiplication by t^n - Division by t - Laplace transforms of derivatives and integrals - Unit step function - Second shifting theorem - Periodic function - Evaluation of integrals by Laplace transforms - Inverse Laplace transforms- Method of partial fractions - Other methods of finding inverse transforms - Convolution theorem - Applications of Laplace transforms to ordinary differential equations.		
UNIT-IV	FOURIER SERIES	
Periodic function-Determination of Fourier Coefficients-Fourier Series-Even and Odd functions-Fourier series in arbitrary interval-Even Odd periodic continuation-Half range Fourier sine and cosine expansions.		
UNIT-V	NUMERICAL TECHNIQUES	
ROOT FINDING TECHNIQUES : Bisection method-Regula falsi method, Iteration method and Newton Raphson method. NUMERICAL INTEGRATION : Trapezoidal rule - Simpson's one-third rule - Simpson's three-eighth rule. NUMERICAL SOLUTION OF ORDINARY DIFFERENTIAL EQUATIONS: Taylor's series method - Euler's - modified Euler's Method - Runge-Kutta method.		
TEXT BOOKS:		

1. Ervin Kreyszig, Advanced Engineering Mathematics, 9th Edition, John Wiley & Sons, 2006.
2. B.S.Grewal, Higher Engineering Mathematics, Khanna publishers, 36th Edition, 2010.

REFERENCE BOOKS:

1. G.B.Thomas, calculus and analytical geometry,9th Edition, Pearson Reprint 2006.
2. N.P Bali and Manish Goyal ,A Text of Engineering Mathematics,Laxmi publications,2008.
3. E.L.Ince, Ordinary differential Equations,Dover publications,1958.

WEB REFERENCES:

1. https://www.efunda.com/math/math_home/math.cfm
2. <https://www.ocw.mit.edu/resources/#Mathematics>
2. <https://www.sosmath.com/>
3. <https://www.mathworld.wolfram.com/>

E -TEXT BOOKS:

1. <https://www.e-booksdirectory.com/details.php?ebook=10166>
2. <https://www.e-booksdirectory.com/details.php?ebook=10166>

MOOCS COURSE:

1. <https://swayam.gov.in/>
2. <https://onlinecourses.nptel.ac.in/>

CHEMISTRY

I B. TECH- II SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4BS11	BSC	L	T	P	C	CIE	SEE	Total
		4	-	-	4	30	70	100

COURSE OBJECTIVES:

Student will be able to:

1. Bring adaptability to the concepts of chemistry and to acquire the required skills to become a perfect engineer.
2. To impart the basic knowledge of atomic, molecular and electronic modifications which makes the student to understand the technology based on them.
3. To acquire the knowledge of electrochemistry, corrosion and water treatment which are essential for the Engineers and in industry?
4. To acquire the skills pertaining to spectroscopy and to apply them for medical and other fields.
5. To impart the knowledge of stereochemistry and synthetic aspects useful for understanding reaction pathways.

COURSE OUTCOMES:

The basic concepts included in this course will help the student to gain:

1. The knowledge of atomic, molecular and electronic changes, band theory related to conductivity.
2. The required principles and concepts of electrochemistry, corrosion and in understanding the problem of water and its treatments.
3. The required skills to get clear concepts on basic spectroscopy and application to medical and other fields.
4. The knowledge of configurationally and conformational analysis of molecules and reaction mechanisms

UNIT-I	MOLECULAR STRUCTURE AND THEORIES OF BONDING	
Atomic and Molecular orbitals. Linear Combination of Atomic Orbitals (LCAO), molecular orbitals of diatomic molecules, molecular orbital energy level diagrams of N ₂ , O ₂ and F ₂ molecules. π molecular orbitals of butadiene and benzene. Crystal Field Theory (CFT): Salient Features of CFT – Crystal Field Splitting of transition metal ion d-orbitals in Tetrahedral, Octahedral and square planar geometries. Band structure of solids and effect of doping on conductance.		
UNIT-II	WATER AND ITS TREATMENT	
Introduction – hardness of water – Causes of hardness - Types of hardness: temporary and permanent – expression and units of hardness – Estimation of hardness of water by complexometric method. Potable water and its specifications. Steps involved in treatment of water – Disinfection of water by chlorination and ozonization. Boiler feed water and its treatment – Calgon conditioning, Phosphate conditioning and Colloidal conditioning. External treatment of water – Ion exchange process. Desalination of water – Reverse osmosis. Numerical problems.		
UNIT-III	ELECTROCHEMISTRY AND CORROSION	
Electro chemical cells – electrode potential, standard electrode potential, types of electrodes – calomel, Quinhydrone and glass electrode. Nernst equation Determination of pH of a solution by using quinhydrone and glass electrode. Electrochemical series and its applications. Numerical problems. Potentiometric titrations. Batteries – Primary (Lithium cell) and secondary batteries (Lead – acid storage battery). Corrosion: Causes and effects of corrosion – theories of chemical and electrochemical corrosion – mechanism of electrochemical corrosion, Types of corrosion: Galvanic, water-line and pitting corrosion. Factors affecting rate of corrosion, Corrosion control methods- Cathodic protection– Sacrificial anode and impressed current cathodic methods. Surface coatings – metallic coatings –Hot dipping , metal cladding and electro plating(copper plating).		

UNIT-IV	STEREOCHEMISTRY, REACTION MECHANISM AND SYNTHESIS OF DRUG MOLECULES	
Introduction to representation of 3-dimensional structures, Structural and stereoisomers, configurations, symmetry and chirality. Enantiomers, diastereomers, optical activity and Absolute configuration. Conformation analysis of n-butane. Substitution reactions: Nucleophilic substitution reactions: Mechanism of SN ₁ , SN ₂ reactions. Electrophilic and nucleophilic addition reactions: Addition of HBr to propene. Markownikoff and anti Markownikoff's additions. Grignard additions on carbonyl compounds. Elimination reactions: Dehydro halogenation of alkylhalides. Saytzeff rule. Oxidation reactions: Oxidation of alcohols using KMnO ₄ and chromic acid. Reduction reactions: reduction of carbonyl compounds using LiAlH ₄ & NaBH ₄ . Hydroboration of olefins. Structure, synthesis and pharmaceutical applications of Aspirin.		
UNIT-V	SPECTROSCOPIC TECHNIQUES AND APPLICATIONS	
Principles of spectroscopy, selection rules and applications of electronic spectroscopy, vibrational and rotational spectroscopy. Basic concepts of Nuclear magnetic resonance Spectroscopy, chemical shift. Introduction to Magnetic resonance imaging. Fluorescence and its applications in medicine.		
TEXT BOOKS:		
1. Engineering Chemistry 14th edition by P.C.Jain & M.Jain; Dhanpat Rai Publishing Company (P) Ltd., New Delhi. 2. Fundamentals of Molecular Spectroscopy 5th edition by C.N. Banwell Mc.Graw-Hills book company.		
REFERENCE BOOKS:		
1. Organic Chemistry 7th Edition, Robert Thornton Morrison, Robert Neilson Boyd, Saibal Kanti Bhattacharjee Pearson Education Singapore Pvt.Ltd. 2. Engineering Chemistry (NPTEL Web-book), 11th edition by B.L. Tembe, Kamaluddin and M.S. Krishnan. 3. Physical Chemistry 11th edition by P.W. Atkins OUP Oxford.		
WEB REFERENCES:		
1. https://www.scribd.com/document/23180395/Engineering-Chemistry-Unit-I-Water-Treatment 2. https://chem.libretexts.org/Core/Inorganic_Chemistry/Descriptive_Chemistry/Periodic_Trends_of_Elemental_Properties/Periodic_Properties_of_the_Elements 3. https://www.khanacademy.org/science/biology/chemistry--of-life/chemical-bonds-and-reactions/v/intermolecular-forces-and-molecular-bonds 4. https://study.com/academy/lesson/the-relationship-between-free-energy-and-the-equilibrium-constant.html		
E-TEXT BOOKS:		
1. http://www.freebookcentre.net/Chemistry/Chemistry-Books-Online.html 2. https://www.sdu.dk/en/om_sdu/institutter_centre/fysik_kemi_og_farmaci		
MOOC COURSE		
1. http://nptel.ac.in/courses/122101001/34 2. https://ocw.mit.edu/courses/chemistry/		

PROGRAMMING FOR PROBLEM SOLVING

I B. TECH- II SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS01	ESC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES

1. To impart basic knowledge about simple algorithms for arithmetic and logical problems.
2. To understand how to write a program, syntax and logical errors.
3. To enable them how to implement conditional branching, iteration and recursion.
4. To understand how to decompose a problem into functions and synthesize a complete program.
5. To enable them to use arrays, pointers, strings and structures in solving problems.
6. To understand how to solve problems related to matrices, Searching and sorting.
7. To make them to understand the use files to perform read and write operations.

COURSE OUTCOMES

At the end of the course, student will be able to:

1. Formulate simple algorithms for arithmetic and logical problems.
2. Test and execute the programs and correct syntax and logical errors.
3. Implement conditional branching, iteration and recursion.
4. Decompose a problem into functions and synthesize a complete program.
5. Use arrays, pointers, strings and structures to formulate algorithms and programs.
6. Apply programming to solve problems related to matrices, Searching and sorting.
7. Use files to perform read and write operations.

UNIT-I INTRODUCTION

Introduction to Programming: Computer system, components of a computer system, computing environments, computer languages, creating and running programs, Algorithms, flowcharts.
Introduction to C language: History of C, basic structure of C programs, process of compiling and running a C program, C tokens, keywords, identifiers, constants, strings, special symbols, variables, data types, I/O statements.

UNIT-II OPERATORS, EXPRESSIONS AND CONTROL STRUCTURES

Operators and expressions: Operators, arithmetic, relational and logical, assignment operators, increment and decrement operators, bitwise and conditional operators, special operators, operator precedence and associativity, evaluation of expressions, type conversions in expressions.
Control structures: Decision statements; if and switch statement; Loop control statements: while, for and do while loops, jump statements, break, continue, goto statements.

UNIT-III ARRAYS AND FUNCTIONS

Arrays: Concepts, One dimensional array, declaration and initialization of one dimensional arrays, two dimensional arrays, initialization and accessing, multi dimensional arrays, Basic Algorithms: Searching, Basic Sorting Algorithms- Bubble sort, Insertion sort and Selection sort.
Functions: User defined and built-in Functions, storage classes, Parameter passing in functions, call by value, Passing arrays to functions: idea of call by reference, Recursion, as a different way of solving problems. Example programs, such as Finding Factorial, Fibonacci series, Ackerman function etc, Quick sort or Merge sort.

UNIT-IV STRINGS AND POINTERS

Strings: Arrays of characters, variable length character strings, inputting character strings, character library functions, string handling functions.
Pointers: Pointer basics, pointer arithmetic, pointers to pointers, generic pointers, array of pointers, functions returning pointers, Dynamic memory allocation.

UNIT-V STRUCTURES AND FILE HANDLING

Structures and unions: Structure definition, initialization, accessing structures, nested structures, arrays of structures, structures and functions, self referential structures, unions, typedef, enumerations.
File handling: command line arguments, File modes, basic file operations read, write and append, example programs

TEXT BOOKS:

1. Byron Gottfried, "Programming with C", Schaum's Outlines Series, McGraw Hill Education, 3rd edition, 2017.
2. E. Balagurusamy, "Programming in ANSI C", McGraw Hill Education, 6th Edition, 2012.

REFERENCE BOOKS:

1. W. Kernighan Brian, Dennis M. Ritchie, "The C Programming Language", PHI Learning, 2nd Edition, 1988.
2. Yashavant Kanetkar, "Exploring C", BPB Publishers, 2nd Edition, 2003.
3. Schildt Herbert, "C: The Complete Reference", Tata McGraw Hill Education, 4th Edition, 2014.
4. R. S. Bichkar, "Programming with C", Universities Press, 2nd Edition, 2012.
5. Dey Pradeep, Manas Ghosh, "Computer Fundamentals and Programming in C", Oxford University Press, 2nd Edition, 2006.
6. Stephen G. Kochan, "Programming in C", Addison-Wesley Professional, 4th Edition, 2014.
7. B. A. Forouzan, R. F. Gillberg, "C Programming and Data Structures", Cengage Learning, India, 3rd Edition, 2014.

WEB REFERENCES:

1. <https://www.bfoit.org/itp/Programming.html>
2. <https://www.khanacademy.org/computing/computer-programming>
3. <https://www.edx.org/course/programming-basics-iitbombayx-cs101-1x-0>
4. <https://www.edx.org/course/introduction-computer-science-harvardx-cs50x>

E-TEXT BOOKS:

1. <http://www.freebookcentre.net/Language/Free-C-Programming-Books-Download.htm>
2. <http://www.imada.sdu.dk/~svalle/courses/dm14-2005/mirror/c/>
3. <http://www.enggnotebook.weebly.com/uploads/2/2/7/1/22718186/ge6151-notes.pdf>

MOOC COURSE

1. https://onlinecourses.nptel.ac.in/noc18_cs33/preview
2. <https://www.alison.com/courses/Introduction-to-Programming-in-c>
3. <http://www.ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-s096-effective-programming-in-c-and-c-january-iap-2014/index.htm>

ENGLISH								
I B. TECH- II SEMESTER								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4HS01	HSMC	L	T	P	C	CIE	SEE	Total
		2	-	-	2	30	70	100
COURSE OBJECTIVES: The course should enable the students to: 1. Improve language proficiency with emphasis on Vocabulary, Grammar, Reading and Writing skills. 2. Apply the theoretical and practical components of English syllabus to study academic subjects more effectively and critically. 3. Analyze a variety of texts and interpret them to demonstrate in writing or speech. 4. Write clearly and creatively, and adjust writing style appropriately to the content, the context, and nature of the subject. 5. Develop language components to communicate effectively in formal and informal situations.								
COURSE OUTCOMES: Students should be able to: 1. Use English Language effectively in spoken and written forms. 2. Comprehend the given texts and respond appropriately. 3. Communicate confidently in various contexts and different cultures. 4. Acquire basic proficiency in English including reading and listening comprehension, writing and speaking skills.								
UNIT-I	THE RAMAN EFFECT							
Chapter entitled ‘The Raman Effect’ from the prescribed textbook, ‘English for Engineers’ published by Cambridge University Press. V- The concept of Word Formation, Root Words and their use in English G- Types of Sentences, Parts of Speech, Nouns, Pronouns, Adjectives R- Reading and its importance W- Sentence Structures, Punctuation, Writing Introductions and Conclusions								
UNIT-II	LETTER WRITING							
Letter Writing. V- Synonyms and Antonyms, Standard Abbreviations G- Verbs, Modal Auxiliaries, Adverbs, Prepositions, Conjunctions R- Improving Comprehension Skills, Reading and Comparing Two Articles W- Writing Paragraphs, Letter Writing- Letters of Request, Apology and Complaint- Letter of Application with Resume.								
UNIT-III	BLUE JEANS							
Chapter entitled ‘Blue Jeans’ from the prescribed textbook, ‘English for Engineers’ published by Cambridge University Press. V- Prefixes and Suffixes, Idioms and Phrasal verbs G- Articles, Tenses and its forms R- Sub skills of Reading- Skimming and Scanning W- Essay writing andDescribing Objects, Places and Events								
UNIT-IV	WHAT SHOULD YOU BE EATING							
Chapter entitled ‘What Should You Be Eating’ from the prescribed textbook, ‘English for Engineers’ published by Cambridge University Press. V- One word Substitutes, Words often confused G- Active and passive Voice Subject Verb Agreement (Concord) R- Reading Comprehension- Intensive and Extensive W- Technical Report Writing. E-mail writing. Picture Essay								

UNIT-V	HOW A CHINESE BILLIONAIRE BUILT HER FORTUNE	
Chapter entitled 'How a Chinese Billionaire Built Her Fortune' from the prescribed textbook, 'English for Engineers' published by Cambridge University Press. V- Technical Vocabulary and their Usage, Misplaced Modifiers, Redundancies G- Direct and Indirect Speech, Degrees of Comparison, Common Errors in English R-Effective Reading and Exercises for Practice W- Memo, Précis and Resume Writing		
TEXT BOOKS:		
1. Sudarshan, N. P. and Savitha, C. (2018). English for Engineers. Cambridge University Press. 2. Wren & Martin. (2017). High School English Grammar and Composition Book. S Chand Publishing.		
REFERENCE BOOKS:		
1. Murphy, R. (2015). Essential Grammar in Use. Cambridge University Press. 2. Wood, F.T. (2007). Remedial English Grammar. Macmillan. 3. Swan, M. (2016). Practical English Usage. Oxford University Press. 4. Kumar, S and Lata, P. (2018). Communication Skills. Oxford University Press. 5. Zinsser, William. (2001). On Writing Well. Harper Resource Book. 6. Hamp-Lyons, L. (2006). Study Writing. Cambridge University Press. 7. Exercises in Spoken English. Parts I –III. CIEFL, Hyderabad. Oxford University Press.		
WEB REFERENCES:		
1. https://www.oxfordlineenglish.com 2. https://www.bbclearningenglish.com 3. https://www.learnenglish.britishcouncil.org 4. https://www.fluentu.com/english		
E-TEXT BOOKS:		
1. https://www.uop.edu.io/download/research/members/oxford_guide_to_english_grammar.pdf 2. http://www.espressoenglish.net/wp-content/uploads/2012/07/Free-Grammar-Ebook-Level-2.pdf 3. https://update24hour.com/2018/03/wren-and-martin-english-grammar-pdf		
MOOC COURSE		
1. http://www.cambridgeenglish.org/learning-english/free-resources/mooc/ 2. https://www.britishcouncil.org.tr/en/english/mooc 3. https://mooc.com		

PROGRAMMING FOR PROBLEM SOLVING LAB

I B. TECH- II SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS02	ESC	L	T	P	C	CIE	SEE	Total
		-	-	4	2	30	70	100

COURSE OBJECTIVES:

1. To understand how to formulate the algorithms for simple problems
2. To be able to translate given algorithms to a working and correct program
3. To make them understand how to correct syntax errors as reported by the compilers
4. To be able to identify and correct logical errors encountered at run time
5. To understand how to write iterative as well as recursive programs
6. To enable them to represent data in arrays, strings and structures
7. To impart the knowledge of declare pointers of different types and their usage.
8. To understand how to create, read and write to and from simple text files.

COURSE OUTCOMES:

At the end of the course, student will be able to

1. Formulate the algorithms for simple problems
2. Translate given algorithms to a working and correct program
3. Correct syntax errors as reported by the compilers
4. Identify and correct logical errors encountered at run time
5. Write iterative as well as recursive programs
6. Represent data in arrays, strings and structures and manipulate them through a program
7. Declare pointers of different types and use them in defining self-referential structures.
8. Create, read and write to and from simple text files.

LIST OF EXPERIMENTS

WEEK-1 INTRODUCTION TO LINUX COMMANDS

1. Basic Linux commands
2. Write a C program to use printf() and scanf() functions
3. Write C programs to implement basic arithmetic operations – sum, average, product, difference, quotient and remainder of given numbers etc.

WEEK-2 OPERATORS AND EVALUATION OF EXPRESSIONS

1. Write a C program to check whether a number is even or odd using ternary operator.
2. Write a C program to perform the addition of two numbers without using +operator.
3. Write a C program to evaluate the arithmetic expression $((a + b / c * d - e) * (f - g))$. Read the values a, b, c, d, e, f, g from the standard input device.
4. Write a C program to find the sum of individual digits of a 3 digit number.
5. Write a C program to read the values of x and y and print the results of the following expressions in one line:

$(x + y) / (x - y)$
 $(x + y)(x - y)$

WEEK-3 CONDITIONAL STATEMENTS

1. Write a C program to find largest and smallest of given numbers.
2. Write a C program to find roots of a quadratic equation.
3. Write a C program which takes two integer operands and one operator from the user(+, -, *, /, % use switch)

WEEK-4 LOOPING STATEMENTS

1. Write a C program to find Sum of individual digits of given integer
2. Write a C program to generate first n terms of Fibonacci series
3. Write a C program to generate prime numbers between 1 and n

WEEK-5 LOOPING STATEMENTS

1. Write a C Program to find the Sum of Series $SUM=1-x^2/2! +x^4/4!-x^6/6!+x^8/8!-x^{10}/10!$ 2. Write a C program to generate Pascal's triangle. 3. Write a C program to generate pyramid of numbers. <pre> 1 1 3 1 1 3 5 3 1 </pre>	
WEEK-6	ARRAYS
1. Write a C Program to implement following sorting methods Bubble sort Selection sort Insertion sort 2. Write a C program to find largest and smallest number in a list of integers	
WEEK-7	ARRAYS
1. Write a C program To add two matrices To multiply two matrices 2. Write a C program to find Transpose of a given matrix	
WEEK-8	FUNCTIONS
1. Write a C program to find the factorial of a given integer using functions 2. Write a C program to find GCD of given integers using functions 3. Write a C Program to find the power of a given number using functions	
WEEK-9	RECURSION
1. Write a C Program to find binary equivalent of a given decimal number using recursive functions. 2. Write a C Program to print Fibonacci sequence using recursive functions. 3. Write a C Program to find LCM of 3 given numbers using recursive functions	
WEEK-10	STRINGS
1. Write a C program using functions to 2. Insert a sub string into a given main string from a given position 3. Delete n characters from a given position in a string 4. Write a C program to determine if given string is palindrome or not	
WEEK-11	POINTERS AND STRUCTURES
1. Write a C program to print 2-D array using pointers 2. Write a C program to allocate memory dynamically using memory allocation functions (malloc, calloc, realloc, free) 3. Write a C Program using functions to 4. Reading a complex number 5. Writing a complex number 6. Add two complex numbers 7. Multiply two complex numbers Note: represent complex number using structure.	
WEEK-12	FILES
1. Write a C program to copy one file to other 2. Write a C program to copy one file to other 3. Write a C Program to merge two files into a third file	
TEXT BOOKS:	

- | |
|--|
| <ol style="list-style-type: none">1. Yashavant Kanetkar, "Let Us C", BPB Publications, New Delhi, 13th Edition, 2012.2. Oualline Steve, "Practical C Programming", O'Reilly Media, 3rd Edition, 1997. |
|--|

REFERENCE BOOKS:

- | |
|--|
| <ol style="list-style-type: none">1. King KN, "C Programming: A Modern Approach", Atlantic Publishers, 2nd Edition, 2015.2. Kochan Stephen G, "Programming in C: A Complete Introduction to the C Programming Language", Sam's Publishers, 3rd Edition, 2004.3. Linden Peter V, "Expert C Programming: Deep C Secrets", Pearson India, 1st Edition, 1994. |
|--|

WEB REFERENCES:

- | |
|---|
| <ol style="list-style-type: none">1. http://www.sanfoundry.com/c-programming-examples2. http://www.geeksforgeeks.org/c3. http://www.cprogramming.com/tutorial/c4. http://www.cs.princeton.edu |
|---|

CHEMISTRY LAB

I B. TECH- II SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks	
A4BS12	BSC	L	T	P	C	CIE	SEE
		-	-	3	1.5	30	70

COURSE OBJECTIVES:

The course should enable the students to:

1. Estimation of hardness and chloride content in water to check its suitability for drinking purpose.
2. To determine the rate constant of reactions from concentrations as a function of time.
3. The measurement of physical properties like adsorption and viscosity.
4. To synthesize the drug molecules and check the purity of organic molecules by thin layer chromatographic (TLC) technique.

COURSE OUTCOMES:

The course should enable the students to:

1. Determination of parameters like hardness and chloride content in water.
2. Estimation of rate constant of a reaction from concentration – time relationships.
3. Determination of physical properties like adsorption and viscosity.
4. Calculation of R_f values of some organic molecules by TLC technique.

LIST OF EXPERIMENTS

WEEK-1	DETERMINATION OF TOTAL HARDNESS OF WATER BY COMPLEXOMETRIC METHOD USING EDTA
WEEK-2	ESTIMATION OF AN HCL BY CONDUCTOMETRIC TITRATIONS
WEEK-3	ESTIMATION OF ACETIC ACID BY CONDUCTOMETRIC TITRATIONS
WEEK-4	ESTIMATION OF HCL BY POTENTIOMETRIC TITRATIONS
WEEK-5	ESTIMATION OF ACETIC ACID BY POTENTIOMETRIC TITRATIONS
WEEK-6	DETERMINATION OF RATE CONSTANT OF ACID CATALYSED HYDROLYSIS OF METHYL ACETATE
WEEK-7	SYNTHESIS OF ASPIRIN
WEEK-8	THIN LAYER CHROMATOGRAPHY CALCULATION OF R_f VALUES. EG ORTHO AND PARA NITRO PHENOLS
WEEK-9	VERIFICATION OF FREUNDLICH ADSORPTION ISOTHERM- ADSORPTION OF ACETIC ACID ON CHARCOAL
WEEK-10	DETERMINATION OF VISCOSITY OF CASTOR OIL AND GROUND NUT OIL BY USING OSTWALD'S VISCOMETER
WEEK-11	DETERMINATION OF SURFACE TENSION OF A GIVEN LIQUID USING STALAGMOMETER
WEEK-12	SYNTHESIS OF THIOL RUBBER
WEEK-13	DETERMINATION OF CHLORIDE CONTENT OF WATER USING ARGENTOMETRIC METHOD

WEEK-14	DETERMINATION OF RATE CONSTANT OF ACID CATALYSED HYDROLYSIS OF METHYL ACETATE
REFERENCE BOOKS:	
1. Senior practical physical chemistry, b.d. khosla, a. Gulati and v. Garg (r. Chand & co., delhi). 2. An introduction to practical chemistry, k.k. sharma and d. S. Sharma (vikas publishing, n. Delhi). 3. Vogel's text book of practical organic chemistry 5th edition. 4. Text book on experiments and calculations in engineering chemistry – s.s. dara.	
WEB REFERENCES:	
1. Http://amrita.olabs.edu.in/?sub=73&brch=8&sim=153&cnt=2 2. https://chem.libretexts.org/textbook_maps/analytical_chemistry_textbook_maps/map%3a_analytical_chemistry_2.0_(harvey)/11_electrochemical_methods/11.2%3a_potentiometric_methods 3. Http://fch.upol.cz/skripta/fcc_and_zvem_english/fch/adsorption%20of%20oxalic%20acid%20on%20activated%20charcoal.htm 4. https://www.askiitians.com/iit-jee-chemistry/physical-chemistry/electrolytic-conductance-molar-conductance-and-specific-conductance.aspx 5. https://owlcation.com/stem/tlc-thin-layer-chromatography-principle-procedure	

ENGLISH LANGUAGE AND COMMUNICATION SKILLS LABORATORY**I B. TECH- II SEMESTER**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4HS02	HS	L	T	P	C	CIE	SEE	Total
		0	0	2	1	30	70	100

COURSE OBJECTIVES:**The course should enable the students to:**

1. Facilitate computer-assisted multi-media instruction enabling individualized and independent language learning.
2. Enhance English language skills, communication skills and to practice soft skills.
3. Improve fluency and pronunciation intelligibility by providing an opportunity for practice in speaking.
4. Train students in different interview and public speaking skills such as JAM, debate, role play, group discussion etc.
5. Instill confidence and make them competent enough to express fluently and neutralize their mother tongue influence.

COURSE OUTCOMES:**The course should enable the students to:**

1. Better perception of nuances of English language through audio- visual experience
2. Neutralization of accent for intelligibility
3. Take part in group activities
4. Speaking skills with clarity and confidence which in turn enhances their employability

LIST OF ACTIVITIES

WEEK-1	GENERAL INTRODUCTIONS AND FORMAL GREETINGS
1. Introductions and greetings in formal and informal situations 2. Worksheets to extract information 3. Questionnaires to enquire about the expressions used during formal introductions 4. Ice Breaking activity by preparing and asking five questions each 5. Creation of dialogues using greetings, leave- taking and introductions	
WEEK-2	JAM- JUST A MINUTE
1. Strengthen the ability to analyze a topic and logical organization of thoughts. 2. Logically starting with introductory sentence, points of discussion and closing sentence. 3. Practicing to speak within one minute 4. Activity based on JAM on a familiar topic 5. Planning and executing JAM considering the parameters	
WEEK-3	PHONETICS
1. Speech sounds and their prominence in pronunciation 2. Understanding and practicing word stress 3. Neutralizing the accent and practicing the right intonation 4. Knowing the differences between different accents 5. Increase fluency with the help of Phonetics	
WEEK-4	LISTENING SKILLS

	1. Developing good listening skills for effective communication 2. Enhancing listening skills through audio tracks and oral conversation 3. Empathizing others point of view while they speak 4. Incorporating non verbal communications while listening 5. Improving overall performance listening to the audio tracks
WEEK-5	SITUATIONAL DIALOGUES AND GIVING DIRECTIONS
	1. Creating dialogues in any given situations 2. Framing and choosing appropriate words to frame the dialogues in any situation 3. Guiding and giving directions using appropriate expressions 4. Activities on how to make polite requests, offers, rejections etc 5. Practicing to speak confidently in different situations
WEEK-6	ROLE PLAY
	1. Understanding a Role play and its procedure 2. Planning and Executing a Role Play accordingly 3. Practicing to get into the role and perform within stipulated time 4. Activities based on Role Play with different situations 5. Performing a Role Play considering the parameters
WEEK-7	GROUP DISCUSSIONS
	1. Understanding a Group Discussion (GD) and its procedure 2. Following the rules of a GD 3. Planning and Executing a GD within the stipulated time 4. Activities based on GD 5. Performing a GD considering the parameters
WEEK-8	DEBATE
	1. Understanding the procedure of a Debate 2. Planning and executing a Debate following its rules 3. Strengthen the ability to analyze a topic and logical organization of thoughts. 4. Logically arranging the arguments 5. Performing a Debate considering the parameters
WEEK-9	TELEPHONIC ETIQUETTES
	1. Understanding basic Telephonic Etiquettes 2. The approach one needs to follow while making and answering a call 3. Making a formal telephonic conversation 4. Activities based on modulating voice and tone 5. Interpersonal skills required to overcome rude and hostile behavior 6. Write a C program to compare two strings using pointers. 7. Write a C program to copy a string from source to destination using pointers. 8. Write a C program to reverse a string using pointers.
Week-10	PRESENTATION SKILLS
	1. Planning a Presentation 2. Enhancing skills required for making effective presentations 3. Usage of different tools that help us to give effective presentations 4. Executing a presentation effectively 5. Activities based on presentations
WEEK-11	ORAL PRESETATIONS AND EXTEMPORE

1. Planning an oral presentation or an Extempore 2. Preparing good PPT 3. Using appropriate body language in public speaking domain 4. Planning and Executing oral presentation 5. Activities based on oral presentations and extempore	
WEEK-12	INTERVIEW SKILLS
1. Preparing to succeed in Interviews 2. Preparing a strong Resume for interviews 3. Practicing different techniques to overcome nervousness in interviews 4. Using appropriate body language in interviews 5. Activities based on Interviews skills	
WEEK-13	INFORMATION TRANSFER
1. Extracting Information Transfer from different kinds of representation 2. Reading and decoding the information given in various types 3. Representing the information in charts or graphs in a written document 4. Developing writing skills from these aspects 5. Activity on transferring given data into graphs or charts for presentation skills	
REFERENCE BOOKS:	
1. E. Suresh Kumar. A Handbook for English Language Laboratories (with CD) Revised Edition 2. Normal Whitby. Business Benchmark. Cambridge University Press(with CD) 2nd Edition 3. Liz Hamp-Lyons and Ben Heasley. Study Writing. Cambridge University Press. 2006. 4. Sanjay Kumar and Pushp Lata. Communication Skills. Oxford University Press. 2011. 5. Exercises in Spoken English. Parts. I-III. CIEFL, Hyderabad. Oxford University Press 6. Raman Sharma, Technical Communications, Oxford Publication, London, 2004. 7. Dale Jungk, Applied Writing for Technicians, McGraw Hill, New York, 2004. 8. T. Balasubramanian, A Text book of English Phonetics for Indian Students, Macmillan, 2008. 9. Edgar Thorpe, Winning at Interviews, Pearson Education, 2006. 10. J. Sethi et al, A Practical Course in English Pronunciation (with CD), Prentice Hall of India, 2005. 11. Hari Mohan Prasad, How to Prepare for Group Discussions and Interviews, Tata McGraw Hill, 2006	
WEBSITES:	
1. https://www.britishcouncil.org 2. https://www.bbc.co.uk 3. https://www.grammarly.com 4. https://www.fluentu.com 5. https://www.cambridgeenglish.org/exams-and-tests/business-preliminary 6. https://www.cambridgeenglish.org/exams-and-tests/business-vantage	

ENGINEERING EXPLORATION

I B. TECH- II SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4HS04	HSMC	L	T	P	C	CIE	SEE	Total
		-	-	2	1	30	70	100

COURSE OBJECTIVES:

The course should enable the students to:

1. Understand the Engineering attributes and Ethics.
2. Identify the community problem and its stakeholder.
3. Examine required specifications and gap in existing and required product.
4. Build sustaining interactions among people that create social value by transforming ideas into tangible products, services, or initiatives.
5. Develop skills to work collaboratively, reports and progress updates throughout the lifecycle of the project.

UNIT-I	INTRODUCTION TO ENGINEERING AND ENGINEERING EXPLORATION	
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Engineering Projects in Community Service, Design Thinking Process-Empathize, Define, Ideate, Prototype, Test.

Engineering Ethics: Introduction to ethics, moral values, significance of professional ethics, code of conduct for engineers, identify ethical dilemmas in different tasks of engineering, applying moral theories and codes of conduct for resolution of ethical dilemmas.

UNIT-II	PROBLEM IDENTIFICATION	
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Authentic need in the community or society. Identify a real user or stake holder, Interaction with Stakeholders, Viewpoints, Interviewing, Scenario.

UNIT-III	SPECIFICATION DEVELOPMENT	
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Clear and measurable requirements, criteria for success, Identifying relevant benchmarks, identifying the gap between the available and required products, requirements documentation.

UNIT-IV	CONCEPTUAL DESIGN	
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Ideation-generated multiple ideas, evaluation of ideas, systems model, Architectural Design, prototype development, testing – real/simulated users, feedback.

UNIT-V	PROJECT MANAGEMENT	
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Importance of team work, importance of project life cycle, project management, tools, various tools used in electronics documentation, importance of communication, usage of communication media.

TEXT BOOKS:

1. Software Engineering: A Practitioner's Approach, Roger S. Pressman, 7th Edition, Mc Graw Hill Education (India) Pvt. Ltd.
2. Software Engineering, Sommerville Ian, 7th Edition, Pearson Education.
3. EPICSDesignProcess
https://sharepoint.ecn.purdue.edu/epics/teams/Public%20Documents/EPICS_Design_Process.pdf
4. Examples of good practice in Special Needs Education & Community Based Programs, UNESCO PRESS.
5. Project Management, GRY r. Heerkens, McGraw-Hill

WEB REFERENCES:

1. <http://www.purdue.edu/epics>

2.	http://epics.ieee.org/
3.	https://www.uninettunouniversity.net/en/epics.aspx
E-TEXT BOOKS:	
1.	http://www.uoitc.edu.iq/images/documents/informatics-institute/exam_materials/Software%20Engineering%20(9th%20Edition)%20by%20Ian%20Somerville.pdf
2.	https://engineering.purdue.edu/EPICS/k12/resources/1.6%20Teacher%20Toolbox%20EPICS%20High%20Design%20Process%20and%20Cycle.pdf
3.	https://launchschool.com/books/agile_planning/read/epics_and_stories
4.	http://www.enggnotebook.weebly.com/uploads/2/2/7/1/22718186/ge6151-notes.pdf
MOOC COURSE	
1.	https://www.mooc-list.com/tags/design-thinking
2.	https://www.class-central.com/tag/design%20thinking

II B.TECH I SEMESTER SYLLABUS

PROBABILITY AND STATISTICS

II B. TECH- I SEMESTER

Course Code:	Category	Hours / Week			Credits	Maximum Marks		
A4BS05	BSC	L	T	P	C	CIE	SEE	Total
		2	1	-	3	30	70	100

COURSE OBJECTIVES:

To learn

1. The concepts of discrete and continuous random variables, the probability distribution and density function.
2. Evaluation of marginal and conditional distribution of multiple random variables.
3. The concept of correlation and regression to find covariance.
4. Evaluation of the given data for appropriate test of hypothesis.
5. Analyzing the data for variance.

COURSE OUTCOMES:

After learning the contents of this paper the student must be able to

1. Describe the continuous random variables and moments of Discrete and continuous random variables.
2. Calculate probabilities and derive the marginal and conditional distribution of bivariate random variables.
3. Apply the concept of correlation and regression to find covariance.
4. Evaluate the given data for appropriate test of hypothesis.
5. Test the data for analysis of variance.

UNIT-I	SINGLE RANDOM VARIABLES	
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Basic definitions of probability- Random Variables – Discrete and Continuous. Probability distributions, mass function/ density function of a probability distribution, mathematical expectation, moments about origin, central moments, skewness, Kurtosis. Moment generating function of probability distribution.

UNIT-II	PROBABILITY DISTRIBUTIONS	
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Binomial, Poisson, Normal, exponential and Gamma distributions -their Properties. Moment generating functions of the above distributions and hence find the mean and variance. Joint probability distributions- Joint probability mass /density function, Marginal probability, mass / density functions.

UNIT-III	CORRELATION & REGRESSIONSAMPLING DISTRIBUTIONS	
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Coefficient of correlation, the rank correlation, Covariance of two random variables. Regression- Regression Coefficient, The lines of regression and multiple correlation & regression

Sampling: Definitions of population, sampling, statistic, parameter. Types of sampling, Expected values of Sample mean and variance, sampling distribution, Standard error, Sampling distribution of means and sampling distribution of variance.

Parameter estimation- Point estimation and interval estimation

UNIT-IV	TESTING OF HYPOTHESIS - I	
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Testing of hypothesis: Null hypothesis, Alternate hypothesis, Type I & Type II errors – critical region, confidence interval, and Level of significance. One sided test, Two sided test.

Large sample tests:

- (i) Test of Equality of means of two samples, equality of sample mean and population mean (cases of known variance & unknown variance, equal and unequal variances)
- (ii) Tests of significance of difference between sample S.D and population S.D.
- (iii) Tests of significance difference between sample proportion and population proportion, difference Between two sample proportions.

UNIT-V	TESTING OF HYPOTHESIS-II	
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Student t-distribution, its properties; Test of significance sample mean and population mean, difference between means of two small samples. Snedecor's F- distribution and its properties. Test of equality of two population variances. Chi-square distribution, its properties, Chi-square test of goodness of fit.

TEXT BOOKS:

1. B.S.Grewal, Higher Engineering Mathematics, Khanna publishers, 36th Edition, 2010.
2. Probability and Statistics for Engineers by Richard Arnold Johnson, Irwin Miller and John E. Freund, New Delhi, Prentice Hall
3. Probability and Statistics for Engineers and Sciences by Jay L. Devore, Cengage Learning

REFERENCE BOOKS:

1. Ervin Kreyszig, Advanced Engineering Mathematics, 9th Edition, John Wiley & Sons, 2006.
2. Fundamentals of Mathematical Statistics by S.C. Gupta & V.K. Kapoor, S. Chand
3. Introduction to Probability and Statistics for Engineers and Scientists by Sheldon M. Ross, Academic Press

WEB REFERENCES:

1. https://www.efunda.com/math/math_home/math.cfm
2. <https://www.ocw.mit.edu/resources/#Mathematics>
3. <https://www.sosmath.com/>
4. <https://www.mathworld.wolfram.com/>

E -TEXT BOOKS:

1. <https://www.e-booksdirectory.com/details.php?ebook=10166>[https://](https://www.e-booksdirectory.com/details.php?ebook=10166)
2. www.e-booksdirectory.com/details.php?ebook=10166

MOOCS COURSE:

1. <https://swayam.gov.in/>
2. <https://onlinecourses.nptel.ac.in/>

DISCRETE STRUCTURES

II B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS03	PCC	L	T	P	C	CIE	SEE	Total
		3	1	-	4	30	70	100

COURSE OBJECTIVES

The course should enable the students to:

1. To help students understand discrete and continuous mathematical structures
2. To impart basics of relations and functions
3. To facilitate students in applying principles of Recurrence Relations to calculate generating functions and solve the Recurrence relations
4. To acquire knowledge in graph theory

COURSE OUTCOMES:

At the end of the course, student will be able to

1. Apply the knowledge of discrete and continuous mathematical structures.
2. Solve various problems on relations and functions.
3. Apply the principles of Recurrence Relations to generate functions and solve various problems on it.
4. Solve problems using the knowledge of graph theory.

UNIT-I MATHEMATICAL LOGIC

Statements and notations, Connectives, Well formed formulas, Truth Tables, Tautology, Equivalence implication, Normal forms, Logical Inference, Rules of inference, Direct Method, Direct Method using CP(Conditional Proof), Consistency, Proof of contradiction, Automatic Theorem Proving. Quantifiers, Universal quantifiers. Predicates: Predicative logic, Free & Bound variables.

UNIT-II RELATIONS

Introduction to set theory, Relations, Properties of Binary Relations, Equivalence Relation, Transitive closure, Compatibility and Partial ordering relations, Lattices, Hasse diagram. Functions: inverse Function, Composition of functions, Recursive Functions

UNIT-III ELEMENTARY COMBINATORICS

Basis of counting, Combinations & Permutations, Enumeration of Combinations and Permutations, Enumeration of Combinations and Permutations With repetitions, Enumerating Permutations with Constrained repetitions, Binomial Coefficients, Binomial and Multinomial theorems, The principles of Inclusion – Exclusion, Pigeon- hole principles and its applications.

UNIT-IV RECURRENCE RELATION

Generating Functions, Function of Sequences, Calculating Coefficient of generating function, Recurrence relations, Solving recurrence relation by substitution and Generating functions, The method of Characteristics roots, Solution of Inhomogeneous Recurrence Relation.

UNIT-V GRAPHS

Basic Concepts, Isomorphism and Sub graphs, Trees and their properties, Spanning Trees- DFS,BFS, Minimal Spanning Trees- Prims, Kruskal's Algorithm, Planar Graphs, Euler's Formula, Multi graph and Euler circuits, Hamiltonian Graphs, Chromatic number.

TEXT BOOKS:

1. T1. Discrete Mathematics for computer scientists & Mathematicians, J.L. Mott, A. Kandel, T.P. Baker PHI
2. Discrete Mathematical Structures With Applications to Computer Science, JP Tremblay, R Manohar

REFERENCE BOOKS:

1. R1. Logic and Discrete Mathematics, Grass Man & Trembley, Pearson Education.

DATA STRUCTURES

II Year I Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS04	PCC	L	T	P	C	CIE	SEE	Total
		3	1	-	4	30	70	100

COURSE OBJECTIVES:

1. Impart the basic concepts of data structures and algorithms.
2. Understand concepts linked lists and their applications.
3. Understand basic concepts about stacks, queues and their applications.
4. Understand basic concepts of trees, graphs and their applications.
5. Enable them to write algorithms for sorting and searching and hashing.
6. Use advanced data structures like B-Trees, AVL-trees etc., for efficient problem solving.

COURSE OUTCOMES

At the end of the course, student will be able to:

1. Evaluate algorithms in terms of time and memory complexity.
2. Formulate new solutions for problems or improve existing code using data structures and algorithms.
3. Implement basic data structures such as arrays, linked lists, stacks and queues.
4. Solve problem involving graphs, trees and heaps
5. Apply Algorithms for solving problems like sorting, searching, and hashing.
6. Implement advanced data structures such as B-Trees, Red-Black, and AVL-Trees

UNIT-I

INTRODUCTION TO DATA STRUCTURES

Basic concepts- Algorithm Specification-Introduction, Recursive algorithms, Data Abstraction, Performance analysis- time complexity and space complexity, Asymptotic Notation-Big O, Omega and Theta notations, **Introduction to Linear and Non Linear data structures**-Singly Linked Lists-Operations-Insertion, Deletion, Concatenating singly linked lists, Circularly linked lists-Operations for Circularly linked lists, Doubly Linked Lists- Operations- Insertion, Deletion. Representation of single, two dimensional arrays, sparse matrices-array and linked representations.

UNIT-II

STACKS AND QUEUES

Stacks-Stack ADT, definition, operations, array and linked implementations in C, applications-infix to postfix conversion, Postfix expression evaluation, recursion implementation,
Queues-Queue ADT, definition and operations, array and linked Implementations in C, Circular queues-Insertion and deletion operations, Dequeue (Double ended queue) ADT, array and linked implementations in C.

UNIT-III

TREES AND GRAPHS

Trees – Terminology, Representation of Trees, Binary tree ADT, Properties of Binary Trees, Binary Tree Representations-array and linked representations, Binary Tree traversals, Threaded binary trees, **Max Priority Queue**-ADT-implementation-Max Heap-Definition, Insertion into a Max Heap, Deletion from a Max Heap. Graphs, Introduction, Definition, Terminology, Graph ADT, Graph Representations-Adjacency matrix, Adjacency lists, Graph traversals- DFS and BFS.

UNIT-IV

SEARCHING AND SORTING

Searching- Linear Search, Binary Search, Static Hashing-Introduction, hash tables, hash functions, Overflow Handling.
Sorting-Insertion Sort, Selection Sort, Radix Sort, Quick sort, Merge Sort, Heap Sort, Comparison of Sorting methods.

UNIT-V

BINARY SEARCH TREES

Search Trees-Binary Search Trees, Definition, Operations- Searching, Insertion and Deletion, AVL Trees-Definition and Examples, Insertion into an AVL Tree, B-Trees, Definition, B-Tree of order m, operations-Insertion and Searching, Introduction to Red-Black and Splay Trees (Elementary treatment-only Definitions and Examples), Comparison of Search Trees.

Pattern matching algorithm- The Knuth-Morris-Pratt algorithm, Tries (examples only).

TEXT BOOKS:

1. "Fundamentals of Data Structures", Illustrated Edition by Ellis Horowitz, Sartaj Sahni, Computer Science Press.
2. Fundamentals of Data structures in C, 2nd Edition, E.Horowitz, S.Sahni and SusanAnderson-Freed, Universities Press.

REFERENCE BOOKS:

1. Algorithms, Data Structures, and Problem Solving with C++", Illustrated Edition by Mark Allen Weiss, Addison-Wesley Publishing Company
2. "How to Solve it by Computer", 2nd Impression by R. G. Dromey, Pearson Education

WEB REFERENCES:

1. <https://hackr.io/tutorials/learn-data-structures-algorithms>
2. <https://www.geeksforgeeks.org/fundamentals-of-algorithms/>
3. <https://www.udemy.com/introduction-to-algorithms-and-data-structures-in-c/>
4. <https://leetcode.com>

E-TEXT BOOKS:

1. <http://www.freetechbooks.com/algorithm-analysis-and-design-t1030.html>
2. <http://www.freetechbooks.com/algorithmic-problem-solving-t373.html>
3. <http://www.freetechbooks.com/algorithms-and-data-structures-the-basic-toolbox-t871.html>

MOOC COURSE

1. <https://www.coursera.org/specializations/data-structures-algorithms>
2. https://onlinecourses.nptel.ac.in/noc16_cs06/preview

DATABASE MANAGEMENT SYSTEMS

II B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS05	PCC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OVERVIEW:

This course introduces the core principles and techniques required in the design and implementation of database systems. This introductory application-oriented course covers the relational database systems RDBMS - the predominant system for business, scientific and engineering applications at present. It includes Entity-Relational model, Normalization, Relational model, Relational algebra, and data access queries as well as an introduction to SQL. It also covers essential DBMS concepts such as: Transaction Processing, Concurrency Control and Recovery. It also provides students with theoretical knowledge and practical skills in the use of databases and database management systems in information technology applications.

COURSE OBJECTIVES:

The course should enable the students to:

1. **Discuss** the basic database concepts, applications, data models, schemas and instances.
2. **Design** Entity Relationship model for a database.
3. **Demonstrate** the use of constraints and relational algebra operations.
4. **Describe** the basics of SQL and construct queries using SQL
5. **Understand** the importance of normalization in databases.
6. **Demonstrate** the basic concepts of transaction processing and concurrency control.
7. **Understand** the concepts of database storage structures and identify the access techniques.

COURSE OUTCOMES:

At the end of the course the students are able to:

1. Use the basic concepts of Database Systems in Database design
2. Apply SQL queries to interact with Database
3. Apply normalization on database design to eliminate anomalies
4. Analyze database transactions and can control them by applying ACID properties
5. Analyze physical database storage system of database.

UNIT-I

Introduction: Database system applications, Database system Vs file systems, Advantage of a DBMS, Describing and storing data in a DBMS, Structure of a DBMS, People who work with databases.

Entity Relationship Model (ER Model): Database Design and ER Diagrams, Entities Attributes and Entity sets, Features of ER Model, Conceptual design with the ER model.

UNIT-II

Introduction to relational model: Structure of Relational Databases, Database Schema, Keys, Schema Diagrams, Relational Query Languages, Relational Operations.

Introduction to SQL: Overview of the SQL Query Language, SQL Data Definition language, Basic Structure of SQL Queries, Basic operations, Set Operations, NULL Values, Aggregate Functions, Nested Sub Queries, JOIN Expressions, Views, Transactions, Integrity Constraints, SQL Data types and Schemas, Functions and Procedures, Triggers.

UNIT-III

Relational Algebra and Calculus: Relational Algebra, Tuple Relational Calculus, Domain Relational Calculus.

Schema Refinement and Normal Forms:

Introduction to schema refinement, Functional Dependencies, Reasoning about FDs, Normal Forms: 1NF, 2NF, 3NF, Boyce Codd Normal Form, Properties of decompositions, Multi valued Dependencies, Fourth Normal Form, Join Dependencies and Fifth Normal Form.

UNIT-IV		
Transaction Management: Transaction Concept, A simple transaction Model, Storage Structure, Transaction Atomicity and Durability, Transaction Isolation, Serializability. Concurrency Control and Recovery System: Lock based protocols, Deadlock handling, Multiple granularity, Time stamp based protocols, Validation based protocols. Failure Classification, Storage, Recovery and Atomicity, Failure with Non-volatile Storage, Remote backup systems.		
UNIT-V		
Storage and File Structure: Overview of Physical Storage Media, Magnetic Disk and Flash Storage, RAID, Tertiary Storage, File Organization, Organization of Records in Files, Data Dictionary storage. Indexing and Hashing: Basic Concepts, Ordered Indices, B+ Tree Index Files, Multiple Key access, Static Hashing, Dynamic Hashing, Comparison of Ordered Indexing and Hashing, Bitmap Indices.		
TEXT BOOKS:		
1. Abraham Silberschatz, Henry F. Korth, S. Sudharshan, "Database System Concepts", Sixth Edition, Tata McGraw Hill, 2011. 2. Raghurama Krishnan, Johannes Gehrke, "Data base Management Systems", TATA McGraw Hill, 3rd Edition, 2007. 3. R.P. Mahapatra & Govind Verma, Database Management Systems, Khanna Publishing House, 2013.		
REFERENCE BOOKS:		
1. Peter Rob, Carlos Coronel, Database Systems Design Implementation and Management, 7th edition, 2009. 2. Scott Urman, Michael McLaughlin, Ron Hardman, "Oracle database 10g PL/SQL programming ", 6th edition, Tata McGraw Hill, 2010 3. .K.Singh, "Database Systems Concepts, Design and Applications", First edition, Pearson Education, 2006. 4. Ramez Elmasri, Shamkant B. Navathe, "Fundamentals of Database Systems", Fourth Edition, Pearson / Addison wesley, 2007.		
WEB REFERENCES:		
1. http://www.learnadb.com/databases/how-to-convert-er-diagram-to-relational-database 2. https://www.w3schools.com/sql/sql_create_table.asp 3. http://www.edugrabs.com/conversion-of-er-model-to-relational-model/?upm_export=print 4. http://ssyu.im.ncnu.edu.tw/course/CSDB/chap14.pdf 5. http://web.cs.ucdavis.edu/~green/courses/ecs165a-w11/8-query.pdf		
E-TEXT BOOKS:		
1. http://www.freebookcentre.net/Database/Free-Database-Systems-Books-Download.html 2. http://www.ddegjust.ac.in/studymaterial/mca-3/ms-11.pdf		
MOOC Course		
1. https://www.mooc-list.com/tags/dbms-extensions 2. https://onlinecourses.nptel.ac.in/noc18_cs15/preview		

ELECTRONIC DEVICES

II B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EC71	ESC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

The course should enable the students to:

1. To learn principle of operation, construction and characteristics of various electronic devices.
2. To study operation and characteristics of Rectifiers with filters.
3. To understand the analysis of (FET) and MOSFET characteristics.
4. To provide the concepts involved in design of electronic Circuits.

COURSE OUTCOMES:

After completing this course, the student will be able to:

1. Understand the operation and characteristics of various electronic devices.
2. Design half wave and full wave rectifiers with filters.
3. Characterize the current flow of a bipolar transistor in CB, CE and CC configurations and Biasing the transistors
4. Analyze Field effect transistor (FET) and MOSFET characteristics.
5. Design and analyze of various Oscillators.

UNIT-I	P-N JUNCTION DIODE :	
PN JUNCTION DIODE: Operation of PN junction Diode: No bias, forward bias and reverse bias, diode current equation (qualitative treatment), volt-ampere (V-I) characteristics, temperature dependence of V-I characteristics, ideal versus practical diode, static and dynamic resistances, diode equivalent circuits, breakdown mechanisms in semiconductor diodes, zener diode characteristics.		
UNIT-II	RECTIFIERS AND FILTERS :	
The P-N junction as a Rectifier, Half wave Rectifier, Full wave Rectifier, Bridge Rectifier, Harmonic components in a Rectifier Circuit, Inductor Filters, Capacitor Filters, L Section Filters, π - Section Filters, Comparison of Filters, Voltage Regulation using Zener Diode.		
UNIT-III	BIPOLAR JUNCTION TRANSISTOR AND ITS BIASING	
BIPOLAR JUNCTION TRANSISTOR (BJT): BJT construction, operation, symbol, transistor current components, input & output characteristics of a transistor in CB, CE and CC configurations TRANSISTOR BIASING AND STABILIZATION: Need for biasing, operating point, DC and AC load lines, stability factor, fixed bias circuit, collector to base bias circuit, self bias circuit.		
UNIT-IV	FIELD EFFECT TRANSISTOR	
FIELD EFFECT TRANSISTOR: Junction field effect transistor (construction, principle of operation, symbol), volt-ampere characteristics of JFET, MOSFETS (construction, principle of operation, symbol), volt-ampere characteristics of MOSFETS in enhancement and depletion modes.		
UNIT-V	OSCILLATORS	
OSCILLATORS: Barkhausen criterion, RC Phase shift oscillator using BJT, General form of LC oscillators, Hartley oscillator, Colpitts oscillator and Crystal oscillator.		
TEXT BOOKS:		
1. Jacob Milliman, Christos C .Halkias, Satyabrata Jit (2015), Electronic Devices and Circuits, 3rd edition, Tata McGraw Hill, New Delhi. 2. G. K. Mittal (1999), Electronic Devices and Circuits, 22nd edition, Khanna Publications, New Delhi. 3. Robert Boylestad, Lowis Nashelsky (1993), Electronic Devices and Circuit Theory, 5th edition,		

Prentice Hall of India, New Delhi, India.

REFERENCE BOOKS:

1. David. A. Bell (1986), *Electronic Devices and Circuits*, 4th edition, Prentice Hall of India, New Delhi.
2. S. Shalivahanan, N. Suresh Kumar, A. Vallavaraj (2007), *Electronic Devices and Circuits*, 3rd edition, McGraw Hill, New Delhi, India.
3. Theodore. F. Bogart Jr, Jeffrey S. Beasley, Guillermo Rico (2004), *Electronic Devices and Circuits*, 6th edition Pearson Education, India.

WEB REFERENCES:

1. <https://unacademy.com/course/electronic-devices-gate-ece/JTIAAKX1>
2. <https://freevideolectures.com/course/2261/basic-electronics-and-lab/2>
3. <https://unacademy.com/lesson/build-in-potential-and-depletion-width/ALHF5QVM>

E-TEXT BOOKS:

1. [http://www.freebookcentre.net/electronics-ebooks-download/Electronic-Devices-and-Circuits-\(PDF-313p\).html](http://www.freebookcentre.net/electronics-ebooks-download/Electronic-Devices-and-Circuits-(PDF-313p).html)
2. <https://www.goodreads.com/book/show/25345857-electronic-devices-and-circuits>
3. <https://thebookee.net/el/electronic-devices-and-circuits-by-jb-gupta-pdf>

MOOC COURSE

1. <https://www.edx.org/>
2. <https://www.coursera.org/learn/electronics>

DATA STRUCTURES LAB

II B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS06	PCC	L	T	P	C	CIE	SEE	Total
		-	-	3	1.5	30	70	100

COURSE OBJECTIVES:

The course should enable the students to:

1. Ability to identify the appropriate data structure for given problem.
2. Design and analyze the time and space complexity of algorithm or program.
3. Effectively use compilers include library functions, debuggers and troubleshooting.
4. Write and execute programs using data structures such as arrays, linked lists to implement stacks, queues.
5. Write and execute programs in C to implement various sorting and searching.
6. Write and execute programs using data structures such as arrays, linked lists to implement trees, graphs, hash tables and search trees.

COURSE OUTCOMES:

The course should enable the students to:

1. Use appropriate data structure for given problem.
2. To analyze the time and space complexity of algorithm or program.
3. Use compilers include library functions, debuggers and troubleshooting.
4. Execute programs using data structures such as arrays, linked lists to implement stacks and queues.
5. Execute write programs in C to implement various sorting and searching.
6. Execute programs using data structures such as arrays, linked lists to implement trees, graphs, hash tables and search trees.

LIST OF EXPERIMENTS

WEEK-1	SINGLE LINKED LIST
Write a C program that uses functions to perform the following: a) Create a singly linked list of integers. b) Delete a given integer from the above linked list. c) Display the contents of the above list after deletion.	
WEEK-2	DOUBLE LINKED LIST
Write a C program that uses functions to perform the following: a) Create a doubly linked list of integers. b) Delete a given integer from the above doubly linked list. c) Display the contents of the above list after deletion.	
WEEK-3	INFIX TO POSTFIX CONVERSION
Write a C program that uses stack operations to convert a given infix expression into its postfix Equivalent, Implement the stack using an array.	
WEEK-4	DOUBLE ENDED QUEUE
Write C programs to implement a double ended queue ADT using i) array and ii) doubly linked list	
WEEK-5	BINARY SEARCH TREES USING RECURSION

Write a C program that uses functions to perform the following: a) Create a binary search tree of characters. b) Traverse the above Binary search tree recursively in Postorder.	
WEEK-6	BINARY SEARCH TREES USING NON-RECURSION
Write a C program that uses functions to perform the following: a) Create a binary search tree of integers. b) Traverse the above Binary search tree non recursively in inorder.	
WEEK-7	SORTING
Write C programs for implementing the following sorting methods to arrange a list of integers in Ascending order : a) Insertion sort b) Merge sort	
WEEK-8	SORTING
Write C programs for implementing the following sorting methods to arrange a list of integers in ascending order: a) Quicksort b) Selection sort	
WEEK-9	AVL-TREES
1. Write a C program to perform the following operations on AVL: a) Insertion into an AVL. b) Display elements of AVL Tree 2. Write a C program for implementing Heap sort algorithm for sorting a given list of integers in ascending order.	
WEEK-10	HASHING
Write a C program to implement all the functions of a dictionary (ADT) using hashing.	
Week-11	PATTERN MATCHING ALGORITHM
Write a C program for implementing Knuth-Morris- Pratt pattern matching algorithm.	
Week-12	GRAPH TRAVERSAL ALGORITHMS
Write C programs for implementing the following graph traversal algorithms: a) Depth first traversal b) Breadth first traversal	
TEXT BOOKS:	
1. C and Data Structures, Prof. P.S.Deshpande and Prof. O.G. Kakde, Dreamtech Press. 2. Data structures using C, A.K.Sharma, 2nd edition, Pearson. 3. Data Structures using C, R.Thareja, Oxford University Press.	
WEB REFERENCES:	
1. http://www.sanfoundry.com/data-structures-examples 2. http://www.geeksforgeeks.org/c 3. http://www.cs.princeton.edu	

DATABASE MANAGEMENT SYSTEMS LAB

II B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS07	PCC	L	T	P	C	CIE	SEE	Total
		-	-	3	1.5	30	70	100

COURSE OBJECTIVES:

The course should enable the students to:

1. Apply the basic concepts of Database Systems and Applications.
2. Use the basics of SQL and construct queries using SQL in database creation and interaction
3. Design a commercial relational database system (Oracle, MySQL) by writing SQL using the system.
4. Analyze and Select storage and recovery techniques of database system.

COURSE OUTCOMES:

Upon successful completion of this course, student will be able to:

1. Apply the basic concepts of Database Systems and Applications.
2. Develop an ER model for a given database.
3. Use the basics of SQL and construct queries using SQL in database creation and interaction.
4. Design a commercial relational database system (Oracle, MySQL) by writing SQL using the system.

LIST OF EXPERIMENTS

WEEK-1 INTRODUCTION TO E R MODEL

Student should select a case study from given ideas and formulate the problem statement.

WEEK-2 CONCEPTUAL DESIGNING USING ER DIAGRAMS

Conceptual Designing using ER Diagrams (Identifying entities, attributes, keys and relationships between entities, cardinalities, generalization, specialization etc.) Note: Student is required to submit a document by drawing ER Diagram to the Lab teacher.

WEEK-3 CONVERTING ER MODEL TO RELATIONAL MODEL

Converting ER Model to Relational Model (Represent entities and relationships in Tabular form, Represent attributes as columns, identifying keys) Note: Student is required to submit a document showing the database tables created from ER Model.

WEEK-4 NORMALIZATION

Normalization -To remove the redundancies and anomalies in the above relational tables, Normalize up to Third Normal Form

WEEK-5 CREATION OF TABLES USING SQL

Creation of Tables using SQL- Overview of using SQL tool, Data types in SQL, Creating Tables (along with Primary and Foreign keys), Altering Tables and Dropping Tables

WEEK-6 DML COMMANDS

Practicing DML commands- Insert, Select, Update, Delete.	
WEEK-7	QUERIES
Practicing Queries using ANY, ALL, IN, EXISTS, NOT EXISTS, UNION, INTERSECT, CONSTRAINTS etc.	
WEEK-8	SUB QUERIES
Practicing Sub queries (Nested, Correlated) and Joins (Inner, Outer and Equi)..	
WEEK-9	AGGREGATE FUNCTIONS
Practice Queries using COUNT, SUM, AVG, MAX, MIN, GROUP BY, HAVING, VIEWS Creation and Dropping.	
WEEK-10	TRIGGERS
Practicing on Triggers - creation of trigger, Insertion using trigger, Deletion using trigger, Updating using trigger.	
WEEK-11	PROCEDURES
Procedures- Creation of Stored Procedures, Execution of Procedure, and Modification of Procedure	
WEEK-12	CURSORS
. Cursors- Declaring Cursor, Opening Cursor, Fetching the data, closing the cursor.	
REFERENCE BOOKS:	
1. Database System Concepts, by Silberschatz, Sudarshan, and Korth, 6th edition. 2. Database management System by RaghuRamaKrishna, 3rd edition.	
WEB REFERENCES:	
1. http://www.learnadb.com/databases/how-to-convert-er-diagram-to-relational-database 2. https://www.w3schools.com/sql/sql_create_table.asp 3. http://www.edugrabs.com/conversion-of-er-model-to-relational-model/?upm_export=print 4. http://ssyu.im.ncnu.edu.tw/course/CSDB/chap14.pdf 5. http://web.cs.ucdavis.edu/~green/courses/ecs165a-w11/8-query.pdf	

ENVIRONMENTAL SCIENCE

II B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4MC01	MC	L	T	P	C	CIE	SEE	Total
		2	-	-	0	30	70	100
COURSE OBJECTIVES: The course should enable the students to: 1. Understanding the importance of ecological balance for sustainable development. 2. Understanding the impacts of developmental activities and mitigation measures. 3. Understanding the environmental policies and regulations								
COURSE OUTCOMES: Upon successful completion of the course, the student should be able to: 1. Identify the natural resources and understand resource management. 2. Identify classification and functioning of Ecosystems. 3. Critique the Importance of biodiversity and its conservation. 4. Analyze the problems related to environmental pollution and management. 5. Apply the role of information technology, Analyze social issues and Acts associated with environment.								
UNIT-I	ECOSYSTEMS							
Definition, Scope, and Importance of ecosystem. Classification, structure, and function of an ecosystem, Food chains, food webs, and ecological pyramids. Flow of energy, Biogeochemical cycles, Bioaccumulation, Biomagnification, ecosystem value, services and carrying capacity, Field visits.								
UNIT-II	NATURAL RESOURCES: CLASSIFICATION OF RESOURCES							
Living and Non-Living resources, water resources: use and over utilization of surface and ground water, floods and droughts, Dams: benefits and problems. Mineral resources: use and exploitation, environmental effects of extracting and using mineral resources, Land resources: Forest resources, Energy resources: growing energy needs, renewable and non renewable energy sources, use of alternate energy source, case studies.								
UNIT-III	BIODIVERSITY AND BIOTIC RESOURCES							
Introduction, Definition, genetic, species and ecosystem diversity. Value of biodiversity; consumptive use, productive use, social, ethical, aesthetic and optional values. India as a mega diversity nation, Hot spots of biodiversity. Field visit. Threats to biodiversity: habitat loss, poaching of wildlife, man-wildlife conflicts; conservation of biodiversity: In-Situ and Ex-situ conservation. National Biodiversity act.								
UNIT-IV	ENVIRONMENTAL POLLUTION AND CONTROL TECHNOLOGIES							
Environmental Pollution: Classification of pollution, Air Pollution: Primary and secondary pollutants, Automobile and Industrial pollution, Ambient air quality standards. Water pollution: Sources and types of pollution, drinking water quality standards. Soil Pollution: Sources and types, Impacts of modern agriculture, degradation of soil. Noise Pollution: Sources and Health hazards, standards, Solid waste: Municipal Solid Waste management, composition and characteristics of e-Waste and its management. Pollution control technologies: Wastewater Treatment methods: Primary, secondary and Tertiary. Overview of air pollution control technologies, Concepts of bioremediation. Global Environmental Issues and Global Efforts: Climate change and impacts on human environment. Ozone depletion and Ozone depleting substances (ODS). Deforestation and desertification. International conventions / Protocols: Earth summit, Kyoto protocol, and Montréal Protocol. NAPCC-Gol Initiatives.								
UNIT-V	ENVIRONMENTAL POLICY, LEGISLATION & EIA							
Environmental Protection act, Legal aspects Air Act- 1981, Water Act, Forest Act, Wild life Act, Municipal solid waste management and handling rules, biomedical waste management and handling rules, hazardous waste management and handling rules. EIA: EIA structure, methods of baseline data								

acquisition. Overview on Impacts of air, water, biological and Socio-economical aspects. Strategies for risk assessment, Concepts of Environmental Management Plan (EMP). **Towards Sustainable Future:** Concept of Sustainable Development Goals, Population and its explosion, Crazy Consumerism, Environmental Education, Urban Sprawl, Human health, Environmental Ethics, Concept of Green Building, Ecological Foot Print, Life Cycle assessment (LCA), Low carbon life style.

TEXT BOOKS:

1. Textbook of Environmental Studies for Undergraduate Courses by Erach Bharucha for University Grants Commission.
2. Environmental Studies by R. Rajagopalan, Oxford University Press.

REFERENCE BOOKS:

1. Environmental Science: towards a sustainable future by Richard T. Wright. 2008 PHL Learning Private Ltd. New Delhi.
2. Environmental Engineering and science by Gilbert M. Masters and Wendell P. Ela. 2008 PHI Learning Pvt. Ltd.
3. Environmental Science by Daniel B. Botkin & Edward A. Keller, Wiley INDIA edition.
4. Environmental Studies by Anubha Kaushik, 4th Edition, New age international publishers.
5. Text book of Environmental Science and Technology - Dr. M. Anji Reddy 2007, BS Publications.
6. Introduction to Environmental Science by Y. Anjaneyulu, BS.Publications.

II B.TECH II SEMESTER SYLLABUS

DIGITAL ELECTRONICS**II B. TECH- II SEMESTER**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EC72	ESC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

The course should enable the students to:

1. Understand numerical and character representations in digital logic
2. Design combinational and sequential logic circuits
3. Optimize combinational and sequential logic circuits.
4. Analyze a memory cell and apply for organizing larger memories

COURSE OUTCOMES

1. Understand the different switching algebra theorems and apply them for logic functions.
2. Define the Karnaugh map for a few variables and perform an algorithmic reduction of logic functions.
3. Define the following combinational circuits: buses, encoders/decoders, (de)multiplexers, exclusive-ORs, comparators, arithmetic-logic units.
4. Understand the bistable element and the different latches and flip-flops.
5. Understand sequential circuits, like counters and shift registers.

UNIT-I NUMBER THEORY

Representation of numbers of different radix, conversion of numbers from one radix to another radix, $r-1$'s complement and r 's complement of unsigned numbers subtraction, problem solving. problem solving for addition and subtraction. 4-bit codes: BCD, EXCESS 3, alphanumeric codes, 9's complement

UNIT-II Boolean algebra LOGIC SIMPLIFICATION

Basic Theorems and Properties of Boolean algebra, Switching Functions, Canonical and Standard Forms, Algebraic Simplification of Digital Logic Gates, Properties of XOR Gates, Universal Logic Gates. Multilevel NAND/NOR realizations. K- Map Method, up to five variable K- Maps, Don't Care Map Entries, Prime and Essential prime Implications.

UNIT-III COMBINATIONAL LOGIC CIRCUITS DESIGN

Design of Half adder, full adder, half subtractor, full subtractor, Design of decoder, demultiplexer, higher order demultiplexing, encoder, multiplexer, higher order multiplexing, realization of Boolean functions using decoders and multiplexers, priority encoder, 4-bit digital comparator.

UNIT-IV SEQUENTIAL LOGIC DESIGN-I

Difference between combinational and sequential circuits, Classification of sequential circuits (synchronous and asynchronous); basic flip-flops, truth tables and excitation tables (nand RS latch, nor RS latch, RS flip-flop, JK flip-flop, T flip-flop, D flip-flop with reset and clear terminals). Conversion of flip-flops to other flip-flops.

UNIT-V SEQUENTIAL LOGIC DESIGN-II

Design of Ripple counters, design of synchronous counters, Johnson counter, ring counter, shift register, bi-directional shift register, universal shift register. Ripple and Synchronous counters, Finite state machines, Design of synchronous FSM, Algorithmic State Machines charts. Designing synchronous circuits like Pulse train generator, Pseudo Random Binary Sequence generator, Clock generation.

TEXT BOOKS:

1. Switching and Finite Automata Theory- Zvi Kohavi & Niraj K. Jha, 3rd Edition, Cambridge.
2. Digital Design- Morris Mano, PHI, 3rd Edition.

REFERENCE BOOKS:

1. Introduction to Switching Theory and Logic Design – Fredriac J. Hill, Gerald R. Peterson, 3rd

Ed, John Wiley & Sons Inc.

2. Digital Fundamentals – A Systems Approach – Thomas L. Floyd, Pearson, 2013.
3. Digital Logic Design - Ye Brian and HoldsWorth, Elsevier
4. Fundamentals of Logic Design Charles H. Roth, Cengage Learning, 5th, Edition, 2004.

COMPUTER ORGANIZATION AND ARCHITECTURE								
II B. TECH- II SEMESTER								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS08	PCC	L	T	P	C	CIE	SEE	Total
		3	1	-	4	30	70	100
COURSE OUTCOMES:								
At the end of the course students will be able to:								
1. Identify various components of computer and their interconnection								
2. Identify basic components and design of the CPU: the ALU and control unit.								
3. Compare and select various Memory devices as per requirement.								
4. Compare various types of IO mapping techniques								
5. Critique the performance issues of cache memory and virtual memory								
UNIT-I								
STRUCTURE OF COMPUTERS: STRUCTURE OF COMPUTERS: Computer types, Functional units, Basic operational concepts, Von Neumann Architecture, Bus Structures, Software, Performance, Multiprocessors and Multicomputer, Data representation, Fixed and Floating point. COMPUTER ARITHMETIC: Addition and Subtraction, Multiplication and Division algorithms, Floating-point Arithmetic Operations, Decimal arithmetic operations								
UNIT-II								
BASIC COMPUTER ORGANIZATION AND DESIGN: BASIC COMPUTER ORGANIZATION AND DESIGN: Instruction codes, Computer Registers, Computer Instructions and Instruction cycle. Timing and Control, Memory-Reference Instructions, Input-Output and interrupt. Central processing unit: Stack organization, Instruction Formats, Addressing Modes, Data Transfer and Manipulation, Complex Instruction Set Computer (CISC) Reduced Instruction Set Computer (RISC), CISC vs RISC								
UNIT-III								
REGISTER TRANSFER AND MICRO-OPERATIONS: REGISTER TRANSFER AND MICRO-OPERATIONS: Register Transfer Language, Register Transfer, Bus and Memory Transfers, Arithmetic Micro-Operations, Logic Micro-Operations, Shift Micro-Operations, Arithmetic logic shift unit. MICRO-PROGRAMMED CONTROL: Control Memory, Address Sequencing, Micro-Program, Design of Control Unit								
UNIT-IV								
MEMORY SYSTEM: MEMORY SYSTEM: Memory Hierarchy, Semiconductor Memories, RAM(Random Access Memory), Read Only Memory (ROM), Types of ROM, Cache Memory, Virtual memory, Secondary Storage, RAID.								
UNIT-V								
MEMORY SYSTEM: INPUT OUTPUT: I/O interface, Programmed IO, Memory Mapped IO, Interrupt Driven IO, DMA. MULTIPROCESSORS: Characteristics of multiprocessors, Interconnection structures, Inter Processor Arbitration, Inter processor Communication and Synchronization, Cache Coherence								
TEXT BOOKS:								
1. M. Morris Mano (2006), Computer System Architecture, 3rd edition, Pearson/PHI, India.								
2. John P. Hayes (1998), Computer Architecture and Organization, 3rd edition, Tata McGrawHill								
REFERENCE BOOKS:								
1. Carl Hamacher, Zvonks Vranesic, SafeaZaky (2002), Computer Organization, 5th edition,McGraw Hill, New Delhi, India.								
2. William Stallings (2010), Computer Organization and Architecture- designing for performance, 8th edition, Prentice Hall, New Jersey.								
3. Anrew S. Tanenbaum (2006), Structured Computer Organization, 5th edition, Pearson Education Inc.								

OBJECT ORIENTED PROGRAMMING								
II B. TECH- II SEMESTER								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS09	PCC	L	T	P	C	CIE	SEE	Total
		3	1	-	4	30	70	100
COURSE OUTCOMES:								
1. Use object oriented programming concepts to solve real worldproblems. 2. Demonstrate the user defined exceptions by exception handling keywords (try, catch, throw, throws andfinally). 3. Use multithreading concepts to develop inter processcommunication. 4. Develop java application to interact with database by using relevant software component(JDBC Driver). 5. Build the internet-based dynamic applications using the concept ofapplets.								
UNIT-I	JAVA BASICS							
JAVA BASICS: Review of Object oriented concepts, History of Java, Java buzzwords, JVM architecture, Data types, Variables, Scope and life time of variables, arrays, operators, control statements, type conversion and casting, simple java program, constructors, methods, Static block, Static Data, Static Method String and String Buffer Classes, Using Java API Document								
UNIT-II	INHERITANCE , POLYMORPHISM, PACKAGES AND FILES							
INHERITANCE AND POLYMORPHISM: Basic concepts, Types of inheritance, Member access rules, Usage of this and Super key word, Method Overloading, Method overriding, Abstract classes, Dynamic method dispatch, Usage of final keyword.								
PACKAGES AND INTERFACES: Defining package, Access protection, importing packages, Defining and Implementing interfaces, and Extending interfaces.								
I / O STREAMS AND FILES: Concepts of streams, Stream classes- Byte and Character stream, Reading console Input and Writing Console output, File Handling.								
UNIT-III	EXCEPTION HANDLING AND MULTI THREADING							
EXCEPTION HANDLING: Exception types, Usage of Try, Catch, Throw, Throws and Finally keywords, Built-in Exceptions, Creating own Exception classes.								
MULTI THREADING: Concepts of Thread, Thread life cycle, creating threads using Thread class and Runnable interface, Synchronization, Thread priorities, Inter Thread communication.								
UNIT-IV	COLLECTION FRAMEWORK AND JDBC							
COLLECTION FRAMEWORK: Introduction to Java Collections, Overview of Java Collection frame work, Generics, Commonly used Collection classes- Array List, Vector, Hash table, Stack, Enumeration, Iterator, String Tokenizer, Random, Scanner, calender and Properties								
JDBC-Connecting to Database - JDBC Type 1 to 4 drives, connecting to a database, querying a database and processing the results, updating data with JDBC.								
UNIT-V	EVENT HANDLING AND APPLETS							
EVENT HANDLING: Events, Event sources, Event classes, Event Listeners, Relationship between Event sources and Listeners, Delegation event model, Examples: handling a button click, handling mouse events, Adapter classes.								
APPLETS - Inheritance hierarchy for applets, differences between applets and applications, life cycle of an applet, passing parameters to applets, applet security issues.								
TEXT BOOKS:								
1. Herbert Schildt and Dale Skrien, "Java Fundamentals – A comprehensive Introduction", McGraw Hill, 1st Edition,2013. 2. Herbert Schildt, "Java the complete reference", McGraw Hill, Osborne, 7th Edition,2011. 3. T.Budd, "Understanding Object- Oriented Programming with Java", Pearson Education, Updated Edition (New Java 2 Coverage),1999.								
REFERENCE BOOKS:								
1. P.J.Dietel and H.M.Dietel , "Java How to program", Prentice Hall, 6th Edition,2005. 2. P.Radha Krishna , "Object Oriented programming through Java", CRC Press, 1st Edition,2007. 3. S.Malhotra and S. Choudhary, "Programming in Java", Oxford University Press, 2nd Edition, 2014.								

DESIGN AND ANALYSIS OF ALGORITHMS

II B. TECH- II SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS10	PCC	L	T	P	C	CIE	SEE	Total
		3	1	-	4	30	70	100

COURSE OUTCOMES:

At the end of this course students will be able to:

1. Identify various Time and Space complexities of various algorithms
2. Understand Tree Traversal method and Greedy Algorithms
3. Apply Dynamic Programming concept to solve various problems
4. Apply Backtracking, Branch and Bound concept to solve various problems
5. Implement different performance analysis methods for non deterministic algorithms

UNIT-I

INTRODUCTION: Algorithm, pseudo code for expressing algorithms, performance analysis-space complexity, time complexity, asymptotic notation- big (O) notation, omega notation, theta notation and little (o) notation, recurrences, probabilistic analysis, disjoint set operations, union and find algorithms.

UNIT-II

DIVIDE AND CONQUER: General method, applications-analysis of binary search, quick sort, merge sort, AND OR Graphs.
GREEDY METHOD: General method, Applications-job sequencing with deadlines, Fractional knapsack problem, minimum cost spanning trees, Single source shortest path problem.

UNIT-III

GRAPHS (Algorithm and Analysis): Breadth first search and traversal, Depth first search and traversal, Spanning trees, connected components and bi-connected components, Articulation points.
DYNAMIC PROGRAMMING: General method, applications - optimal binary search trees, 0/1 knapsack problem, All pairs shortest path problem, Travelling sales person problem, Reliability design.

UNIT-IV

BACKTRACKING: General method, Applications- n-queen problem, Sum of subsets problem, Graph coloring and Hamiltonian cycles.
BRANCH AND BOUND: General method, applications - travelling sales person problem, 0/1 knapsack problem- LC branch and bound solution, FIFO branch and bound solution.

UNIT-V

NP-HARD AND NP-COMPLETE PROBLEMS: Basic concepts, non-deterministic algorithms, NP-hard and NP-complete classes.

TEXT BOOKS:

1. Ellis Horowitz, Satraj Sahni, Rajasekharam (2007), Fundamentals of Computer Algorithms, 2nd edition, University Press, New Delhi.

REFERENCE BOOKS:

1. R. C. T. Lee, S. S. Tseng, R.C. Chang and T. Tsai (2006), Introduction to Design and Analysis of Algorithms A strategic approach, McGraw Hill, India.
2. Allen Weiss (2009), Data structures and Algorithm Analysis in C++, 2nd edition, Pearson education, New Delhi.
3. Aho, Ullman, Hopcroft (2009), Design and Analysis of algorithms, 2nd edition, Pearson education, New Delhi

BUSINESS ECONOMICS AND FINANCIAL ANALYSIS								
II Year IISemester								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4HS06	HSMC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100
COURSE OUTCOMES:								
At the end of the course, the student will								
- Understand the market dynamics namely, demand and supply, demand forecasting, elasticity of demand and supply, pricing methods and pricing in different market structures. - Gain an insight into how production function is carried out to achieve least cost combination of inputs and cost analysis. - Develop an understanding of - Analyse how capital budgeting decisions are carried out. - Understanding the framework for both manual and computerised accounting process - Know how to analyse and interpret the financial statements through ratio analysis.								
UNIT-I								
Introduction & Demand Analysis: Definition, Nature and Scope of Managerial Economics. Demand Analysis: Demand Determinants, Law of Demand and its exceptions. Elasticity of Demand: Definition, Types, Measurement and Significance of Elasticity of Demand. Demand Forecasting, Factors governing demand forecasting, methods of demand forecasting.								
UNIT-II								
Production & Cost Analysis: Production Function - Isoquants and Isocosts, MRTS, Least Cost Combination of Inputs, Cobb-Douglas Production function, Laws of Returns, Internal and External Economies of Scale. Cost Analysis: Cost concepts. Break-even Analysis (BEA)-Determination of Break-Even Point (simple problems) - Managerial Significance.								
UNIT-III								
Markets & New Economic Environment: Types of competition and Markets, Features of Perfect competition, Monopoly and Monopolistic Competition. Price-Output Determination in case of Perfect Competition and Monopoly. Pricing: Objectives and Policies of Pricing. Methods of Pricing. Business: Features and evaluation of different forms of Business Organisation: Sole Proprietorship, Partnership, Joint Stock Company, Public Enterprises and their types, New Economic Environment: Changing Business Environment in Post-liberalization scenario.								
UNIT-IV								
Capital Budgeting: Capital and its significance, Types of Capital, Estimation of Fixed and Working capital requirements, Methods and sources of raising capital - Trading Forecast, Capital Budget, Cash Budget. Capital Budgeting: features of capital budgeting proposals, Methods of Capital Budgeting: Payback Method, Accounting Rate of return (ARR) and Net Present Value Method (simple problems).								
UNIT-V								
Introduction to Financial Accounting & Financial Analysis: Accounting concepts and Conventions - Introduction IFRS - Double - Entry Book Keeping, Journal, Ledger, Trial Balance - Final Accounts (Tracing Account, Profit and Loss Account and Balance Sheet with simple adjustments). Financial Analysis: Analysis and Interpretation of Liquidity Ratios, Activity Ratios, and Capital structure Ratios and Profitability ratios. Du Pont Chart.								
TEXT BOOKS:								
- Varshney & Maheswari: Managerial Economics, Sultan Chand, 2009. - S.A. Siddiqui & A.S. Siddiqui, Managerial Economics and Financial Analysis, New Age international Publishers, Hyderabad 2013. M. Kasi Reddy & Saraswathi, Managerial Economics and Financial Analysis, PHI New Delhi, 2012.								
REFERENCE BOOKS:								
- Ambrish Gupta, Financial Accounting for Management, Pearson Education, New Delhi, 2012. H. Craig Peterson & W. Cris Lewis, Managerial Economics, Pearson, 2012. - Lipsey & Chrystel, Economics, Oxford University Press, 2012. - Domnick Salvatore: Managerial Economics In a Global Economy, Thomson, 2012. - Narayanaswamy: Financial Accounting - A Managerial Perspective, Pearson, 2012.								

6. S.N. Maheswari & S.K. Maheswari, Financial Accounting, Vikas, 2012.
7. Truet and Truet: Managerial Economics: Analysis, Problems and Cases, Wiley, 2012.
8. Dwivedi: Managerial Economics, Vikas, 2012.
9. Shailaja & Usha: MEFA, University Press, 2012.
10. Aryasri: Managerial Economics and Financial Analysis, TMH, 2012.
11. Vijay Kumar & Appa Rao, Managerial Economics & Financial Analysis, Cengage 2011.
12. J.V. Prabhakar Rao & P.V. Rao, Managerial Economics & Financial Analysis, Maruthi Publishers, 2011.

ELECTRONIC DEVICES AND DIGITAL LOGIC LAB

II B. TECH- II SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EC73	ESC	L	T	P	C	CIE	SEE	Total
		-	-	3	1.5	30	70	100

COURSE OBJECTIVES:

The course should enable the students to:

1. Design the semiconductor devices based circuits and analyze its parameters
2. Design of oscillators and amplifiers
3. Study of logic gates and verify their truth tables.
4. Implementation of the given Boolean function using logic gates.
5. Verification of K-Map.
6. Design the combinational and sequential circuits

COURSE OUTCOMES:

The course should enable the students to:

1. Calculate various parameters of devices from their characteristics.
2. Understanding of the fundamental concepts and techniques used in digital electronics.
3. To understand and examine the structure of various number systems and its application in digital design.
4. The ability to understand, analyze and design various combinational and sequential circuits.
5. Ability to identify basic requirements for a design application and propose a cost effective solution.
6. To develop skill to build, and troubleshoot digital circuits.

LIST OF EXPERIMENTS

WEEK-1

- a. Characteristics of PN Junction diode
- b. Characteristics of Zener diode.

WEEK-2

- a. Ripple Factor and Load Regulation of Half-wave Rectifier with and without filters
- b. Ripple Factor and Load Regulation of Full-wave Rectifier with and without filters

WEEK-3

Input and Output characteristics of Transistor in Common Emitter configuration

WEEK-4

a Drain and Transfer Characteristics of Junction Field Effect Transistor (JFET)

WEEK-5

Heartley and Colpitts Oscillator

WEEK-6

RC phase shift Oscillator

WEEK-7

- a. Realization of Logic gates using discrete components
- b. Binary Adders

WEEK-8

Subtractions and Comparators
WEEK-9
a. Multiplexers
b. Decoders
WEEK-10
Flip-Flops
WEEK-11
Counters
WEEK-12
Shift Registers
REFERENCE BOOKS:
1. R.P. Jain, "Modern digital Electronics", Tata McGraw Hill, 4th edition, 2009.
2. W.H. Gothmann, "Digital Electronics- An introduction to theory and practice", PHI, 2nd edition ,2006.
3. D.V. Hall, "Digital Circuits and Systems", Tata McGraw Hill, 1989
WEB REFERENCES:
1. https://www.youtube.com/watch?v=M0mx8S05v60&list=PLBlnK6fEyqRjMH3mWf6kwqiTbT798eA Om
2. https://www.youtube.com/watch?v=CeD2L6KbtVM&list=PL018B3BB2E6FE781D

OBJECT ORIENTED PROGRAMMING LAB**II B. TECH- II SEMESTER**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS11	PCC	L	T	P	C	CIE	SEE	Total
		-	-	3	1.5	30	70	100

COURSE OUTCOMES:

1. Implement Object Oriented programming concept using basic syntaxes of control Structures, strings and function for developing skills of logic building activity.
2. Understand the use of different exception handling mechanisms and concept of multithreading for robust and efficient application development.
3. Understand and implement concepts on file streams and operations in java programming for a given application programs.
4. Develop java application to interact with database by using relevant software component (JDBC Driver).

LIST OF EXPERIMENTS**WEEK-1: JAVA BASICS**

- a. Write a java program that prints all real solutions to the quadratic equation $ax^2+bx+c=0$. Read in a, b, c and use the quadratic formula.
- b. The Fibonacci sequence is defined by the following rule. The first two values in the sequence are 1 and 1. Every subsequent value is the sum of the two values preceding it. Write a java program that uses both recursive and non recursive functions.

WEEK-2: ARRAYS

- a. Write a java program to sort given list of integers in ascending order.
- b. Write a java program to multiply two given matrices.

WEEK-3: STRINGS

- a. Write a java program to check whether a given string is palindrome.
- b. Write a java program for sorting a given list of names in ascending order.

WEEK-4: OVERLOADING & OVERRIDING

- a. Write a java program to implement method overloading and constructor overloading.
- b. Write a java program to implement method overriding.

WEEK-5: INHERITANCE

Write a java program to create an abstract class named Shape that contains two integers and an empty method named print Area (). Provide three classes named Rectangle, Triangle and Circle such that each one of the classes extends the class Shape. Each one of the classes contains only the method print Area () that prints the area of the given shape.

WEEK-6: I/O STREAMS

- a. Write a java program that reads a file name from the user, and then displays information about whether the file exists, whether the file is readable, whether the file is writable, the type of file and the length of the file in bytes.
- b. Write a java program that displays the number of characters, lines and words in a textfile.

WEEK-7: EXCEPTION HANDLING

Write a program that creates a user interface to perform integer division. The user enters two numbers in the text fields, Num1 and Num2. The division of Num1 and Num2 is displayed in the Result field when the Divide button is clicked. If Num1 and Num2 were not integers, the program would throw a Number Format Exception. If Num2 were zero, the program would throw an Arithmetic Exception Display the exception in a message dialog box.

WEEK-8: MULTI THREADING

Write a java program that implements a multi-thread application that has three threads. First thread

generates random integer every 1 second and if the value is even, second thread computes the square of the number and prints. If the value is odd, the third thread will print the value of cube of the number.

WEEK-9 :COLLECTIONS

- a. Create a doubly linked list of elements.
- b. Delete a given element from the above list.
- c. Display the contents of the list after deletion.

WEEK-10: CONNECTING TO DATABASE

Write a java program that connects to a database using JDBC and does add, delete, modify and retrieve operations.

WEEK-11: APPLET

- a. Develop an applet that displays a simple message
- b. Develop an applet to perform integer addition. Pass Parameters from applet

WEEK-12 :EVENT HANDLING

- a. Write a java program that handles all mouse events .
- b. Write a java program to demonstrate the key event handlers.

TEXT BOOKS:

1. P. J. Deitel, H. M. Deitel, "Java for Programmers", Pearson Education, PHI, 4th Edition, 2007.
2. P. Radha Krishna, "Object Oriented Programming through Java", Universities Press, 2nd Edition, 2007
3. Bruce Eckel, "Thinking in Java", Pearson Education, 4th Edition, 2006.
4. Sachin Malhotra, Saurabh Chaudhary, "Programming in Java", Oxford University Press, 5th Edition, 2010.

GENDER SENSITIZATION

II B. TECH- II SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4MC02	MC	L	T	P	C	CIE	SEE	Total
		-	--	2	Nil	30	70	100

COURSE OUTCOMES:

- Students will have developed a better understanding of important issues related to gender in contemporary India.
- Students will be sensitized to basic dimensions of the biological, sociological, psychological and legal aspects of gender. This will be achieved through discussion of materials derived from research, facts, everyday life, literature and film.
- Students will attain a finer grasp of how gender discrimination works in our society and how to counter it.
- Students will acquire insight into the gendered division of labour and its relation to politics and economics.
- Men and women students and professionals will be better equipped to work and live together as equals.
- Students will develop a sense of appreciation of women in all walks of life. Through providing accounts of studies and movements as well as the new laws that provide protection and relief to women, the textbook will empower students to understand and respond to gender violence.

UNIT-I

UNDERSTANDING GENDER: Gender: Why Should We Study It? (*Towards a World of Equals*: Unit -1)
Socialization: Making Women, Making Men (*Towards a World of Equals*: Unit -2) Introduction. Preparing for Womanhood. Growing up Male. First lessons in Caste. Different Masculinities.

UNIT-II

GENDER AND BIOLOGY: Missing Women: Sex Selection and Its Consequences (*Towards a World of Equals*: Unit -4) Declining Sex Ratio. Demographic Consequences. **Gender Spectrum:** Beyond the Binary (*Towards a World of Equals*: Unit -10) Two or Many? Struggles with Discrimination

UNIT-III

GENDER AND LABOUR: Housework: the Invisible Labour (*Towards a World of Equals*: Unit -3) "My Mother doesn't Work." "Share the Load."
Women's Work: Its Politics and Economics (*Towards a World of Equals*: Unit -7) Fact and Fiction. Unrecognized and Unaccounted work. Additional Reading: Wages and Conditions of Work.

UNIT-IV

ISSUES OF VIOLENCE: Sexual Harassment: Say No! (*Towards a World of Equals*: Unit -6) Sexual Harassment, not Eve-teasing- Coping with Everyday Harassment- Further Reading: "Chupulu".
Domestic Violence: Speaking Out (*Towards a World of Equals*: Unit -8) Is Home a Safe Place? -When Women Unite [Film]. Rebuilding Lives. Additional Reading: New Forums for Justice.
 Thinking about Sexual Violence (*Towards a World of Equals*: Unit -11) Blaming the Victim- "I Fought for my Life..." - Additional Reading: The Caste Violence.

UNIT-V

GENDER: CO – EXISTENCE: Just Relationships: Being Together as Equals (*Towards a World of Equals*: Unit -12) Mary Kom and Onler. Love and Acid just do not Mix. Love Letters. Mothers and Fathers Additional Reading: Rosa Parks-The Brave Heart

TEXT BOOKS:

- All the five Units in the Textbook, "***Towards a World of Equals: A Bilingual Textbook on Gender***" written by A. Suneetha, Uma Bhargubanda, Duggirala Vasanta, Rama Melkote, Vasudha Nagaraj, Asma Rasheed, Gogu Shyamala, Deepa Sreenivas and Susie Tharu and published by **Telugu Akademi, Hyderabad**, Telangana State in the year **2015**.

REFERENCE BOOKS:

1. Menon, Nivedita. Seeing like a Feminist. New Delhi: Zubaan-Penguin Books, 2012
2. Abdulali Sohaila. "*I Fought For My Life...and Won.*" Available online at: <http://www.thealternative.in/lifestyle/i-fought-for-my-lifeand-won-sohaila-abdulal/>

III B.TECH I SEMESTER SYLLABUS

WEB TECHNOLOGIES

III B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4IT04	PCC	L	T	P	C	CIE	SEE	Total
		3	1	-	4	30	70	100

COURSE OBJECTIVES:

To learn

1. Teach students the basics of server side scripting using PHP
2. Explain web application development procedures
3. Impart servlet technology for writing business logic
4. Facilitate students to connect to databases using JDBC
5. Familiarize various concepts of application development using JSP

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Create web pages using PHP
2. Identify the difference between the HTML PHP and XML documents.
3. Identify the engineering structural design of XML and parse tree
4. Understand the concept of JAVA SCRIPT.
5. Identify the difference and choose between the JSP and Servlet.

UNIT-I

XHTML: Basic Syntax, Standard structure, Basic Text markup, Images, Hypertext links, Lists, Tables, Frames

XML: Introduction to XML, Defining XML tags, their attributes and values, Document type definition, XML Schemas, Document Object model.

UNIT-II

Introduction to PHP: Declaring variables, data types, arrays, strings, operations, expressions, control structures, functions, Reading data from web form controls like Text Boxes, radio buttons, lists etc., Handling File Uploads, Connecting to database (My SQL as reference), executing simple queries, handling results, Handling sessions and cookies.

UNIT-III

Client side Scripting: Introduction to JavaScript: JavaScript language – declaring variables, scope of variables functions, event handlers (on click, on submit etc.), Document Object Model, Form validations.

UNIT-IV

Introduction to Servlets: Common Gateway Interface (CGI), Lifecycle of a Servlets, deploying a Servlets, The Servlets API, Reading Servlets parameters, Reading initialization parameters, Handling Http Request & Responses, Using Cookies and sessions, connecting to a database using JDBC.

UNIT-V

Introduction to JSP: The Anatomy of a JSP Page, JSP Processing, Declarations, Directives, Expressions, Code Snippets, implicit objects, Using Beans in JSP Pages, Using Cookies and session tracking, connecting to database in JSP.

TEXT BOOKS:

1. Web Technologies, Uttam K Roy, Oxford University Press
2. The Complete Reference PHP – Steven Holzner, Tata McGraw-Hill

REFERENCE BOOKS:

1. Web Programming, building internet applications, Chris Bates 2nd edition, Wiley Dremtech
2. Java Server Pages – Hans Bergsten, SPD O'Reilly
3. Java Script, D.Flanagan, O'Reilly, SPD.
4. Beginning Web Programming-Jon Duckett WROX.
5. Programming world wide web, R.W. Sebesta. Fourth Edition, Pearson.
6. Internet and World Wide Web – How to program, Dietel and Nieto, Pearson.

FORMAL LANGUAGE AND AUTOMATA THEORY

III B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS12	PCC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OVERVIEW:

Formal languages and automata theory deals with the concepts of automata, formal languages, grammar, computability and decidability. The reasons to study Formal Languages and Automata Theory are Automata Theory provides a simple, elegant view of the complex machine that we call a computer. Automata Theory possesses a high degree of permanence and stability, in contrast with the ever-changing paradigms of the technology, development, and management of computer systems. Further, parts of the Automata theory have direct bearing on practice, such as Automata on circuit design, compiler design, and search algorithms; Formal Languages and Grammars on compiler design; and Complexity on cryptography and optimization problems in manufacturing, business, and management. Last, but not least, research oriented students will make good use of the Automata theory studied in this course.

COURSE OBJECTIVES:

1. To present the theory of finite automata as the first step towards learning advanced topics such as compiler design.
2. To discuss the applications of finite automata towards text processing.
3. To develop an understanding of Regular expressions and context free grammars and how these concepts are used in lexical analyzer.
4. To develop an understanding of finite automata through Turing machines

COURSE OUTCOMES:

At the end of the course students will be able to:

1. To use basic concepts of formal languages of finite automata techniques
2. To Design Finite Automata's for different Regular Expressions and Languages
3. To Construct context free grammar for various languages
4. To solve various problems of applying normal form techniques, push down automata and Turing Machines.

UNIT-I	INTRODUCTION:FINITE AUTOMATA	
FINITE AUTOMATA (FA): Introduction, Deterministic Finite Automata (DFA) -Formal definition, simpler notations (state transition diagram, transition table), language of a DFA. Nondeterministic Finite Automata (NFA)- Definition of NFA, language of an NFA, Equivalence of Deterministic and Nondeterministic Finite Automata, Applications of Finite Automata, Finite Automata with Epsilon Transitions, Eliminating Epsilon transitions, Minimization of Deterministic Finite Automata, Finite automata with output (Moore and Mealy machines) and Inter conversion.		
UNIT-II	REGULAR EXPRESSIONS (RE)	
REGULAR EXPRESSIONS (RE): Introduction, Identities of Regular Expressions, Finite Automata and Regular Expressions- Converting from DFA to Regular Expressions, Converting Regular Expressions to Automata, applications of Regular Expressions. REGULAR GRAMMARS: Definition, regular grammars and FA, FA for regular grammar, Regular grammar for FA. Proving languages to be non-regular - Pumping lemma, applications, Closure properties of regular languages.		
UNIT-III	CONTEXT FREE GRAMMER (CFG):	

CONTEXT FREE GRAMMER (CFG): Derivation Trees, Sentential Forms, Rightmost and Leftmost derivations of Strings. AMBIGUITY IN CFG: Elimination of ambiguity, Minimization of CFG's, CNF, GNF, Pumping Lemma for CFL's,		
UNIT-IV	PUSHDOWN AUTOMATA	
PUSHDOWN AUTOMATA: Definition, Model, Acceptance of CFL, Acceptance by Final State and Acceptance by Empty stack and its Equivalence, Equivalence of CFG and PDA. TURING MACHINES (TM): Formal definition and behavior, Languages of a TM, TM as accepters, and TM as a computer of integer functions, Types of TM's.		
UNIT-V	RECURSIVELY ENUMERABLE LANGUAGES	
RECURSIVELY ENUMERABLE LANGUAGES (REL): Properties of Recursively Enumerable Languages, Universal Turing machine, The Halting problem, Context sensitive language and linear bounded automata (LBA), Chomsky hierarchy, Decidability, Post's correspondence problem (PCP), Undecidability of PCP.		
TEXT BOOKS:		
1. John E. Hopcroft, Rajeev Motwani, Jeffrey D. Ullman (2007), Introduction to Automata Theory Languages and Computation, 3rd Edition, Pearson Education, India 2. Theory of Computer Science (Automata, Languages and Computation), 2nd Edition by K. L. P. Mishra and N. Chandrasekharan, PHI.		
REFERENCE BOOKS:		
1. Finite Automata and Formal Languages: A Simple Approach, A. M. Padma Reddy, Pearson Education India. 2. An Introduction to FORMAL LANGUAGES and AUTOMATA (Fifth Edition), PETER LINZ.		
WEB REFERENCES:		
1. https://nptel.ac.in/courses/111103016/ 2. https://www.ics.uci.edu/~goodrich/teach/cs162/notes/ 3. https://www.geeksforgeeks.org/theory-of-computation-automata-tutorials/		
E-Text Books:		
1. https://mcdtu.files.wordpress.com/2017/03/introduction-to-automata-theory.pdf 2. https://www.amazon.in/Formal-Languages-Automata-Theory-agpal/dp/019807106X		
MOOC COURSE		
1. https://swayam.gov.in/course/3831-theory-of-computation 2. https://www.udemy.com/theory-of-automata/ 3. https://nptel.ac.in/courses/111103016/		

OPERATING SYSTEMS

III B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS13	PCC	L	T	P	C	CIE	SEE	Total
		3	1	-	4	30	70	100

COURSE OBJECTIVES:

To learn

1. To explain main components of OS and their working
2. To familiarize the operations performed by OS as a resource Manager
3. To impart various scheduling policies of OS
4. To teach the different memory management techniques.

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Identify and analyze the different structures and services of operating system.
2. Compare various algorithms used for OS services with respect to defined/chosen criteria.
3. Solve the resource allocation and sharing problems.
4. Assess different methods to solve OS problems.
5. Analyze the memory management approaches of operating systems.

UNIT-I

OPERATING SYSTEMS OVERVIEW: Introduction, operating system operations, process management, memory management, storage management, protection and security, distributed systems.

OPERATING SYSTEMS STRUCTURES: Operating system services and systems calls, system programs, operating system structure, operating systems generations

UNIT-II

PROCESS MANAGEMENT: Process concepts, process state, process control block, scheduling queues, process scheduling, multithreaded programming, threads in UNIX, comparison of UNIX and windows.

CONCURRENCY AND SYNCHRONIZATION: Process synchronization, critical section problem, Peterson's solution, synchronization hardware, semaphores, classic problems of synchronization, readers and writers problem, dining philosophers problem, monitors, synchronization examples(Solaris), atomic transactions, Comparison of UNIX and windows.

UNIT-III

DEADLOCKS: System model, deadlock characterization, deadlock prevention, detection and avoidance, recovery from deadlock banker's algorithm.

MEMORY MANAGEMENT: Swapping, contiguous memory allocation, paging, structure of the page table, segmentation, virtual memory, demand paging, page-replacement algorithms, allocation of frames, thrashing, case study - UNIX.

UNIT-IV

FILE SYSTEM: Concept of a file, access methods, directory structure, file system mounting, file sharing, protection. **File system implementation:** file system structure, file system implementation, directory implementation, allocation methods, free-space management, efficiency and performance, comparison of UNIX and windows

UNIT-V

MASS STORAGE STRUCTURE: overview of mass storage structure, disk structure, disk attachment, disk scheduling algorithms, swap space management, stable storage implementation, tertiary storage structure.

I/O: Hardware, application I/O interface, kernel I/O subsystem, transforming I/O requests to hardware operations, streams, performance

TEXT BOOKS:

1. Abraham Silberschatz, Peter Baer Galvin, Greg Gagne (2006), Operating System Principles, 7th edition, Wiley India Private Limited, New Delhi.

REFERENCE BOOKS:

1. Stallings (2006), Operating Systems, Internals and Design Principles, 5th edition, Pearson Education, India.
2. Andrew S. Tanenbaum (2007), Modern Operating Systems, 2nd edition, Prentice Hall of India, India
3. Deitel & Deitel (2008), Operating systems, 3rd edition, Pearson Education, India.

ARTIFICIAL INTELLIGENCE

(Professional Elective – 1)

III B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS21	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

To learn

1. The difference between optimal reasoning vs human like reasoning
2. Understand the notions of state space representation, exhaustive search, heuristic search along with the time and space complexities
3. Different knowledge representation techniques
4. Understand the applications of AI: namely Game Playing, Theorem Proving, Expert Systems, Machine Learning and Natural Language Processing

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Possess the ability to formulate an efficient problem space for a problem expressed in English.
2. Possess the ability to select a search algorithm for a problem and characterize its time and space complexities.
3. Possess the skill for representing knowledge using the appropriate technique
4. Possess the ability to apply AI techniques to solve problems of Game Playing, Expert Systems, Machine Learning and Natural Language Processing

UNIT-I

Introduction: History, Intelligent Systems, Foundations of AI, Sub areas of AI, Applications. Problem Solving – State-Space Search and Control Strategies: Introduction, General Problem Solving, Characteristics of Problem, Exhaustive Searches, Heuristic Search Techniques, Iterative-Deepening A*, Constraint Satisfaction, Game Playing, Bounded Look-ahead Strategy and use of Evaluation Functions, Alpha-Beta Pruning

UNIT-II

Logic Concepts and Logic Programming: Introduction, Propositional Calculus, Propositional Logic, Natural Deduction System, Axiomatic System, Semantic Tableau System in Propositional Logic, Resolution Refutation in Propositional Logic, Predicate Logic, Logic Programming.

Knowledge Representation: Introduction, Approaches to Knowledge Representation, Knowledge Representation using Semantic Network, Extended Semantic Networks for KR, Knowledge Representation using Frames.

UNIT-III

Expert System and Applications: Introduction, Phases in Building Expert Systems, Expert System Architecture, Expert Systems Vs Traditional Systems, Truth Maintenance Systems, Application of Expert Systems, List of Shells and Tools.

Uncertainty Measure – Probability Theory: Introduction, Probability Theory, Bayesian Belief Networks, Certainty Factor Theory, Dempster - Shafer Theory.

UNIT-IV

Machine-Learning Paradigms: Introduction, Machine Learning Systems, Supervised and Unsupervised Learning, Inductive Learning, Learning Decision Trees, Deductive Learning, Clustering, Support Vector Machines.

Artificial Neural Networks: Introduction, Artificial Neural Networks, Single-Layer Feed Forward Networks, Multi-Layer Feed-Forward Networks, Radial-Basis Function Networks, Design Issues of Artificial Neural Networks, Recurrent Networks.

UNIT-V		
Advanced Knowledge Representation Techniques: Case Grammars, Semantic Web Natural Language Processing: Introduction, Sentence Analysis Phases, Grammars and Parsers, Types of Parsers, Semantic Analysis, and Universal Networking Knowledge.		
TEXT BOOKS:		
1. Saroj Kaushik. Artificial Intelligence. Cengage Learning. 2011 2. Russell, Norvig: Artificial intelligence, A Modern Approach, Pearson Education, Second Edition. 2004		
REFERENCE BOOKS:		
1. Rich, Knight, Nair: Artificial intelligence, Tata McGraw Hill, Third Edition 2009. 2. Introduction to Artificial Intelligence by Eugene Charniak, Pearson. 3. Introduction to Artificial Intelligence and expert systems Dan W.Patterson. PHI. 4. Artificial Intelligence by George Fluger Pearson fifth edition.		

PRINCIPLES OF PROGRAMMING LANGUAGE

(Professional Elective – 1)

III B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4IT06	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

To learn

1. To familiarize various programming methodologies
2. To explain merits and demerits of various programming patterns
3. To gain knowledge of new programming practices
4. To familiarize various debugging techniques.

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Describe syntax and semantics of programming language design and implementation.
2. Compare programming languages and assess programming languages.
3. Use formal description for a programming language and assess using evaluators.
4. Apply different programming paradigms: analyze the principles of imperative, object-oriented, functional and logic programming.
5. Design a new programming language in principle.

UNIT-I

Preliminary Concepts: concepts of programming languages, Programming domains, Language Evaluation Criteria, influences on Language design, Language categories, Programming Paradigms – Imperative, Object Oriented, functional Programming, Logic Programming, Programming Language Implementation – Compilation and Virtual Machines, programming environments.

UNIT-II

Syntax and Semantics: General Problem of describing Syntax and Semantics, formal methods of describing syntax - BNF, EBNF for common programming languages features, parse trees, ambiguous grammars, attribute grammars, denotation semantics and axiomatic semantics for common programming language features.

UNIT-III

Data types: Introduction, primitive, character, user defined, array, associative, record, union, pointer and reference types, design and implementation uses related to these types, Names, Variable, concept of binding, type checking, strong typing, type compatibility, named constants, variable initialization.

UNIT-IV

Expressions and Statements: Arithmetic relational and Boolean expressions, Short circuit evaluation mixed mode assignment, Assignment Statements, Control Structures – Statement Level, Compound Statements, Selection, Iteration, Unconditional Statements, guarded commands.

UNIT-V

Subprograms and Blocks: Fundamentals of sub-programs, Scope and lifetime of variable, static and dynamic scope, Design issues of subprograms and operations, local referencing environments, parameter passing methods, overloaded sub-programs, generic sub-programs, parameters that are sub-program names, design issues for functions user defined overloaded operators, co routines.

TEXT BOOKS:

1. Concepts of Programming Languages Robert .W. Sebesta 6/e, Pearson Education.
2. Programming Languages – Louden, Second Edition, Thomson.

REFERENCE BOOKS:

1. Programming languages –Ghezzi, 3/e, John Wiley
2. Programming Languages Design and Implementation – Pratt and Zelkowitz, Fourth Edition
3. PHI/Pearson Education
4. Programming languages –Watt, Wiley Dreamtech
5. LISP Patric Henry Winston and Paul Horn Pearson Education.
6. Programming in PROLOG Clocksin, Springer

DISTRIBUTED DATABASES

(Professional Elective – 1)

III B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS22	PEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES

1. To understand the theoretical and practical aspects of the database technologies.
2. To understand the need for distributed database technology to tackle deficiencies of the centralized database systems.
3. To introduce the concepts and techniques of distributed database including principles, architectures, design, implementation and major domain of application.

COURSE OUTCOMES

At the end of the course, student will be able to:

1. Identify the introductory distributed database concepts and its structures.
2. Describe terms related to distributed object database design and management.
3. Produce the transaction management and query processing techniques in DDBMS.
4. Relate the importance and application of emerging database technology

UNIT-I INTRODUCTION

Features of Distributed versus Centralized Databases, Principles of Distributed Databases, Levels Of Distribution Transparency, Reference Architecture for Distributed Databases, Types of Data Fragmentation, Integrity Constraints in Distributed Databases, Distributed Database Design

UNIT-II QUERY PROCESSING

Translation of Global Queries to Fragment Queries, Equivalence transformations for Queries, Transforming Global Queries into Fragment Queries, Distributed Grouping and Aggregate Function Evaluation, Parametric Queries.

Optimization of Access Strategies, A Framework for Query Optimization, Join Queries, General Queries

UNIT-III TRANSACTION MANAGEMENT AND CONCURRENCY CONTROL

The Management of Distributed Transactions, A Framework for Transaction Management, Supporting Atomicity of Distributed Transactions, Concurrency Control for Distributed Transactions, Architectural Aspects of Distributed Transactions

Concurrency Control, Foundation of Distributed Concurrency Control, Distributed Deadlocks, Concurrency Control based on Timestamps, Optimistic Methods for Distributed Concurrency Control.

UNIT-IV RELIABILITY AND SECURITY IN THE DISTRIBUTED DATABASES

Reliability, Basic Concepts, Non blocking Commitment Protocols, Reliability and concurrency Control, Determining a Consistent View of the Network, Detection and Resolution of Inconsistency, Checkpoints and Cold Restart, Distributed Database Administration, Catalog Management in Distributed Databases, Authorization and Protection

UNIT-V DISTRIBUTED OBJECT DATABASE MANAGEMENT SYSTEMS

Architectural Issues, Alternative Client/Server Architectures, Cache Consistency, Object Management, Object Identifier Management, Pointer Swizzling, Object Migration, Distributed Object Storage, Object Query Processing, Object Query Processor Architectures, Query Processing Issues, Query Execution, Transaction Management, Transaction Management in Object DBMSs, Transactions as Objects

TEXT BOOKS:

1. Distributed Databases - Principles and Systems; Stefano Ceri; Guiseppe Pelagatti; Tata McGraw Hill; 1985.
2. Fundamental of Database Systems; Elmasri & Navathe; Pearson Education; Asia Database System Concepts; Korth & Sudarshan; TMH
3. Principles of Distributed Database Systems; M. Tamer Özsu; and Patrick Valduriez Prentice Hall

REFERENCE BOOKS:

1. Data Base Management System; Leon & Leon; Vikas Publications
2. Introduction to Database Systems; Bipin C Desai; Galgotia

WEB REFERENCES:

1. https://www.tutorialspoint.com/distributed_dbms/distributed_dbms_databases.htm
2. <https://www.geeksforgeeks.org/distributed-database-system/>

E-TEXT BOOKS:

1. <https://biblio.com.au/distributed-databases-by-ceri-stefano-pelagatti/work/245682>
2. <https://www.amazon.in/Distributed-Database-Systems-Chhanda-Ray-ebook/dp/B009NEMZ0W>

MOOC COURSE

1. <https://nptel.ac.in/syllabus/106106107/>
2. <https://www.coursera.org/learn/distributed-database>

Human Computer Interaction

(Professional Elective – 1)

III B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS23	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

To learn

1. The human components functions.
2. The Computer components functions.
3. The Interaction between the human and computer components.
4. Paradigms
5. Interaction design basics
6. HCI in the software process
7. Design rules
8. Implementation supports
9. Evaluation techniques

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Explain the human components functions regarding interaction with computer
2. Explain Computer components functions regarding interaction with human
3. Demonstrate Understanding of Interaction between the human and computer components.
4. Use Paradigms
5. Implement Interaction design basics
6. Use HCI in the software process

UNIT-I	Introduction	
Introduction: Importance of user Interface – definition, importance of good design, Benefits of good design, A brief history of Screen design, The graphical user interface – popularity of graphics, the concept of direct manipulation, graphical system, Characteristics, Web user – Interface popularity, characteristics-Principles of user interface.		
UNIT-II	Design process	
Design process – Human interaction with computers, importance of human characteristics human consideration, Human interaction speeds, and understanding business junctions. Screen Designing : Design goals – Screen planning and purpose, organizing screen elements, ordering of screen data and content – screen navigation and flow – Visually pleasing composition – amount of information – focus and emphasis – presentation information simply and meaningfully – information retrieval on web – statistical graphics – Technological consideration in interface design.		
UNIT-III	Screen Designing	
Screen Designing: Windows – New and Navigation schemes selection of window, selection of devices based and screen based controls. Components – text and messages, Icons and increases – Multimedia, colors, uses problems, choosing colors.		
UNIT-IV	Windows& Components	
Windows & Components: Software tools – Specification methods, interface – Building Tools.		
UNIT-V	Software tools	

Software tools: Interaction Devices – Keyboard and function keys – pointing devices – speech recognition digitization and generation – image and video displays – drivers.

TEXT BOOKS:

1. The essential guide to user interface design, Wilbert O Galitz, Wiley Dreamtech.
2. Designing the user interface. 3rd Edition Ben Shneidermann , Pearson Education Asia.

REFERENCE BOOKS:

1. Human – Computer Interaction. ALAN DIX, JANET FINCAY, GRE GORYD, ABOWD, RUSSELL BEALG, PEARSON.
2. Interaction Design PRECE, ROGERS, SHARPS. Wiley Dreamtech,
3. User Interface Design, Soren Lauesen , Pearson Education.

WEB TECHNOLOGIES LAB

III B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4IT05	PCC	L	T	P	C	CIE	SEE	Total
		-	-	3	1.5	30	70	100

COURSE OBJECTIVES:

To learn

1. Teach students the basics of server side scripting using PHP
2. Explain web application development procedures
3. Impart Servlets technology for writing business logic
4. Facilitate students to connect to databases using JDBC
5. Familiarize various concepts of application development using JSP

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Create web pages using PHP
2. Identify the difference between the HTML PHP and XML documents.
3. Identify the engineering structural design of XML and parse tree
4. Analyze the difference between and PHP and XML.
5. Understand the concept of JAVA SCRIPTS.
6. Identify the difference between the JSP and Servlets.
7. Design web application using MVC architecture
8. Understand the JSP and Servlet concepts.

SYLLABUS

Note:

1. Apache, MySQL and PHP for the Lab Experiments. Though not mandatory, encourage the use of Eclipse platform wherever applicable
 2. The list suggests the minimum program set. Hence, the concerned staff is requested to add more problems to the list as needed
1. Install the following on the local machine
 - i. Apache Web Server (if not installed)
 - ii. Tomcat Application Server locally
 - iii. Install MySQL (if not installed)
 - iv. Install PHP and configure it to work with Apache web server and MySQL (if not already configured)
 2. Write an HTML page including any required Javascript that takes a number from one text field in the range of 0 to 999 and shows it in range of 0 to 999 and shows it in another text field in words. If the number is out of range, it should show "out of range" and if it is not a number, it should show "not a number" message in the result box.
 3. Write an HTML page that has one input, which can take multi-line text and a submit button. Once the user clicks the submit button, it should show the number of characters, words and lines in the text entered using an alert message. Words are separated with while space and lines are separated with new line character.

4. Write an HTML page that contains a selection box with a list of 5 countries. When the user selects a country, its capital should be printed next to the list. Add CSS to customize the properties of the font of the capital (color, bold and font size).
 5. Create an XML document that contains 10 users information. Write a Java program, which takes User Id as input and returns the user details by taking the user information from the XML document using (a) DOM Parser and (b) SAX parser.
 6. Implement the following web applications using (a) PHP, (b) Servlets and (c) JSP:
 - i. A user validation web application, where the user submits the login name and password to the server. The name and password are checked against the data already available in Database and if the data matches, a successful login page is returned. Otherwise a failure message is shown to the user.
 - ii. Modify the above program to use an xml file instead of database.
 - iii. Modify the above program to use AJAX to show the result on the same page below the submit button.
 - iv. A simple calculator web application that takes two numbers and an operator (+, -, /, * and %) from an HTML page and returns the result page with the operation performed on the operands.
 - v. Modify the above program such that it stores each query in a database and checks the database first for the result. If the query is already available in the DB, or it computes the result and returns it after storing the new query and result on DB.
 - vi. A web application takes a name as input and on submit it shows a hello<name>page where<name> is taken from the request. It shows the start time at the right top corner of the page and provides a logout button. On clicking this button, it should show a logout page with Thank You <name> message with the duration of usage (hint: Use session to store name and time).
 - vii. A web application that takes name and age from an HTML page. If the age is less than 18, it should send a page with "Hello <name>, you are not authorized to visit this site" message, where <name> should be replaced with the entered name. Otherwise it should send "Welcome <name> to the site" message.
 - viii. A web application for implementation: The user is first served a login page which takes user's name and password. After submitting the details the server checks these values against the data from a database and takes the following decisions. If name and password doesn't match, then serves "password mismatch" page. If name is not found in the database, serves a registration page, where user's full name is asked and on submitting the full name, it stores, the login name, password and full name in the database (hint: use session for storing the submitted login name and password)
 - ix. A web application that lists all cookies stored in the browser on clicking "List Cookies" button.
- Add cookies if necessary.

TEXT BOOKS:

1. Web Technologies, Uttam K Roy, Oxford University Press
2. The Complete Reference PHP – Steven Holzner, Tata McGraw-Hill

REFERENCE BOOKS:

1. Web Programming, building internet applications, Chris Bates 2nd edition, Wiley Dremtech
2. Java Server Pages – Hans Bergsten, SPD O'Reilly
3. Java Script, D.Flanagan, O'Reilly, SPD.
4. Beginning Web Programming-Jon Duckett WROX.
5. Programming world wide web, R.W. Sebesta. Fourth Edition, Pearson.
6. Internet and World Wide Web – How to program, Dietel and Nieto, Pearson

ADVANCED ENGLISH COMMUNICATION SKILLS LAB

III B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4HS07	HSMC	L	T	P	C	CIE	SEE	Total
		-	-	3	1.5	30	70	100

INTRODUCTION

The introduction of the Advanced Communication Skills Lab is considered essential at 3rd year level. At this stage, the students need to prepare themselves for their careers which may require them to listen to, read, speak and write in English both for their professional and interpersonal communication in the globalised context.

The proposed course is a laboratory course to enable students to use 'good' English and perform the following:

- Gather ideas and information, to organize ideas relevantly and coherently.
- Engage in debates.
- Participate in group discussions.
- Face interviews.
- Write project/research reports/technical reports.
- Make oral presentations.
- Write formal letters.
- Transfer information from non-verbal to verbal texts and vice versa.
- To take part in social and professional communication.

COURSE OUTCOMES

By the end of the course students will be able to

1. Organize the ideas coherently from the text.
2. Participate in debates, group discussions.
3. Write project/research reports/technical reports/formal letters.
4. Make oral presentations.

SYLLABUS:

The following course content to conduct the activities is prescribed for the Advanced Communication Skills (ACS) Lab:

Week -1

Activities on Fundamentals of Inter-personal Communication

Starting a conversation - responding appropriately and relevantly - using the right body language - Role Play in different situations & Discourse Skills- using visuals.

Week -2

Activities on Building Vocabulary

Synonyms and antonyms, word roots, one-word substitutes, prefixes and suffixes, study of word origin, business vocabulary, analogy, idioms and phrases, collocations & usage of vocabulary

Week -3

Activities on Reading Comprehension

General Vs Local Comprehension, Reading for facts, guessing meanings from context, Scanning and Skimming.

Week -4

Activities on Reading for Specific Purposes

Inferring meaning, Critical reading & Effective goggling.

Week-5

Activities on Writing Skills- Technical Reports

Structure and presentation of different types of writing - letter writing/ Resume writing/ e-correspondence

Week-6

Activities on Writing Skills

Technical report writing/ Portfolio writing - planning for writing - improving one's writing.

Week- 7

Activities on Presentation Skills

Oral presentations (individual and group) through JAM sessions and Seminars.

Week- 8

Activities on Presentation Skills Using ICT

PPTs and written presentations through posters/ projects/ reports/ e-mails/ assignments etc.

Week- 9

Activities on Group Discussion

Dynamics of group discussion, intervention, summarizing, modulation of voice, body language, relevance, fluency and organization of ideas and rubrics for evaluation- Concept and process

Week-10

Interview Skills

Pre-interview planning, opening strategies, answering strategies, interview through tele-conference & video-conferencing and Mock Interviews.

BOOKS RECOMMENDED:

1. Raman, M & Sharma, S. (2009). Technical Communication. Oxford University Press.
2. Rani. S. (2011). Advanced Communication Skills Laboratory Manual. Pearson Education.
3. Anderson, V. (2007). Technical Communication. Cengage Learning pvt. Ltd.
4. Kelly M. Quintanilla & Shawn T. Wahl. (2011). Business and Professional Communication: Keys for Workplace Excellence. Sage South Asia Edition. Sage Publications.
5. Stev. D & David T. Mc Mahan. (2012). The Basics of Communication: A Relational Perspective. Sage South Asia Edition. Sage Publications.
6. Mc Murrey. D & Buckley. J. (2012). Handbook for Technical Communication Cengage Learning.
7. Sen. L. (2009). Communication Skills. PHI Learning Pvt Ltd.
8. Vishvamohan, A. (2009). English for Technical Communication for Engineering Students. Tata Mc Graw Hill.
9. Books on TOFEL/ GRE/ GMAT/ CAT/ IELTS by Barron's/ DELTA/ Cambridge University Press.
10. Tomalin, B & Thomas, B. (2009). International English for Call Centers. Macmillan Publishers

CONSTITUTION OF INDIA

III B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4MC03	MC	L	T	P	C	CIE	SEE	Total
		2	-	-	-	30	70	100

COURSE OBJECTIVES:

Students will be able to:

1. Understand the need for constitution
2. Appreciate the fundamental duties and rights of the citizens of India.
3. Explain the role and amendments of constitution in a democratic society.
4. Describe the directive principles of state policy and their significance.
5. List the key features of the constitution, union government and state government.

COURSE OUTCOMES:

Students will be able to:

1. Create awareness about the constitutional values and objectives written in the Indian constitution.
2. List fundamental rights and fundamental duties of Indian citizens.
3. Identify the division of legislative, executive and financial powers between the union and state governments.
4. Understand the working of Indian democracy ,its institutions and processes at the local,state and union levels.
5. Explain the functions and responsibilities of election commission of india and union public service commission.

UNIT-I

History of Making of the Indian Constitution: Introduction to the constitution of India, the making of the constitution and salient features of the constitution.

UNIT-II

Philosophy of the Indian Constitution: Preamble Salient Features, Contours of Constitutional Rights & Duties: Fundamental Rights, Right to Equality, Right to Freedom, Right against Exploitation, Right to Freedom of Religion, Cultural and Educational Rights, Right to Constitutional Remedies, Directive Principles of State Policy, Fundamental Duties, Amendment of the constitutional powers and procedures.

UNIT-III

Union Government: Union Government, Union Legislature (Parliament), Lok Sabha and Rajya Sabha (with powers and functions), president of India (with powers and functions), Prime minister of India (With powers and functions), Union judiciary (Supreme court), Jurisdiction of the supreme court.

UNIT-IV

State Government: State Government, State legislature (Legislative Assembly/ Vidhan Sabha, Legislative council/ Vidhan parishad), powers and functions of the state legislature, State executive, Governor of the state (with powers and functions), The chief Minister of the state (with powers and functions), State Judiciary (High courts)

UNIT-V

Election Commission: Election Commission: Role and Functioning, Chief Election Commissioner and Election Commissioners, State Election Commission: Role and Functioning, Institute and Bodies for the welfare of SC/ST/OBC and women.

TEXT BOOKS:

1. M.V. Pylee, Indian Constitution Durga Das Basu, Human Rights in Constitutional Law, Prentice – Hall of India Pvt. Ltd. New Delhi
2. Noorani, A.G., (South Asia Human Rights Documentation Centre), Challenges to Civil Right), Challenges to Civil Rights Guarantees in India, Oxford University Press 2012
3. The constitution of India, P.M.Bakshi, Universal Law Publishing Co.,
4. The Constitution of India, 1950 (Bare Act), GovernmentPublication.
5. Dr. S. N. Busi, Dr. B. R. Ambedkar framing of Indian Constitution, 1st Edition,2015.

REFERENCE BOOKS:

1. M. P. Jain, Indian Constitution Law, 7th Edn., Lexis Nexis,2014.
2. D.D. Basu, Introduction to the Constitution of India, Lexis Nexis,2015.
3. Indian constitution at work, NCERT
4. SubashKashyap, Indian Constitution, National Book Trust
5. J.A. Siwach, Dynamics of Indian Government & Politics
6. D.C. Gupta, Indian Government and Politics
7. H.M.Sreevai, Constitutional Law of India, 4th edition in 3 volumes (Universal Law Publication)
8. J.C. Johari, Indian Government and Politics Hans
9. J. Raj Indian Government and Politics

E-RESOURCES:

1. nptel.ac.in/courses/109104074/8
2. nptel.ac.in/courses/109104045/
3. nptel.ac.in/courses/101104065/
4. www.hss.iitb.ac.in/en/lecture-details
5. www.iitb.ac.in/en/event/2nd-lecture-institute-lecture-series-indian-constitution

III B.TECH II SEMESTER SYLLABUS

COMPILER DESIGN

III B. TECH- II SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS14	PCC	L	T	P	C	CIE	SEE	Total
		3	1	-	4	30	70	100

COURSE OVERVIEW:

This course deals with the basic techniques of Compiler Construction and tools that can be used to perform Syntax-directed translation of a high-level programming language into an executable code. This will provide deeper insights into the more advanced semantics aspects of programming languages, code generation, machine independent optimizations, dynamic memory allocation, types and their inferences, object orientation. The course is presented to the students by using power point projections, lecture notes, subjective and objective tests, assignments, and laboratory

COURSE OBJECTIVES:

1. To teach concepts of language translation and phases of compiler design
2. To describe the common forms of parsers
3. To inculcate knowledge of parser by parsing LL parser and LR parser
4. To demonstrate intermediate code using technique of syntax directed translation
5. To illustrate the various optimization techniques for designing various optimizing compilers

COURSE OUTCOMES:

At the end of the course students will be able to:

1. Use compiler construction tools and describes the Functionality of each stage of compilation process
2. Construct Grammars for Natural Languages and find the Syntactical Errors/Semantic errors during the compilations using parsing techniques
3. Analyze different representations of intermediate code.
4. Construct new compiler for new languages.
5. Participate in GATE, PGECET and other competitive examinations

UNIT-I

INTRODUCTION TO COMPILERS: Definition of compiler, interpreter and its differences, the phases of a compiler, role of lexical analyzer, regular expressions, finite automata, conversion from regular expressions to finite automata, pass and phases of translation, bootstrapping, LEX-lexical analyzer generator.

PARSING: Parsing, role of parser, context free grammar, derivations, parse trees, ambiguity, elimination of left recursion, left factoring, eliminating ambiguity from dangling-else grammar, classes of parsing, top down parsing - backtracking, recursive descent parsing, predictive parsers, LL(1) grammars.

UNIT-II

BOTTOM UP PARSING: Definition of bottom up parsing, handles, handle pruning, stack implementation of shift-reduce parsing, conflicts during shift-reduce parsing, LR grammars, LR parsers-simple LR, canonical LR(CLR) and Look Ahead LR (LALR) parsers, error recovery in parsing, parsing ambiguous grammars, YACC-automatic parser generator.

UNIT-III

SYNTAX DIRECTED TRANSLATION: Syntax directed definition, construction of syntax trees, S-attributed and L-attributed definitions, translation schemes, emitting a translation.

INTERMEDIATE CODE GENERATION: intermediate forms of source programs- abstract syntax tree, polish notation and three address code, types of three address statements and its implementation, syntax directed translation into three-address code, translation of simple statements, Boolean expressions and flow-of-control statements.

UNIT-IV		
TYPE CHECKING: Definition of type checking, type expressions, type systems, static and dynamic checking of types, specification of a simple type checker, equivalence of type expressions, type conversions, overloading of functions and operators. RUN TIME ENVIRONMENTS: Source language issues, Storage organization, storage-allocation strategies, access to non-local names, parameter passing, symbol tables and language facilities for dynamic storage allocation.		
UNIT-V		
CODE OPTIMIZATION: Organization of code optimizer, basic blocks and flow graphs, optimization of basic blocks, the principal sources of optimization, the directed acyclic graph (DAG) representation of basic block, global data flow analysis. CODE GENERATION: Machine dependent code generation, object code forms, the target machine, a simple code generator, register allocation and assignment, peephole optimization.		
TEXT BOOKS:		
1. Alfred V. Aho, Ravi Sethi, Jeffrey D. Ullman (2007), Compilers Principles, Techniques and Tools, 2nd edition, Pearson Education, New Delhi, India.		
REFERENCE BOOKS:		
1. Alfred V. Aho, Jeffrey D. Ullman (2001), Principles of compiler design, Indian student edition, Pearson Education, New Delhi, India. 2. Kenneth C. Loudon (1997), Compiler Construction– Principles and Practice, 1st edition, PWS Publishing. 3. K. L. P Mishra, N. Chandrashekar (2003), Theory of computer science- Automata Languages and computation, 2nd edition, Prentice Hall of India, New Delhi, India. 4. Andrew W. Appel (2004), Modern Compiler Implementation C, Cambridge University Press, UK.		

COMPUTER NETWORKS

III B. TECH- II SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS15	PCC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

Student will be able

1. To introduce the fundamental s of various types of computer networks
2. To demonstrate the TCP/IP and OSI models with merits and demerits
3. To explore the various layers of OSI model
4. To introduce UDP and TCP models

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Identify computer networks and its components.
2. Identify the different types of network topologies and protocols.
3. Enumerate the layers of the OSI model and TCP/IP. Explain the function(s) of each layer.
4. Select and use various sub netting and routing mechanisms.
5. Design a network diagram for a given scenario.

UNIT-I

INTRODUCTION: Network applications, network hardware, network software, reference models: OSI, TCP/IP, Internet, Connection oriented network - X.25, frame relay.

THE PHYSICAL LAYER: Theoretical basis for communication, guided transmission media, wireless transmission, the public switched telephone networks, mobile telephone system.

UNIT-II

THE DATA LINK LAYER: Design issues, error detection and correction, elementary data link protocols, sliding window protocols.

THE MEDIUM ACCESS SUBLAYER: Channel allocations problem, multiple access protocols, Ethernet, Data Link Layerswitching, Wireless LAN, Broadband Wireless, Bluetooth

UNIT-III

THE NETWORK LAYER: Network layer design issues, routing algorithms, Congestion control algorithms, Internet networking, the network layer in the internet (IPv4 and IPv6), Quality of Service.

UNIT-IV

THE TRANSPORT LAYER: Transport service, elements of transport protocol, Simple Transport Protocol, Internet transport layer protocols: UDP and TCP.

UNIT-V

THE APPLICATION LAYER: Domain name system, electronic mail, World Wide Web: architectural overview, dynamic web document and http.

APPLICATION LAYER PROTOCOLS: Simple Network Management Protocol, File Transfer Protocol, Simple Mail Transfer Protocol, Telnet.

TEXT BOOKS:

1. Computer networks-Andrew S Tanenbaum, 4th edition, pearson education Data communication and networking-Behrouz. A. Forouzan , fifth edition, TMH, 2013

REFERENCE BOOKS:

1. Behrouz A. Forouzan (2006), Data communication and Networking, 4th Edition, Mc Graw-Hill, India.
2. Kurose, Ross (2010), Computer Networking: A top down approach, Pearson Education, India.

LINUX PROGRAMMING

III B. TECH- II SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS16	PCC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

The course should enable the students to:

1. Understand basic principles of Linux Internals.
2. To learn Linux process control and shell programming.
3. Explain the basic Methods on which the Linux kernel is built upon.
4. To facilitate students in understanding process.
5. To facilitate students in understanding Inter process communication.

COURSE OUTCOMES:

After completion of the course the student is able to

1. Understand how to work with Linux commands and how to write Shell Scripts.
2. Apply fundamental knowledge of Linux Internals in Real time scenarios.
3. Demonstrate tools and interfaces to successfully develop new features of the kernel.
4. Design client server application to support communication interfaces.

UNIT-I INTRODUCTION & LINUX UTILITIES

Linux Utilities: File handling utilities, Security by file permissions, process utilities, Disk utilities, networking commands, filters, text processing utilities, back up utilities.

Sed- scripts, operation, addresses, commands, applications, **awk-** execution, fields and records, scripts, operations, patterns, actions, functions, using system commands in awk.

Working with bourne again shell(bash): Introduction, shell responsibilities, pipes and redirection, output redirection, here documents, running a shell script, the shell as programming language, shell meta characters, file name substitution, shell variables, command substitution, shell commands, the environment, quoting, test command, control structures, arithmetic in shell, shell script examples, interrupt processing, functions, debugging shell scripts

UNIT-II FILES

Files: File Concept, File System Structure, Inodes, File Attributes, File types, Library functions, the standard I/O and formatted I/O in C, stream errors, kernel support for files, System calls, file descriptors, low level file access – File structure related system calls(File APIs), file and record locking, file and directory management – Directory file APIs, Symbolic links & hard links

UNIT-III PROCESS & SIGNALS

Process: Process concept, process identifiers, process structure, process table, viewing processes, system processes, process scheduling, Process creation: waiting for a process, process termination, zombie processes, orphan process, process APIs.

Signals: Introduction to signals, signal generation and handling, kernel support for signals, signal function, unreliable signals, reliable signals, kill, raise, alarm, pause, abort, sleep functions.

UNIT-IV IPC

Inter Process Communication: Introduction to IPC, IPC between processes on a single computer system, IPC between processes on different systems, pipes-creation, IPC between related processes using unnamed pipes, FIFOs-creation, IPC between unrelated processing using FIFOs(named pipes), differences between unnamed and named pipes, popen and pclose library functions.

Message queues: kernel; support for message queues, unix system V APIs for messages, client/server example.

Semaphores- kernel support for semaphores, unix system V APIs for semaphores, file locking with semaphores.

UNIT-V**SOCKETS**

Sockets: Introduction to Berkeley Sockets, IPC over a network, Client-Server model, Socket address structures (Unix domain and Internet domain), socket system calls for connection oriented protocols and connectionless protocol, Example-client/server programs-Single server-client connection, Multiple simultaneous clients, Socket options-setsocket and fcntl system calls, Comparison of IPC mechanisms

TEXT BOOKS:

1. W. Richard. Stevens (2005), Advanced Programming in the UNIX Environment, 3rd edition, Pearson Education, New Delhi, India.
2. Unix and shell Programming Behrouz A. Forouzan, Richard F. Gilberg.Thomson

REFERENCE BOOKS:

1. Linux System Programming, *Robert Love, O'Reilly, SPD.*
2. Advanced Programming in the UNIX environment, 2nd Edition, *W.R.Stevens*, Pearson Education.
3. UNIX Network Programming, *W.R. Stevens*, PHI.
4. UNIX for Programmers and Users, 3rd Edition, *Graham Glass, King Ables*, Pearson Education
5. B. A. Forouzan, R. F. Gillberg, "C Programming and Data Structures", Cengage Learning, India,

WEB REFERENCES:

1. <https://www.bfoit.org/itp/Programming.html>
2. <https://www.khanacademy.org/computing/computer-programming>
3. <https://www.edx.org/course/programming-basics-iitbombayx-cs101-1x-0>
4. <https://www.edx.org/course/introduction-computer-science-harvardx-cs50x>

E-TEXT BOOKS:

1. <http://www.freebookcentre.net/Language/Free-C-Programming-Books-Download.htm>
2. <http://www.imada.sdu.dk/~svalle/courses/dm14-2005/mirror/c/>
3. <http://www.enggnotebook.weebly.com/uploads/2/2/7/1/22718186/ge6151-notes.pdf>

MOOC COURSE

1. <https://www.alison.com/courses/Introduction-to-Programming-in-c>
2. <http://www.ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-s096-effective-programming-in-c-and-c-january-iap-2014/index.htm>

PYTHON PROGRAMMING

(Professional Elective – 2)

III B. TECH- II SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4IT01	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

To learn

1. To describe the basic elements of the Python language and the Python interpreter and discuss the differences between Python and other modern languages.
2. To Describe Python dictionaries and demonstrate the use of dictionary methods.
3. Define, analyze and code the basic Python conditional and iterative control structures and explain how they can be nested and how exceptions can be used.
4. To Explain and demonstrate methods of error handling and Python exceptions.
5. To demonstrate the understanding of —magic methodsll through use of these in the context of a Python application.

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Write and debug Python programs which make use of the fundamental control structures and method-building techniques common to all programming languages. Specifically, the student will use data types, input, output, iterative, conditional, and functional components of the language in his or her programs.
2. Use object-oriented programming techniques to design and implement a clear, well-structured Python program. Specifically, the student will use and design classes and objects in his or her programs.
3. Outline the specific features of Python which made it more powerful programming language.

UNIT-I

Introduction, python overview. Basic data types. Control Flows, Tuples, Lists More advanced data types (dictionary, string), file I/O, functions

UNIT-II

Module and Packages, Object oriented programming, multi-processing and multi-threading

UNIT-III

Serialization with pickle, Network programming with python, debugging with pdb, python unit testing, Exceptions, sorting, advanced function: map, filter, and reduce, Introduction to standard libraries (pandas, Turtle, numpy, os).

UNIT-IV

Serialization with pickle, Network programming with python, debugging with pdb, python unit testing

UNIT-V

Selected topics: DB programming (Basic operation and connectivity), Web development (Django with python), Performance optimizations.

TEXT BOOKS:

1. Learning Python, by Shroff Pub& Dist., O'relly publications, Publication Year: 2013

REFERENCE BOOKS:

1. Python Programming for Beginners: Python Programming Languageby Joseph Joyner

MULTIMEDIA COMPUTING

(Professional Elective – 2)

III B. TECH- II SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS24	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

The course should enable the students to:

1. This course aims to further develop students' competency in producing dynamic and creative graphic solutions for multimedia productions.
2. To provide the foundation knowledge of multimedia computing, e.g. media characteristics, compression standards, multimedia representation, data formats, multimedia technology development

COURSE OUTCOMES:

1. Understand the characteristics of different media; understand the representations of different multimedia data;
2. Understand different data formats; be able to take into considerations in multimedia system designs;
3. Understand the characteristics of human's visual system; understand the characteristics of human's audio system; be able to take into considerations in multimedia techniques design and implementation;
4. Understand different compression principles; understand different compression techniques; understand different multimedia compression standards; be able to design and develop multimedia systems according to the requirements of multimedia applications.

UNIT-I

Fundamental concepts in Text and Image: Multimedia and hypermedia, World Wide Web, overview of multimedia software tools. Graphics and image data representation graphics/image data types, file formats, Color in image and video: color science, color models in images, color models in video.

UNIT-II

Fundamental concepts in video and digital audio: Types of video signals, analog video, digital video, digitization of sound, MIDI, quantization and transmission of audio.

UNIT-III

Multimedia data compression II: Lossy compression algorithm: Quantization, Transform Coding, Wavelet-Based Coding.
MPEG Video Coding: Embedded Zerotree of Wavelet Coefficients Set Partitioning in Hierarchical Trees (SPIHT)..

UNIT-IV

Basic Video Compression Techniques: Introduction to video compression, video compression based on motion compensation, search for motion vectors, MPEG, Basic Audio Compression Techniques.

UNIT-V

Multimedia Networks: Basics of Multimedia Networks, Multimedia Network Communications and Applications: Quality of Multimedia Data Transmission, Multimedia over IP, Multimedia over ATM Networks, Transport of MPEG-4, Media-on-Demand (MOD)..

TEXT BOOKS:

- | |
|---|
| <ol style="list-style-type: none">1. Fundamentals of Multimedia by Ze-Nian Li and Mark S. Drew PHI Learning, 20042. Professional Adobe Flex 3, Joseph Balderson, Peter Ent, et al, Wrox Publications, Wiley India, 2009. |
|---|

REFERENCE BOOKS:

- | |
|--|
| <ol style="list-style-type: none">1. Digital Multimedia, Nigel chapman and jenny chapman, Wiley-Dreamtech2. Macromedia Flash MX Professional 2004 Unleashed, Pearson.3. Multimedia and communications Technology, Steve Heath, Elsevier (Focal Press).4. Multimedia Applications, Steinmetz, Nahrstedt, Springer.5. Multimedia Basics by Weixel Thomson6. Multimedia Technology and Applications, David Hilman , Galgotia |
|--|

DATA MINING

(Professional Elective – 2)

III B. TECH- II SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS25	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

1. Learn data mining concepts understand association rules mining.
2. Discuss classification algorithms learn how data is grouped using clustering techniques.
3. To develop the abilities of critical analysis to data mining systems and applications.
4. To implement practical and theoretical understanding of the technologies for data mining
5. To understand the strengths and limitations of various data mining models

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Perform the preprocessing of data and apply mining techniques on it.
2. Identify the association rules, classification and clusters in large data sets.
3. Solve real world problems in business and scientific information
4. Classify web pages, extracting knowledge from the web

UNIT-I

INTRODUCTION TO DATA MINING: Introduction, What is Data Mining, Definition, Knowledge Discovery from Data (KDD), What Kinds of Data can be Mined, Data Mining Tasks, Integration of Data Mining System With A Database or Data Warehouse System, Types of Data Sets and Attribute Values
PREPROCESSING: Data Quality, Major Tasks in Data Pre-processing, Data Cleaning and Data Integration, Data Reduction, Data Transformation and Data Discretization, Measures of Similarity and Dissimilarity- Basics

UNIT-II

ASSOCIATION RULES: Problem Definition, Frequent Item Set Generation, The APRIORI Principle, Support and Confidence Measures, Association Rule Generation; APRIORI Algorithm, The Partition Algorithms, FP-Growth Algorithms, Compact Representation of Frequent Item Set- Maximal Frequent Item Set, Closed Frequent Item Set.

UNIT-III

CLASSIFICATION: Problem Definition, General Approaches to solving a classification problem ,Evaluation of Classifiers , Classification techniques, Decision Trees-Decision tree Construction , Methods for Expressing attribute test conditions, Measures for Selecting the Best Split, Algorithm for Decision tree Induction ; Naive-Bayes Classifier, Bayesian Belief Networks; K- Nearest neighbor classification-Algorithm and Characteristics.

UNIT-IV

CLUSTERING: Problem Definition, Clustering Overview, Evaluation of Clustering Algorithms, Partitioning Clustering-K-Means Algorithm, K-Means Additional issues, PAM Algorithm; Hierarchical Clustering- Agglomerative Methods and divisive methods, Basic Agglomerative Hierarchical Clustering Algorithm, Specific techniques, Key Issues in Hierarchical Clustering, Strengths and Weakness; Outlier Detection.

UNIT-V

WEB AND TEXT MINING: Introduction, web mining, web content mining, web structure mining, we usage mining, Text mining –unstructured text, episode rule discovery for texts, hierarchy of categories, text clustering.

TEXT BOOKS:

1. Data Mining: Concepts and Techniques - Jiawei Han, Micheline Kamber, Jian Pei (2012), 3rd edition, Elsevier, United States of America.
2. Introduction to Data Mining, Pang-Ning Tan, Vipin Kumar, Michael Steinbach, Pearson Education
3. Data mining Techniques and Applications, Hongbo Du Cengage India Publishing

REFERENCE BOOKS:

1. Data Mining Techniques, Arun K Pujari, 3rd Edition, Universities Press.
2. Data Mining Principles & Applications – T.V Suresh Kumar, B.Esware Reddy, Jagadish S Kalimani, Elsevier.
3. Data Mining, Vikram Pudi, P Radha Krishna, Oxford University Press

SOFTWARE ENGINEERING

(Professional Elective – 2)

III B. TECH- II SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS26	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

To learn

1. To familiarize with basic Software engineering methods and practices, and its applications.
2. To explain layered technology in software engineering
3. To teach software metrics and software risks.
4. To familiarize with software requirements and the SRS documents.
5. To facilitate students in software design

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Understand software development life cycle and select appropriate model suited for diverse software application.
2. Analyze the customer's requirements for a project to be developed and formulate the software requirements document
3. Conceptualize the system through design with emphases on architectural modeling and user interface
4. Classify software testing strategies and recommend testing techniques during the construction of software.
5. Examine the application of metrics and software tools during software development

UNIT-I

INTRODUCTION TO SOFTWARE ENGINEERING: The Evolving nature of software engineering, Changing nature of software engineering, Software engineering Layers, The Software Processes, Software Myths.

PROCESS MODELS: A Generic Process Model, Waterfall Model, Incremental Process Models, Evolutionary Process Models, Spiral Model, The Unified Process, Personal and Team Process Models, the Capability Maturity Model Integration (CMMI).

UNIT-II

REQUIREMENTS ENGINEERING: Functional and Non-Functional Requirements, The Software requirements Document, Requirements Specification, requirements Engineering, Requirements Elicitation and Analysis, Requirement Validation, Requirement Management, System Modeling: Context Models, Interaction Models, Structural Models, Behavioral Model, Model-Driven Engineering.

DESIGN CONCEPTS: The Design Process, Design Concepts, The Design Models, Architectural Design: Software Architecture, Architectural Genres, Architectural Styles.

UNIT-III

DESIGN AND IMPLEMENTATION: Design Patterns, Implementation Issues, Open Source Development. User Interface Design: The Golden Rules, User Interface Analysis and Design, Interface Analysis, Interface Design Steps, Design Evaluation.

SOFTWARE TESTING STRATEGIES: A Strategic approach to Software Testing, Strategic Issues, Test Strategies for Conventional Software, Validation Testing, System Testing, The Art of Debugging, White-Box Testing, Black Box Testing.

UNIT-IV

PRODUCT METRICS: A Frame Work for Product Metrics, Metrics for the Requirements Model, Metrics for DesignModel, Metrics for Source Code, Metrics for Testing.

PROCESS AND PROJECT METRICS: Metrics in the Process and Project Domains, Software Measurements, Metrics for Software Quality, Risk Management: Risk versus Proactive Risk Strategies, Software Risks, Risk Identification, Risk Projection, Risk Refinements, Risk Mitigation Monitoring and Management (RMMM), The RMMM Plan.

UNIT-V

OVERVIEW OF QUALITY MANAGEMENT AND PROCESS IMPROVEMENT: Overview of SEI -CMM, ISO 9000, CMMI, PCMM, TQM and Six Sigma.

OVERVIEW OF CASE TOOLS: Software tools and environments: Programming environments; Project management tools; Requirements analysis and design modeling tools; testing tools; Configuration management tools;

TEXT BOOKS:

1. Roger S. Pressman (2011), Software Engineering, A Practitioner's approach, 7th edition, McGraw Hill International Edition, New Delhi.
2. Sommerville (2001), Software Engineering, 9th edition, Pearson education, India.

REFERENCE BOOKS:

1. K. K. Agarwal, Yogesh Singh (2007), Software Engineering, 3rd edition, New Age International Publishers, India.
2. Lames F. Peters, Witold Pedrycz(2000), Software Engineering an Engineering approach, John Wiely & Sons, New Delhi, India.
3. Shely Cashman Rosenblatt (2006), Systems Analysis and Design, 6th edition, Thomson Publications, India.

INTERNET OF THINGS

(Professional Elective – 3)

III B. TECH- II SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS27	PEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

To learn

1. To understand the fundamentals of Internet of Things.
2. To learn about the basics of IOT protocols.
3. To build a small low cost embedded system using Raspberry Pi and Arduino board.
4. To apply the concept of Internet of Things in the real world scenario.

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Design a portable IoT using Arduino/ equivalent boards and relevant protocols.
2. Develop web services to access/control IoT devices.
3. Analyze various protocols for IoT
4. Deploy an IoT application.
5. Analyze applications of IoT in real time scenario

UNIT-I	INTRODUCTION TO IoT	
Introduction: Definition, Characteristics of IOT, IOT Architectural view, Physical design of IOT, Logical design of IOT, IOT Enabling technologies, Application of IOT.		
UNIT-II	IoT ARCHITECTURE	
Domain specific IOT's: Introduction, Home Automation, Cities, Environment, Energy, retail, Logistics, Agriculture, Industry, Health. Machine-to-machine (M2M): Introduction, Difference between IOT and M2M, SDN (software defined networking) and NFV (network function virtualization) for IOT.		
UNIT-III	IoT PROTOCOLS & BUILDING IoT WITH ARDUINO	
Developing Internet of things: Introduction, IoT platforms design methodology, Need for IoT systems management		
UNIT-IV	BUILDING IoT WITH RASPBERRY PI	
IoT Physical devices & End points: Basic building blocks of IoT device, About Board, Raspberry Pi Interfaces, Other IoT devices.		
UNIT-V	CASE STUDIES AND REAL – WORLD APPLICATIONS	
Case Studies Illustrating IoT design: Home automation, Weather Monitoring, Agriculture, Cities		
TEXT BOOKS:		
1. Vijay Madiseti and Arshdeep Bahga, “Internet of things(A-Hand-on-Approach)” 1st Edition, Universal Press		
REFERENCE BOOKS:		

1. Rajkamal,"Internet of Things", Tata McGraw Hill publication
2. Hakima Chaouchi "The Internet of Things: Connecting Objects", Wiley pcation.
3. Charless Bell "MySQL for the Internet of things", Apress publications.
4. Francis dacosta "Rethinking the Internet of things:A scalable Approach to
5. Connecting everything", 1st edition, Apress publications 2013.
6. Donald Norris"The Internet of Things: Do-It-Yourself at Home Projects for Arduino, Raspberry Pi and BeagleBone Black", McGraw Hill publication.

CLOUD COMPUTING

(Professional Elective – 3)

III B. TECH- II SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS28	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-				

COURSE OBJECTIVES:

To learn

1. To inculcate the concepts of distributed computing
2. To familiarize the concepts of cloud computing and services
3. To explain cloud platform and types of cloud
4. To explain resource management in cloud

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Understand the fundamental principles of distributed computing
2. Create virtual machines and virtual templates.
3. Create Cloud platform using Virtual machines
4. Identify suitable business models of cloud computing

UNIT-I

INTRODUCTION TO VIRTUALIZATION AND TECHNOLOGIES: Introduction to Virtualization: Definition, Objectives, Characteristics, Benefits of virtualization, Taxonomy of virtualization technologies, Pros and cons of virtualization. Virtualization Technologies: VMware, Hyper-V, Zen and virtual iron.

UNIT-II

FUNDAMENTAL CLOUD COMPUTING AND MODELS: Cloud Computing: Origin and influences, Basic concepts and terminology, Goals and benefits, Risks and challenges. Cloud Models, roles and boundaries, Cloud characteristics, Cloud delivery models, Cloud deployment models.

UNIT-III

CLOUD COMPUTING MECHANISMS AND ARCHITECTURE: Cloud-Enabling Technology: Broadband networks and internet architecture, Data center technology, Virtualization technology, Web technology, Multitenant technology, Service technology. Cloud Architectures: Architecture - Workload distribution, Resource pooling, Dynamic scalability, Elastic resource capacity, Service load balancing, Cloud bursting, Elastic disk provisioning, Redundant storage.

UNIT-IV

CLOUD SECURITY AND DISASTER RECOVERY: Cloud Security: Data, Network and host security, Cloud security services and cloud security possible solutions. Cloud Disaster Recovery: Disaster recovery planning, Disasters in the cloud, Disaster management, Capacity planning and cloud scale.

UNIT-V

CLOUD CASE STUDIES: Case Studies: Software-as-a-Service (SaaS) - Salesforce.com, Facebook; Platform-as-a-Service (PaaS) - Google App Engine, MS-Azure and IBM Bluemix; Infrastructure-as-a-Service (IaaS) - Amazon EC2, Amazon S3 and Netflix.

TEXT BOOKS:

1. Thomas Erl and RicardoPuttini Cloud ComputingConcepts, Technology and Architecture, Pearson, 2013.
2. Ivanka Menken and Gerard Blokdijk, Cloud Computing Virtualization Specialist Complete Certification Kit-Study Guide Book, Lightning Source, 2009

REFERENCE BOOKS:

1. Barrie Sosinsky, Cloud Computing Bible, Wiley India Pvt Ltd, 2011.
2. Rajkumar Buyya, James Broberg and Andrzej Goscinski, Cloud Computing Principles and Paradigms, John Wiley and Sons, 2011.
3. John W. Rittinghouse and James F. Ransome, Cloud Computing Implementation, Management and Security, CRC Press, Taylor & Francis Group, 2010.

NATURAL LANGUAGE PROCESSING (Professional Elective – 3)								
III B. TECH- II SEMESTER								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS29	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100
COURSE OBJECTIVES: 1. To learn the fundamentals of natural language processing 2. To understand the use of CFG and PCFG in NLP 3. To understand the role of semantics of sentences and pragmatics 4. To apply the NLP techniques to IR applications								
COURSE OUTCOMES: 1. To tag a given text with basic Language features 2. To design an innovative application using NLP components 3. To implement a rule based system to tackle morphology/syntax of a language 4. To design a tag set to be used for statistical processing for real-time applications 5. To compare and contrast the use of different statistical approaches for different types of NLP applications.								
UNIT-I								
Introduction: Origins and challenges of NLP – Language Modeling: Grammar-based LM, Statistical LM – Regular Expressions, Finite-State Automata – English Morphology, Transducers for lexicon and rules, Tokenization, Detecting and Correcting Spelling Errors, Minimum Edit Distance.								
UNIT-II								
Word Level Analysis: Unsmoothed N-grams, Evaluating N-grams, Smoothing, Interpolation and Backoff – Word Classes, Part-of-Speech Tagging, Rule-based, Stochastic and Transformation-based tagging, Issues in PoS tagging – Hidden Markov and Maximum Entropy models.								
UNIT-III								
Syntactic Analysis: Context-Free Grammars, Grammar rules for English, Treebanks, Normal Forms for grammar – Dependency Grammar – Syntactic Parsing, Ambiguity, Dynamic Programming parsing – Shallow parsing – Probabilistic CFG, Probabilistic CYK, Probabilistic Lexicalized CFGs – Feature structures, Unification of feature structures.								
UNIT-IV								
Semantics and Pragmatics : Requirements for representation, First-Order Logic, Description Logics – Syntax-Driven Semantic analysis, Semantic attachments – Word Senses, Relations between Senses, Thematic Roles, selectional restrictions – Word Sense Disambiguation, WSD using Supervised, Dictionary & Thesaurus, Bootstrapping methods – Word Similarity using Thesaurus and Distributional methods.								
UNIT-V								
Discourse Analysis and Lexical Resources: Discourse segmentation, Coherence – Reference Phenomena, Anaphora Resolution using Hobbs and Centering Algorithm – Coreference Resolution – Resources: Porter Stemmer, Lemmatizer, Penn Treebank, Brill's Tagger, WordNet, PropBank, FrameNet, Brown Corpus, British National Corpus (BNC).								
TEXT BOOKS:								

1. Daniel Jurafsky, James H. Martin—Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics and Speech, Pearson Publication, 2014.
2. Steven Bird, Ewan Klein and Edward Loper, —Natural Language Processing with Python, First Edition, O'Reilly Media, 2009.

REFERENCE BOOKS:

1. Breck Baldwin, —Language Processing with Java and LingPipe Cookbook, Atlantic Publisher, 2015.
2. Richard M Reese, —Natural Language Processing with Java, O'Reilly Media, 2015.
3. Nitin Indurkha and Fred J. Damerau, —Handbook of Natural Language Processing, Second Edition, Chapman and Hall/CRC Press, 2010.
4. Tanveer Siddiqui, U.S. Tiwary, —Natural Language Processing and Information Retrieval, Oxford University Press, 2008.

MOBILE APPLICATION DEVELOPMENT (Professional Elective – 3)								
III B. TECH- II SEMESTER								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS30	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100
COURSE OBJECTIVES: The course should enable the students to: 1. To facilitate students to understand android SDK 2. To help students to gain a basic understanding of Android application development 3. To inculcate working knowledge of Android Studio development tool								
COURSE OUTCOMES: Upon successful completion of this course, student will be able to: 1. Describe the components and compare different mobile application models/architectures and patterns. 2. Apply the mobile development framework knowledge to develop an application for android devices. 3. Use the various layout structures to design an Android application with rich uses interactive interfaces 4. Develop an application that focus on user experience design, native data handling and background tasks and notification 5. Develop applications for Android Devices and deploy in mobile stores(ex:google playstore)								
UNIT-I	INTRODUCTION TO ANDROID							
Setting up Android Studio Development Environment, Creating an Example Android App in Android Studio, Creating and Configuring the Android Studio AVD Emulator, Android Architecture, The Anatomy of an Android Application								
UNIT-II	ANDROID VIEW, VIEWGROUP AND LAYOUTS							
Android Views: TextView, EditText, Button, CheckBox, RadioButton, ImageButton, ProgressBar, Spinner, View Groups and Layouts: LinearLayout, RelativeLayout, TableLayout, FrameLayout, ScrollView, WebView and GridView, Event Handling,								
UNIT-III	ACTIVITY AND INTENTS							
Understanding Activities, Activity Lifecycle and Intents: Explicit Intent and Implicit Intents, Animating User Interfaces: Image Rotation and Interpolators, Working with Menu								
UNIT-IV	ANDROID USER INTERFACE DESIGN ESSENTIALS							
Fragments, Android Floating Action Button, Working with AppBar Creating a Tabbed Interface, An Android ListView and RecyclerView, Implementing an Android Navigation Drawer, Android Broadcast Intents and Broadcast Receivers, Services								
UNIT-V	ANDROID DATABASE, TESTING AND PUBLISHING APPS							
Android SQLite Database, Introduction to Firebase, Firebase User Authentication, reading and writing data to Firebase database, Shared Preferences, Content Providers, Testing and Debugging, Publishing and Deploying Android application to the world.								
TEXT BOOKS:								

1. Neil Smyth, "Android Studio 3.0 Development Essentials – Android 8 Edition", Payload Media, 3rd Edition, 2017.
2. Lauren Darcey and Shane Conder, "Android Wireless Application Development", Pearson Education, 2nd ed. (2011)
3. Neil Smyth, "Firebase Essentials – Android Edition", Payload Media, 1st Edition, 2017.

REFERENCE BOOKS:

1. Reto Meier, —Professional Android 2 Application Developmentll, Wiley India Pvt Ltd
2. Mark L Murphy, —Beginning Androidll, Wiley India Pvt Ltd
3. Android Application Development All in one for Dummies by Barry Burd, Edition: IB. A. Forouzan, R. F. Gillberg, "C Programming and Data Structures", Cengage Learning, India, 3rd Edition, 2014.

WEB REFERENCES:

1. <https://developer.android.com/>
2. <https://google-developer-training.gitbooks.io/android-developer-fundamentals-course-concepts/content/en/>
3. <https://google-developer-training.gitbooks.io/android-developer-fundamentals-course-practicals/content/en/>

E-TEXT BOOKS:

1. <https://developer.android.com/>
2. <http://enos.itcollege.ee/~jpoial/allalaadimised/reading/Android-Programming-Cookbook.pdf>

MOOC COURSE

1. <https://www.coursera.org/specializations/android-app-development>

NETWORK SIMULATION LAB

III B. TECH- II SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS17	PCC	L	T	P	C	CIE	SEE	Total
		-	-	3	1.5	30	70	100

COURSE OVERVIEW:

The course is designed for the students to develop their network simulation skills using NS3. It will help the students to work with their networking and research projects.

COURSE OBJECTIVES:

1. Understand the importance and usage of NS3
2. To inculcate working knowledge among Node to Node communication in networks.
3. Understand and implement various networking topology.
4. To familiarize with the Net Anim

COURSE OUTCOMES:

By the end of this course, the student will be able to

1. Install NS3 and its supporting packages.
2. Implement connecting various nodes
3. Implement Star and bus and Hybrid topologies using NS3
4. Use and develop applications using NetAnim.

SYLLABUS

EXPERIMENTS:

1. Introduction about discrete events simulation and its tools.
2. Installation of NS3 in Linux Operating System.
3. Program in NS3 to connect two nodes.
4. Program in NS3 for connecting three nodes considering one node as a central node.
5. Implement physical star topology and simulate using NS3.
6. Implement physical bus topology and simulate using NS3.
7. Program for connecting multiple routers and nodes and building a hybrid topology.
8. Installation and configuration of NetAnim
9. Program in NS3 to implement FTP using TCP bulk transfer.
10. Implement AODV Routing Protocol
11. Implement DSDV Routing Protocol.
12. Implement DSR Routing Protocol.

TEXT BOOKS:

1. Andrew S Tanenbaum David J Wetherall , Computer Networks, 5th Edition, Pearson Education.
2. Kurose, Ross (2010), Computer Networking: A top down approach, Sixth Edition, Pearson Education, India.

REFERENCES:

1. Behrouz A. Forouzan, Data communication and Networking, 4th Edition, Mc Graw-Hill, India.
2. Larry L. Peterson, Bruce S. Davie, Computer Networks: A Systems Approach, MK Publishers

WEB LINKS:

1. <https://www.nsnam.org/releases/>
2. <https://www.nsnam.org/documentation/>

LINUX PROGRAMMING LAB

III B. TECH- II SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS18	PCC	L	T	P	C	CIE	SEE	Total
		-	-	3	1.5	30	70	100

OBJECTIVES:

The course should enable the students to:

1. Apply the basic commands to handle the Linux Environment
2. Use the Shell/C scripting constructs to modify the file system content.
3. Implement the process management concepts using C language.
4. Design Client-Server Applications using Sockets and TCP/UDP protocols.

COURSE OUTCOMES:

After completion of the course Students is able to:

1. Write and use the Shell Scripts in managing Linux Environment
2. Construct C Scripts to handle the File system in Linux.
3. Implement the IPC Mechanisms in Linux operating system using C language
4. Build and Analyze the Client-Server Environment using various protocols.

LIST OF EXPERIMENTS

1. Linux Commands & Introduction of Shell
2. Write a shell script that accepts a file name, starting and ending line numbers as arguments and displays all the lines between the given line numbers.
3. Write a shell script that deletes all lines containing a specified word in one or more files supplied as arguments to it.
4. Write a shell script that displays a list of all the files in the current directory to which the user has read, write and execute permissions.
5. Write a shell script that receives any number of file names as arguments checks if every argument supplied is a file or a directory and reports accordingly. Whenever the argument is a file, the number of lines on it is also reported.
6. Write a shell script that accepts a list of file names as its arguments, counts and reports the occurrence of each word that is present in the first argument file on other argument files.
7. Write a shell script to list all of the directory files in a directory.
8. Write an awk script to count the number of lines in a file that do not contain vowels.
9. Write an awk script to find the number of characters, words and lines in a file.
10. Write a c program that makes a copy of a file using standard I/O and system calls.
11. Implement in C the following UNIX commands using System calls.
 - i. A. cat B. mv
12. Write a C Program to list files in a directory.
13. Write a C program to list for every file in a directory, its inode number and file name.
14. Write a C program to create a child process and allow the parent to display "parent" and the child to display "child" on the screen.
15. Write a C program to create a Zombie process.
16. Write a C program that illustrates how an orphan is created.
17. Write C programs that illustrate communication between two unrelated processes using named pipe(FIFO).
18. Write a C program in which a parent writes a message to a pipe and the child reads the message.
19. Write Client server program in C for connection oriented communication between server and client process using unix domain sockets to perform the following
20. Write Client server program in C for connection oriented communication between server and client process using internet domain sockets.

TEXT BOOKS:

1. Unix System Programming using C++, T.Chan, PHI.
2. Unix Concepts and Applications, 4th Edition, Sumitabha Das, TMH, 2006.

REFERENCES:

1. Beginning Linux Programming, 4th Edition, N.Matthew, R.Stones,Wrox, Wiley India Edition,rp-2008
2. Linux System Programming, Robert Love, O'Reilly, SPD.
3. Advanced Programming in the Unix environment, 2nd Edition, W.R.Stevens, Pearson Education.
4. Unix Network Programming ,W.R.Stevens,PHI.
5. Unix for programmers and users, 3rd Edition, Graham Glass, King Ables, Pearson Education.

ESSENCE OF INDIAN TRADITIONAL KNOWLEDGE

III B. TECH- II SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4MC06	MC	L	T	P	C	CIE	SEE	Total
		2	-	-	-	30	70	100

COURSE OBJECTIVES:

To facilitate the students with the concepts of Indian traditional knowledge and to make them understand the Importance of roots of knowledge system.

COURSE OUTCOMES:

After completion of the course, students will be able to:

1. Upon completion of the course, the students are expected to:
2. Understand the concept of Traditional knowledge and its importance
3. Know the need and importance of protecting traditional knowledge.
4. Know the various enactments related to the protection of traditional knowledge.
5. Understand the concepts of Intellectual property to protect the traditional knowledge.

UNIT-I

Introduction to traditional knowledge: Define traditional knowledge, nature and characteristics, scope and importance, kinds of traditional knowledge, the physical and social contexts in which traditional knowledge develop, the historical impact of social change on traditional knowledge systems. Indigenous Knowledge (IK), characteristics, traditional knowledge vis-à-vis indigenous knowledge, traditional knowledge Vs western knowledge traditional knowledge vis-à-vis formal knowledge

UNIT-II

Protection of traditional knowledge: the need for protecting traditional knowledge Significance of TK Protection, value of TK in global economy, Role of Government to harness TK.

UNIT-III

Legal frame work and TK: A: The Scheduled Tribes and Other Traditional Forest Dwellers (Recognition of Forest Rights) Act, 2006, Plant Varieties Protection and Farmer's Rights Act, 2001 (PPVFR Act); B: The Biological Diversity Act 2002 and Rules 2004, the protection of traditional knowledge bill, 2016. Geographical indicators act 2003.

UNIT-IV

Traditional knowledge and intellectual property: Systems of traditional knowledge protection, Legal concepts for the protection of traditional knowledge, Certain non IPR mechanisms of traditional knowledge protection, Patents and traditional knowledge, Strategies to increase protection of traditional knowledge, global legal FORA for increasing protection of Indian Traditional Knowledge.

UNIT-V

Traditional knowledge in different sectors: Traditional knowledge and engineering, Traditional medicine system, TK and biotechnology, TK in agriculture, Traditional societies depend on it for their food and healthcare needs, Importance of conservation and sustainable development of environment, Management of biodiversity, Food security of the country and protection of TK. 139

TEXT BOOKS:

1. Traditional Knowledge System in India, by Amit Jha, 2009.
2. Traditional Knowledge System and Technology in India by Basanta Kumar Mohanta and Vipin Kumar Singh, Pratibha Prakashan 2012.

REFERENCE BOOKS:

1. Traditional Knowledge System in India by Amit Jha Atlantic publishers, 2002
2. "Knowledge Traditions and Practices of India" Kapil Kapoor¹, Michel Danino²

E-RESOURCES:

1. <https://www.youtube.com/watch?v=LZP1StpYEPM>
2. <http://nptel.ac.in/courses/121106003>

IV B.TECH I SEMESTER SYLLABUS

BIG DATA ANALYTICS

III B. TECH- II SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS19	PCC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

To learn

1. To introduce the terminology, technology and its applications
2. To introduce the concept of Analytics and Visualization
3. To demonstrate the usage of various Big Data tools and Data Visualization tools

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. **Compare** various file systems and **use** an appropriate file system for storing different types of data.
2. **Demonstrate** the concepts of Hadoop ecosystem for storing and processing of unstructured data.
3. **Apply** the knowledge of programming to process the stored data using Hadoop tools and generate reports.
4. **Connect** to web data sources for data gathering, **Integrate** data sources with hadoop components to process streaming data.
5. **Tabulate** and **examine** the results generated using hadoop components

UNIT-I

INTRODUCTION TO BIG DATA: Data and its importance, Big Data - definition, implications of Big Data, addressing Big Data implications using Hadoop, Hadoop Ecosystem

HADOOP ARCHITECTURE:

Hadoop Storage : HDFS, Hadoop
 Processing : Map Reduce Framework
 Hadoop Server Roles : Name Node, Secondary Name Node and Data Node, Job Tracker, TaskTracker

HDFS-HADOOP DISTRIBUTED FILE SYSTEM: Design of HDFS, HDFS Concepts, HDFS Daemons, HDFS High Availability, Block Abstraction, FUSE: File System in User Space. HDFS Command Line Interface (CLI), Concept of File Reading and Writing in HDFS.

UNIT-II

MAPREDUCE PROGRAMMING MODEL: Introduction to Map Reduce Programming model to process Big Data, key features of Map Reduce, Map Reduce Job skeleton, Introduction to Map Reduce API, Hadoop Data Types, Develop Map Reduce Job using Eclipse, built a Map Reduce Job export it as a java archive(.jar file).

MAPREDUCE JOB LIFE CYCLE: Understanding Mapper, Combiner, Partitioner, Shuffle & Sort and Reduce phases of Map Reduce Application, Developing Map Reduce Jobs based on the requirement using given datasets like weather dataset.

UNIT-III

INTRODUCTION TO PIG: Understanding pig and pig Platform, introduction to Pig Latin Language and Execution engine, running pig in different modes, Pig Grunt Shell and its usage.

PIG LATIN LANGUAGE –SEMANTICS –DATA TYPES IN PIG: Pig Latin Basics, Key words, Pig Data types, Understanding Pig relation, bag, tuple and writing pig relations or statements using Grunt Shell, expressions, Data processing operators, using Built in functions.

WRITING PIG SCRIPTS USING PIG LATIN: Writing pig scripts and saving them text editor, running pig scripts from command line.

UNIT-IV		
INTRODUCTION TO HIVE: Understanding Hive Shell, Running Hive, Understanding Schema on read and Schema on write. HIVE QL DATA TYPES, SEMANTICS: Introduction to Hive QL (Query Language), Language semantics, Hive Data Types. HIVE DDL, DML AND HIVE SCRIPTS: Hive Statements, Understanding and working with Hive Data Definition Languages and Manipulation Language statements, Creating Hive Scripts and running them from hive terminal and command line.		
UNIT-V		
SQOOP: Introduction to Sqoop tool, commands to connect databases and list databases and tables, command to import data from RDBMS into HDFS, Command to export data from HDFS into required tables of RDBMS. FLUME: Introduction to Flume agent, understanding Flume components Source, Channel and Sink. OOZIE: Introduction to Oozie, Understanding work flow Management.		
TEXT BOOKS:		
1. Hadoop: The Definitive Guide, 4th Edition - O'Reilly Media 2. Chris Eaton, Dirk deRoos et al. , "Understanding Big data ", McGraw Hill, 2012. 3. Seema Acharya, Subhasini Chellappan, "Big Data Analytics" Wiley 2015.		
REFERENCE BOOKS:		
1. Michael Berthold, David J. Hand, "Intelligent Data Analysis", Springer, 2007. 2. Paul Zikopoulos ,Dirk DeRoos , Krishnan Parasuraman , Thomas Deutsch , James Giles , David Corgan , "Harness the Power of Big Data The IBM Big Data Platform ", Tata McGraw Hill Publications, 2012.		

SOFTWARE TESTING

(PROFESSIONAL ELECTIVE – 4)

IV B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS31	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

To learn

1. An ability to design and conduct a software test process for a software testing project and identify the needs of software test automation.
2. An ability understand and identify various software testing problems, and solve these problems by designing and selecting software test models, criteria, strategies, use software testing methods and modern software testing tools for their testing projects and basic understanding and knowledge of contemporary issues in software testing, such as component-based software testing problems.
3. An ability to use various communication methods and skills to communicate with their teammates to conduct their practice-oriented software testing projects.
4. Analyze different approaches to software testing and quality assurance, and select optimal solutions for different situations and projects.

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Apply fundamental knowledge of Testing in Real time scenarios.
2. Test a simple application.
3. Understand and Applying the Techniques in Software Development Life cycle
4. Implement various communication methods and skills to communicate with their teammates to conduct their practice-oriented software testing projects.

UNIT-I

INTRODUCTION: Purpose of testing-Dichotomies-Software Testing Principles- Bugs, consequences of bugs, Taxonomy of bugs -The Tester's Role in a Software Development Organization-Black box testing and white box testing- Defects -Cost of defects- Defect Classes- Defect Examples, software testing life cycle.

UNIT-II

Flow graphs and Path testing: Basics concepts of path testing-predicates-path predicates and achievable paths- path sensitizing- path instrumentation, application of path testing. Transaction Flow Testing: Transaction flows- transaction flow testing techniques, Dataflow testing: Basics of dataflow testing - strategies in data flow testing – application of data flow testing.

UNIT-III

Test Case Design Strategies: Using Black Box Approach to Test Case Design - Random Testing – Requirements based testing – Boundary Value Analysis – Equivalence Class Partitioning– Compatibility testing – User documentation testing – Domain testing. Alpha, Beta Tests – Usability and Accessibility testing – Configuration testing - Compatibility testing – Testing the documentation.

UNIT-IV

Paths, Path products and Regular expressions: Path products & path expression- reduction procedure- applications- regular expressions & flow anomaly detection. Logic Based Testing: Overview decision tables-path expressions, k-v charts. State, state graphs and transition testing: state graphs, good and bad state graphs, state testing, testability tips.

UNIT-V		
Graph Matrices and applications: motivational over view, matrix of graph, relations, power of matrix, node reduction algorithm. People and organizational issues in testing: Organization structures for testing teams – testing services - Test Planning – Test Plan Components – Test Plan Attachments – Locating Test Items – test management – test process- bug detection life cycle.		
TEXT BOOKS:		
1. “Software Testing – Principles and Practices”, Srinivasan Desikan and Gopalaswamy Ramesh, Pearson education, 2006. 2. “Software Testing Techniques”, Boris Beizer, 2nd Edition, Van Nostrand Reinhold, New York, 1990		
REFERENCE BOOKS:		
1. “Software Testing”, Ron Patton, Second Edition, Sams Publishing, Pearson education, 2007 2. “Software Testing – Effective Methods, Tools and Techniques”, Renu Rajani, Pradeep Oak, TataMcGraw Hill, 2004. 3. “Software Testing in the Real World – Improving the Process”, Edward Kit, Pearson Education, 1995. 4. “Foundations of Software Testing – Fundamental algorithms and techniques”, Aditya P. Mathur, Dorling Kindersley (India) Pvt. Ltd., Pearson Education, 2008		

CRYPTOGRAPHY AND NETWORK SECURITY

(PROFESSIONAL ELECTIVE – 4)

II Semester: Computer Science and Engineering

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS32	PEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

To learn

1. To provide deeper understanding into cryptography, its application to network security, threats/vulnerabilities to networks and countermeasures.
2. To explain various approaches to Encryption techniques, strengths of Traffic Confidentiality, Message Authentication Codes.
3. To familiarize Digital Signature Standard and provide solutions for their issues.
4. To familiarize with cryptographic techniques for secure (confidential) communication of two parties over an insecure (public) channel; verification of the authenticity of the source of a message

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Identify basic security attacks and services
2. Use symmetric and asymmetric key algorithms for cryptography
3. Design a security solution for a given application
4. Analyze Key Management techniques and importance of number Theory.
5. Understanding of Authentication functions the manner in which Message Authentication Codes and Hash Functions works.
6. To examine the issues and structure of Authentication Service and Electronic Mail Security

UNIT-I

INTRODUCTION: Security trends, The OSI Security Architecture, Security Attacks, Security Services and Security Mechanisms, A model for Network security.
CLASSICAL ENCRYPTION TECHNIQUES: Symmetric Cipher Modes, Substitute Techniques, Transposition Techniques, Stenography

UNIT-II

BLOCK CIPHER AND DATA ENCRYPTION STANDARDS: Block Cipher Principles, Data Encryption Standards, the Strength of DES, Block Cipher Design Principles.
ADVANCED ENCRYPTION STANDARDS: Evaluation Criteria for AES, the AES Cipher.
MORE ON SYMMETRIC CIPHERS: Multiple Encryption, Triple DES, Block Cipher Modes of Operation, Stream Cipher and RC4.

UNIT-III

PUBLIC KEY CRYPTOGRAPHY AND RSA: Principles Public key crypto Systems the RSA algorithm, Key Management, Diffie Hellman Key Exchange.
MESSAGE AUTHENTICATION AND HASH FUNCTIONS: Authentication Requirement, Authentication Function, Message Authentication Code, Hash Function, Security of Hash Function and MACs.
HASH AND MAC ALGORITHM: Secure Hash Algorithm, Whirlpool, HMAC, CMAC. **DIGITAL SIGNATURE:** Digital Signature, Authentication Protocol, Digital Signature Standard

UNIT-IV

AUTHENTICATION APPLICATION: Kerberos, X.509 Authentication Service, Public Key Infrastructure. **EMAIL SECURITY:** Pretty Good Privacy (PGP) and S/MIME.
IP SECURITY: Overview, IP Security Architecture, Authentication Header, Encapsulating Security Payload, Combining Security Associations and Key Management.

UNIT-V		
WEB SECURITY: Requirements, Secure Socket Layer (SSL) and Transport Layer Security (TLS), Secure Electronic Transaction (SET), Intruders, Viruses and related threats. FIREWALL: Firewall Design principles, Trusted Systems.		
TEXT BOOKS:		
1. William Stallings (2006), Cryptography and Network Security: Principles and Practice, 4th edition, Pearson Education, India. 2. William Stallings (2000), Network Security Essentials (Applications and Standards), Pearson Education, India.		
REFERENCE BOOKS:		
1. Charlie Kaufman (2002), Network Security: Private Communication in a Public World, 2nd edition, Prentice Hall of India, New Delhi. 2. Atul Kahate (2008), Cryptography and Network Security, 2 nd edition, Tata Mc Grawhill, India. 3. Robert Bragg, Mark Rhodes (2004), Network Security: The complete reference, Tata Mc Grawhill, India.		

WEB SERVICES AND SERVICE ORIENTED ARCHITECTURE

(PROFESSIONAL ELECTIVE – 4)

IV B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS33	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

1. To understand the details of web services technologies like WSDL,UDDI, SOAP
2. To learn how to implement and deploy web service client and server
3. To explore interoperability between different frameworks
4. To Understand Web Services and implementation model for SOA
5. To Understand XML concepts
6. To explore different Test Strategies for SOA-based applications

COURSE OUTCOMES:

1. Identify various Web Services with benefits and challenges.
2. Analyze SOAP and in the Web Services.
3. Analyze the WSDL in the world of web services with bindings, tools and limitations.
4. Identify the Importance of UDDI Registry and Security issues related to Web Services.

UNIT-I	INTRODUCTION TO WEB SERVICES	
Evolution and Emergence of Web Services – Evolution of distributed computing, Core distributed computing technologies — client/server, CORBA, JAVA RMI, Microsoft DCOM, MOM, Challenges in Distributed Computing, role of J2EE and XML in distributed computing, emergence of Web Services and Service Oriented Architecture (SOA). Introduction to Web Services — The definition of web services, basic operational model of web services, tools and technologies enabling web services, benefits and challenges of using web services. Web Services Architecture — Web services Architecture and its characteristics, core building blocks of web services, standards and technologies available for implementing web services, web services communication models, basic steps of implementing web services.		
UNIT-II	FUNDAMENTALS OF SOAP	
Fundamentals of SOAP — SOAP Message Structure, SOAP encoding, Encoding of different data types, SOAP message exchange models, SOAP communication and messaging, Java and Axis, limitations of SOAP		
UNIT-III	FUNCTIONALITY OF WEB SERVICES	
Describing Web Services — WSDL — WSDL in the world of Web Services, Web Services life cycle, anatomy of WSDL definition document, WSDL bindings, WSDL Tools, limitations of WSDL.		
UNIT-IV	FUNDAMENTALS OF UDDI	
Discovering Web Services — Service discovery, role of service discovery in a SQA, service discovery mechanisms, UDDI — UDDI registries, uses of UDDI Registry, Programming with UDDI, UDDI data structures, Publishing API, Publishing, searching and deleting information in a UDDI Registry, limitations of UDDI.		
UNIT-V	WEB SERVICES INTEROPERABILITY & SECURITY	
Web Services Interoperability — Means of ensuring Interoperability, Overview of .NET, Creating a .NET client for an Axis Web Service, creating Java client for a Web service, Challenges in Web Services Interoperability. Web Services Security — XML security frame work, Goals of Cryptography,Digital signature, Digital Certificate, XML Encryption		

TEXT BOOKS:

1. Developing Java Web Services, R. Nagappan, R. Skoczylas, R.P. Sriganesh, Wiley India.
2. Web Services & SOA Principles and Technology, Second Edition, Michael P. Papazoglou.
3. Developing Enterprise Web Services, S. Chatterjee, J. Webber, Pearson Education.

REFERENCE BOOKS:

1. Java Web Service Architecture, James McGovern, Sameer Tyagi et al., Elsevier
2. Building Web Services with Java, 2 Edition, S. Graham and others, Pearson Edn.
3. Java Web Services, D.A. Chappell & T. Jewell, O'Reilly, SPD.
4. Web Services, G. Alonso, F. Casati and others, Springer.Outcomes
5. Basic details of WSDL, UDDI, SOAP
6. Implement WS client and server with interoperable systems.

AD-HOC AND SENSOR NETWORKS

(PROFESSIONAL ELECTIVE – 4)

IV B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS34	PEC	L	T	P	L	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

The course should enable the students to:

1. Describe the concepts of adhoc wireless networks.
2. Analyze different routing protocols of mobile adhoc networks.
3. Apply the energy management policies in routing algorithms.
4. Implement protocols for location based QoS.
5. Design and simulate sensor networks and evaluate performance.

COURSE OUTCOMES:

After completion of the course, students will be able to

1. Identify the different types of Ad-hoc networks.
2. Differentiate various types of Routing Protocols and apply right protocol wherever it is necessary.
3. Analyze energy management issues in Ad-hoc networks for better network formation.
4. Identify the Sensor network Applications and differentiate various MAC Protocols.

UNIT-I	Mobile Ad-Hoc Networking with a View of 4G Wireless	
Mobile Ad-Hoc Networking with a View of 4G Wireless: Imperatives and Challenges, Off-the-Shelf Enables of Ad Hoc Networks, IEEE 802.11 in Ad Hoc Networks: Protocols, Performance and Open Issues, Scatternet Formation in Bluetooth Networks, Antenna Beam forming and Power Control for Ad Hoc Networks.		
UNIT-II	Routing Protocols in Ad Hoc Networks	
Routing Protocols in Ad Hoc Networks: Topology Control in Wireless Ad Hoc Networks, Broadcasting and Activity Scheduling in Ad Hoc Networks, Location Discovery, Mobile Ad Hoc Networks (MANETs): Routing Technology for Dynamic, Wireless Networking, Routing Approaches in Mobile Ad Hoc Networks.		
UNIT-III	Energy Management in Ad hoc Wireless Networks	
Energy Management in Ad hoc Wireless Networks: Energy-Efficient Communication in Ad Hoc Wireless Networks, Ad Hoc Networks Security, Self-Organized and Cooperative Ad Hoc Networking, Simulation and Modeling of Wireless, Mobile, and Ad Hoc Networks, Modeling Cross-Layering Interaction Using Inverse Optimization, Algorithmic Challenges in Ad Hoc Networks. Energy Management in Ad hoc Wireless Networks: Introduction, need for energy management in ad hoc networks, battery management schemes-overview of battery characteristic, device dependent schemes.		
UNIT-IV	Introduction and Overview of Wireless Sensor Networks	
Introduction and Overview of Wireless Sensor Networks: Applications of Wireless Sensor Networks, Examples of Category 1 WSN Applications, Another Taxonomy of WSN Technology. Basic Wireless Sensor Technology: Sensor Node Technology, Sensor Taxonomy, WN Operating Environment, WN Trends.		
UNIT-V	Wireless Sensor Networks Technology and Medium Access Control Protocols	
Wireless Transmission Technology and Systems: Radio Technology Primer, Available Wireless Technologies. Medium Access Control Protocols for Wireless Sensor Networks: Fundamentals of MAC Protocols, MAC Protocols for WSNs, Sensor-MAC Case Study, IEEE 802.15.4 LR-WPANs Standard Case Study.		
TEXT BOOKS:		

1. **“Adhoc and Sensor Networks”** by Stefano Basagni, Silvia Giordano, Ivan Stojmenvic. IEEE Press, A John Wiley & Sons, Inc., Publication 2004.
2. **Wireless Sensor Networks**, Kazem Sohraby, Daniel Minoli, Taieb Znati. A John Wiley & Sons, Inc., Publication 2007
3. C. Siva Ram Murthy and B. S. Manoj, Ad Hoc Wireless Networks-Architectures and Protocols, New Delhi: Pearson Education, 2013.

REFERENCE BOOKS:

1. Feng Zhao and Leonidas Guibas, Wireless Sensor Networks. Noida: Morgan Kaufman Publishers, 2004.
2. C. K. Toh, Ad Hoc Mobile Wireless Networks. New Delhi: Pearson Education, 2002.
3. Thomas Krag and Sebastin Buettrich, Wireless Mesh Networking. Mumbai: O'Reilly Publishers, 2007

WEB REFERENCES:

1. <https://www.elsevier.com/journals/ad-hoc-networks/1570-8705/guide-for-authors>
2. https://en.wikipedia.org/wiki/Wireless_ad_hoc_network
3. www.oldcitypublishing.com/journals/ahsw-n-home/

E-TEXT BOOKS:

1. <https://ebooks.benthamscience.com/book/9781608050185/>
2. <https://books.google.co.in> › Computers › Networking › General

MOOC COURSE

1. nptel.ac.in/courses/106105160/
2. https://onlinecourses.nptel.ac.in/noc18_cs09/
3. <https://swayam.gov.in/course/4408-wireless-adhoc-and-sensor-networks>

DISTRIBUTED SYSTEMS

(PROFESSIONAL ELECTIVE – 5)

IV B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS35	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

To learn

1. To layout foundations of Distributed Systems.
2. To introduce the idea of middleware and related issues
3. To understand in detail the system level and support required for distributed system
4. To understand the issues involved in studying data and design of distributed algorithms

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Identify the advantages and challenges in designing distributed algorithms for different primitives like mutual exclusion, deadlock detection, agreement, etc.
2. Design and develop distributed programs using sockets and RPC/RMI.
3. Differentiate between different types of faults and fault handling techniques in order to implement fault tolerant systems.
4. Analyze different algorithms and techniques for the design and development of distributed systems subject to specific design and performance constraints.

UNIT-I

Introduction: Definition of Distributed Systems, Goals, Types of Distributed Systems.

Characterization of Distributed Systems: Introduction, Examples of Distributed Systems, Resource Sharing and the Web, Challenges.

System Models: Introduction, Architectural Models, Fundamental Models.

UNIT-II

Time and Global States: Introduction, Clocks Events and Process States, Synchronizing Physical Clocks, Logical Time and Logical Clocks, Global States, Distributed Debugging.

Coordination and Agreement: Introduction, Distributed Mutual Exclusion, Elections, Multicast Communication, Consensus and Related Problems.

UNIT-III

Inter Process Communication: Introduction, the API for the Internet Protocols, External Data Representation and Marshalling, Client-Server Communication, Group Communication, and Case Study: IPC in UNIX.

Distributed Objects and Remote Invocation: Introduction, Communication between Distributed Objects, Remote Procedure Call, Events and Notifications, Case Study: JAVA RMI.

UNIT-IV

Distributed File Systems: Introduction, File Service Architecture, and Case Study 1: Sun Network File System, Case Study 2: The Andrew File System.

Name Services: Introduction, Name Services and the Domain Name System, Directory Services, Case Study of the Global Name Services.

Distributed Shared Memory: Introduction, Design and Implementation Issues, Sequential Consistency and IVY case study, Release Consistency, Munin Case Study, Other Consistency Models.

UNIT-V

Transactions and Concurrency Control: Introduction, Transactions, Nested Transactions, Locks, Optimistic Concurrency Control, Timestamp Ordering, Comparison of Methods for Concurrency Control.

Distributed Transactions: Introduction, Flat and Nested Distributed Transactions, Atomic Commit Protocols, Concurrency Control in Distributed Transactions, Distributed Deadlocks, Transaction Recovery.

TEXT BOOKS:

1. Distributed Systems, Concepts and Design, George Koulouris, J Dollimore and Tim Kind berg, Pearson Education, 41 Edition. 2009.

REFERENCE BOOKS:

1. Distributed Systems, Principles and Paradigms, Andrew S. Tanenbaum, Maarten Van Steen, 2nd Edition, PHI.
2. Distributed Systems, an Algorithm Approach, Sukumar Ghosh, Chapman Hall/CRC, Taylor & Fransis Group, 2007.

SOFT COMPUTING (PROFESSIONAL ELECTIVE – 5)

IV B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS36	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVE

To learn

1. To teach basic neural networks, fuzzy systems, and optimization algorithms concepts and their relations
2. To provide knowledge of Neuron model, and Applications of NN to discuss their work
3. To provide the graduate the better understanding of Fuzzy Logic and Evolutionary Computations

COURSE OUTCOMES

Upon successful completion of the course, the student is able to

1. To understand neural network (NN) paradigms
2. Apply different supervised learning algorithms on given data
3. Understand feedback neural networks & self organizing feature map
4. To learn fuzzy logic To have a knowledge of evolutionary computations, genetic algorithm(GA), evolutionary programming, classifier systems, genetic programming parse trees, mathematical foundation of GA variants of GA

UNIT-I

BASICS OF ARTIFICIAL NEURAL NETWORK: Characteristics of Neural Networks, Structure and working of a biologicalneural network, Artificial neural network: terminology, models of neurons: McCulloch Pitts model, Perceptron model, Adaline model, topology, Basic learning laws.

FUNCTIONAL UNITS FOR ANN FOR PATTERN RECOGNITION TASK: Pattern recognition problem, Basic functionalunits, PR by functional units.

UNIT-II

FEEDFORWARD NEURAL NETWORKS:

SUPERVISED LEARNING - I: Perceptrons - Learning and memory, Learning algorithms, Error correction and gradientdescent rules, Perceptron learning algorithms.

SUPERVISED LEARNING-II: Back propagation, Multilayered network architectures, Back propagation learningalgorithm, Example applications of feed forward neural networks

UNIT-III

FEEDBACK NEURAL NETWORKS & SELF ORGANIZING FEATURE MAP: Introduction, Associative learning, Hopfieldnetwork, Error performance in Hopfield networks, simulated annealing, Boltzmann machine and Boltzmann learning, state transition diagram and false minima problem, stochastic update, simulated annealing, Boltzmann machine, bidirectional associative memory, bam stability analysis. Self organization, generalized learning laws, competitive learning, vector quantization, self organizing feature map, applications of self organizing feature map.

UNIT-IV

FUZZY LOGIC: Fuzzy set theory, crisp sets, operations on crisp set, fuzzy sets, fuzzy versus crisp, operations, fuzzyrelations, crisp relations, properties. Fuzzy logic Application: Fuzzy Control of Blood Pressure.

UNIT-V

FUZZY LOGIC IN DATABASE AND INFORMATION SYSTEMS: Fuzzy Information, Fuzzy Logic in database Systems, FuzzyRelational data Models, operations in Fuzzy Relational data Models, Design

theory for Fuzzy Relational databases, Fuzzy information Retrieval and Web search, Fuzzy Object Oriented databases.

GENETIC ALGORITHMS: Introduction to Genetic Algorithms, Evolutionary Algorithms.

TEXT BOOKS:

1. Satish Kumar (2004), Neural Networks A classroom Approach, Tata McGraw Hill Publication, New Delhi.
2. Lotfi A. Zadeh(1997), Soft computing and Fuzzy Logic, World Scientific Publishing Co., Inc. River Edge, NJ, USA.

REFERENCE BOOKS:

1. B. Yegnanarayana (2006), Artificial Neural Networks, Prentice Hall of India, New Delhi, India.
2. John Yen, Reza Langari(2006), Fuzzy Logic, Pearson Education, New Delhi, India.
3. S. Rajasekaran, Vijaylakshmi Pari (2003), Neural networks, Fuzzy Logic and Genetic Algorithms Synthesis andApplications, Prentice Hall of India, New Delhi, India.

AGILE SOFTWARE DEVELOPMENT

(PROFESSIONAL ELECTIVE – 5)

IV B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS37	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

To learn

1. To familiarize the students with the features of agile development.
2. To familiarize the students with basic Agile Estimation & Planning.
3. To facilitate students in understanding the Tracking Agile Projects.
4. To facilitate students in differentiating tools.

To felicitate students in Testing

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Define the common characteristics of an agile development process.
2. List and contrast state of the practice agile methodologies.
3. Contrast agile software development process models and plan driven process models.
4. Identify software project characteristics that would be suitable for an agile process.
5. Identify software project characteristics that would not be suitable for an agile process.
6. Plan and implement a small scale software project using the Scrum process methodology.

UNIT-I

History of Agile Methodologies, Agile and Lean Software Development: Basics and Fundamentals, Extreme Programming, Scrum, Agile and Scrum Principles Agile Manifesto, Twelve Practices of XP

UNIT-II

Agile Estimation & Planning, Agile Requirements, Agile techniques to improve requirements, User Stories, Backlog Management: scrum product backlog, product backlog, Agile Architecture

UNIT-III

Tracking Agile Projects: Time tracking agile projects, Tracking agile development project progress. Lean Software Development, Agile model driven development, Agile Risk management

UNIT-IV

Agile Project Tools, Enterprise Agility, Dynamic Systems Development Method, Continuous Integration (CI).

UNIT-V

Team Dynamics and Collaboration, Enterprise agile project team, Agile Testing, Scaling Agile for Large Projects, Best Practices for Achieving Agile at Scale

TEXT BOOKS:

1. Agile Development with Scrum, Ken Schwaber & Mike Beedle, Prentice Hall, 2001
2. Integrating Agile Development in the Real World, Peter Schuh, Charles River Media, 2005

REFERENCE BOOKS:

1. Agile Software Development – The Cooperative Game (2nd Edition), Alistair Cockburn, 2007
2. Succeeding With Agile, Software Development Using Scrum, Mike Cohn, Addison Wesley, 2010

E-COMMERCE

PROFESSIONAL ELECTIVE – 5

IV B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS38	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OVERVIEW

This course is designed to understand issues that companies need to address when moving from being offline to online and learn how companies can craft business models where IT is embedded as an integral part of products, processes and customer interactions. Student would get to appreciate how IT is changing the way companies create value through networks and competition.

COURSE OBJECTIVES:

The objectives of the course are to introduce the concept of electronic commerce, and to understand how electronic commerce is affecting business enterprises, governments, consumers and people in general. In addition, we will study the development of websites using relevant software tools.

1. Acquaint students with a fundamental understanding of the environment and strategies in the New Economy.
2. Provide analytical tools to understand opportunities in unserved or underserved New Economy markets.
3. Provide a fundamental understanding of the different types and key components on business models in the New Economy.
4. Provide guiding principles behind the design and strategy of the customer web interface.
5. Understand the traditional and new communication/marketing approaches that create competitive advantage in the New Economy.
6. Provide insights on how to implement strategy in the New Economy.
7. Understand the metrics that New Economy firms to use to measure progress, customer satisfaction, and financial performance.
8. Understand the fundamentals of financially valuing New Economy companies.
9. Provide an overview of the hardware, software, servers, and the parts that make up the enabling "railroad" for the New Economy.

COURSE OUTCOMES:

After completion of this course, the students would be able to

1. Defining and analyzing the principles of E-commerce and basics of World Wide Web.
2. Defining and analyzing the concept of electronic data interchange and its legal, social and technical aspects.
3. Defining and analyzing the security issues over the web, the available solutions and future aspects of e-commerce security.
4. Defining and analyzing the concept of E-banking, electronic payment system

UNIT-I

Introduction Overview of Electronic Commerce - Definition of Electronic Commerce - E Business - Potential Benefits of E Commerce - The Internet as enablers of E Commerce - Impact of E Commerce on Business Models - E Commerce Applications - Market Forces influencing highway - The global information distribution networks - The regulatory environment for E Commerce.

UNIT-II

Electronic Data Interchange (EDI), Electronic Commerce and the Internet Introduction - traditional EDI systems - Benefits - Data Transfer and Standards - Financial EDI – EDI Systems and the Internet - Legal, Security and Private Concerns - Authentication - Internet Trading Relationships Consumer to Business (B2C) Business to Business (B2C), Consumer to Consumer (C2C) Government to Citizen - Features and Benefits - Portal Vs. Website. Impact. - Intra Organizational E Commerce - Supply Chain Management.

UNIT-III		
Cryptography and Authentication Introduction - Messaging Security Issues - Confidentiality - Integrity - Authentication. Encryption Techniques - Integrity Check Values and Digital Signatures - Good Encryption Practices – Key Management - key management tasks - Additional Authentication Methods. Firewalls - Definition - component - Functionality - securing the firewall - factors considered in securing the firewall - Limitations.		
UNIT-IV		
Electronic Payment Mechanisms Introduction r- The SET protocol - SET vs. SSL - Payment Gateway - Certificate Issuance – Certificate Trust Chain - Cryptography Methods - Dual Signatures - SET Logo - Compliance Testing - Status of Software - Magnetic Strip Cards - Smart Cards - Electronic Cheques - Electronic Cash - Third party Processors and Credit Cards - Risk and electronic payment system - Designing Electronic payment systems.		
UNIT-V		
E-Commerce Strategy and Implementation Strategic Planning for E-Commerce, Strategy in Action, Competitive Intelligence on the Internet, Implementation; Plans and Execution, Project and Strategy assessment - Managerial Issues – Global Electronic Commerce - Future of Electronic Commerce.		
TEXT BOOKS:		
1. Electronic Commerce - Security, Risk Management and Control, Greenstein and Feinman, Irwin Mc.Gra-Hill, 2000.		

BIG DATA ANALYTICS LAB

IV B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS20	PCC	L	T	P	C	CIE	SEE	Total
		-	-	4	2	30	70	100

COURSE OBJECTIVES:

To learn

1. To introduce the terminology, technology and its applications
2. To introduce the concept of Analytics for Business
3. To introduce the tools, technologies & programming languages this is used in day to day analytics cycle.

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Connect to Hadoop cluster, experiment with various Linux and HDFS commands to store data.
2. Apply the knowledge of Map Reduce programming to process the stored data in HDFS.
3. Make use of database operations to store results in tables and generate reports.
4. Connect to web data sources for data gathering, Integrate data sources with Hadoop components to process streaming data.
5. Generate reports using data visualization tools.

SYLLABUS

Week 1

- i) Perform setting up and installing Vmware for Hadoop and Linux.
- ii) Basic Linux Commands

Week 2

Run basic HDFS shell commands

Week 3

Implement the following file management tasks in Hadoop:

- Adding files and directories
- Retrieving files
- Deleting files and directories.

Hint: A typical Hadoop workflow creates data files (such as log files) elsewhere and copies them into HDFS using one of the above command line utilities.

Week 4

Write the steps to export JAR using eclipse.

Run a basic Word Count Map Reduce program to understand Map Reduce Paradigm

Week 5

Write a Map Reduce program that mines weather data.

(Weather sensors collecting data every hour at many locations across the globe gather a large volume of log data, which is a good candidate for analysis with MapReduce, since it is semi structured and record-oriented).

Week 6

Run Pig and perform basic PIG commands.

Week 7

Write Pig Latin scripts to sort, group, join, project, and filter your data.

Week 8

Run HIVE and perform basic HIVE commands to create a table and enter data into tables.

Week 9

Use Hive to create, alter, and drop databases, tables, views, functions, and indexes

Week 10

Use CDH and HUE to analyze data and generate reports for sample datasets

Week 11

Importing and exporting Data in HDFS using Sqoop from MySql database

Week 12

Use data visualization tool to generate reports on sample datasets.

IV B.TECH II SEMESTER SYLLABUS

SOFTWARE PROJECT MANAGEMENT

(PROFESSIONAL ELECTIVE – 6)

IV B. TECH- II SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS39	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

To learn

1. The main goal of software development projects is to create a software system with a predetermined functionality and quality in a given time frame and with given costs.
2. For achieving this goal, models are required for determining target values and for continuously controlling these values.
3. This course focuses on principles, techniques, methods & tools for model-based engagement of software projects.
4. Assurance of product quality and process adherence (quality assurance), as well as experience-based creation & improvement of models (process management).

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Understanding the specific roles within a software organization as related to project and process management
2. Understanding the basic infrastructure competences (e.g., process modeling and measurement)
3. Understanding the basic steps of project planning, project management. Quality assurance, and process management and their relationships
4. Describe and determine the purpose and importance of project management from the perspectives of planning, tracking and completion of project.
5. Compare and differentiate organization structures and project structures.
6. Implement a project to manage project schedule, expenses and resources with the application of suitable project management tools.

UNIT-I

CONVENTIONAL SOFTWARE MANAGEMENT: The waterfall model, conventional software Management performance. Evolution of Software Economics: Software Economics. Pragmatic software cost estimation.

IMPROVING SOFTWARE ECONOMICS: Reducing Software product size, Improving software processes, improving team effectiveness. Improving automation, Achieving required quality, peer inspections.

UNIT-II

THE OLD WAY AND THE NEWWAY - The principles of conventional software engineering. Principles of modern software management, transitioning to an iterative process.

ARTIFACTS OF THE PROCESS: The artifact sets, Management artifacts, Engineering artifacts, programmatic artifacts.

UNIT-III

WORK FLOWS OF THE PROCESS: Software Process Workflow, Inter Trans Workflows.

CHECKPOINTS OF THE PROCESS: Major Mile Stones, Minor Milestones, Periodic status assessments.

ITERATIVE PROCESS PLANNING: Work breakdown structures, planning guidelines, cost and scheduled estimating, Interaction, planning process, Pragmatic planning.

UNIT-IV

PROCESS AUTOMATION: Automation Building blocks, Project Control and Process instrumentation: The seven core Metrics, Management indicators, quality indicators, life cycle expectations, pragmatic

Software Metrics, Metrics automation, Tailoring the Process: Process discriminants.

UNIT-V

PROJECT CONTROL AND PROCESS INSTRUMENTATION: The server care Metrics, Management indicators, and quality indicators. life cycle expectations pragmatic Software Metrics, Metrics automation. Tailoring the Process: Process discriminates, Example.

FUTURE SOFTWARE PROJECT MANAGEMENT: Modern Project Profiles Next generation Software economics modern Process transitions.

Case Study: The Command Centre Processing and Display System, Replacement (CCPDS. R).

TEXT BOOKS:

1. Software Project Management. Walker Royce, Pearson Education.
2. Software Project Management, Bob Hughes & Mike Cotterell, fourth edition, Tate McGraw Hd.

REFERENCE BOOKS:

1. Applied Software Project Management, Andrew Stelbian & Jennifer Greene, O'Reilly. 2006
2. Head First PMP, Jennifer Greene & Andrew Stelman, O'Reilly. 2007
3. Software Engineering Project Management. Richard H. Thayer & Edward Yourdon, second edition, Wiley India, 2004.
4. A Project Management, Jim Highsmith. Pearson education, 2004
5. The art of Project management. Scott Berkun. O'Reilly, 2005.
6. Software Project Management in Practice. Pankaj Jalote. Pearson Education, 2002.

MACHINE LEARNING (PROFESSIONAL ELECTIVE – 6)

IV B. TECH- II SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS40	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

To learn

1. To introduce the fundamental concepts of machine learning and its applications
2. To learn the classification, clustering and regression machine learning algorithms
3. To understand the methods of solving real life problems using the machine learning techniques

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Illustrate the basic concepts of machine learning
2. Implement the classification, clustering and regression algorithms
3. Design and implement a method for solving real life problem using a suitable machine learning technique
4. Combine the evidence from two or more models/methods for designing a system.

UNIT-I

INTRODUCTION AND BAYESIAN DECISION THEORY: Machine perception - feature extraction - classification, clustering and regression - design cycle - types of learning, Bayesian decision theory - classifiers, discriminant functions, and decision surfaces - univariate and multivariate normal densities - Bayesian belief networks.

UNIT-II

COMPONENT ANALYSIS AND HIDDEN MARKOV MODELS: Principal component analysis - Linear discriminant analysis - Independent component analysis. Expectation-maximization algorithm - hidden Markov models: Evaluation - decoding - learning.

UNIT-III

CLASSIFICATION ALGORITHMS: Perceptron and back propagation neural network - radial basis function neural network - probabilistic neural network - k-nearest-neighbour rule. Support vector machine: Training - multicategory generalizations. Decision trees: classification and regression tree - random forest.

UNIT-IV

CLUSTERING AND REGRESSION ALGORITHM: k-means clustering - fuzzy k-means clustering - Gaussian mixture models - auto associative neural network. Regression analysis - support vector regression.

UNIT-V

COMBINING MULTIPLE LEARNERS: Generating diverse learners - model combination schemes - voting - error-correcting output codes - bagging - boosting - mixture of experts revisited - stacked generalization - fine-tuning an ensemble - cascading.

TEXT BOOKS:

1. R. O. Duda, E. Hart, and D.G. Stork, "Pattern classification", Second edition, John Wiley & Sons, Singapore, 2003.
2. Ethem Alpaydin, "Introduction to Machine Learning", MIT Press, Third Edition, 2014.
3. C. M. Bishop, "Pattern Recognition and Machine Learning", Springer, 2006.

REFERENCE BOOKS:

1. M. Mohri, A. Rostamizadeh, and A. Talwalkar, "Foundations of Machine Learning", MIT Press, 2012.
2. Kevin P. Murphy, "Machine Learning: A Probabilistic Perspective", MIT Press, 2012.

INFORMATION RETRIEVAL SYSTEM

(PROFESSIONAL ELECTIVE – 6)

IV B. TECH- II SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS41	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

To Learn

1. To learn about the relationships to data base management systems and libraries.
2. To understand the important concepts, algorithms, and data/file structures
3. To learn about clusters, searching techniques and visualization.
4. To expose the students to find the searching algorithms and Evaluation systems.
5. To facilitate design, and implementation of Information Retrieval (IR) systems.

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Understand the retrieval information systems and its capabilities.
2. Analyze and apply appropriate data structures and indexing concepts.
3. Use the various searching techniques for improving the information visualization.
4. Apply the searching algorithms to evaluate information systems.
5. Understand and use the multimedia information retrieval of query languages

UNIT-I

INTRODUCTION TO INFORMATION RETRIEVAL SYSTEMS: Definition, Objectives, Functional Overview, Relationship to Database Management Systems, Digital Libraries and Data Warehouses.

INFORMATION RETRIEVAL SYSTEM CAPABILITIES: Search, Browse and Miscellaneous.

UNIT-II

CATALOGING AND INDEXING: Objectives, Indexing Process, Automatic Indexing, Information Extraction. **DATA STRUCTURES:** Introduction, Stemming Algorithms, Inverted file structures, N-gram data structure, and PAT data structure, Signature file structure, Hidden Markov Models.

AUTOMATIC INDEXING: Classes of Automatic Indexing, Statistical Indexing, Natural Language, concept , Indexing, Hypertext Linkages.

UNIT-III

DOCUMENT AND TERM CLUSTERING: Introduction, Thesaurus Generation, Item Clustering,

Hierarchy of Clusters.

USER SEARCH TECHNIQUES: Search Statements and Binding, Similarity Measures and Ranking, Relevance Feedback, Selective Dissemination of Information Search, Weighted Searches of Boolean Systems, Searching the Internet and Hypertext.

INFORMATION VISUALIZATION: Introduction, Cognition and Perception, Information Visualization Technologies.

UNIT-IV

TEXT SEARCH ALGORITHMS: Introduction, Software Text Search Algorithms, Hardware Text Search Systems.

INFORMATION SYSTEM EVALUATION: Introduction, Measures used in System Evaluation, Measurement Example -TREC results.

UNIT-V

MULTIMEDIA INFORMATION RETRIEVAL: Models and Languages, Data Modeling, Query Languages, Indexing and Searching.

LIBRARIES AND BIBLIOGRAPHICAL SYSTEMS: Online IR Systems, OPACs, Digital Libraries.

TEXT BOOKS:

1. Gerald J. Kowalski, Mark T. Maybury (2000), Information Storage and Retrieval Systems: Theory and Implementation, 2nd edition, Springer International Edition, USA.
2. Information Retrieval: Algorithms and Heuristics By David A Grossman and Ophir Frieder, 2nd Edition, Springer.

REFERENCE BOOKS:

1. Database Systems, 6th edition, RamezElmasri, ShamkantB.Navathe, Pearson Education, 2013.
2. Introduction to Database Systems, C.J.Date, Pearson Education.
3. Fundamentals of Relational Database Management Systems, S. Sumathi, S. Esakkirajan, Springer.

PREDICTIVE ANALYTICS (PROFESSIONAL ELECTIVE – 6)								
IV B. TECH- II SEMESTER								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS42	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100
COURSE OBJECTIVES: To learn 1. To introduce the terminology, technology and its applications 2. To introduce the concept of predictive analysis 3. To introduce linear regression, time series concepts 4. To introduce the tools, technologies, programming languages which are used in day to day analytics cycle.								
COURSE OUTCOMES: Upon successful completion of the course, the student is able to 1. Identify basic terminology of Data Analytics 2. Compare learning with classification 3. Develop the knowledge skill and competences using tools and training 4. Analyze the importance of Analytics in business perspective.								
UNIT-I								
INTRODUCTION TO PREDICTIVE ANALYTICS & LINEAR REGRESSION (NOS 2101): What and Why Analytics, Introduction to Tools and Environment, Application of Modelling in Business, Databases & Types of data and variables, Data Modelling Techniques, Missing imputations etc. Need for Business Modelling, Regression— Concepts, Blue property- assumptions- Least Square Estimation, Variable Rationalization, and Model Building etc.								
UNIT-II								
LOGISTIC REGRESSION (NOS 2101): Model Theory, Model fit Statistics, Model Conclusion, Analytics applications to various Business Domains etc. Regression Vs Segmentation --- Supervised and Unsupervised Learning , Tree Building --- Regression, Classification, Over fitting, pruning and complexity, Multiple Decision Trees etc.								
UNIT-III								
DEVELOP KNOWLEDGE, SKILL AND COMPETENCES (NOS 9005): Introduction to Knowledge skills & competences, Training & Development, Learning & Development, Policies and Record keeping, etc.								
UNIT-IV								
TIME SERIES METHODS / FORECASTING, FEATURE EXTRACTION (NOS 2101): ARIMA, Measures of Forecast Accuracy, STL approach, Extract features from generated model as Height, Average, Energy etc and Analyze for prediction.								
UNIT-V								
WORKING WITH DOCUMENTS (NOS 0703): Standard Operating Procedures for documentation and Knowledge sharing, Defining purpose and scope documents, Understanding structure of documents – case studies, articles, white papers, technical reports, minutes of meeting etc., Style and format, Intellectual Property and Copyright, Document preparation tools – Visio, PowerPoint, Word, Excel etc., Version Control, Accessing and updating corporate knowledge base, Peer review and feedback.								
TEXT BOOKS:								
1. Student’s Handbook for Associate Analytics-III								
REFERENCE BOOKS:								

1. Gareth James Daniela Witten Trevor Hastie Robert Tibshirani. An Introduction to Statistical Learning with Applications in R

OPERATIONS RESEARCH

IV B. TECH- II SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4HS05	HSMC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100
Contact Classes: 42	Tutorial Classes: 00	Practical Classes:--			Total Classes: 42			

COURSE OBJECTIVES:

To learn

1. Explain optimum utilization of resources, effort and implement the decision effectively.
2. Understand scientific systematic approach involved and provide a good intellectual support for decision making.
3. Discuss substantial experience in taking timely management decisions and for corrective measures.
4. Analyzing the behavior of the system for the purpose of improving its performance.
5. Demonstrate reliability of solution obtained from a model depends on the validity of the model in representing the real systems.

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Applying a different linear programming problem technique which has a broad experience in finding the optimum profit.
2. Apply the knowledge of the course in solving real life problems.
3. Calculate the knowledge that tries to optimize total return by maximizing the profit and minimizing the cost or loss
4. Analyze the best (optimum) decisions relative to largest possible portion of the total organization.
5. Analyze queuing models and characteristic features of a queuing system.

UNIT-I	LINEAR PROGRAMMING	
Introduction to Linear Programming – Graphical method, Simplex methods, Primal & Dual Problem Sensitivity Analysis.		
UNIT-II	TRANSPORTATION AND ASSIGNMENT PROBLEMS	
Transportation problem – Methods of finding Initial Basic Feasible Solution, Test for optimality (MODI method), Assignment Problem –Hungarian Method ,Maximization case in assignment problem, unbalanced assignment problem.		
UNIT-III	GAME THEORY	
Decision theory-Decision under various conditions. Game Theory-Minimum & maximum strategies. Application of linear programming		
UNIT-IV	INVENTORY MODELS	
Stochastic inventory models-Single & multi period models with continuous & discrete demands. Cartesian co-ordinates).		
UNIT-V	QUEING MODELS	
Simulations-Monte Carlo simulation, generation of random numbers & simulation. Queuing models-M.M.1 & M/M/S system cost consideration		

TEXT BOOKS:

- | |
|--|
| <ol style="list-style-type: none">1. H.A.Taha, — “Operations Research ”, Prentice Hall of India, 1999, Sixth Edition.2. J.K.Sharma, — “Operations Research Theory and applications”, Macmillian |
|--|

REFERENCE BOOKS:

- | |
|---|
| <ol style="list-style-type: none">1. R.Panner Selvam – “Operation Research”, PHI Learning Private Limited.2. Hillier and Lieberman, — “Operations Research ”, Holden Day, 1986 |
|---|

**OPEN ELECTIVES
OFFERED BY
AERO DEPARTMENT**

FABRICATION PROCESSES

V Semester: OPEN ELECTIVE - I

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE17	OEC	L	T	P	C	CIA	SEE	Total
		3	1	-	3	25	75	100

COURSE OBJECTIVES

- To develop competency in understanding major fabrication processes used for aircraft production.
- To have a broad knowledge in choosing a suitable fabrication process for manufacturing aircraft components.

COURSE OUTCOMES:

The students should be able to

- Exemplify the major metal casting processes used in aircraft industry.
- Exemplify the major metal joining processes used in aircraft industry.
- Exemplify hot & cold working of metals including high energy rate forming.
- Exemplify the major surface finishing processes used in aircraft industry.
- Exemplify the major sheet metal forming processes used in aircraft industry.
- Suggest suitable molding process for plastics for the given application

UNIT-I	METAL CASTING PROCESSES	
Sand casting – Sand moulds – Types of patterns – Pattern materials – Pattern allowances – Types of Moulding sand- Properties – Core making – Moulding machines – Types of moulding machines – Melting furnaces – Working principle of Special casting processes – Shell, investment casting – Lost Wax process –Centrifugal casting –Sand Casting defects.		
UNIT-II	METAL JOINING TECHNIQUES	
Fusion welding processes – Types of Gas welding – Equipment's used – Flame characteristics – Filler and Flux materials – Arc welding equipment – Electrodes – Coating and specifications – Principles of Resistance welding – Spot/butt, seam welding – Percussion welding – Gas metal arc welding – Flux cored – Submerged arc welding –Electro slag welding – TIG welding – Principle and application of special welding processes – Plasma arc welding – Thermit welding – Electron beam welding – Friction welding – Weld defects – Brazing and soldering process.		
UNIT-III	BULK DEFORMATION AND SURFACE FINISHING PROCESSES	
Hot working and cold working of metals – Forging processes – Open and close die forging – Characteristics of the process – Types of Forging Machines – Typical forging operations – Rolling of metals –Types of Rolling mills –Tube piercing – Defects in rolled parts – Principles of Extrusion – Types of Extrusion – Hot and Cold extrusion – Principle of rod and wire drawing – Equipments used - surface roughness and measurements - grinding - honing - lapping - polishing - burnishing - thermal spray coating.		
UNIT-IV	SHEET METAL FORMING PROCESSES	
Sheet metal characteristics – Typical shearing operations, bending and drawing operations –Formability of sheet metal –Working principle and application of special forming processes – Hydro forming – Explosive forming – Magnetic pulse forming – Peen forming – Super plastic forming – Process characteristics		
UNIT-V	FORMING AND SHAPING OF PLASTICS	

Types of plastics – Characteristics of forming and shaping processes – Moulding of Thermoplastics – Working principles– Injection moulding – Plunger and screw machines – Blow moulding –Extrusion – Typical industrial applications – Thermoforming – Processing of Thermosets – Working principles and typical applications – Compression moulding – Transfer moulding – Bonding of Thermoplastics – Induction and Ultrasonic methods.

TEXT BOOKS:

1. Hajra Choudhury, Elements of Workshop Technology, Vol. I and II, Media Promoters Pvt. Ltd., Mumbai, 2007
2. Serope Kalpak Jain, Steven R. Schmid, Manufacturing Engineering and Technology, Pearson Education, Inc. 4th Edition, 2009

REFERENCE BOOKS:

1. Elements of Manufacturing Processes, B.S. Magendran Parashar & R.K. Mittal, Prentice Hall of India, 2008 Manufacturing Technology, P.N. Rao, Tata Mc Graw-Hill Publishing Limited, 2010.
2. A text book of production technology, P.c. Sharma, S. Chand and Company, 2010
3. Manufacturing Process – Begman, John Wiley & Sons, VIII Edition, 1999

WEB REFERENCES:

1. <https://www.velaction.com/fabrication/>
2. <https://www.boconline.co.uk/en/.../fabrication-processes/fabrication-processes.html>

E-TEXT BOOKS

:

1. Manufacturing Technology, P.N. Rao, Tata Mc Graw-Hill Publishing Limited, 2010.
2. Manufacturing Process – Begman, John Wiley & Sons, VIII Edition, 1999

MOOC COURSE

1. <https://nptel.ac.in/courses/112107144/>
2. <https://nptel.ac.in/courses/112105127/>
3. <https://nptel.ac.in/courses/112104195/>
4. <https://nptel.ac.in/courses/112107078/>

FUNDAMENTALS OF AVIONICS

V Semester: OPEN ELECTIVE - I

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE18	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES

The purpose of this subject is to provide the students with the theoretical background and engineering applications.

1. Overview on Aviation using Electronics
2. Basic understanding about major electronics systems used for communication
3. Basic understanding about major devices, display and flight controls used in aircraft

COURSE OUTCOMES:

At the end of the course the students are able to:

1. To explain the instrumentation used in avionics.
2. To classify various ranges of the communication techniques used in aircraft.
3. To distinguish between network systems, controlling parts & surfaces
4. To compare various principles of navigation systems
5. To build phenomena of auto pilot control system

UNIT-I

BASICS & FLIGHT DECK AND DISPLAY SYSTEMS

BASICS: Basic principles of Avionics, Typical avionics sub system in civil/ military aircraft and space vehicles.

FLIGHT DECK AND DISPLAY SYSTEMS: Flight deck display technologies, CRT, LED, LCD, Touch screen, Head up display, Electronic instrumentation systems.

UNIT-II

COMMUNICATION SYSTEMS

AUDIO AND COMMUNICATION SYSTEMS: Aircraft audio systems, basic audio transmitter and receiver principles, VHF communication system, UHF communication systems.

UNIT-III

FREQUENCY RANGING SYSTEM

RANGING AND LANDING SYSTEMS: VHF Omnirange, VOR receiver principles, distance maturity equipment, principles of operation, Instrument landing system, and localizer and glide slope.

POSITIONING SYSTEM: Global positioning system principles, triangulation, position accuracy, applications in aviation

UNIT-IV

NAVIGATION SYSTEM

INERTIAL NAVIGATION SYSTEM: Principle of Operation of INS, navigation over earth, components of inertial Navigation systems, accelerometers, gyros and stabilized platform.

SURVEILLANCE SYSTEM: ATC surveillance systems principles and operation interrogation and replay standards, Collision avoidance system, ground proximity warning system

UNIT-V

AUTO FLIGHT SYSTEM

AUTO FLIGHT SYSTEM: Automatic flight control systems fly by wire and fly by light technologies, flight director systems, flight management systems..

TEXT BOOKS:

1. N. S. Nagaraja(1996), Elements of electronic navigation, 2nd edition, Tata McGraw Hill, New Delhi.
2. Janes W. Wasson, Jeppesen Sandersen(1994), Avionic systems Operation and maintenance,

REFERENCE BOOKS:

1. Albert Hel Frick (2010), Principle of Avionics, 6th edition, Avionics Communications Inc, India.
2. H. J. Pallet (2010), Aircraft Instrumentation and Integrated systems, Pearson Education, New Delhi.
3. J. Powell (1998), Aircraft Radio Systems, Pitman publishers, London

WEB REFERENCES:

1. <https://www.britannica.com/technology/avionics>
2. <https://www.pdc.com/aviation/>

E-TEXT BOOKS:

1. Advanced Avionics Handbook (FAA)
2. Digital Avionics: A Computing Perspective - Elisabeth A. Strunk John C. Knight

MOOC COURSE

1. <https://www.canvas.net/browse/erau/courses/aviation-101>

INTRODUCTION TO JETS AND ROCKETS

V Semester: OPEN ELECTIVE - I

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE19	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	25	75	100

COURSE OBJECTIVES

The course is intended to serve as an introduction to air breathing propulsion systems and Rocket Propulsion Systems.

- 1 Illustrate an overview of aerospace propulsion system.
- 2 Identify the foundation in fundamentals of thermodynamics.
- 3 Compare the ideal components and characteristics of jet engine
- 4 Interpret the performance of nozzles
- 5 Simplify the ideal performance analysis of rocket engines.
- 6 Select appropriate fuel for aerospace application.

COURSE OUTCOMES:

At the end of the course the students are able to:

- 1 Explain the complexity in working of various engines
- 2 Interpret the elementary principles of thermodynamic cycles as applied to propulsion analysis
- 3 Analyze the process involved in individual components
- 4 Compare the nozzles with various operating conditions.
- 5 Determine Equations of motion in free space, Tsiolkovsky's equation.
- 6 Classify the types of fuel in aviation and aerospace engineering.

UNIT-I	INTRODUCTION TO AEROSPACE PROPULSION	
What is Propulsion system? Propulsive Systems – Evolution, Development, Growth and Challenges. Fundamentals of Thermodynamics – Variables, Thermodynamic Process, Introduction to IC Engines and Reciprocating Engines, Propellers and Working of Propellers.		
UNIT-II	PRINCIPLES OF JET PROPULSION	
Fundamentals of jet propulsion, Working Principle, Analysis of Ideal Jet Engine cycle, Engine components- merit- significance- ideal component characteristics, Classification – turbo jet, turbo fan, turbo prop and Ramjet engines. Basic Problems based on Engine Cycle.		
UNIT-III	RAMJET, SCRAMJET ENGINES AND NOZZLES	
Speed limitations of gas turbines, Basics of Ramjets, Combustors for liquid fuel ramjet engines, Combustion Instability and its Suppression, Solid fuel Ramjet Engines, SCRAM jet engines, Applications of RAM Jet and SCRAM Jet Engines to Missiles with Examples, Nozzles- Types of Nozzles, Converging-Diverging Nozzle, Variable Nozzle and Effects of Pressure Ratios on Engine Performance.		
UNIT-IV	ROCKET THEORY	
Applications of Rockets, Types of Rockets, Basics of Thermal Rocket Engine-Thermodynamics and Ideal Performance Analysis, Equations of motion-Rocket Motion in free space, Tsiolkovsky's equation, Rocket Parameters, Burnout range, Burnout Velocity. Practical Problems		
UNIT-V	PROPELLANT ROCKETS	

Solid Propulsion-Solid Propellant Rockets, Basic Configuration and Performance, Propellant Grain and Configuration, Propellant Characteristics Combustion Chamber, Ignition Process Liquid Propulsion - Design consideration of liquid rocket combustion chamber, injector, and propellant feed lines, valves, propellant tank outlet and helium pressurized and turbine feed systems-.BIO Fuels and Impact on the Atmosphere, Aviation turbine fuels - Requirements of aviation fuels of kerosene type.

TEXT BOOKS:

1. Mechanics and Thermodynamics of Propulsion – Philip G Hill & Carl R Peterson , Pearson Publication – 2nd Edt
2. Rocket Propulsion Elements, Sutton, G.P., John Wiley, 1993.
3. Fuels and Combustion, Sharma, S.P. and Mohan .C., Tata McGraw Hill Publishing Co, Ltd., 1984

REFERENCE BOOKS:

1. The Jet Engine – Rolls Royce
2. Gas Turbines and Jet and Rocket Propulsion, M. L. Mathur, R. P. Sharma, Standard Publishers Distributors.

WEB REFERENCES:

1. <http://engineeringvideolectures.com/course/24>
2. <http://engineeringvideolectures.com/course/28>

E-TEXT BOOKS:

1. Jet Propulsion (3rd ed. - Nicholas Cumpsty
2. A Simple Guide To Understanding Jet Engines

MOOC COURSE

1. <https://online.stanford.edu/courses/aa283-aircraft-and-rocket-propulsion>

INTRODUCTION TO AIRCRAFT INDUSTRY

VI Semester: OPEN ELECTIVE - II

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE25	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	25	75	100

COURSE OBJECTIVES

The aim is to introduce students the overview of the Aircraft Industry. The course covers basic design process, Aircraft Stability & control and different mechanical systems.

- 1 Familiarize students with major types of aircraft industry.
- 2 Illustrate the role of regulatory bodies and its business context
- 3 Exemplify the organisational structure of airlines and airports
- 4 Illustrate the importance of air navigation services and its types.
- 5 Familiarize students with air safety and security as a means for ensuring passenger safety.

COURSE OUTCOMES:

At the end of the course the students are able to:

- 1 Discuss the types of aerospace industry and evolution of flying machine to airline alliances.
- 2 Explain standardisation and the importance of regulatory bodies.
- 3 Outline the organisational structure of airlines and airports.
- 4 Demonstrate major types of air navigation services and air traffic control services.
- 5 Illustrate the major processes involved in air safety and security.

UNIT-I	AIRCRAFT INDUSTRY OVERVIEW	
Flying machine to airline alliances, Types of Aerospace Industry, Introduction to ages of engineering, Aerospace Manufacturing, Introduction to the space environment & human space exploration.		
UNIT-II	REGULATORY AND BUSINESS CONTEXT	
Regulatory bodies, Introduction to standardisation, Standards and recommended practices (SARPS), Freedoms of the air, Role of governments, Major agreements and treaties, Business side of the industry – Key performance indicators (revenue passenger kilometre, yield etc.), Industry characteristics of passenger airlines.		
UNIT-III	AIRLINES AND AIRPORTS	
Introduction to airlines, Organisational structure, Types of airline personnel, Flight crew and cabin crew, Role of training with airlines, Organisational culture of an airline Introduction to airports – Airport personnel, Processing passengers and freight, Security		
UNIT-IV	AIR NAVIGATION SERVICES	
Introduction to air navigation services, Types of navigation services, Air traffic control (ATC) services, Structure of controlled airspace, Air traffic and control services, Classifications of airspace, Flight information services and alerting services, Air navigation facilities and air navigation service (ANS) providers		
UNIT-V	AIR SAFETY AND SECURITY	
Introduction to air safety and security, Manufacturer responsibilities, Maintenance procedures, Airside safety, Safety in the air, Issues in air safety, Accident and incident investigation, Security – Response of International Civil Aviation Organization (ICAO).		
TEXT BOOKS:		

1. Peter Belobaba, Amedeo Odoni and Cynthia Barnhart (2015), The Global Airline Industry (Aerospace Series), John Willey & Sons, Ltd, United Kingdom.
2. Connor R Walsh, Airline Industry – Strategies, Operations and Safety, Nova Science Publishers, New York.
3. Shevell (2004), Fundamentals of Flight, 2nd edition, Pearson Education Limited, New Delhi
4. Allan Seabridge, Ian Moir (2008), Aircraft Systems: Mechanical, Electrical and Avionics Subsystems Integration, 3rd edition, John Willey & Sons, New York.

REFERENCE BOOKS:

1. Andrew R Thomas (2011), Soft Landing: Airline Industry Strategy, Service and Safety, Apress, New York
2. Ahmed Abdelghany, Khaled Abdelghany (2009), Modeling Applications in the Airline Industry, Ashgate Publishing Ltd, England.
- 3.. Kermode, A.C. (1989), Flight without Formulae, 5th edition, Pearson Education Limited, New Delhi.
4. Raymer Daniel (2002), Aircraft Design: A Conceptual Approach, AIAA Publisher, USA.

WEB REFERENCES:

1. <https://www.investopedia.com/features/industryhandbook/airline.asp>
2. http://web.mit.edu/airlines/analysis/analysis_airline_industry.html
3. <https://www.ukessays.com/essays/commerce/an-introduction-to-the-airline-industry-business-essay.php>

E-TEXT BOOKS:

1. Introduction to the Airline Industry (IATA Aviation Distance Learning) Kindle Edition, IATA Commercial Aviation 101, Greg Gayden.

MOOC COURSE

1. <https://www.iata.org/training/courses/Pages/airline-industry-introduction-talg50.aspx>

NON DESTRUCTIVE TESTING METHODS

VI Semester: OPEN ELECTIVE - II

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE26	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	25	75	100

COURSE OBJECTIVES:

1. To impart knowledge about the non-destructive testing methods.
2. To provide knowledge on the selection of NDT methods for application in engineering industries.

COURSE OUTCOMES:

1. Recognize various non-destructive techniques for engineering industries.
2. Select appropriate non-destructive technique for defects detection in manufactured/operating parts.
3. Perform inspection using major non-destructive testing methods.

UNIT-I SURFACE TECHNIQUES

Introduction to non-destructive testing (NDT) - importance of NDT techniques - types of NDT techniques - ASME, ASTM, AWS, BIS, SAE standard sample specifications - visual testing (direct and remote visual inspection) - principle and types of liquid penetrant tests (LPT) - properties of liquid penetrants and developers - advantages and limitations of LPT - applications of LPT.

UNIT-II MAGNETIC PARTICLE TESTING

Introduction to magnetic particle testing (MPT) - magnetization methods - dry particle and wet fluorescent particle techniques - demagnetization - advantages and limitations of MPT - magnetic flux leakage testing - principle, instrumentation and applications of electromagnetic induction techniques and eddy current testing (ECT) method.

UNIT-III ULTRASONIC TESTING

Introduction to ultrasonic testing (UT) - characteristics of ultrasonic waves - principle of UT – UT probes - UT inspection methods (pulse echo, transmission and phased array techniques, PAUT) - types of scanning and displays - application of UT for welded parts.

UNIT-IV RADIOGRAPHY TESTING

Introduction to radiography testing (RT) - sources of X-rays and Gamma rays - characteristics of Xrays and Gamma rays (absorption, scattering) - filters and screens - film radiography and digital radiography (shadow formation, exposure factors, film handling and storage) - inverse square law - exposure charts - penetrameters - safety issues.

UNIT-V SPECIAL TECHNIQUES

Acoustic emission testing (AET) principle, advantages, limitations - instrumentation and application of AET - infra red thermography (IRT) - contact and non-contact inspection methods - pressure and leak detection - LASER shearography - acoustic holography.

TEXT BOOKS:

1. Baldev Raj, T. Jayakumar, M. Thavasimuthu, "Practical Non-Destructive Testing", Narosa Publishing, London, 2012.
2. Paul E. Mix, "Introduction to Non Destructive Testing", A Training Guide, Wiley- Interscience, New Jersey, USA, June 2005.

REFERENCE BOOKS:

1. ASM Metals Handbook, V-17, "Non-Destructive Evaluation and Quality Control", American Society of Metals, Metals Park, Ohio, USA, 2001
2. W.T. McGonagle, "Non-Destructive Testing", McGraw Hill Book Co., USA, 2013.
2. Louis Cartz, "Non-Destructive Testing", ASM International, Metals Park Ohio, US, 2007.
3. 4. Barry Hull and Vernon John, "Non Destructive Testing", ELBS/Macmillan, Hampshire, UK, 2015.

WEB REFERENCES:

1. <https://inspectioneering.com/tag/nondestructive+testing>
2. <https://www.element.com/materials-testing-services/non-destructive-testing-and-inspection>
3. <https://www.twi-global.com/technical-knowledge/faqs/what-is-non-destructive-testing>

E-TEXT BOOKS:

1. Non destructive Testing Methods and New Applications by Mohammed Omar, Intech, 2012.
2. Non-Destructive Testing by S Ramachandran, A Anderson and T Rajasanthosh Kumar, Airwalk Publications, 2017, Kindle Edition.

MOOC COURSE

1. <https://nptel.ac.in/courses/113106070/>

FUNDAMENTALS OF FINITE ELEMENT METHODS

VI Semester: OPEN ELECTIVE - II

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE27	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	25	75	100

COURSE OBJECTIVES:

1. To understand the concept of numerical analysis of structural components.
2. To understand the basic knowledge of Isoparametric Element Formulation.
3. To impart knowledge on both ANSYS, NASTRAN.

COURSE OUTCOMES:

1. Demonstrate an understanding of the principles and theories of finite element method.
2. Understand the concepts of discretization, approximations, mathematical tools, FEM tools and error control
3. Analyze 2D problems using bar, beam and truss elements.
4. Demonstrate the basic concepts of isoparametric elements and its application in aircraft.
5. Apply Finite element methods for heat transfer and dynamic loading problems.

UNIT-I

INTRODUCTION TO FEM & 1-D PROBLEMS

Basic concept, historical background, application of FEM, general description, comparison of FEM with other methods. Basic equations of elasticity, Stress, strain relations, Strain, Displacement relations. ONE DIMENSIONAL PROBLEM: Finite element modeling coordinates and shape functions. Potential Energy approach: Assembly of Global stiffness matrix and load vector. Finite element equations, Treatment of boundary conditions.

UNIT-II

ANALYSIS OF BEAMS

Hermite shape functions-Element stiffness matrix for two nodes, two degrees of freedom per node beam element, load vector, deflection, stresses.

UNIT-III

2-D PROBLEMS

CST-Stiffness matrix and load vector, Isoparametric element representation, Shape functions, convergence requirements, Problems.

UNIT-IV

STEADY STATE HEAT TRANSFER ANALYSIS

One dimensional analysis of a fin and two dimensional analysis of thin plate.

UNIT-V

DYNAMIC ANALYSIS

Formulation of finite element model, element matrices, Lumped and consistent mass matrices-evaluation of Eigen values and Eigen vectors for a stepped bar.

TEXT BOOKS:

1. R. Tirupathi Chandrapatla (2011), Introduction to Finite Elements in Engineering, 4th edition, Pearson Education, India.
2. S. S. Rao (2012), The Finite Element Methods in Engineering, 5th edition, Elsevier, USA.
3. V. David. Hutton (2010), Fundamentals of finite elements analysis, 1st edition, Tata McGraw-Hill education (P) Ltd, New Delhi, India

REFERENCE BOOKS:

1. Chennakesava R. Alavala (2009), Finite elements methods, 1st edition, second reprint, Prentice Hall of India publishers, New Delhi, India.
2. J. N. Reddy (2010), Anintroduction to Finite Element Method, 3rd edition, Tata McGraw hill education (P) Ltd, New Delhi, India.
3. Kenneth H. Huebner, Donald L. rDd ewhirst, Douglas E. Smith , Ted G. Byrom (2009), The Finite
4. Element Methodfor Engineers, 3 edition, John Wiley & sons (ASIA) Pvt. Ltd., New York.

WEB REFERENCES:

<https://nptel.ac.in/courses/112104116/>

E-TEXT BOOKS:

http://web.mit.edu/kjb/www/Books/FEP_2nd_Edition_4th_Printing.pdf

MOOC COURSE

<https://nptel.ac.in/courses/112104193/>

GUIDANCE AND CONTROL OF AEROSPACE VEHICLES

VII Semester: OPEN ELECTIVE - III

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE40	OEC	L	T	P	C	CIA	SEE	Total
		3	1	-	3	25	75	100

COURSE OBJECTIVES

The purpose of this subject is to provide the students with the theoretical background and engineering applications.

1. To introduce the concepts of Navigation, guidance and control
2. To familiarize with various ways in which aerospace vehicles are guided and controlled
3. The dynamic objectives which students also learn to achieve by designing flight control systems.
4. Familiarize with the control principles of rockets and missiles
5. To give Insight into the manoeuvres of the space craft

COURSE OUTCOMES:

At the end of the course the students are able to:

1. Formulate the navigational equations of the space vehicle
2. Describe the guidance of the vehicle with state feed back
3. Explain the automatic control and guidance of the aircraft
4. Evaluate the control techniques of the rockets and missiles
5. Describe major manoeuvres of the space aircraft.

UNIT-I	NAVIGATION	
Introduction, Basic Principles and Definitions; Dead reckoning and Position Fixing, Celestial, Radio, Inertial Navigation; Principle and Construction of Accelerometers, Mechanical Gyros and Ring Laser Gyros, Inertial Measurement Units, Navigation Equations, Sensor Error Models, Kalman Filter, Attitude Heading Reference System, GPS, Terrain Reference Navigation.		
UNIT-II	GUIDANCE	
Optimal Terminal Guidance of Interceptors, Optimal Terminal Guidance - planar and non-planar, Robust and Adaptive Guidance, Guidance with State Feedback , Guidance with Normal Acceleration Input , Minimum Energy Orbital Transfer.		
UNIT-III	GUIDANCE AND CONTROL OF AIRCRAFT	
Powered Flying Controls, Helicopter Flight Controls, Fly-by-Wire Flight Control, Control laws, Redundancy and Failure Survival, Digital Implementation, Fly-by-Light Flight Control, Auto Pilot, Flight Management Systems, Unmanned Aerial Vehicle.		
UNIT-IV	CONTROL TECHNIQUES/ CONTROL OF ROCKETS AND MISSILES	Classes: 12
Open-loop and Closed Loop Control Systems, Multi-variable Optimization, Optimal Control of Dynamic Systems, Hamiltonian and Minimum Principle and Jacobi-Bellman Equation, Linear Time-Varying System with Quadratic Performance Index..		
UNIT-V	CONTROL OF SPACECRAFT	
Launch of Satellite/ Spacecraft, Terminal Control of Spacecraft Attitude, Optimal Single-Axis Rotation of Spacecraft, Multi-axis Rotational Manoeuvres of Spacecraft, Spacecraft Control Torques, Rocket Thrusters, Reaction Wheels, Momentum Wheels and Control Moment Gyros, Torque.		
TEXT BOOKS:		

1. Tewari, A.—Advanced Control of Aircraft, Spacecraft and RocketsII, John Wiley & Sons, Ltd, Chichester, UK, 2011
2. Nelson R. C - Flight Stability and Automatic Control, SIE edition, McGraw Hill, New York, 2007.
3. R.P.G Collinson —Introduction to Avionics SystemsII, Springer; 3rd ed. edition, 2011.

REFERENCE BOOKS:

1. Noton,M. —Spacecraft navigation and Guidancell, Springer-Verlag, Germany, 1998
2. Mc. Cormic 2. B. W - Aerodynamics, Aeronautics and Flight Mechanics, Wiley India Pvt. Ltd, USA, 2010.
3. Richard H. Battin —An Introduction to the Mathematics and Methods of AstrodynamicsII, AIAA, 1999.

WEB REFERENCES:

1. <https://nptel.ac.in/courses/101106043/6>
2. <https://nptel.ac.in/courses/101106042/>

E-TEXT BOOKS:

1. <https://www.amazon.com/Guidance-Control-Aerospace-Vehicles-Cornelius/dp/1258807521>

MOOC COURSE

1. https://swayam.gov.in/nd1_noc19_ae10
2. <https://www.mooc-list.com/course/abc-flight-mechanics-how-do-planes-fly-futurelearn>

WIND TUNNEL TECHNIQUES

VII Semester: OPEN ELECTIVE - III

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE41	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	25	75	100

COURSE OBJECTIVES

1. To interpret the basic concepts of measurement of forces and moments on models during the wind tunnel testing.
2. To understand the application of various types of wind tunnels.
3. To learn the basic measurement procedure involving wind tunnel testing.

COURSE OUTCOMES:

At the end of the course, student will be able to

1. Understand basics of aerodynamics and to identify the type of wind tunnel
2. Develop and understand flow visualization techniques over model
3. Understand concepts of low speed and high speed wind tunnels
4. Understand measurement and balancing of loads on model
5. Understand the different types of equipment's for measuring pressure and velocity

UNIT-I WIND TUNNELS

Wind Tunnel, layouts and nomenclature, Types of Wind Tunnels – continuous and intermittent -closed circuit and open circuit - closed jet and open jet test section – application.
Special purpose tunnels - Smoke Tunnels – Water Tunnels – Spin tunnel, automobile wind tunnel and environmental wind tunnel Important parameters of flow similarity. types of flow similarities for compressible and incompressible flows Model power consideration.

UNIT-II FLOW VISUALIZATION TECHNIQUES

Path – Streak – Stream and Timelines; Techniques: Smoke, Tuft, Streaks, Surface oil flow. Pressure measurements: Manometers – U-Tube, Inclined and Precession. Bourdon Gauge and Pressure Transducer – Strain Gauge, Semi conductor – Absolute and Differential. Velocity Measurements: Pivot Tube – Static and Total. Calibration of test section: Test section flow calibration and Boundary Layers.

UNIT-III MEASUREMENTS OF FORCES AND MOMENTS

Forces, moments and Reference Frames – Balances – Internal and External - Requirements and Specifications – Fundamentals of Model Installations. Boundary correction, types of blockages: 2-d.

UNIT-IV HIGH SPEED WIND TUNNELS

Supersonic Wind Tunnels and - Classification - Runtime - Compressors - Charging Times - nozzle Mass Flows - Starting Loads - Model Size – Calibration.
Hypersonic Wind Tunnels: Classification – Runtime – Vacuum Tanks – Vacuum pumps – Evacuation Times. Shock Tube: Driver – driven – Vacuum Pumps – Diaphragm.

UNIT-V HIGH SPEED FLOW VISUALIZATIONS AND MEASUREMENTS

Schlieren and Shadow Graph – Pressure sensitive Paints – Temperature sensitive Paints – Force Measurements – Strain Gauge Balances – Pressure Measurements.

TEXT BOOKS:

1. Rae, W.H. and Pope, A. —Low Speed Wind Tunnel TestingII, John Wiley Publication, 1999
2. Pope, A., and Goin, L., —High Speed wind Tunnel TestingII, John Wiley Publication , 1999
3. Pope, J B Barlow —low speed wind tunnel testing — 3 edition j.w publication

REFERENCE BOOKS:

1. John D. Anderson, Jr., "Fundamentals of Aerodynamics", Third edition, McGraw-Hill publications, 2001
2. E L Houghton and PW Carpenter, "Aerodynamics for Engineering students", Fourth edition, Edward Arnold publications, 1993.
3. L.M Milne Thomson, —Theoretical AerodynamicsII, 1996 McGraw-Hill, New Delhi.
4. R. Halmshaw (1991), Non-Destructive Testing, 2nd edition, Edward Arnold, New York

WEB REFERENCES:

1. wingsofaero.in/introduction-to-wind-tunnel-techniques
<https://www.keyshone.com/wind-tunnel-techniques-lecture-notes-for-aero/>

E-TEXT BOOKS:

1. <https://www.scribd.com/document/324894849/wind-tunnel-technique-notes-for-aeronautical-engineers-tp-1-pdf>
2. https://www.academia.edu/.../john_d_anderson_jr_fundamentals_of_aerodynamics

MOOC COURSE

<https://nptel.ac.in/courses/101106040>

INTRODUCTION TO AEROSPACE TECHNOLOGY

VII Semester: OPEN ELECTIVE - III

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE42	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES

1. To introduce the basic concepts of Aerospace technology and the current developments in the field.
2. To provide knowledge on the basic principles on which the development of aerodynamics, Structures, propulsion and satellite systems.

COURSE OUTCOMES:

Students should be able to

1. Compare the atmosphere conditions of different altitudes for spacecraft system
2. Analyze how lift, drag and thrust are generated and understand which components constitute them
3. Analyze the flight performance parameters with respective stability condition
4. Distinguish the different systems used in a satellite mission
5. Design lighter than air vehicle using Catia software

UNIT-I	HISTORY OF FLIGHT- THE AEROSPACE ENVIRONMENT	
Balloons and dirigibles, heavier than air aircraft, commercial air transport, introduction of jet aircraft, helicopters, missiles, conquest of space, commercial use of space, exploring solar system and beyond. Earth's atmosphere, the temperature extremes of space, laws of gravitation, low earth orbit, microgravity, benefits of microgravity. The near earth radiative environment. The magnetosphere. Environmental impact on spacecraft. Meteoroids and micrometeoroids, space debris. Planetary environments		
UNIT-II	AERODYNAMICS AND FLIGHT VEHICLE PROPULSION	
Anatomy of the airplane, helicopter, launch vehicles and missiles, space vehicles. Static forces and moments on the vehicle. Understanding engineering models. Aerodynamics of wings and bodies. Generation of lift. Sources of drag. Force and moment coefficients, centre of pressure. Thrust for flight, the propeller, the jet engine, rocket engines- description, principles of operation. Governing equations.		
UNIT-III	FLIGHT VEHICLE PERFORMANCE AND STABILITY	
Performance parameters. Performance in steady flight, cruise, climb, range, endurance; accelerated flight- symmetric manoeuvres, turns, sideslips, take off and landing. Flight vehicle stability- longitudinal, lateral and directional- static, dynamic; trim, control. Handling qualities of airplanes		
UNIT-IV	SATELLITE SYSTEMS ENGINEERING- HUMAN SPACE EXPLORATION	
Satellite missions, an operational satellite system, elements of satellite, satellite subsystems. Satellite structures, mechanisms and materials. Power systems. Communication and telemetry. Thermal control. Attitude determination and control. Propulsion and station keeping. Space missions. Mission objectives. Case studies. Human space flight missions- goals, historical background. The Soviet and US missions. The Mercury, Gemini, Apollo (manned flight to the moon), Skylab, Apollo-Soyuz, Space Shuttle. International Space Station, extravehicular activity		
UNIT-V	INTRODUCTION TO ENGINEERING DESIGN, AIR TRANSPORTATION	

Design as a critical component of engineering education- as a skill- the process, design thinking, design drawing. Design for mission, performance and safety requirements. Concurrent engineering. Computer aided engineering, design project. Example: the lighter-than – air vehicle student design project at MIT. Air Transportation Systems- civil, military- objectives- principal constituents- the vehicle, the ground facilities, the organization- role. Regulation- national and international. Indian effort- civil and military- in the field of Aerospace Engineering.

TEXT BOOKS:

1. Newman, D., Interactive Aerospace Engineering and Design, (with software and reference material on CD), McGraw-Hill, 2002, ISBN 0-07-112254-0.
2. Anderson, J.D., Introduction to Flight, fifth edition, Tata McGraw-Hill, 2007, ISBN: 0-07-006082

REFERENCE BOOKS:

1. Russell Mikel, Aerospace and Aeronautical Engineering, Willford press, 2017.
2. Ajoy kumar Kundu, Mark A Price and David Riordan, Conceptual Design: An Industrial Approach, Wiley-Blackwell, 2019.

WEB REFERENCES:

1. <https://www.grc.nasa.gov/www/k-12/airplane/index.html>

E-TEXT BOOKS:

1. https://www.researchgate.net/publication/259576416_Fundamentals_of_Aerospace_Engineering
2. http://www.adl.gatech.edu/extrovert/Ebooks/ebook_Intro.pdf

MOOC COURSE

1. https://onlinecourses.nptel.ac.in/noc18_ae07/

**OPEN ELECTIVES
OFFERED BY
CSE DEPARTMENT**

FUNDAMENTALS OF DATABASES

V SEMESTER - OPEN ELECTIVE – I

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS43	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

The course should enable the students to

1. Understand the basic database concepts, data models including the architecture and operation of the relational data model.
2. Make a study of Structured Query Language (SQL) and construct simple and moderately advanced database queries using SQL.
3. Conceptualize relational database design and depict a database system using ERdiagram.
4. Know the fundamental concepts of transaction processing- concurrency control techniques and recovery procedure.
5. Develop a database application using commercial databases

COURSE OUTCOMES:

At the end of the course students are able to

1. Design effective database schema for realistic applications.
2. Design a database using ER modelling.
3. Make use the basic concepts of database systems in database design.
4. Apply SQL queries to interact with database.
5. Apply normalization on database design to eliminate anomalies.

UNIT-I	INTRODUCTION	
Introduction :Relational model, File System vs. Database System -- Data Models – Database System Architecture – Relational Databases - Relational Model - Keys -- Integrity Constraints - Relational Algebra		
UNIT-II	SQL FUNDAMENTALS	
SQL FUNDAMENTALS :SQL fundamentals - SQL Data Definition - Basic Structure of SQL Queries - Additional Basic Operations - Set Operations - Aggregate Functions - Nested Subqueries - Join Operations - Views		
UNIT-III	DATABASE DESIGN	
DATABASE DESIGN : Entity-Relationship Model – ER Diagrams - Mapping ER Model to Relational Model - Functional Dependencies Normalization - First Normal Form - Second Normal Form - Third Normal Form		
UNIT-IV	TRANSACTION MANAGEMENT	
TRANSACTION MANAGEMENT: Transaction Concepts - Transaction Recovery – ACID Properties - Concurrency Control – Need for Concurrency Control – Serializability - Locking Protocols – Two Phase Locking – Deadlock – Recovery		
UNIT-V	APPLICATION DEVELOPMENT	
APPLICATION DEVELOPMENT :Commercial Databases - Oracle/MySQL - PL/SQL – Stored Procedures - Functions - Triggers - Database Security using SQL commands		
TEXT BOOKS:		
1. Abraham Silberschatz, Henry F. Korth, S. Sudharshan, “Database System Concepts”, Sixth Edition, Tata McGraw Hill, 2010.		

REFERENCE BOOKS:

1. C.J.Date, A.Kannan, S.Swamynathan, "An Introduction to Database Systems", Eighth Edition, Pearson Education, 2006.
2. Ramez Elmasri, Shamkant B. Navathe, "Fundamentals of Database Systems", Sixth Edition, Pearson Education, 2010.
3. Raghu Ramakrishnan, Johannes Gehrke "Database Management Systems", Fourth Edition, Tata McGraw Hill, 2010.
4. G.K.Gupta, "Database Management Systems", Tata McGraw Hill, 2011.
5. Carlos Coronel, Steven Morris, Peter Rob, "Database Systems: Design, Implementation and Management", Ninth Edition, Cengage Learning, 2011
6. B. A. Forouzan, R. F. Gillberg, "C Programming and Data Structures", Cengage Learning, India, 3rd Edition, 2014.

WEB REFERENCES:

1. <http://iips.icci.edu.iq/images/exam/databases-ramaz.pdf>
2. https://www.w3schools.com/sql/sql_intro.asp

E-TEXT BOOKS:

1. <http://www.freebookcentre.net/Database/dbCategory.html>

MOOC COURSE

1. <https://swayam.gov.in/course/1322-fundamentals-of-database-system>

INTRODUCTION TO C++								
V SEMESTER - OPEN ELECTIVE – I								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS44	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100
COURSE OBJECTIVES: 1. To familiarize the students with the different techniques pertaining to problem solving skills. 2. To familiarize the students with basic concepts of object oriented programming. 3. To facilitate students in understanding the syntax and semantics of the “C++” language as well as basic constructs offered by the language 4. To facilitate students in solving real world problems using Object Oriented approach. 5. To impart the concepts of File handling techniques 6. To familiarize students with Generic Programming								
COURSE OUTCOMES: Students will be able to 1. Demonstrate different basic concepts of object oriented programming. 2. Analyze the performance of OOP system with POP in terms of the complexity. 3. Apply OOP concepts and design a new system. Analyze various Inheritance & polymorphism techniques to design a system. 4. Use file handling mechanism in C++ and design data base applications. 5. Design C++ programs using Templates and analyze the performance of OOP & POP program in terms of reusability.								
UNIT-I	OBJECT ORIENTED PROGRAMMING							
Introduction to Object Oriented Programming, Structure of C++ program, Tokens, Keywords, Identifiers and Constants, Basic Data Types, Operators and Type Conversions. Functions: Returning values from functions. Reference arguments, Overloaded function, Inline function, Default arguments, returning by reference.								
UNIT-II	OBJECT, CLASSES, CONSTRUCTORS AND DESTRUCTORS							
Object and Classes: Specifying a Class, Member function, Nesting of Member functions, Arrays in a Class, Memory Allocation for Objects, Static Members and Member Functions, Array of Objects, Friend Functions, Pointers to Members. Constructors and Destructors: Parameterized Constructor, Copy Constructor, Dynamic Constructor, Destructors.								
UNIT-III	OPERATOR OVERLOADING AND INHERITANCE							
Operator Overloading: Overloading unary, binary operators, data conversion, pitfalls of operators overloading and conversion keywords, Explicit and Mutable. Inheritance: Derived class and Base class, Single Inheritance, Multiple Inheritance, Multilevel Inheritance, Hierarchical Inheritance, Virtual Base Classes, Abstract Classes, Constructor in Derived Classes.								
UNIT-IV	POINTER, STREAMS AND FILES							
Pointer: Introduction, Pointers to Objects, this Pointer, Pointers to Derived Class, Virtual Functions, Pure Virtual Functions. Streams and Files: C++ Streams, Stream Classes, Unformatted IO Operations, Formatted Console IO Operations, Managing Output with Manipulators.								

UNIT-V	FILES, TEMPLATES AND EXCEPTION HANDLING	
Files: Classes for File Handling, File Pointer, Updating a File, Error Handling in Files. Templates and Exception Handling: Class Templates, Function Templates, Exception Handling Mechanism, And Components of STL: Containers, Algorithms, Iterators.		
TEXT BOOKS:		
1. Object Oriented Programming with C++, <i>E Balaguruswamy</i> , Third Edition, and McGraw Hill Publications.		
REFERENCE BOOKS:		
1. Object Oriented Programming in C++ by <i>Robert Lafore</i> Techmedia Publication. 2. Object Oriented Programming in C++ <i>Saurav Sahay</i> Oxford University Press.		
WEB REFERENCES:		
1. https://www.mycplusplus.com/tutorials/cplusplus-programming-tutorials/introduction-cplusplus-lecture-notes/ 2. https://lecturenotes.in/notes/6206-notes-for-object-oriented-programming-using-cpp-oop-by-swarnalata-rath		
E-TEXT BOOKS:		
1. https://books.google.co.in/books?id=KbiDKrvExnsC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=rMjcBAAQBAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false		
MOOC COURSE		
1. https://www.my-mooc.com/en/mooc/introduction-c-microsoft-dev210x-0/ 2. https://www.coursera.org/learn/c-plus-plus-a		

CORE JAVA PROGRAMMING								
V SEMESTER - OPEN ELECTIVE – I								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS45	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100
COURSE OBJECTIVES: 1. To introduce the object oriented programming concepts. 2. To understand object oriented programming concepts, and apply them in solving problems. 3. To introduce the principles of inheritance and polymorphism; and demonstrate how they relate to the design of abstract classes 4. To introduce the implementation of packages and interfaces 5. To introduce the concepts of exception handling and multithreading. 6. To introduce the design of Graphical User Interface using applets and swing controls.								
COURSE OUTCOMES: At the end of the course students are able to 1. Demonstrate object oriented programming concepts, and apply them in solving problems. 2. Demonstrate principles of inheritance and polymorphism and how they relate to the design of abstract classes 3. Explain concepts of packages, exception handling and multithreading. 4. Develop multithreaded applications with synchronization 5. Design GUI based applications								
UNIT-I	OPEN CONCEPTS AND JAVA PROGRAMMING							
Oops concepts- Procedural and objects oriented programming paradigms, classes and objects,Data abstraction, encapsulation, polymorphism, inheritance, benefits of inheritance. Java Programming- History of java, data types, variables, constants, scope and life time of variables, operators, operator hierarchy, expression, type conversion and casting , control flow- block scope, condition statements, loops, break, and continue statements, simple java stand alone programs, arrays, constructors, methods, parameter passing, static keyword, access control, this pointer, overloading methods and constructor, recursion, garbage collection, exploring string class								
UNIT-II	INHERITANCE AND POLYMORPHISM							
Inheritance- Inheritance hierarchies super and sub classes, member access rules, super keyword, final keyword, the Object class. Polymorphism- dynamic binding, method overriding, abstract classes and methods. Interface- Interfaces vs. Abstract classes, defining an interface, implementing interfaces, implementing multiple inheritance using interfaces, extending interfaces. Inner classes- Uses of inner classes, local inner classes, anonymous inner classes, static inner classes.								
UNIT-III	PACKAGES, EXCEPTION HANDLING AND FILES							
Packages- Defining, Creation and Accessing Packages, Understanding CLASSPATH, importing packages. Exception handling- Types of errors, benefits of exception handling, classification of exceptions- exception hierarchy, checked exceptions and unchecked exceptions, usage of try, catch, throw, throws and finally, creating own exception sub classes. Files- streams- byte streams, character streams, text input/output, binary input/output, File management								

UNIT-IV	MULTITHREADING AND AWT CONTROLS	
Multithreading- Difference between multiprocessing and multithreading, thread states, creating threads, interrupting threads, thread priorities, synchronizing threads, inter-thread communication, producer consumer problem. AWT CONTROLS: The AWT class hierarchy, user interface components- Labels, Button, Text Components, Check Box, Check Box Group, Choice, List Box, Panels – Scroll Pane, Menu, ScrollBar. Working with Frame class, Colour, Fonts and layout managers.		
UNIT-V	GUI PROGRAMMING WITH JAVA AND EVENT HANDLING	
GUI Programming with Java-Introduction to Swing, Hierarchy for Swing components, Swing vs.AWT, Containers- JFrame, JApplet, JDialog, JPanel, JButton, JLabel, JTextField, JTextArea Event Handling- Events, Event sources, Event classes, Event Listeners, Relationship between Event sources and Listeners, Delegation event model, Examples: handling a button click, handling mouse events, Adapter classes.		
TEXT BOOKS:		
1. Java Fundamentals – A comprehensive Introduction, Herbert Schildt and Dale Skrien, TMH 2. Java A Beginner's Guide Sixth Edition , Herbert Schildt		
REFERENCE BOOKS:		
1. Java for Programmers, P.J. Deitel and H.M. Deitel, Pearson education (OR) Java: How to Program P.J.Deitel and H.M. Deitel, PHI. 2. Object Oriented Programming through Java, P. Radha Krishna, Universities Press. 3. Thinking in Java, Bruce Eckel, Pearson Education 4. Programming in Java, S. Malhotra and S. Choudary, Oxford Univ. Press.		
WEB REFERENCES:		
1. http://www.math.hcmuns.edu.vn/~hvtbao/courses/java_programming/lecture_notes/ 2. http://people.alari.ch/derino/Teaching/Java/JavaLectureNotes-Derin.pdf		
E-TEXT BOOKS:		
1. https://books.google.co.in/books?id=pnwTLvCJk0C&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=8qFDDAAQBAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false		
MOOC COURSE		
1. http://moocfi.github.io/courses/2013/programming-part-1/ 2. https://www.edx.org/learn/java		

INTRODUCTION TO COMPUTER NETWORKS								
VI SEMESTER - OPEN ELECTIVE – II								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS46	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100
COURSE OBJECTIVES: 1. To introduce the fundamental various types of computer networks. 2. To demonstrate the TCP/IP and OSI models with merits and demerits. 3. To explore the various layers of OSI Model.								
COURSE OUTCOMES: By the end of this course, student will be able to 1. Explore the basic concepts of Computer Networks. 2. Analyze the internal implementation of Physical Layer, Data link layer and Network Layer. 3. Analyze the internal implementation of Transport Layer and Application layer. 4. Solve problems with respect to various layers of reference model. 5. Administrate a network and flow of information								
UNIT-I	INTRODUCTION TO COMPUTER NETWORKS AND PHYSICAL LAYER							
INTRODUCTION TO COMPUTER NETWORKS: Uses of Computer Networks, Network Hardware, Network Software, Reference Models (OSI, TCP/IP), Example Networks. PHYSICAL LAYER- Basis for Data Communication, Guided Transmission Media, Wireless Transmission, Communication Satellites, Digital Modulation And Multiplexing, The Public Switched Telephone Network.								
UNIT-II	DATA LINK LAYER							
DATA LINK LAYER- Data Link Layer Design Issues, Error Detection and Correction, Elementary Data Link Protocols, Sliding Window Protocols. The Channel Allocation Problem, Multiple Access Protocols.								
UNIT-III	NETWORK LAYER							
NETWORK LAYER- Virtual Circuit and Datagram Networks, Inside A Router, The Internet Protocol (IP): Forwarding and Addressing in The Internet, Routing Algorithms, Routing in The Internet, Broadcast And Multicast Routing.								
UNIT-IV	TRANSPORT LAYER							
TRANSPORT LAYER: Transport-Layer Services, Multiplexing and Demultiplexing, Principles of Reliable Data Transfer, Connection-Oriented Transport: TCP, Principles of Congestion Control, TCP Congestion Control.								
UNIT-V	APPLICATION LAYER							
APPLICATION LAYER: Principles of Network Applications, The Web and HTTP, File Transfer: FTP, Electronic Mail In The Internet, DNS—The Internet's Directory Service, Peer-To-Peer Application.								
TEXT BOOKS: 1. Andrew S Tanenbaum David J Wetherall, Computer Networks, 5th Edition, Pearson Education. 2. Kurose, Ross (2010), Computer Networking: A top down approach, Sixth Edition, Pearson Education. India.								

REFERENCE BOOKS:

1. Behrouz A. Forouzan, Data communication and Networking, 4th Edition, Mc Graw-Hill, India.
2. Larry L. Peterson, Bruce S. Davie, Computer Networks: A Systems Approach, MK Publishers.

WEB REFERENCES:

1. <https://www.cse.iitk.ac.in/users/dheeraj/cs425/>
2. <http://www2.kios.ucy.ac.cy/christos/courses/ece360/lecture-notes.html>

E-TEXTBOOKS:

1. https://books.google.co.in/books?id=KpOb37EHETcC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false
2. https://books.google.co.in/books?id=b2HyGSu46lQC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false

MOOC COURSE

1. <https://www.classcentral.com/subject/computer-networking>
2. <https://www.udacity.com/course/computer-networking--ud436>

INTRODUCTION TO LINUX PROGRAMMING								
VI SEMESTER - OPEN ELECTIVE – II								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS47	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100
COURSE OBJECTIVES:								
1. To understand principles of operating system including File handling utilities, Security by file permissions, Process utilities, Disk utilities, Networking Commands, Basic Linux commands, Scripts and filters. 2. To familiarize fundamentals of the Bourne again shell (bash), shell programming, pipes, input and output redirection Control structures, arithmetic in shell interrupt processing, functions, debugging shell scripts. 3. To impart fundamentals of file concepts kernel support for file, File structure related system calls (file API's). 4. To facilitate students in understanding Inter process communication. 5. To facilitate students in understanding semaphore and shared memory.								
COURSE OUTCOMES:								
At the end of the course students are able to								
1. Use various Linux commands that are used to manipulate system operations at admin level and a prerequisite to pursue job as a Network administrator. 2. Write Shell Programming using Linux commands. 3. Design and write application to manipulate internal kernel level Linux File System. 4. Develop IPC-API's that can be used to control various processes for synchronization. 5. Develop Network Programming that allows applications to make efficient use of resources available on different machines in a network.								
UNIT-I	INTRODUCTION TO LINUX AND LINUX UTILITIES							
INTRODUCTION TO LINUX AND LINUX UTILITIES: A brief history of LINUX, architecture of LINUX, features of LINUX, introduction to vi editor. Linux commands- PATH, man, echo, printf, script, passwd, uname, who, date, stty, pwd, cd, mkdir, rmdir, ls, cp, mv, rm, cat, more, wc, lp, od, tar, gzip, file handling utilities, security by file permissions, process utilities, disk utilities, networking commands, unlink, du, df, mount, umount, find, unmask, ulimit, ps, w, finger, arp, ftp, telnet, rlogin. Text Processing utilities and backup utilities , tail, head , sort, nl, uniq, grep, egrep, fgrep, cut, paste, join, tee, pg, comm, cmp, diff, tr, awk, cpio								
UNIT-II	INTRODUCTION TO SHELLS AND FILTERS							
Introduction to Shells: Linux Session, Standard Streams, Redirection, Pipes, Tee Command, Command Execution, Command-Line Editing, Quotes, Command Substitution, Job Control, Aliases, Variables, Predefined Variables, Options, Shell/Environment Customization. Filters: Filters and Pipes, Concatenating files, Display Beginning and End of files, Cut and Paste, Sorting, Translating Characters, Files with Duplicate Lines, Count Characters, Words or Lines, Comparing Files								
UNIT-III	GREP, SED, UNIX FILE STRUCTURE AND FILE MANAGEMENT							
GREP:Operation, grep Family, Searching for File Content. Sed :Scripts, Operation, Addresses, commands, Applications, grep and sed. UNIX FILE STRUCTURE: Introduction to UNIX file system, inode (Index Node), file descriptors, system calls and device drivers. FILE MANAGEMENT :File Structures, System Calls for File Management – create, open, close, read, write, lseek, link, symlink, unlink, stat, fstat, lstat, chmod, chown, Directory API – opendir, readdir, closedir, mkdir, rmdir, umask.								

UNIT-IV	PROCESS AND SIGNALS	
PROCESS AND SIGNALS: Process, process identifiers, process structure: process table, viewing processes, system processes, process scheduling, starting new processes: waiting for a process, zombie processes, orphan process, fork, vfork, exit, wait, waitpid, exec, signals functions, unreliable signals, interrupted system calls, kill, raise, alarm, pause, abort, system, sleep functions, signal sets. File locking: creating lock files, locking regions, use of read and write with locking, competing locks, other lock commands, deadlocks.		
UNIT-V	INTER PROCESS COMMUNICATION AND SOCKETS	
INTER PROCESS COMMUNICATION: Pipe, process pipes, the pipe call, parent and child processes, and named pipes: fifos, semaphores: semget, semop, semctl, message queues: msgget, msgsnd, msgrcv, msgctl, shared memory: shmget, shmat, shmdt, shmctl, ipc status commands. INTRODUCTION TO SOCKETS: Socket, socket connections - socket attributes, socket addresses, socket, connect, bind, listen, accept, socket communications.		
TEXT BOOKS:		
1. W. Richard. Stevens (2005), Advanced Programming in the UNIX Environment, 3rd edition, Pearson Education, New Delhi, India. 2. Unix and shell Programming Behrouz A. Forouzan, Richard F. Gilberg.Thomson		
REFERENCE BOOKS:		
1. Linux System Programming, Robert Love, O'Reilly, SPD. 2. Advanced Programming in the UNIX environment, 2nd Edition, W.R.Stevens, Pearson Education. 3. UNIX Network Programming, W.R. Stevens, PHI. 4. UNIX for Programmers and Users, 3rd Edition, Graham Glass, King Ables, Pearson Education		
WEB REFERENCES:		
1. https://www.smartzworld.com/notes/linux-programming-pdf-notes-lp-pdf-notes/ 2. https://lecturenotes.in/subject/455/linux-programming-lp		
E-TEXTBOOKS:		
1. https://books.google.co.in/books?id=vvuzDziOMeMC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=TKjATVAXNlWC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false		
MOOC COURSE		
1. https://www.classcentral.com/course/edx-c-programming-using-linux-tools-and-libraries-11538 2. https://www.udacity.com/course/linux-command-line-basics--ud595		

FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE								
VI SEMESTER - OPEN ELECTIVE – II								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS48	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100
COURSE OBJECTIVES: 1. To learn the difference between optimal reasoning vs human like reasoning 2. To understand the notions of state space representation, exhaustive search, heuristic search along with the time and space complexities 3. To learn different knowledge representation techniques 4. To understand the applications of AI viz Game Playing, Theorem Proving, Expert Systems, Machine Learning and Natural Language Processing								
COURSE OUTCOMES: At the end of the course students are able to 1. Demonstrate fundamental concepts of artificial intelligence. 2. Select a search algorithm for a problem and characterize its time and space complexities. 3. Discuss various approaches to knowledge representation. 4. Apply AI techniques to solve problems of Game Playing, Expert Systems, Machine Learning and Natural Language Processing 5. Outline advanced knowledge representation techniques.								
UNIT-I	INTRODUCTION							
Introduction, History, Intelligent Systems, Foundations of AI, Sub areas of AI, Applications. Problem Solving – State-Space Search and Control Strategies: Introduction, General Problem Solving, Characteristics of Problem, Exhaustive Searches, Heuristic Search Techniques, Iterative-Deepening A*, Constraint Satisfaction. Game Playing, Bounded Look-ahead Strategy and use of Evaluation Functions, Alpha-Beta Pruning								
UNIT-II	LOGIC CONCEPTS AND LOGIC PROGRAMMING & KNOWLEDGE REPRESENTATION							
Logic Concepts and Logic Programming: Introduction, Propositional Calculus, Propositional Logic, Natural Deduction System, Axiomatic System, Semantic Tableau System in Propositional Logic, Resolution Refutation in Propositional Logic, Predicate Logic, Logic Programming. Knowledge Representation: Introduction, Approaches to Knowledge Representation, Knowledge Representation using Semantic Network, Extended Semantic Networks for KR, Knowledge Representation using Frames.								
UNIT-III	EXPERT SYSTEM AND APPLICATIONS & UNCERTAINTY MEASURE – PROBABILITY THEORY							
Expert System and Applications: Introduction, Phases in Building Expert Systems, Expert System Architecture, Expert Systems Vs Traditional Systems, Truth Maintenance Systems, Application of Expert Systems, List of Shells and Tools. Uncertainty Measure – Probability Theory: Introduction, Probability Theory, Bayesian Belief Networks, Certainty Factor Theory, Dempster-Shafer Theory.								
UNIT-IV	MACHINE-LEARNING PARADIGMS & ARTIFICIAL NEURAL NETWORKS							
Machine-Learning Paradigms: Introduction. Machine Learning Systems. Supervised and Unsupervised Learning. Inductive Learning. Learning Decision Trees, Deductive Learning. Clustering, Support Vector Machines. Artificial Neural Networks: Introduction, Artificial Neural Networks, Single-Layer Feed Forward Networks, Multi-Layer Feed-Forward Networks, Radial-Basis Function Networks, Design Issues of Artificial Neural Networks, Recurrent Networks.								

UNIT-V	ADVANCED KNOWLEDGE REPRESENTATION TECHNIQUES	
Advanced Knowledge Representation Techniques: Case Grammars, Semantic Web Natural Language Processing: Introduction, Sentence Analysis Phases, Grammars and Parsers, Types of Parsers, Semantic Analysis, Universal Networking Knowledge.		
TEXT BOOKS:		
1. Saroj Kaushik. Artificial Intelligence. Cengage Learning. 2011 2. Russell, Norvig: Artificial intelligence, A Modern Approach, Pearson Education, Second Edition. 2004		
REFERENCE BOOKS:		
1. Rich, Knight, Nair: Artificial intelligence, Tata McGraw Hill, Third Edition 2009. 2. Introduction to Artificial Intelligence by Eugene Charniak, Pearson. 3. Introduction to Artificial Intelligence and expert systems Dan W.Patterson. PHI. 4. Artificial Intelligence by George Fluger Pearson fifth edition.		
WEB REFERENCES:		
1. http://zsi.tech.us.edu.pl/~nowak/bien/BIEN_introduction.pdf 2. https://epub.uni-regensburg.de/13629/1/ubr06078_ocr.pdf 3. https://lecturenotes.in/subject/128/artificial-intelligence-ai		
E-TEXTBOOKS:		
1. https://books.google.co.in/books?id=DDNHZcN6jasC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=YmH1tXFA14MC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false		
MOOC COURSE		
1. https://www.edx.org/course/artificial-intelligence-1 2. https://www.appliedaicourse.com/		

SOFT COMPUTING								
VII SEMESTER - OPEN ELECTIVE – III								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS49	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100
COURSE OBJECTIVES: Students will 1. Learn the various soft computing frame works 2. Be familiar with design of various neural networks 3. Be exposed to fuzzy logic 4. Learn genetic programming.								
COURSE OUTCOMES: At the end of the course students are able to 1. Demonstrate fundamental concepts of Neural Networks & Feed Forward Networks 2. Use fuzzy logic in computing and data analytics. 3. Use Genetic Algorithms to solve optimization problems. 4. Design hybrid systems to suit a given real life problem 5. Familiarize with neural network toolbox, Fuzzy logic toolbox etc.								
UNIT-I	NEURAL NETWORKS							
Fundamentals of Neural Networks–History-Architectures-Learning methods-XOR problem-Delta rule-derivation-Back propagation-applications-parameters in BPN-Associative memory–Hetero associative-BAM-energy function-problems-applications of associative memories-ART1-ART2-applications of adaptive Networks.								
UNIT-II	FUZZY LOGIC							
Fuzzy set theory–crisp sets–fuzzysets–crisp relations–Fuzzy relations–Fuzzy systems-Crisp logic–predicate logic–fuzzy logic-fuzzy based systems-Defuzzification methods–applications.								
UNIT-III	GENETIC ALGORITHMS							
Fundamentals of GA–creation of off springs–encoding–fitness function-reproduction–crossover-Insertion & deletion-mutation-bitwise operators–applications.								
UNIT-IV	HYBRID SYSTEMS							
Hybrid systems–Neuro Fuzzy–Neuro Genetic–fuzzy Genetic hybrids-GA based weight determination and applications-fuzzy BPN–simplified fuzzy ARTMAP.								
UNIT-V	FPROGRAMMING USING MATLAB							
Using Neural Network toolbox–Using Fuzzy Logic toolbox-Using Genetic Algorithm & directed search toolbox.								
TEXT BOOKS:								
1. Rajasekaran.S and VijayalakshmiPai.G.A, “Neural Networks, Fuzzy Logic and Genetic Algorithms”, PHI, 2011								

REFERENCE BOOKS:

1. Timothy J. Ross, "Fuzzy Logic with Engineering applications ", John Wiley and Sons, 2010.
2. Jang.J.S.R, Sun.C.T, Mizutani.E, "Neuro fuzzy and Soft Computing ", PHI Learning Pvt. Ltd., 2012.
3. Davis E. Goldberg, "Genetic Algorithms: Search, Optimization and Machine Learning ", Addison Wesley, N.Y., 1989.

WEB REFERENCES:

1. http://www.myreaders.info/html/soft_computing.html
2. https://en.wikipedia.org/wiki/Soft_computing

E-TEXT BOOKS:

1. <http://freecomputerbooks.com/Introduction-to-Soft-Computing.html>
2. <https://www.shroffpublishers.com/books/9789351106159/>
3. <https://bookboon.com/en/introduction-to-soft-computing-ebook>

MOOC COURSE

1. https://onlinecourses.nptel.ac.in/noc18_cs13/course
2. <https://swayam.gov.in/course/4574-introduction-to-soft-computing>

INTRODUCTION TO CLOUD COMPUTING

VII SEMESTER - OPEN ELECTIVE – III

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS50	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

1. To understand the Virtualization paradigms
2. To learn the Cloud Computing fundamentals and its importance to various organizations.
3. To analyze the concepts of IaaS, PaaS, SaaS, Public and Private Clouds.
4. To develop applications in cloud security.

COURSE OUTCOMES:

At the end of the course, student will be able to:

1. Demonstrate concepts of system modeling, clustering and virtualization.
2. Describe the infrastructure of cloud computing including SaaS, PaaS, IaaS, public cloud, private cloud, hybrid cloud, etc.
3. Discuss about monitoring, management and applications of cloud computing.
4. Analyze the core issues of cloud computing such as security, privacy, and interoperability.
5. Identify problems, analyze, and evaluate various cloud computing solutions.

UNIT-I INTRODUCTION

System Modeling, Clustering And Virtualization: Distributed system models and enabling technologies, computer clusters for scalable parallel computing, virtual machines and virtualization of clusters and data centers.

UNIT-II FOUNDATIONS

Foundations: Introduction to cloud computing, migrating into cloud, enriching the integration of service paradigm for cloud era, the enterprise cloud computing paradigm.

UNIT-III INTER PROCESS COMMUNICATION

Inter Process Communication: Infra Structure As Service (IAAS) & Platform And Software Service(PAAS/SAAS): Virtual machine provisioning and migration services, on the management of virtual machines for cloud infrastructure, enhancing cloud computing environments using a cluster as service. Secure distributed data storage in cloud computing. Aneka, comet cloud, T-systems, work flow engine for clouds, understanding scientific applications for cloud environments

UNIT-IV MONITORING, MANAGEMENT AND APPLICATIONS

Monitoring, Management And Applications : An Architecture for federated cloud computing ,SLA management in cloud computing, performance prediction for HPC on clouds, best practices in Architecting cloud applications in the AWS cloud, building content delivery networks using clouds, resource cloud mashups.

UNIT-V GOVERNANCE AND CASE STUDIES

Governance and case studies: organizational readiness and change management in cloud age, data security in cloud, legal issues in cloud computing, achieving production readiness for cloud services.

TEXT BOOKS:

1. Cloud computing: principles and paradigms by rajkumar buyya, James Broberg and Andrzej M.Goscinski,Wiley,2011.

Distributed and cloud computing , kai Hwang, Geoffery C.fox, jack J.dongarra, Elsevier, 2012 .

REFERENCE BOOKS:

1. 1. Cloud Computing :A practical approach, Anthony T.velte, Toby J.velte, Robert Isenpeter, Tata McGraw Hill,2011.
2. Enterprise Cloud Computing, Gautam Shroff, Cambridge University press,2010.
3. Cloud computing: implementation, management and security, john W .Ritting house ,James F. Ransome ,CRC press,rp2012.
4. Cloud applications architectures: building Applications and infrastructure in the cloud,George Reese , O reilly,SPD,rp2011.
5. 5. Cloud security and privacy: An Enterprise perspective on Risks and compliance,im Mather, Subra Kumaraswamy, Shahed Latif,Oreilly,SPD,rp2011

WEB REFERENCES:

1. https://cs.uwaterloo.ca/~a78khan/courses-offered/cs446/2010_05/lecture-slides/16_CloudComputing.pdf
2. <https://www.smartworld.com/notes/cloud-computing-pdf-notes-cc/>
3. http://michaelkona.blogspot.com/p/introduction-to-cloud-computing_4.html

E-TEXTBOOKS:

1. https://books.google.co.in/books?id=F8VoBAAQBAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false
2. https://books.google.co.in/books?id=8es62w1MOVcC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false

MOOC COURSE

1. <https://www.coursera.org/learn/cloud-computing>
2. <https://www.classcentral.com/subject/cloud-computing>

WEB PROGRAMMING								
VII SEMESTER - OPEN ELECTIVE – III								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS51	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100
COURSE OBJECTIVES: 1. To learn the basics of server side scripting using HTML 2. To explain web application development procedures 3. To impart servlet technology for writing business logic 4. To facilitate students to connect to databases using JDBC 5. To familiarize various concepts of application development using XML								
COURSE OUTCOMES: At the end of the course students are able to 1. Use HTML, CSS to develop web pages. 2. Apply concepts of Java. 3. Build JDBC connectivity. 4. Identify the basic building blocks of XML and describe the applications of web services. 5. Write server side programming using applets.								
UNIT-I	SCRIPTING							
Web page Designing using HTML, Scripting basics- Client side and server side scripting. Java Script- Object, names, literals, operators and expressions- statements and features-events - windows - documents - frames - data types - built-in functions- Browser object model - Verifying forms.-HTML5- CSS3- HTML 5 canvas - Web site creation using tools.								
UNIT-II	JAVA							
Introduction to object oriented programming-Features of Java – Data types, variables and arrays – Operators – Control statements – Classes and Methods – Inheritance. Packages and Interfaces – Exception Handling – Multithreaded Programming –Input/Output – Files – Utility Classes – String Handling.								
UNIT-III	JDBC							
JDBC Overview – JDBC implementation – Connection class – Statements – Catching Database Results, handling database Queries. Networking– Inet Address class – URLclass- TCP sockets - UDP sockets, Java Beans –RMI.								
UNIT-IV	APPLETS							
Java applets- Life cycle of an applet – Adding images to an applet – Adding sound to an applet, Passing parameters to an applet, Event Handling, Introducing AWT: Workingwith Windows Graphics and Text. Using AWT Controls, Layout Managers and Menus.Servlet – life cycle of a servlet, The Servlet API, Handling HTTP Request and Response, using Cookies, Session Tracking. Introduction to JSP.								
UNIT-V	XML AND WEB SERVICES							
Xml – Introduction-Form Navigation-XML Documents- XSL – XSLT- Web services-UDDI-WSDL-Java web services – Web resources.								
TEXT BOOKS: 1. Java™ : The Complete Reference, Seventh Edition, Herbert Schildt 2. J2EE: The complete Reference by James Keogh 3. Java EE and HTML5 Enterprise Application Development (Oracle Press) by John Brock								

(Author), Arun Gupta (Author), Geertjan Wielenga (Author) Struts: The Complete Reference, 2nd Edition by James Holmes
REFERENCE BOOKS:
1. ASP.NET 4 Unleashed by Stephen Walther (Author), Kevin Scott Hoffman• (Author), Nate Dudek (Author) Microsoft Visual C# 2013 Step by Step by John Sharp
WEB REFERENCES:
1. http://cgi.csc.liv.ac.uk/~martin/teaching/comp519/notes.php 2. https://lecturenotes.in/notes/24802-note-for-web-programming-wp-by-ashutosh-jaiswal 3. http://web.stanford.edu/class/cs193x/lectures/01/lecture01.pdf
E-TEXTBOOKS:
1. https://books.google.co.in/books?id=PZhIDwAAQBAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=fj1Pd8uxNLEc&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false
MOOC COURSE
1. https://www.classcentral.com/subject/web-development 2. https://www.coursera.org/courses?query=web%20programming

INTRODUCTION TO NETWORK SECURITY								
VIII SEMESTER - OPEN ELECTIVE – IV								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS52	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100
COURSE OBJECTIVES: 1. To understand the basic components of security and mathematics of cryptography. 2. To demonstrate various traditional symmetric key ciphers 3. To Analyze DES and AES its variants. 4. To demonstrate private and public-key cryptosystem, Digital signature and attacks on it.								
COURSE OUTCOMES: At the end of the course students able to 1. Identify and explain various types of security threats/vulnerabilities attacks, consequences and security mechanisms 2. Discuss concepts of advanced and data encryption standards. 3. Apply various Cryptographic and Authentication techniques for secure transmission of data. 4. Compare different security algorithms with respect feasibility, cost and security service 5. Choose the most suitable security mechanisms based on the nature of data and complexity of algorithm.								
UNIT-I	INTRODUCTION							
Introduction: Security Goals, Attacks, Services and Mechanism, Techniques, Mathematics of Cryptography, Integer arithmetic, Modular Arithmetic, Matrices, Linear Congruence, Primes, Primality Testing, Factorization, Chinese Remainder Theorem. Traditional Symmetric key Ciphers: Introduction, Substitution Ciphers, Transposition Ciphers, Streams and Block Ciphers.								
UNIT-II	SYMMETRIC CIPHERS							
Advanced Encryption Standards: AES Structure, AES Transformation Function, AES Key Expansion, AES Implementation. Data Encryption Standards: Introduction, DES Structure, DES Analysis, Multiple DES, Security of DES, Differential Cryptanalysis, Linear Cryptanalysis of DES, Encipherment Using Modern Symmetric Key Ciphers Use of Modern Block Ciphers, Use of Stream Ciphers								
UNIT-III	ASYMMETRIC CIPHERS, DATA INTEGRITY ALGORITHMS							
Asymmetric Key Cryptography and Message Integrity: Introduction, RSA Cryptosystem, Message Integrity, Message Authentication Cryptographic Hash Functions, Introduction, SHA-512, Digital Signature Comparison, Attacks on Digital Signature. User Authentication Protocols: Remote User Authentication Principles, Remote User Authentication using Symmetric Encryption, Kerberos								
UNIT-IV	MUTUAL TRUST							
Key Management And Distribution : Symmetric key Distribution Using Symmetric and Asymmetric Encryption, Distribution of Public Key, X.509 Certificate , Public Key Infrastructure. Network Access Control : Network Access Control, Extensible Authentication Protocol, IEEE802.1,								
UNIT-V	NETWORK SECURITY							
Transport-Level Security: Transport-Level Security, HTTPS, Secure Shell(SSH) Wireless Network Security: Web Security ,Mobile Device Security, IEEE802.11 Wireless LANOverview, IEEE802.11i Wireless LAN Security								
TEXT BOOKS:								

1.	Cryptography and Network Security"by BehrouzA.Forouzan, Special Indian Edition, 2008
2.	Cryptography and Network Security"by William Stallings, 6th Edition, Pearson Education/PHI, 2014
REFERENCE BOOKS:	
1.	"Cryptography and Network Security"by AtulKahate, Tata McGraw Hill, 2003.
2.	"RSA Security's official guide to cryptography" by Steve Burnett, Stephene Paine, RSA Pren, Tata McGraw Hill Edition, 2001
WEB REFERENCES:	
1.	www.open.edu/openlearn/science.../network-security/content-section---references www.cse.iitm.ac.in/~chester/courses/16e_cns/slides/01_Introduction.pdf williamstallings.com/Crypto/Crypto4e.html
2.	pubs.sciepub.com/iteces/3/1/1/
3.	https://www.sanfoundry.com/best-reference-books-cryptography-network-security/
4.	https://dl.acm.org/citation.cfm?id=1824151
5.	web.mit.edu/6.857/OldStuff/Fall03/references.html
E-TEXT BOOKS:	
1.	http://williamstallings.com/Crypto3e.html
2.	http://www.cacr.math.uwaterloo.ca/hac/
MOOC COURSE	
1.	https://www.mooc-list.com/tags/network-security
2.	nptel.ac.in/courses/106105031/
3.	https://onlinecourses.nptel.ac.in/noc18_cs07/course

DATA ANALYTICS								
VIII SEMESTER - OPEN ELECTIVE – IV								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS53	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100
COURSE OBJECTIVES: 1. To explore the fundamental concepts of data analytics 2. Formulate and use appropriate models of data analysis to solve hidden solutions to business-related challenges.								
COURSE OUTCOMES: At the end of the course students are able to 1. Demonstrate the ability to use technical skills in Descriptive and prescriptive modelling to support business decision-making 2. Apply the knowledge of classification and regression techniques for various data analysis. 3. Examine the results of Classification for further predictions. 4. Make use of regression techniques for predicting future outcomes on various data 5. Illustrate the concepts of data analytics for data visualization and bigdata analysis.								
UNIT-I	DESCRIPTIVE STATISTICS							
Descriptive Statistics, Introduction to the course, Descriptive Statistics, Probability Distributions, Inferential Statistics, Inferential Statistics through hypothesis tests, Permutation & Randomization Test								
UNIT-II	REGRESSION AND ANOVA							
Regression & ANOVA, Regression , ANOVA(Analysis of Variance), Machine Learning: Introduction and Concepts, Differentiating algorithmic and model based frameworks. Regression: Ordinary Least Squares, Ridge Regression, Lasso Regression, K Nearest Neighbours Regression & Classification								
UNIT-III	SUPERVISED LEARNING WITH REGRESSION AND CLASSIFICATION TECHNIQUES - I							
Supervised Learning with Regression and Classification techniques -1, Bias-Variance Dichotomy, Model Validation Approaches, Logistic Regression, Linear Discriminant Analysis, Quadratic Discriminant Analysis Regression and Classification Trees, Support Vector Machines								
UNIT-IV	SUPERVISED LEARNING WITH REGRESSION AND CLASSIFICATION TECHNIQUES - 2							
Supervised Learning with Regression and Classification techniques -2, Ensemble Methods: Random Forest Neural Networks, Deep learning, Unsupervised Learning and Challenges for Big Data Analytics Clustering Associative Rule Mining Challenges for big data analytics.								
UNIT-V	PRESCRIPTIVE ANALYTICS							
Prescriptive analytics, Creating data for analytics through designed experiments, Creating data for analytics through Active learning, Creating data for analytics through Reinforcement learning								
TEXT BOOKS: 1. Hastie, Trevor, et al. The elements of statistical learning. Vol. 2. No. 1. New York: springer, 2009. 2. Montgomery, Douglas C., and George C. Runger. Applied statistics and probability for engineers. John Wiley & Sons, 2010								
REFERENCE BOOKS:								

1. Data Analytics- Models and Algorithms for Intelligent Data Analysis, Author: Thomas Runkler ISBN-10: 3834825883
2. Data Analytics, Anil Maheshwari, McGraw Hill Education; First edition, 9352604180.

WEB REFERENCES:

1. <https://www.ee.columbia.edu/~cylin/course/bigdata/EECS6893-BigDataAnalytics-Lecture4.pdf>
2. <https://intellipaat.com/blog/what-is-data-analytics/>

E-TEXTBOOKS:

1. https://books.google.co.in/books?id=PaylsdmgkklC&printsec=frontcover&source=gb_s_ge_summary_r&cad=0#v=onepage&q&f=false
2. https://books.google.co.in/books?id=_xhADQAAQBAJ&printsec=frontcover&source=gb_s_ge_summary_r&cad=0#v=onepage&q&f=false

MOOC COURSE

1. <https://www.edx.org/learn/data-analysis>
2. <https://www.mooc-list.com/categories/statistics-data-analysis>

.NET & C# PROGRAMMING								
VIII SEMESTER - OPEN ELECTIVE – IV								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS54	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100
COURSE OBJECTIVES: 1. To impart basic concepts of .NET Framework Components. 2. To familiarize the C# Core Concepts. 3. To facilitate students in understanding OOP Concepts. 4. To facilitate students in understanding Exception Handling. 5. To impart the concepts of Delegates, Events.								
COURSE OUTCOMES: At the end of the course, students will 1. Explain the purpose of the .net framework, and use c# and visual studio 2010 to build .net framework applications. 2. Outline the syntax of basic c# programming constructs a, use new types (enumerations, classes, and structures), and enumerate the differences between reference types and value types. 3. Create new reference types using inheritance; control the visibility and lifetime of members in a type. 4. Manage the lifetime of objects and control the use of resources, catch, handle and throw exceptions. 5. Exemplify the .net delegate type & event handling.								
UNIT-I	INTRODUCTION TO .NET							
Introduction - The Philosophy of .NET, An Initial Look at the .NET Platform, Introducing the Building Blocks of the .NET Platform (the CLR, CTS, and CLS), An Overview of .NET Assemblies, Understanding the Common Type System, Understanding the Common Language Specification, Understanding the Common Language Runtime, The Assembly/Namespace/Type Distinction, The Platform-Independent Nature of .NET.								
UNIT-II	CORE C# PROGRAMMING CONSTRUCTS							
Core C# Programming Constructs, Part I - The Anatomy of a Simple C# Program, An Interesting Aside: Some Additional Members of the System.Environment Class, The System.Console Class, System Data Types and Corresponding C# Keywords, Working with String Data, Narrowing and Widening Data Type Conversions, Understanding Implicitly Typed Local Variables, C# Iteration Constructs, Decision Constructs and the Relational/Equality Operators. Core C# Programming Constructs, Part II- Methods and Parameter Modifiers, Understanding C# Arrays, Understanding the enum Type, Understanding the Structure Type, Understanding Value Types and Reference Types, Understanding C# Nullable Types.								
UNIT-III	OBJECT-ORIENTED PROGRAMMING WITH C#							
Understanding Encapsulation - Introducing the C# Class Type, Understanding Constructors, The Role of the this Keyword, Understanding the static Keyword, Defining the Pillars of OOP, C# Access Modifiers, The First Pillar: C#'s Encapsulation Services, Understanding Automatic Properties, Understanding Object Initialization Syntax , Working with Constant Field Data, Understanding Partial Types. Understanding Inheritance and Polymorphism - The Basic Mechanics of Inheritance, The Second Pillar of OOP: The Details of Inheritance, Programming for Containment/Delegation, The Third Pillar of OOP: C#'s Polymorphic Support, Understanding Base Class/Derived Class Casting Rules, The Master Parent Class: System.Object.								
UNIT-IV	STRUCTURED EXCEPTION HANDLING							
Understanding Structured Exception Handling - Ode to Errors, Bugs, and Exceptions, The Role of .NET Exception Handling, The Simplest Possible Example, Configuring the State of an Exception, System-Level Exceptions (System.SystemException), Application								

Level Exceptions (System.ApplicationException), Processing Multiple Exceptions, Who Is Throwing What?, The Result of Unhandled Exceptions.

UNIT-V**WORKING WITH INTERFACES, DELEGATES AND EVENTS****Classes: 14**

Working with Interfaces - Understanding Interface Types, Defining Custom Interfaces, Implementing an Interface, Invoking Interface Members at the Object Level, Interfaces As Parameters, Interfaces As Return Values, Arrays of Interface Types, Implementing Interfaces Using Visual Studio, Explicit Interface Implementation, Designing Interface Hierarchies, The IEnumerable and IEnumerator Interfaces, The ICloneable Interface, The IComparable Interface.

Delegates, Events - Understanding the .NET Delegate Type, The Simplest Possible Delegate Example, Sending Object State Notifications Using Delegates, Understanding Generic Delegates, Understanding C# Events, Understanding C# Anonymous Methods.

TEXT BOOKS:

1. Andrew Troelsen, Pro C-Hash and the .NET 4 Platform, 5th edition, Springer (India) Private Limited, New Delhi, India., 2010

REFERENCE BOOKS:

1. E. Balagurusamy (2004), Programming in C#, 5 th edition, Tata McGraw-Hill, New Delhi, India.
2. Herbert Schildt (2004), The Complete Reference: C#, Tata McGraw-Hill, New Delhi, India.
3. Simon Robinson, Christian Nagel, Karli Watson, Jay GI (2006), Professional C#, 3rd edition, Wiley & Sons, India.

WEB REFERENCES:

1. <https://lecturenotes.in/notes/19333-note-for-c-and-net-c-net-by-anna-superkings>
2. <https://www.guru99.com/c-sharp-tutorial.html>

E-TEXTBOOKS:

1. https://books.google.co.in/books?id=JUNqMPIZ4yMC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false
2. https://books.google.co.in/books?id=ZmBzoG--oNAC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false

MOOC COURSE

1. <https://www.qatraininghub.com/c-net-online-training.php>
2. <https://www.abhishekluv.in/online-one-on-one-csharp-training-and-coaching>

**OPEN ELECTIVES
OFFERED BY
ECE DEPARTMENT**

LOGIC DESIGN**V Semester: OPEN ELECTIVE -I**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EC59	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Explain how digital circuit of large complexity can be built in a methodological way, starting from Boolean logic and applying a set of rigorous techniques.
2. Create minimal realizations of single and multiple output Boolean functions.
3. Design and analyze combinational circuits using medium scale integrated (MSI) components, including arithmetic logic units.
4. Derive state diagrams and state transition tables for synchronous systems.
5. Study the characteristics and performance of digital circuits built using various MOS technologies

UNIT-I BOOLEAN ALGEBRA

Boolean Algebra: Axiomatic Definition of Boolean Algebra, Basic Theorems and Properties of Boolean Algebra, Boolean Functions, Canonical and Standard Forms, Digital Logic Gates, IC Digital Logic Families

UNIT-II SIMPLIFICATION OF BOOLEAN FUNCTIONS

Simplification of Boolean Functions: The Map Method, Two & Three Variable Maps, Four Variable Map, Five & Six Variable Maps, Product of Sum Simplification, NAND & NOR Implementation, Two-Level Implementations, Don't-Care Conditions, Tabulation Method, Determination of Prime Implicants, Selection of Prime Implicants.

UNIT-III COMBINATIONAL LOGIC

Combinational Logic: Design Procedure, Adders, Subtractors, Code Conversion, Multilevel NAND Circuits, Multilevel NOR Circuits, Exclusive-Or and Equivalence Functions.
Combinational Logic with MSI And LSI: Binary Parallel Adder, Decimal Adder, Magnitude Comparator, Decoders, Multiplexers, ROM, PLA.

UNIT-IV SEQUENTIAL CIRCUIT DESIGN AND ANALYSIS

Sheet metal characteristics – Typical shearing operations, bending and drawing operations – Formability of sheet metal – Working principle and application of special forming processes – Hydro forming – Explosive forming – Magnetic pulse forming – Peen forming – Super plastic forming – Process characteristics

UNIT-V DIGITAL INTEGRATED CIRCUITS

Digital Integrated Circuits: Introduction, Bi-Polar Transistor Characteristics, Integrated Injection Logic, Transistor-Transistor Logic (TTL), Emitter-Coupled Logic (ECL), MOS, Complementary MOS.

TEXT BOOKS:

1. M.Morris Mano, —Digital Design II, 2nd Edition, 1997, PHI. (Unit I, II, III & IV).

REFERENCE BOOKS:

1. Zvi Kohavi, —Switching and Automata Theory II, 2nd Edition, 1978, McGraw-Hill.
2. Thomas L. Floyd —Digital Fundamentals II, 10th Edition, Pearson Education India.
3. Stephen Brown, Zvonko Vrsanec, II Fundamentals of Digital Logic with Verilog Design II, 2nd Edition, McGraw-Hill.

WEB REFERENCES:

1. <http://www.freebookcentre.net/electronics-ebooks-download/logic design.html>
2. <http://coen.boisestate.edu/sml00/sml00-courses/ece-332-logic designs-fall07/lecture-notes/>
3. <http://www.freebookcentre.net/electronics-ebooks-download/logic design-Lecture-Notes.html>

E-TEXT BOOKS:

1. <http://gen.lib.rus.ec/book/logic design>
2. <http://www.faadooengineers.com/threads/logicdesign -by-RAMESH-GANOKAR-ebook-pdf-download>
3. <https://e.edim.co/123389964/logic design.pdf>

MOOC COURSE

1. <https://www.mooc-list.com/tags/logic design>
2. <https://www.coursera.org/courses?query=logicdesign>

PRINCIPLES OF COMMUNICATIONS								
V Semester: OPEN ELECTIVE -I								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EC60	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100
COURSE OUTCOMES:								
Upon successful completion of the course, the student is able to								
1. Describe the basic concepts of Communication technology 2. Explain the Models of Communication 3. Discourse Analog and Digital Signal transmission 4. Discourse Computer Networking 5. Explain the types of Communication Media 6. Explain the Application of computer in data transmission and Security.								
UNIT-I	FUNDAMENTALS OF ANALOG COMMUNICATION							
Principles of amplitude modulation, AM envelope, frequency spectrum and bandwidth, modulation index and percent modulation, AM Voltage distribution, AM power distribution, Angle modulation FM and PM waveforms, phase deviation and modulation index, frequency deviation and percent modulation, Frequency analysis of angle modulated waves. Bandwidth requirements for Angle modulated waves.								
UNIT-II	DIGITAL COMMUNICATION							
Introduction, Shannon limit for information capacity, digital amplitude modulation, frequency shift keying, FSK bit rate and baud, FSK transmitter, BW consideration of FSK, FSK receiver, phase shift keying binary phase shift keying QPSK, Quadrature Amplitude modulation, bandwidth efficiency, carrier recovery – squaring loop, Costas loop, DPSK.								
UNIT-III	DIGITAL TRANSMISSION							
Introduction, Pulse modulation, PCM,PCM sampling, sampling rate, signal to quantization noise rate, companding analog and digital percentage error, delta modulation, adaptive delta modulation, differential pulse code modulation, pulse transmission – Inter symbol interference, eye patterns.								
UNIT-IV	SPREAD SPECTRUM AND MULTIPLE ACCESS TECHNIQUES:							
Introduction, Pseudo noise sequence, DS spread spectrum with coherent binary PSK, processing gain, FH spread spectrum, multiple access techniques wireless communication, TDMA and CDMA in wireless communication systems, source coding of speech for wireless communications.								
UNIT-V	SATELLITE AND OPTICALCOMMUNICATION							
Satellite Communication Systems ,Keplers Law, LEO and GEO Orbits, footprint, Link model Optical Communication Systems-Elements of Optical Fiber Transmission link, Types, Losses, Sources and Detectors								
TEXT BOOKS:								
1. Wayne Tomasi, —Advanced Electronic Communication Systemsll, 6/e, PearsonEducation, 2007. 2. Simon Haykin, —Communication Systemsll, 4th Edition, John Wiley & Sons., 2001.								
REFERENCE BOOKS:								

1. H.Taub,D L Schilling, G Saha ,||Principles of Communication||3/e,2007.
2. B.P.Lathi,||Modern Analog And Digital Communication systems||, 3/e, Oxford UniversityPress, 2007
3. Blake, —Electronic Communication Systems||, Thomson Delmar Publications, 2002.
4. Martin S.Roden, —Analog and Digital Communication System||, 3rd Edition, PHI, 2002.
5. B.Sklar,||Digital Communication Fundamentals and Applications||2/e PearsonEducation 2007.

WEB REFERENCES:

- 1.https://books.google.co.in/books/about/Principles_Of_Communication.html?id=6Zunu4Acfg8C

E-TEXT BOOKS:

1. https://books.google.co.in/books/about/Principles_Of_Communication.html?id=6Zunu4Acfg8C

MOOC COURSE

1. https://onlinecourses.nptel.ac.in/noc18_ee26/preview

MEASUREMENTS AND INSTRUMENTATION

V Semester: OPEN ELECTIVE -I

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EC61	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Apply knowledge of electronic instruments for measurement of electrical quantities.
2. Understand the principles and operations of instruments.
3. Select and use latest hardware for measurements.
4. Identify the various instruments for various measurements
5. Use the instruments in laboratory and real life

UNIT-I MEASUREMENT CONCEPTS

Measurement concepts : Quantities of Measurement- Performance Characteristics – Static and dynamic characteristics – units and standards of measurements – errors in measurements- accuracy and precision, statistical analysis – moving coil, moving iron types Instruments –DC Voltmeter multi-meter & Calibrations

UNIT-II DIGITAL INSTRUMENTS

Digital instruments: Digital multimeters – Digital frequency meter-digital tacho meter-digital PH meter-digital phase meter-digital capacitance meter-Microprocessor based instruments- IEEE 488 bus

UNIT-III SIGNAL GENERATORS CATHODE RAY OSCILLOSCOPES

Signal generators: Function generators – pulse and square wave generators, RF signal generators – Sweep generators – Frequency synthesizer

Cathode ray oscilloscopes: Basic block diagram – operation-applications

special Oscilloscopes: delayed time base oscilloscopes, sampling oscilloscope

UNIT-IV MEASURING INSTRUMENTS AND BRIDGES

Measuring instruments: Introduction-field strength meter –Q- meter- LCR Bridge-Transistor tester

Bridge: Wheatstone's bridges for resistance measurements-Maxwell's bridge for inductance measurements-Schering's bridge for capacitance measurements measurement

UNIT-V TRANSDUCERS & DATA ACQUISITION SYSTEMS

Transducer: Introduction-electrical transducers-selecting a transducers-active and passive transducers with examples

Data acquisition systems: Elements of a digital data acquisition system – interfacing of transducers – multiplexing –data loggers –computer controlled instrumentation

TEXT BOOKS:

1. Albert D.Helfrick and William D.Cooper – Modern Electronic Instrumentation and Measurement Techniques, Pearson / Prentice Hall of India, 2007.
2. Ernest O. Doebelin, Measurement Systems- Application and Design, TMH, 2007.

REFERENCE BOOKS:

1. Joseph J.Carr, Elements of Electronics Instrumentation and Measurement, Pearson Education, 2003.
2. Alan. S. Morris, Principles of Measurements and Instrumentation, 2nd Edition,Prentice Hall of

India, 2003. - David A. Bell, Electronic Instrumentation and measurements, Prentice Hall of India Pvt Ltd, 2003. - B.C. Nakra and K.K. Choudhry, Instrumentation, Measurement and Analysis, 2nd Edition, TMH, 2004. - James W. Dally, William F. Riley, Kenneth G. McConnell, Instrumentation for Engineering Measurements, 2nd Edition, John Wiley, 2003.
WEB REFERENCES:
https://www.sciencedirect.com/science/book/9780123819604 https://www.schneider-electric.com/en/.../86485-measurement-and-instrumentation/
E-TEXT BOOKS:
https://books.google.co.in/books/about/Electronics_Measurements_And_Instrumenta.html?id=mxVRJRYCi-QC https://sway.com/V1Uclj1JCiLhqcCb
MOOC COURSE
https://www.schneider-electric.com/en/.../86485-measurement-and-instrumentation/ https://www.schneider-electric.com/en/.../86485-measurement-and-instrumentation/ https://www.coursera.org/courses?query=measurements%20and%20instrumentat

FUNDAMENTALS OF INTEGRATED CIRCUITS

VI Semester: OPEN ELECTIVE -II

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EC62	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Analyze the basic fabrication steps of semiconductor devices.
2. Analyze silicon oxidation and photolithography.
3. Analyze various etching techniques.
4. Analyze the Ion implantation process and metallization.
5. Evaluate various IC technologies such as Bipolar, MOSFET and MESFET.

UNIT-I INTEGRATED CIRCUITS AND OPERATIONAL AMPLIFIER

Semiconductor materials, Semiconductor Devices, Semiconductor process technology, Basic fabrication steps. Crystal Growth: Silicon Crystal Growth from melt, Silicon Float-Zone Process, GaAs Crystal Growth Techniques, Material Characterization.

UNIT-II SILICON OXIDATION

Thermal oxidation, Impurity Redistribution during oxidation, masking properties of silicon dioxide, oxide quality, oxide thickness characterization, Photolithography.

UNIT-III ETCHING

Wet chemical etching, Dry etching. Diffusion: Basic Diffusion Process, Extrinsic Diffusion, Lateral Diffusion.

UNIT-IV ION IMPLANTATION

Range of Implanted ions, Implant Damage and annealing, Implantation-related Process. Film Deposition: Epitaxial growth techniques, Structures and defects in epitaxial layers, Dielectric deposition, Poly-silicon deposition, Metallization.

UNIT-V PROCESS INTEGRATION

Passive Components, Bipolar Technology, MOSFET technology, MESFET technology, MEMS technology. IC Manufacturing: Electrical testing, Packaging.

TEXT BOOKS:

1. Gary S. May, Simon M. Sze, "Fundamentals of Semiconductor Fabrication", John Wiley Inc., 2004.

REFERENCE BOOKS:

1. C.Y. Chang and S.M.Sze (Ed), "ULSI Technology", McGraw Hill Companies Inc, 1996.
2. S.K. Ghandhi, "VLSI Fabrication Principles", John Wiley Inc., New York, 1983.
3. S.M. Sze (Ed), "VLSI Technology", 2nd Edition, McGraw Hill, 1988
4. The Science and Engineering of Microelectronic Fabrication, Stephen Cambell, Oxford University Press, 2001

WEB REFERENCES:

- | |
|--|
| 1. https://banavathvalyaitmadras.files.wordpress.com/2015/12/vlsi-fabrication-principles-silicon-and-gallium-arsenide-1994-ghandi-wiley-2e.pdf |
| E-TEXT BOOKS: |
| 1. rku.ac.in/soe/wp-content/uploads/2015/04/VL204_CMOSDeviceFabrication.pdf |
| 2. https://books.google.co.in/books/.../The_Science_and_Engineering_of_Microelectronics |
| MOOC COURSE |
| 1. nptel.ac.in/courses/117103066/35 |
| 2. https://www.edx.org/course/circuits-electronics-1-basic-circuit-mitx-6-002-1x-0 |

SIGNALS TRANSMISSION THROUGH LINEAR SYSTEMS

VI Semester: OPEN ELECTIVE -II

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EC63	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Analyze various signals and systems by applying Laplace and Z-transforms.
2. Apply Fourier series, Fourier transform and inverse Fourier transform techniques to find magnitude and phase spectrum of various signals.
3. Apply convolution and correlation operations to various signals and systems.
4. Design IIR and FIR analog and digital filters.
5. Analyze multirate Digital Signal Processing.

UNIT-I TIME DOMAIN SIGNALS

Introduction: Definitions of signals and systems, Elementary Continuous-time and Discrete time signals: Unit impulse, Unit step, Ramp, Exponential, Sinusoidal, Sync, Signum, and Parabolic. Classification of various signals and systems, Sampling and types of Sampling, Laplace and Z-transforms.

UNIT-II SPECTRAL DOMAIN REPRESENTATION OF SIGNALS

Introduction, Fourier series, Fourier transform, Properties, Magnitude Spectrum and Phase spectrum. Discrete Fourier transforms, properties, Interrelationships between Fourier, Laplace and Z-Transforms, problems.

UNIT-III CONVOLUTION AND CORRELATION

Introduction, System transfer function, Convolution in continuous time domain, Convolution in discrete time domain: Linear convolution and Circular Convolution, System Response, Auto correlation function and properties, Cross Correlation function and properties, Problems.

UNIT-IV DIGITAL FILTERS

IIR Filters: IIR Analog filter approximations - Butterworth and Chebyshev, Design of IIR Digital filters from analog filters: Impulse invariant techniques, Bilinear transformation method.
FIR Filters: Characteristics of FIR Digital Filters, Frequency response, Design of FIR Filters: Fourier Method. Window Techniques, Comparison of IIR & FIR filters.

UNIT-V MULTI RATE DIGITAL SIGNAL PROCESSING

Introduction. Down sampling, Decimation. Up sampling, Interpolation, Sampling Rate Conversion, conversion of band pass signals. Concept of re-sampling. Applications of multi rate signal processing.

TEXT BOOKS:

1. John G. Proakis, Dimitris G. Manolakis, Digital Signal Processing, Pearson Prentice Hall, 2007.
2. B.P. Lathi, Signals, Systems & Communications, BS Publications 2013.
3. Sophocles J. Orfanidis, Introduction to Signal Processing, Pearson Prentice Hall, 1998.

REFERENCE BOOKS:

1. A.V. Oppenheim, Signals and Systems - A.S. Willsky and S.H. Nawab, 2 Ed.,
2. Li Tan, Digital Signal Processing - Fundamentals and Applications - Elsevier. 2008.
3. Robert J. Schilling. Sandra L, Harris, Fundamentals of Digital Signal Processing using Matlab - Thomson. 2007

WEB REFERENCES:

1. [http://www.freebookcentre.net/special-books-download/An-Introduction-to-Signal-Processing-\(T.-O'Haver\).html](http://www.freebookcentre.net/special-books-download/An-Introduction-to-Signal-Processing-(T.-O'Haver).html)
2. <https://www.sciencedirect.com/science/article/pii/B9780123965028000012>
3. Applied Digital Signal Processing (1st Edition)

E-TEXT BOOKS:

1. <http://gen.lib.rus.ec/book/index.php?md5=69531D2C9E6A3A5111E2CC1977F70F59>
2. [http://www.freebookcentre.net/special-books-download/Introduction-to-DSP-\(Digital-Signal-Processing\).html](http://www.freebookcentre.net/special-books-download/Introduction-to-DSP-(Digital-Signal-Processing).html)

MOOC COURSE

1. <https://www.mooc-list.com/course/digital-signal-processing-coursera>
2. <http://nptel.ac.in/courses/117102060/>

FUNDAMENTALS OF VLSI DESIGN

VI Semester: OPEN ELECTIVE -II

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EC64	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Understand the Basic NMOS, CMOS & BiCMOS circuits and their process technology.
2. Designing of stick diagrams and layouts for MOS transistors.
3. Learn the concepts of modeling of Delay techniques and MOS layers.
4. Learn the concepts of Technology Scaling of MOS transistors.
5. Understand the concepts of testing of combinational and sequential circuits and also the scan of design techniques

UNIT-I REVIEW OF MICROELECTRONICS AND INTRODUCTION TO MOS TECHNOLOGY

Introduction MOS and related VLSI technology – NMOS-CMOS-BICMOS-GaAs Technologies – thermal aspects of processing – production of E beam masks.

UNIT-II MOS AND BICMOS CIRCUIT DESIGN PROCESS

MOS layers – stick diagrams – design rules and layout – 2m meter – 1.2 m meter CMOS rules – Layout diagrams – Symbolic diagrams.

UNIT-III BASIC CIRCUIT CONCEPTS

Sheet resistance – Area capacitance of layers – delay unit – wiring capacitances – choice of layers.

UNIT-IV SCALING OF MOS CIRCUITS

Scaling of MOS circuits: Scaling modes I – Scaling function for device parameters – Limitation of Scaling Subsystem design process: Architectural issues – switch logic – examples of structural design (Combinational logic)– design of ALU subsystem – commonly used storage elements – aspects of design rules.

UNIT-V TEST AND TESTABILITY

Test and Testability: Design for testability built in self test (BIST) – teaching combinational logic – testing sequential logic – practical design for test guide lines – scan design techniques – etc.

TEXT BOOKS:

1. Basic VLSI design by Douglas A, Pucknell, Kamran Eshraghian, Prantice Hall, 1996 3rd edition.

REFERENCE BOOKS:

1. Mead, C.A and Conway, L.A., Introduction to VLSI Systems, Wesley – Wesley.

WEB REFERENCES:

1. <http://www.freebookcentre.net/electronics-ebooks-download/vlsi .html>
2. <http://coen.boisestate.edu/sml00/sml00-courses/ece-332-vlsi-fall07/lecture-notes/>
3. <http://www.freebookcentre.net/electronics-ebooks-download/vlsi design-Lecture-Notes.html>

E-TEXT BOOKS:

1. <http://gen.lib.rus.ec/book/vlsi design>
2. <http://www.faadooengineers.com/threads/vlsi design -by-kamran eshragghan-ebook-pdf-download>
3. <https://e.edim.co/123389964/vlsi design.pdf>

MOOC COURSE

1. <https://www.mooc-list.com/tags/vlsi design>
2. <https://www.coursera.org/courses?query=vlsi design>

INTRODUCTION OF MICROPROCESSORS AND MICROCONTROLLERS

VII Semester: OPEN ELECTIVE -III

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EC65	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Understand the basic 8&16 bit microprocessor architectures and its functionalities.
2. Understand Assembly language programming skills of processor.
3. Interface different peripheral devices with Micro processors or Micro controllers.
4. Develop an assembly language program for specified application
5. Develop systems using different micro processors and micro controllers.

UNIT-I MICROPROCESSOR ARCHITECTURE

Overview of 8085, 8086 architecture- functional diagram, Register organization, memory segmentation, Memory addresses, physical memory organization, Signal descriptions of 8086, timing diagrams, Interrupt structure of 8086, Vector interrupt table, Interrupt service routine.

UNIT-II INSTRUCTION SET AND ASSEMBLY LANGUAGE PROGRAMMING OF 8086

Instruction set, Assembler directives, Addressing modes, programming model, Programs involving logical, branch and call instructions, Sorting.

UNIT-III INTERFACING WITH 8086

8255 PPI, various modes of operation and interfacing to 8086, Stepper motor interfacing, D/A & A/D converter, Memory interfacing to 8086, DMA controller (8257).

UNIT-IV INTRODUCTION TO MICROCONTROLLERS

Overview of 8051 microcontroller, Architecture, I/O ports, Memory organization, addressing modes and instruction set of 8051, Simple programs.

UNIT-V 8051 REAL TIME CONTROL

Programming Timer interrupts, programming external hardware interrupts, Programming the serial communication interrupts, programming 8051 timers and Counters.

TEXT BOOKS:

1. D.V.Hall, Microprocessors and Interfacing. TMGH, 2nd edition 2006.
2. Kenneth.J.Ayala. The 8051 microcontroller, 3rd edition, Cengage learning,2010
3. Advanced microprocessors and peripherals-A.K ray and K.M.Bhurchandani, TMH, 2nd edition 2006.
4. R. S. Gaonkar, Microprocessor Architecture: Programming and Applications with the 8085/8080A, Penram International Publishing, 1996.

REFERENCE BOOKS:

1. D A Patterson and J H Hennessy, "Computer Organization and Design The hardware and software interface. Morgan Kaufman Publishers.
2. Micro computer system 8086/8088 family architecture, programming and design- By Liu and GA Gibson, PHI, 2nd Ed.,
3. Microcontrollers and application, Ajay. V. Deshmukh, TMGH, 2005.

WEB REFERENCES:

1. <http://www.freebookcentre.net/electronics-ebooks-download/Microprocessor-and-Microcontroller.html>
2. <http://coen.boisestate.edu/sml00/sml00-courses/ece-332-microprocessors-fall07/lecture-notes/>
3. <http://www.freebookcentre.net/electronics-ebooks-download/Introduction-to-Microcontrollers-Lecture-Notes.html>

E-TEXT BOOKS:

1. <http://gen.lib.rus.ec/book/index.php?md5=67C5AC79DC8180A7F0641609D0C7800C>
2. <http://www.faadooengineers.com/threads/9039-8085-microprocessor-by-RAMESH-GANOKAR-ebook-pdf-download>
3. https://e.edim.co/123389964/The_8051_Microcontroller_Architecture_Programming_And_Applications.pdf
4. https://e.edim.co/123389964/A.K._Ray_and_K.M._Bhurchandi-Advanced_Microprocessors_and_Peripherals_3e-Tata_Mcgraw_Hill.pdf

MOOC COURSE

1. <https://www.mooc-list.com/tags/microprocessors>
2. <https://www.coursera.org/courses?query=microprocessor>

FUNDAMENTALS OF IMAGE PROCESSING

VII Semester: OPEN ELECTIVE -III

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EC66	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Know the Fundamentals of Digital Image Processing.
2. Apply the concepts in real time applications
3. Develop a code for various image processing concepts

UNIT-I INTRODUCTION

Digital Image Fundamentals-Elements of visual perception, image sensing and acquisition, image sampling and quantization, basic relationships between pixels – neighborhood, adjacency, connectivity, distance measures.

UNIT-II IMAGE ENHANCEMENT

Image Enhancements and Filtering-Gray level transformations, histogram equalization and specifications, pixel-domain smoothing filters – linear and order-statistics, pixel-domain sharpening filters – first and second derivative, two-dimensional DFT and its inverse, frequency domain filters – low-pass and high-pass

UNIT-III IMAGE SEGMENTATION

Image Segmentation- Detection of discontinuities, edge linking and boundary detection, thresholding – global and adaptive, region-based segmentation. Wavelets and Multi-resolution image processing- Uncertainty principles of Fourier Transform, Time-frequency localization, continuous wavelet transforms, wavelet bases and multi-resolution analysis, wavelets and Subband filter banks, wavelet packets.

UNIT-IV IMAGE COMPRESSION

Image Compression-Redundancy–inter-pixel and psycho-visual; Lossless compression – predictive, entropy; Lossy compression- predictive and transform coding; Discrete Cosine Transform; Still image compression standards – JPEG and JPEG-2000.

UNIT-V VIDEO PROCESSING

Fundamentals of Video Coding- Inter-frame redundancy, motion estimation techniques – full-search, fast search strategies, forward and backward motion prediction, frame classification – I, P and B; Video sequence hierarchy – Group of pictures, Elements of a video encoder and decoder; Video coding standards – MPEG and H.26X. Video Segmentation- Temporal segmentation–shot boundary detection, hard-cuts and soft-cuts; spatial segmentation – motion-based; Video object detection and tracking.

TEXT BOOKS:

1. R.C. Gonzalez and R.E. Woods, Digital Image Processing, Second Edition, Pearson Education 3rd edition 2008
2. Anil Kumar Jain, Fundamentals of Digital Image Processing, Prentice Hall of India.2nd edition 2004

REFERENCE BOOKS:

1. 1. Murat Tekalp , Digital Video Processing" Prentice Hall, 2nd edition 2015

WEB REFERENCES:

1. <http://homepages.inf.ed.ac.uk/rbf/BOOKS/VERNON/Chap004.pdf>

E-TEXT BOOKS:

1. https://books.google.co.in/books/about/Digital_Image_Processing.html?id=a62xQ2r_f8wC

MOOC COURSE

1. <https://nptel.ac.in/courses/117105079/>

TELEVISION ENGINEERING

VII Semester: OPEN ELECTIVE -III

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EC67	OEC	L	T	P	C	CIA	SEE	Total
		4	-	-	3	30	70	100

COURSE OUTCOMES:
Upon successful completion of the course, the student is able to

1. Know the fundamentals of Television transmitters and receivers
2. Understand different signal processing step monochrome television,
3. Introduce color television transmitters and receivers.
4. Remember various transmission and receiving antennas
5. Know the function of the TV

UNIT-I	INTRODUCTION	
TV transmitter and receivers, synchronization. Geometric form and aspect ratio, image continuity, interlaced scanning, picture resolution Composite video signal, TV standards. Camera tubes: image Orthicon Plumbicon, vidicon, silicon Diode Array vidicon, Comparison of camera tubes Monochrome TV camera, TV Signal Transmission and Propagation: Picture Signal transmission positive and negative modulation, VSB transmission, sound signal transmission, standard channel BW, TV transmitter, TV signal propagation interference, TV broadcast channels, TV transmission Antennas.		
UNIT-II	MONOCHROME TV RECEIVER	
RF tuner, IF subsystem, video amplifier, sound section, sync separation and processing, deflection circuits, scanning circuits, AGC, noise cancellation, video and inter carrier sound signal detection, vision IF subsystem of Black and White receivers, Receiver sound system: FM detection, FM Sound detectors, and typical applications.		
UNIT-III	SYNC SEPARATION AND DETECTION	
TV Receiver Tuners, Tuner operation, VHF and UHF tuners, digital tuning techniques, remote control of receiver functions. Sync Separation, AFC and Deflection Oscillators: Synchronous separation, k noise in sync pulses, separation of frame and line sync pulses. AFC, single ended AFC circuit, Deflection Oscillators, deflection drive IOs, Receiver Antennas, Picture Tubes.		
UNIT-IV	SPEECH RECOGNITION	
Large Vocabulary Continuous Speech Recognition: Architecture of large vocabulary continuous speech recognition system – acoustics and language models – n-grams, context dependent sub-word units; Applications and present status.		
UNIT-V	SPEECH SYNTHESIS	

Color Receiver: Introduction to colour receiver, Electron tuners, IF subsystem, Y-signal channel, Chroma decoder, Separation of U & V Color, Phasors, synchronous demodulators, Sub carrier generation, raster circuits.

Digital TV: Introduction to Digital TV, Digital Satellite TV, Direct to Home Satellite TV, Digital TV Transmitter, Digital TV Receiver, Digital Terrestrial TV, LCD TV, LED TV, CCD Image Sensors, HDTV.

TEXT BOOKS:

1. Television and Video Engineering- A.M.Dhake, 2nd Edition.
2. Modern Television Practice Principles, Technology and Service R. R. Gallatin, New Age International Publication, 2002.
3. Monochrome and Colour TV- R.R. Gulati, New Age International Publication, 2002.

REFERENCE BOOKS:

1. Colour Television Theory and Practice-S.P.Bal, TMH, 1994.
2. Basic Television and Video Systems-B.Grob and C.E.Herndon, McGraw Hill, 1999.

WEB REFERENCES:

1. <http://www.sasurieengg.com/e-course-material/ECE/IV-Year%20Sem%207/EC2034%20Television%20and%20Video%20Engineering.pdf>

E-TEXT BOOKS:

1. https://books.google.co.in/books?id=AvQAa5Zfuj0C&printsec=frontcover&dq=tv+by+rr+gulati&source=gbs_similarbooks_s&cad=1#v=onepage&q=tv%20by%20rr%20gulati&f=false

MOOC COURSE

1. <https://nptel.ac.in/courses/109103018/34>

INTRODUCTION TO SIGNAL PROCESSING

VIII Semester: OPEN ELECTIVE -IV

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EC68	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Identify various signals and systems.
2. Understand various transformation techniques and apply them to find spectral domain representation of various signals.
3. Apply convolution and correlation operations between various signals and systems.
4. Develop the digital filter designs.
5. Understand Multirate digital signal processing concepts.

UNIT-I TIME DOMAIN SIGNALS

Introduction, Definitions of signal and system, Elementary Continuous-time and Discrete time signals: Unit impulse, Unit step, Ramp, Exponential, Sinusoidal, Sinc, Signum, Parabolic. Classification of various signals and systems, Sampling and types of Sampling, Laplace and Z-transforms.

UNIT-II SPECTRAL DOMAIN REPRESENTATION OF SIGNALS

Introduction, Fourier series, Fourier transform, Properties, Magnitude Spectrum and Phase spectrum. Discrete Fourier transform, properties, Interrelationships between Fourier, Laplace and Z-Transforms, problems.

UNIT-III CONVOLUTION AND CORRELATION

Introduction, System transfer function, Convolution in continuous time domain, Convolution in discrete time domain: Linear convolution and Circular Convolution, System Response, Auto correlation function and properties, Cross Correlation function and properties, Problems.

UNIT-IV DIGITAL FILTERS

IIR Filters: IIR Analog filter approximations - Butterworth and Chebyshev, Design of IIR Digital filters from analog filters: Impulse invariant techniques, Bilinear transformation method.
FIR Filters: Characteristics of FIR Digital Filters, Frequency response, Design of FIR Filters: Fourier Method. Window Techniques, Comparison of IIR & FIR filters.

UNIT-V MULTIRATE DIGITAL SIGNAL PROCESSING

Introduction. Down sampling, Decimation. Up sampling, Interpolation, Sampling Rate Conversion, conversion of band pass signals. Concept of re-sampling. Applications of multi rate signal processing.

TEXT BOOKS:

1. John G. Proakis, Dimitris G. Manolakis, Digital Signal Processing, Pearson Prentice Hall, 2007.
2. B.P. Lathi, Signals, Systems & Communications, BS Publications 2013.
3. Sophocles J. Orfanidis, Introduction to Signal Processing, Pearson Prentice Hall, 1998.

REFERENCE BOOKS:

1. A.V. Oppenheim, Signals and Systems - A.S. Willsky and S.H. Nawab, 2 Ed.,
2. Li Tan, Digital Signal Processing - Fundamentals and Applications - Elsevier. 2008.
3. Robert J. Schilling. Sandra L, Harris, Fundamentals of Digital Signal Processing using Matlab - Thomson. 2007

WEB REFERENCES:

1. [http://www.freebookcentre.net/special-books-download/An-Introduction-to-Signal-Processing-\(T.-OHaver\).html](http://www.freebookcentre.net/special-books-download/An-Introduction-to-Signal-Processing-(T.-OHaver).html)
2. <https://www.sciencedirect.com/science/article/pii/B9780123965028000012>
3. Applied Digital Signal Processing (1st Edition)

E-TEXT BOOKS:

1. <http://gen.lib.rus.ec/book/index.php?md5=69531D2C9E6A3A5111E2CC1977F70F59>
2. [http://www.freebookcentre.net/special-books-download/Introduction-to-DSP-\(Digital-Signal-Processing\).html](http://www.freebookcentre.net/special-books-download/Introduction-to-DSP-(Digital-Signal-Processing).html)

MOOC COURSE

1. <https://www.mooc-list.com/course/digital-signal-processing-coursera>
2. <http://nptel.ac.in/courses/117102060/>

INTRODUCTION TO MOBILE COMMUNICATION

VIII Semester: OPEN ELECTIVE -IV

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EC69	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Understand impairments due to multi path fading channel.
2. Understand the fundamental techniques to overcome the different fading effects.
3. Understand Co-channel and Non Co channel interference
4. Familiar with cell coverage for signal and traffic, diversity techniques and mobile antennas.

UNIT-I INTRODUCTION TO CELLULAR MOBILE RADIO SYSTEMS

Limitations of Conventional Mobile Telephone Systems, Basic Cellular Mobile System, First, Second, Third and Fourth Generation Cellular Wireless Systems, Uniqueness of Mobile Radio Environment- Fading -Time Dispersion Parameters, Coherence Bandwidth, Doppler Spread and Coherence Time. Fundamentals of Cellular Radio System Design: Concept of Frequency Reuse, Co-Channel Interference, Co-Channel Interference Reduction Factor, Desired C/I From a Normal Case in a Omni Directional Antenna System, System Capacity, Trunking and Grade of Service, Improving Coverage and Capacity in Cellular Systems- Cell Splitting, Sectoring, Microcell Zone Concept.

UNIT-II CO-CHANNEL INTERFERENCE

Measurement Of Real Time Co-Channel Interference, Design of Antenna System, Antenna Parameters and Their Effects, Diversity Techniques-Space Diversity, Polarization Diversity, Frequency Diversity, Time Diversity. Non-Co-Channel Interference: Adjacent Channel Interference, Near End Far End Interference, Cross Talk, Effects on Coverage and Interference by Power Decrease, Antenna Height Decrease, Effects of Cell Site Components.

UNIT-III MULTIPLE ACCESS & MODULATION SCHEMES

Signal Reflections in Flat And Hilly Terrain, Effect of Human Made Structures, Phase Difference Between Direct and Reflected Paths, Constant Standard Deviation, Straight Line Path Loss Slope, General Formula for Mobile Propagation Over V'later and Flat Open Area, Near and Long Distance Propagation, Path Loss From a Point to Point Prediction Model in Different Conditions, Merits of Lee Model. Cell Site and Mobile Antennas: Space Diversity Antennas, Umbrella Pattern Antennas, Minimum Separation of Cell Site Antennas, Mobile Antennas.

UNIT-IV FREQUENCY MANAGEMENT AND CHANNEL ASSIGNMENT

Numbering And Grouping, Setup Access And Paging Channels, Channel Assignments to Cell Sites and Mobile Units, Channel Sharing and Borrowing, Sectorization, Overlaid Cells, Non Fixed Channel Assignment.

UNIT-V HANDOFFS AND DROPPED CALLS

Handoff Initiation, Types of Handoff, Delaying Handoff, Advantages of Handoff, Power Difference Handoff, Forced Handoff, Mobile Assisted and Soft Handoff, inter system Handoff, Introduction to Dropped Call Rates and their Evaluation.

TEXT BOOKS:

1. Mobile Cellular Telecommunications — W.C.Y. Lee, Mc Graw Hill, 2nd Edn., 1989.
2. Wireless Communications – Theodore. S. Rapport, Pearson Education, 2nd Edn., 2002.
3. Mobile Cellular Communication – Gottapu sashibhushana Rao, Pearson, 2012.

REFERENCE BOOKS:

1. Principles of Mobile Communications — Gordon L. Stuber, Springer International, 2nd Edn., 2001
2. Modern Wireless Communications-Simon Haykin, Michael Moher, Pearson Education, 2005
3. Wireless Communications Theory and Techniques, Asrar U. H .Sheikh, Springer, 2004.

WEB REFERENCES:

1. https://web.fe.up.pt/~mricardo/08_09/cmouv-mieic/slides/introduction.pdf

E-TEXT BOOKS:

1. <http://ee.sharif.edu/~pr.wireless.comm/references/Schwartz.pdf>

MOOC COURSE

1. <http://nptel.ac.in/courses/117102062/>

BASIC EMBEDDED SYSTEMS DESIGN

VIII Semester: OPEN ELECTIVE -IV

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EC70	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Explains about the Basic functions, Structure, Concepts and Applications of Embedded systems.
2. Build simple programs to solve well-defined problems on an Embedded platform
3. Understanding various communication interfaces available.
4. Identify different processors for various applications.
5. Explains about the tools used to develop in an Embedded environment and Implement Real time applications on Embedded platform

UNIT-I INTRODUCTION TO EMBEDDED SYSTEMS

Definition and Classification – Overview of Processors and hardware units in an embedded system – Software embedded into the system – Exemplary Embedded Systems – Embedded Systems on a Chip (SoC) and the use of VLSI designed circuits.

UNIT-II INTRODUCING THE 8051 MICROCONTROLLER FAMILY

Introduction, What's in a name, The external interface of the Standard 8051, Reset requirements, Clock frequency and performance, Memory issues, I/O pins, Timers, Interrupts, Serial interface, Power consumption, Conclusions

UNIT-III DEVICES AND BUSES FOR DEVICES NETWORK

I/O Devices - Device I/O Types and Examples – Synchronous - Iso-synchronous and Asynchronous Communications from Serial Devices - Examples of Internal Serial-Communication Devices - UART and HDLC - Parallel Port Devices - Sophisticated interfacing features in Devices/Ports- Timer and Counting Devices - '12C', 'USB', 'CAN' and advanced I/O Serial high speed buses- ISA, PCI, PCI-X, cPCI and advanced buses.

UNIT-IV PROGRAMMING CONCEPTS AND EMBEDDED PROGRAMMING IN C, C++

Programming in assembly language (ALP) vs. High Level Language - C Program Elements, Macros and functions -Use of Pointers - NULL Pointers - Use of Function Calls – Multiple function calls in a Cyclic Order in the Main Function Pointers – Function Queues and Interrupt Service Routines Queues Pointers – Concepts of EMBEDDED PROGRAMMING in C++ - Objected Oriented Programming – Embedded Programming in C++, 'C' Program compilers – Cross compiler – Optimization of memory codes.

UNIT-V REAL TIME OPERATING SYSTEMS – PART - 1

Tasks and task states, tasks and data, semaphores, and shared data; message queues, mailboxes and pipes, timer functions, events, memory management, interrupt routines in an RTOS environment. Semaphores, queues, hard real-time scheduling considerations, saving memory and power, an example RTOS like uC-OS (open source).

TEXT BOOKS:

1. Rajkamal, Embedded Systems Architecture, Programming and Design, TATA McGraw-Hill, First reprint Oct. 2003
2. Shibu K.V, Introduction to Embedded Systems, Mc Graw Hill.

REFERENCE BOOKS:

1. Steve Heath, Embedded Systems Design, Second Edition-2003, Newnes,
2. David E.Simon, An Embedded Software Primer, Pearson Education Asia, First Indian Reprint 2000.

WEB REFERENCES:

1. https://www.bharathuniv.ac.in/colleges1/downloads/courseware_ece/notes/BEI605-%20Embedded-System.pdf

E-TEXT BOOKS:

1. <http://read.pudn.com/downloads158/ebook/707037/Embedded%20Systems%20Design%20-%202ed%20-%200750655461.pdf>

MOOC COURSE

1. <https://nptel.ac.in/courses/108102045/>

**OPEN ELECTIVES
OFFERED BY
EEE DEPARTMENT**

ELECTRICAL WIRING AND SAFETY MEASURES

V Semester: OPEN ELECTIVE -I

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EE52	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES

- To Study the wiring diagram of residential.
- To understand the Safety measures of Electrical wiring

COURSE OUTCOMES:

The students should be able to

- Know safety measures & state safety precautions.
- Test single phase, three phase transformer, DC & AC machine as per IS.
- Ascertain the condition of insulation & varnishing if necessary
- Identify faults & measures to repair faults.

UNIT-I	BASICS OF ELECTRICAL INSTALLATIONS	
Electric Supply System, Three phase four wire distribution system, Protection of Electric Installation against over load, short circuit and Earth fault, General requirements of electrical installations, testing of installations, Types of loads, Systems of wiring, Service connections, Service Mains, Sub-Circuits, Location of Outlets, Location of Control Switches, Location of Main Board and Distribution board, Guide lines for Installation of Fittings, Load Assessment, Permissible voltage drops and sizes of wires, estimating and costing of Electric installations.		
UNIT-II	EARTHING	
Introduction & importance, Factors affecting Earth Resistance, Methods of earthing Substation and Transmission tower earthing, Neutral and Earth wire, Transformer Neutral Earthing.		
UNIT-III	SAFETY & PREVENTION OF ACCIDENTS	
Definition of terminology used in safety , I.E. Act & statutory regulations for safety of persons & equipments working with electrical installation . Dos & don'ts for substation operators as listed in IS. Meaning & causes of electrical accidents factors on which severity of shock depends.		
UNIT-IV	RESIDENTIAL BUILDING ELECTRIFICATION	
General rules guidelines for wiring of Residential Installation and positioning of equipments. Principles of circuit design in lighting and power circuits. Procedures for designing the circuits and deciding the number of sub- circuits. Method of drawing single line diagram & wiring diagram.		
UNIT-V	R ENDS& SUBSTATION AND METERS	
Rule 28 : Voltage level definitions. Rule 30: Service lines & apparatus on consumer premises. Rule 31: Cut-out on consumer's premises. Rule 46: Periodical inspection & testing of consumer's installation. Rule 47: Testing of consumer's installation. Rule 54: Declared voltage of supply to consumer. Rule 55: Declared frequency of supply to consumer. Rule 56: Sealing of meters & cut-outs. Rule 77: Clearances above ground of the lowest conductor. Rule 79: Clearances between conductors & trolley wires. Rule 87: Lines crossing or approaching each other. Rule 88: Guarding.		
TEXT BOOKS:		

1. K.B. Raina, S.K.Bhattacharya Electrical Design;Estimating and costing New Age International (p) Limited, 2. New Delhi Surjit Singh. 3. Electrical Estimating and costing Dhanpat Rai and company, New Delhi .J.B.Gupta 4. A course in Electrical Installation, Estimating & costing S.K.Kataria & sons,S.L. Uappal . 5. Electrical wiringEstimating and costing Khanna Publication. ,A.K.Sawhney
REFERENCE BOOKS:
1. Electrical Machine Design Danpat Rai & co. 2. The Electricity Rule 2005Universal Law Publishing Co. Pvt. Ltd. N. AlagapanS. Ekambaram 3. Electrical Estimating and costing Tata Mc Graw Hill Publication, New Delhi ,Surjit Singh 4. Tarlok Sibgh Installation, Commissioning & Maintenance of Electrical Equipment S.K.Kataria & Sons 5. B.V.S.Rao Operation & Maintenance of Electrical Machines Vol I & II Media Promoters & Publisher Ltd. Mumbai
WEB REFERENCES:
1. https://electrical-engineering-portal.com › Technical Articles 2. https://www.st-andrews.ac.uk/staff/policy/healthandsafety/publications/electricalsafety/ 3. https://www.cpwd.gov.in/Publication/Internal2013.pdf
E-TEXT BOOKS:
1. https://books.google.co.in/books?isbn=0323170064 2. https://www.jove.com/science.../electrical-safety-precautions-and-basic-equipment
MOOC COURSE
1. https://nptel.ac.in/courses/103106071/5 2. https://nptel.ac.in/courses/108108099/28 3. https://nptel.ac.in/courses/124107001/

ELECTRICAL MATERIALS								
V Semester: OPEN ELECTIVE -I								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EE53	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100
COURSE OBJECTIVES To understand about various electrical engineering material								
COURSE OUTCOMES: The students should be able to 1. Analyze the various engineering materials. 2. Application of various engineering materials.								
UNIT-I	CONDUCTORS							
Classification: High conductivity, high resistivity materials, fundamental requirements of high conductivity materials and high resistivity materials, mobility of electron in metals, commonly used high conducting materials, copper, aluminum, bronze brass, properties, characteristics, constantan, platinum, nichrome, properties, characteristics and applications, materials used for contacts.								
UNIT-II	SEMICONDUCTORS							
General concepts, energy bands, types of semiconductors, Fermi Dirac distribution, intrinsic Semiconductors, extrinsic Semi-conductors, hall effect, drift, mobility, diffusion in Semiconductors, Semiconductors and their applications, superconductors.								
UNIT-III	DIELECTRICS AND INSULATORS							
Properties of gaseous, liquid and solid dielectric, dielectric as a field medium, electric conduction in gaseous, liquid and solid dielectric, breakdown in dielectric materials, mechanical and electrical properties of dielectric materials, effect of temperature on dielectric materials, polarization, loss angle and dielectric loss, petroleum based insulating oils, transformer oil, capacitor oils, properties, solid electrical insulating materials, fibrous, paper boards, yarns, cloth tapes, sleeving wood, impregnation, plastics, filling and bonding materials, fibrous, film, mica, rubber, mica based materials, ceramic materials, classification of insulation (solid) and application in AC and DC machines.								
UNIT-IV	MAGNETIC MATERIALS							
Soft and hard magnetic materials, diamagnetic, paramagnetic and ferromagnetic materials, electric steel, sheet steel, cold rolled grain oriented silicon steel, hot rolled grain oriented silicon steel, hot rolled silicon steel sheet, hysteresis loop, hysteresis loss, magnetic susceptibility, coercive force, curie temperature, magneto-striction.								
UNIT-V	OPTICAL PROPERTIES OF SOLIDS							
Photo emission, photo emission materials, electro luminescence junction diode, photo emitters, photo transistor, photo resistors, injection lasers, optical properties of semiconductors, application of photo sensitive materials (CRT, Tube light, photo panels etc.).								
TEXT BOOKS: 1. "Electrical Engineering Materials", Dekker, PHI Pbs. 2. "Electrical Engineering Materials", Indulkar, S.Chand								
REFERENCE BOOKS:								

1. "Electrical Engineering Materials", Tareev
2. "Electrical Engineering Materials", Yu. Koritsky.
3. "Electrical Engineering Materials", R.K.Rajput, Laxmi Pbs

WEB REFERENCES:

- <https://physics.info/dielectrics/>
2. <https://www.oxfordreference.com/view/10.1093/oi/authority.20110803095631265>
 3. web.mit.edu/course/6/6.732/www/6.732-pt2.pdf

E-TEXT BOOKS:

1. <https://easyengineering.net/electrical-engineering-materials-by-dekker/>
2. <https://www.oreilly.com/library/view/dielectric-materials-for/9781118619780/>

MOOC COURSE

1. <https://nptel.ac.in/courses/108108076/>
2. <https://nptel.ac.in/courses/112104203/3>
3. https://onlinecourses.nptel.ac.in/noc18_ee14/

NEW TRENDS IN ELECTRICAL ENERGY

V Semester: OPEN ELECTIVE -I

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EE54	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES

1. The fundamental principles of wireless power transfer (WPT) for cable-free transfer of power.
2. Theories for inductive power transfer (IPT) based on the coupled inductor model and low-order circuit compensation.
3. Specific converter topologies for lighting and battery charging applications.
4. Technology trends in the adoption of WPT for key consumer applications.

COURSE OUTCOMES:

The students should be able to

1. Understand the characteristics of power transfer through coupled inductors and the significance of leakage inductance.
2. Analyze and design appropriate compensation circuits and efficient power converters for WPT applications.
3. Understand technical requirements for applications involving solid-state loads and battery loads using WPT technologies.
4. Appreciate the factors affecting adoption of WPT in consumer applications including lightings, charging of smartphones and electric vehicles.

UNIT-I	BASIC CIRCUIT THEORY	
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Review of transformers. Leakage inductance. Circuit compensation principles. Low-order compensations; series and parallel compensations. Resonance and operating frequency. Efficiency equation.

UNIT-II	POWER CONVERTERS FUNDAMENTALS	
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DC-DC converters. AC-DC converters and inverters. PWM and soft switching principles. Basic topologies with transformers. Input, output and transfer characteristics of power converters. Incorporation of leaky transformer. Control methods.

UNIT-III	COMPENSATION CONFIGURATIONS	
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Types of compensation for inductor power transfer. Characteristics for various termination requirements. Design for load-independence output voltage and output current. Efficiency optimization.

UNIT-IV	APPLICATIONS	
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Circuit requirements for various loading conditions. Characteristics of LED loads, resistors and battery loads. Appropriate compensation design. Lighting systems. Battery charging profiles. Electric vehicle charging. Energy efficiency metric for charging.

UNIT-V	TECHNOLOGY TRENDS	
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Demand for safe power transfer and durable operation. Portable and smart devices. Mobile communication devices. IoT devices and systems. Sensors. Solid-state lighting development. Battery technologies. Electric vehicle development. Renewable source integration trends. Future trends and demand for wireless power transfer.

TEXT BOOKS:

1. C. T. Rim and C. Mi, Wireless Power Transfer for Electric Vehicles and Mobile Devices, New York: IEEE Press-Wiley, 2017.
2. J. I. Agbinya, Wireless Power Transfer, River Publishers, 2015

REFERENCE BOOKS:

1. Z. Huang, S. C. Wong, and C. K. Tse, "Design of a single-stage inductive power-transfer converter for efficient EV battery charging," IEEE Transactions on Vehicular Technology, vol. 66, no. 7, pp. 5808-5821, July 2017.
2. L. Xu, Q. Chen, X. Ren, S. C. Wong, and C. K. Tse, "Self-oscillating resonant converter with contactless power transfer and integrated current sensing transformer," IEEE Transactions on Power Electronics, vol. 32, no. 6, pp. 4839-4851, June 2017.
3. W. Zhang, S. C. Wong, C. K. Tse, and Q. Chen, "Load-independent duality of current and voltage outputs of a series or parallel compensated inductive power transfer converter with optimized efficiency," IEEE Journal of Emerging and Selected Topics in Power Electronics, vol. 3, no. 1, pp. 137-146, March 2015.
4. J. Hou, Q. Chen, X. Ren, X. Ruan, S. C. Wong, and C. K. Tse, "Precise characteristics analysis of series/series-parallel compensated contactless resonant converter," IEEE Journal of Emerging and Selected Topics in Power Electronics, vol. 3, no. 1, 101-110, March 2015.

WEB REFERENCES:

- https://www.researchgate.net/.../267512972_Wireless_Power_Transmission_Trends
2. <https://ieeexplore.ieee.org/document/6403386>
 3. <https://www.cse.wustl.edu/~jain/cse574-14/ftp/power/>

E-TEXT BOOKS:

1. https://books.google.co.in/books/about/Wireless_Power_Transfer.html?id...
2. <https://www.intechopen.com/books/wireless-power-transfer-principles-and-engineering-explorations>

MOOC COURSE:

- <https://ocw.mit.edu/.../6-452-principles-of-wireless-communications-spring-2006/>
www.iitg.ac.in/e_mobility/WPT.html
<https://nptel.ac.in/courses/117102062/>
<https://nptel.ac.in/courses/117104099/2>
 5. <https://nptel.ac.in/courses/117106108/150>

POWER PLANT ENGINEERING

VI Semester: OPEN ELECTIVE -II

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EE55	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES

1. To Study the wiring diagram of residential.
2. To understand the Safety measures of Electrical wiring

COURSE OUTCOMES:

The students should be able to

1. Know safety measures & state safety precautions.
2. Test single phase, three phase transformer, DC & AC machine as per IS.
3. Ascertain the condition of insulation & varnishing if necessary
4. Identify faults & measures to repair faults.

UNIT-I INTRODUCTION TO THE SOURCES OF ENERGY

Resources and Development of Power in India. Steam Power Plant: Plant Layout, Working of different Circuits, Fuel and handling equipment, types of coals, coal handling, choice of handling equipment, coal storage, Ash handling systems. Combustion Process: Properties of coal — overfeed and underfeed fuel beds, traveling grate stokers, spreader stokers, retort stokers, pulverized fuel burning system and its components, combustion needs and draught system, cyclone furnace, design and construction, Dust collectors, cooling towers and heat rejection. Corrosion and feed water treatment.

UNIT-II INTERNAL COMBUSTION ENGINE PLANT

DIESEL POWER PLANT: Introduction — IC Engines, types, construction— Plant layout with auxiliaries fuel supply system, air starting equipment, lubrication and cooling system-super charging. Gas Turbine Plant: Introduction — classification — construction — Layout with auxiliaries — Principles of working of closed and open cycle gas turbines. Combined Cycle Power Plants and comparison. Direct Energy Conversion: Solar energy, Fuel cells, Thermo electric and Thermo ionic, MHD generation.

UNIT-III HYDRO ELECTRIC POWER PLANT

Water power — Hydro logical cycle / flow measurement — drainage area characteristics — Hydro graphs — storage and Pounding — classification of dams and spill ways. Hydro Projects And Plant: Classification — Typical layouts — plant auxiliaries — plant operation pumped storage plants. Power From Non-Conventional Sources: Utilization of Solar- Collectors- Principle of Working, Wind Energy — types — HAWT, VAWT Tidal Energy.

UNIT-IV NUCLEAR POWER STATION

Nuclear fuel — breeding and fertile materials — Nuclear reactor — reactor operation. Types of Reactors: Pressurized water reactor, Boiling water reactor, sodium-graphite reactor, fast Breeder Reactor, homogeneous Reactor, Gas cooled Reactor, Radiation hazards and shielding radioactive waste disposal.

UNIT-V POWER PLANT ECONOMICS AND ENVIRONMENTAL CONSIDERATIONS

Capital cost, investment of fixed charges, operating costs, general arrangement of power distribution, Load curves, load duration curve. Definitions of connected load, Maximum demand, demand factor, average load, load factor, diversity factor — related exercises. Effluents from power plants and Impact on environment — pollutants and pollution standards — Methods of Pollution control.

TEXT BOOKS:

1. Power Plant Engineering! P.C.Sharma / S.K.Kataria Pub.
2. A Course in Power Plant Engineering: I Arora and S. Domkundwar.

REFERENCE BOOKS:

1. A Text Book of Power Plant Engineering I Rajput I Laxmi Publications.
2. Power Plant Engineering: P.K.Nag/ II Edition /TMH.
3. An Introduction to Power Plant Technology I G.D. Rai/Khanna Publishers.
4. Power plant Engg I Elanchezhian/ I.K. International Pub.
5. Power plant Engineering! Ramalingam / Sciotech Publishers.

WEB REFERENCES:

1. www.cbspd.co.in/power-plant-engineering-9788123919461-black-veatch.html
2. <https://www.semcomaritime.com/en-en/references/power-generation>

E-TEXT BOOKS:

<https://www.intechopen.com/books/subject/power-plant-engineering>
<https://books.google.co.in/books?isbn=1482221519>

MOOC COURSE

- www.nptelvideos.in/2https://nptel.ac.in/courses/108105058/8012/11/energy-resources-and-technology.html
3. www.ignou.ac.in/upload/Unit-2-58.pdf

ANALYSIS OF LINEAR SYSTEMS

VI Semester: OPEN ELECTIVE -II

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EE56	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES

1. To develop ability to analyze linear systems and signals
2. To develop critical understanding of mathematical methods to analyze linear systems and signals

COURSE OUTCOMES:

The students should be able to

1. Use mathematical modeling tools to represent linear systems
2. Use mathematical modeling tools to analyze linear systems

UNIT-I	STATE VARIABLE ANALYSIS	
Choice of state variables in Electrical networks-Formulation of state equations for Electrical networks Equivalent source method. Network topological method – Solution of state equations-Analysis of simple networks with state variable approach.		
UNIT-II	FOURIER SERIES AND FOURIER TRANSFORM REPRESENTATION	
Introduction, Trigonometric form of Fourier series, Exponential form of Fourier series, Wave symmetry, Fourier integrals and transforms, Fourier transform of a periodic function, Properties of Fourier Transform, Parseval's theorem, Fourier transform of some common signals, Fourier transform relationship with Laplace Transform. Applications of Fourier series and Fourier Transform Representation: Introduction, Effective value, and average values of non sinusoidal periodic waves, currents, Power Factor, Effects of harmonics, Application in Circuit Analysis, Circuit Analysis using Fourier Series.		
UNIT-III	LAPLACE TRANSFORM APPLICATIONS	
Application of Laplace transform Methods of Analysis – Response of RL, RC, RLC Networks to Step, Ramp, and impulse functions, Shifting Theorem – Convolution Integral – Applications Testing of Polynomials: Elements of realisability – Hurwitz polynomials-positive real functions-Properties-Testing-Sturm's Test, examples. Network Synthesis: Network synthesis: Synthesis of one port LC networks-Foster and Cauer methods-Synthesis of RL and RC one port networks-Foster and Cauer methods		
UNIT-IV	SAMPLING	
Sampling theorem – Graphical and Analytical proof for Band Limited Signal impulse sampling, natural and Flat top Sampling, Reconstruction of signal from its samples, effect of under sampling – Aliasing, introduction to Band Pass sampling, Cross correlation and auto correlation of functions, properties of correlation function, Energy density spectrum, Power density spectrum, Relation between auto correlation function and Energy / Power spectral density function.		
UNIT-V	Z-TRANSFORMS	
Fundamental difference between continuous and discrete time signals, discrete time complex, exponential and sinusoidal signals, periodicity of discrete time complex exponential, concept of Z Transform of a discrete sequence. Distinction between Laplace, Fourier, and Z-Transforms. Region of convergence in Z-Transforms, constraints on ROC for various classes of signals, Inverse Z-Transform properties of Z-Transforms.		

TEXT BOOKS:

1. B. P. Lathi, "Signals, Systems and Communications", BS Publications 2003.
2. "Umesh Sinha" "Network Analysis and Synthesis", Satya Prakashan Publications, 2013.

REFERENCE BOOKS:

1. "A. N. Tripathi", "Linear System Analysis", New Age International, 2nd Edition 1987.
2. "D. Roy Chowdhary", "Network and Systems", New Age International, 2005.
3. "Gopal G Bhise, Prem R. Chadha", Engineering Network Analysis and Filter Design, Umesh Publications 2009.
4. "A. Cheng", linear system analysis, Oxford publishers, 1999

WEB REFERENCES:

<https://ieeexplore.ieee.org/iel5/9/24171/01101971.pdf>

E-TEXT BOOKS:

1. www.cds.caltech.edu/~murray/books/AM08/pdf/am08-complete_04Mar10.pdf
2. <https://www.springer.com/gp/book/9780387975733>

MOOC COURSE:

1. www.nptelvideos.in/2012/11/estimation-of-signals-and-systems.html
2. <https://nptel.ac.in/courses/108104100/6>

NEURAL NETWORKS AND FUZZY LOGIC

VI Semester: OPEN ELECTIVE -II

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EE57	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES

1. To understand the fundamental theory and concepts of neural networks, neuro-modeling, several neural network paradigms and its applications.
2. To understand the concepts of fuzzy sets, knowledge representation using fuzzy rules, approximate reasoning, fuzzy inference systems, and fuzzy logic control and other machine intelligence applications of fuzzy logic.
3. To understand the basics of an evolutionary computing paradigm known as genetic algorithms and its application to engineering optimization problems.

COURSE OUTCOMES:

The students should be able to

1. Comprehend the concepts of feed forward neural networks
2. Analyze the various feedback network.
3. Understand the concept of fuzziness involved in various systems and fuzzy set theory.
4. Comprehend the fuzzy logic control and adaptive fuzzy logic and to design the fuzzy control using genetic algorithm.
5. Analyze the application of fuzzy logic control to real time systems.

UNIT-I	INTRODUCTION & ESSENTIALS TO NEURAL NETWORKS	
Introduction, Humans and Computers, Organization of the Brain, Biological Neuron, Biological and Artificial Neuron Models, Hodgkin-Huxley Neuron Model, Integrate-and-Fire Neuron Model, Spiking Neuron Model, Characteristics of ANN, McCullochP iUs Model, Historical Developments, Potential Applications of ANN. Artificial Neuron Model, Operations of Artificial Neuron, Types of Neuron Activation Function, ANN Architectures, Classification Taxonomy of ANN — Connectivity, Neural Dynamics (Activation and Synaptic), Learning Strategy (Supervised, Unsupervised, Reinforcement), Learning Rules, Types of Application		
UNIT-II	SINGLE & MULTI LAYER FEED FORWARD NEURAL NETWORKS	
Introduction, Perceptron Models: Discrete, Continuous and Multi-Category, Training. Algorithms: Discrete and Continuous Perceptron Networks, Perceptron Convergence theorem, Limitations of the Perceptron Model, Applications. Credit Assignment Problem, Generalized Delta Rule, and Derivation of Back-propagation (BP) Training, Summary of Back-propagation Algorithm, Kolmogorov Theorem, Learning Difficulties and Improvements.		
UNIT-III	ASSOCIATIVE MEMORIES-I	
Paradigms of Associative Memory, Pattern Mathematics, Hebbian Learning, General Concepts of Associative Memory (Associative Matrix, Association Rules, Hamming Distance, The Linear Associator, Matrix Memories, Content Addressable Memory).		
UNIT-IV	ASSOCIATIVE MEMORIES-II	
Bidirectional Associative Memory (BAM) Architecture, BAM Training Algorithms: Storage and Recall Algorithm, BAM Energy Function, Proof of BAM Stability Theorem. Architecture of Hopfield Network: Discrete and Continuous versions, Storage and Recall Algorithm, Stability Analysis, Capacity of the Hopfield Network Summary and Discussion of Instance/Memory Based Learning Algorithms, Applications		

UNIT-V	FUZZY LOGIC	
Classical & Fuzzy Sets: Introduction to classical sets – properties, Operations and relations; Fuzzy sets, Membership, Uncertainty, Operations, properties, fuzzy relations, cardinalities, membership functions. Fuzzy Logic System Components: Fuzzification, Membership value assignment, development of rule base and decision making system, Defuzzification to crisp sets, Defuzzification methods.		
TEXT BOOKS:		
1. Neural Networks, Fuzzy logic, Genetic algorithms: synthesis and applications, Rajasekharan and Pal, PHI. 2. Neural Networks and Fuzzy Logic, C. Naga Bhaskar, G. Vijay Kumar, BS Publications.		
REFERENCE BOOKS:		
1. Artificial Neural Networks, B. Yegnanarayana, PHI. 2. Artificial Neural Networks, Zaruda, PHI. 3. Neural Networks and Fuzzy Logic System, Bail Kosko, PHI. 4. Fuzzy Logic and Neural Networks, M. Amirthavalli, Scitech Publications India Pvt. Ltd. 5. Neural Networks, James A Freeman and Davis Skapura, Pearson Education. 6. Neural networks by satish Kumar, TIVIH, 2004 7. Neural Networks, Simon Haykin, Pearson Education. 8. Neural Engineering, C. Elasmith and CH. Anderson, PHI.		
WEB REFERENCES:		
1. users.monash.edu/~app/CSE5301/Lnts/LaD.pdf 2. https://engineering.purdue.edu/~tsoukala/rational.html 3. https://pdfs.semanticscholar.org/5e31/c55a00eb3945e3e483caa2e146a95c12f5aa.pdf		
E-TEXT BOOKS:		
1. https://www.mheducation.co.in/computer.../neural-networks-fuzzy-systems/text-book www.crectirupati.com/sites/default/files/lecture_notes/NNFL.pdf 2. www.vssut.ac.in/lecture_notes/lecture1423723637.pdf		
MOOC COURSE:		
1. https://nptel.ac.in/noc/individual_course.php?id=noc19-ge07 https://nptel.ac.in/courses/108104049/16 3. https://nptel.ac.in/courses/117105084/ 4. https://nptel.ac.in/syllabus/127105006/		

ELECTRICAL ENGINEERING MATERIALS

VII Semester: OPEN ELECTIVE -III

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EE58	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES

To understand about various electrical engineering material

COURSE OUTCOMES:

The students should be able to

1. Analyze the various engineering materials.
2. Application of various engineering materials.

UNIT-I	CONDUCTORS	
Classification: High conductivity, high resistivity materials, fundamental requirements of high conductivity materials and high resistivity materials, mobility of electron in metals, commonly used high conducting materials, copper, aluminum, bronze brass, properties, characteristics, constantan, platinum, nichrome, properties, characteristics and applications, materials used for contacts.		
UNIT-II	SEMICONDUCTORS	
General concepts, energy bands, types of semiconductors, Fermi Dirac distribution, intrinsic Semiconductors, extrinsic Semiconductors, hall effect, drift, mobility, diffusion in Semiconductors, Semiconductors and their applications, superconductors.		
UNIT-III	DIELECTRICS AND INSULATORS	
Properties of gaseous, liquid and solid dielectric, dielectric as a field medium, electric conduction in gaseous, liquid and solid dielectric, breakdown in dielectric materials, mechanical and electrical properties of dielectric materials, effect of temperature on dielectric materials, polarization, loss angle and dielectric loss, petroleum based insulating oils, transformer oil, capacitor oils, properties, solid electrical insulating materials, fibrous, paper boards, yarns, cloth tapes, sleeving wood, impregnation, plastics, filling and bonding materials, fibrous, film, mica, rubber, mica based materials, ceramic materials, classification of insulation (solid) and application in AC and DC machines.		
UNIT-IV	MAGNETIC MATERIALS	
Soft and hard magnetic materials, diamagnetic, paramagnetic and ferromagnetic materials, electric steel, sheet steel, cold rolled grain oriented silicon steel, hot rolled grain oriented silicon steel, hot rolled silicon steel sheet, hysteresis loop, hysteresis loss, magnetic susceptibility, coercive force, curie temperature, magnetostriction.		
UNIT-V	OPTICAL PROPERTIES OF SOLIDS	
Photo emission, photo emission materials, electro luminescence junction diode, photo emitters, photo transistor, photo resistors, injection lasers, optical properties of semiconductors, application of photo sensitive materials (CRT, Tube light, photo panels etc.).		
TEXT BOOKS:		
1. "Electrical Engineering Materials", Dekker, PHI Pbs. 2. "Electrical Engineering Materials", Indulkar, S.Chand		

REFERENCE BOOKS:

1. "Electrical Engineering Materials", Tareev
2. "Electrical Engineering Materials", Yu. Koritsky.
3. "Electrical Engineering Materials", R.K.Rajput, Laxmi Pbs

WEB REFERENCES:

1. <https://physics.info/dielectrics/>
2. <https://www.oxfordreference.com/view/10.1093/oi/authority.20110803095631265>
3. web.mit.edu/course/6/6.732/www/6.732-pt2.pdf

E-TEXT BOOKS:

1. <https://easyengineering.net/electrical-engineering-materials-by-dekker/>
2. <https://www.oreilly.com/library/view/dielectric-materials-for/9781118619780/>

MOOC COURSE

1. <https://nptel.ac.in/courses/108108076/>
2. <https://nptel.ac.in/courses/112104203/3>
1. https://onlinecourses.nptel.ac.in/noc18_ee14/

NON CONVENTIONAL POWER GENERATION

VII Semester: OPEN ELECTIVE -III

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EE59	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES

1. Demonstrate power generation systems including major subsystems.
2. Understand basic working principles of nuclear power generation systems.
3. Apply knowledge of solar power generation systems in design and implementation to obtain clean energy.

COURSE OUTCOMES:

The students should be able to

1. Understand basic working principles of nuclear power generation systems.
2. Apply knowledge of solar power generation systems in design and implementation to obtain clean energy.

UNIT-I PRINCIPLES OF SOLAR RADIATION

Role and potential of new and renewable source, the solar energy option, Environmental impact of solar power, physics of the sun, the solar constant, extraterrestrial and terrestrial solar radiation, solar radiation on tilted surface, instruments for measuring solar radiation and Sun shine, solar radiation data.

UNIT-II SOLAR ENERGY COLLECTORS

Flat plate and concentrating collectors, classification of concentrating collectors, orientation and thermal analysis, advanced collectors.

STORAGE AND APPLICATIONS: Different methods of solar energy storage, Sensible, latent heat and stratified storage, solar ponds. Solar Applications- solar heating /cooling technique, solar distillation and drying.

UNIT-III PHOTO VOLTAICS (PV)

Fundamentals of solar cells, types of solar cells, semiconducting materials, band gap theory, absorption of photons, excitations and photoemission of electrons, band engineering.

PV CELL PROPERTIES: Solar cell properties and design, p-n junction photodiodes, depletion region, electrostatic field across the depletion layer, electron and holes transports, device physics, charge carrier generation, recombination and other losses, I-V characteristics, output power.

UNIT-IV SOLAR CELL APPLICATIONS

PV cell interconnection, module structure and module fabrication, Equivalent circuits, load matching, efficiency, fill factor and optimization for maximum power, Design of stand-alone PV systems, system sizing, device structures, device construction, DC to AC conversion, inverters, on-site storage and grid connections.

UNIT-V COST ANALYSIS AND ENVIRONMENTAL ISSUES

Cost analysis and pay back calculations for different types of solar panels and collectors, installation and operating costs, Environmental and safety issues, protection systems, performance monitoring.

ALTERNATIVE ENERGY SOURCES: Solar Energy: Types of devices for Solar Energy Collection, Thermal Storage System. Control Systems, Wind Energy, Availability, Wind Devices, Wind Characteristics, Performance of Turbines and systems.

TEXT BOOKS:

1. G. D. Rai (2009), *Non-Conventional Energy Sources*, 4th edition, Khanna Publishers, New Delhi.
2. Martin A. Green (2008), *Solar Cells: Operating Principles, Technology and system Applications*, 1st edition, Prentice Hall, New Delhi.

REFERENCE BOOKS:

- 1 Sukatme (2008), *Solar Energy*, 3rd Edition, McGraw Hill Companies, New Delhi.
2. D. Yogi gosuami, Frank Kreith, Jan F. Kreider (2000), *Principles of Solar Engineering*, 2nd edition, Taylor & Francis, USA.

WEB REFERENCES:

1. <https://nptel.ac.in/courses/Webcourse-contents/IISc-BANG/notused/Non-Conventional%20Energy%20Systems-/Learning%20Materail%20-%20NCES.pdf>
2. <https://www.ukessays.com/essays/environmental-sciences/promoting-non-conventional-energy-resource>

E-TEXT BOOKS:

1. <bie.telangana.gov.in/files/Nonconventionalenergysources.pdf>
2. https://www.researchgate.net/.../236857095_Non_Conventional_Energy_Resources

MOOC COURSE

1. <https://nptel.ac.in/courses/108102047/>
2. https://onlinecourses.nptel.ac.in/noc19_ge11/course
3. <https://nptel.ac.in/downloads/108108078/>

SOLAR ENERGY AND APPLICATIONS

VII Semester: OPEN ELECTIVE -III

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EE60	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES

1. To Study about solar modules and PV system design and their applications
2. To Deal with grid connected PV systems
3. To Discuss about different energy storage systems

COURSE OUTCOMES:

The students should be able to

1. Understanding on solar energy storage systems
2. Basic knowledge on standalone PV system
3. Understand the issues in grid connected PV systems
4. Study about the modelling of different energy storage systems and their performances
5. Attain more on different applications of solar energy

UNIT-I INTRODUCTION

Characteristics of sunlight – semiconductors and P-N junctions –behavior of solar cells – cell properties – PV cell interconnection

UNIT-II STAND ALONE PV SYSTEM

Solar modules – storage systems – power conditioning and regulation - MPPT- protection – stand alone PV systems design – sizing

UNIT-III GRID CONNECTED PV SYSTEMS

PV systems in buildings – design issues for central power stations – safety – Economic aspect – Efficiency and performance - International PV programs

UNIT-IV ENERGY STORAGE SYSTEMS

Impact of intermittent generation – Battery energy storage – solar thermal energy storage – pumped hydroelectric energy storage

UNIT-V APPLICATIONS

Water pumping – battery chargers – solar car – direct-drive applications –Space – Telecommunications.

TEXT BOOKS:

1. Solanki C.S., "Solar Photovoltaics: Fundamentals, Technologies And Applications", PHI Learning Pvt. Ltd.,2015.
2. Stuart R.Wenham, Martin A.Green, Muriel E. Watt and Richard Corkish, "Applied Photovoltaics", 2007,Earthscan, UK. Eduardo Lorenzo G. Araujo, "Solar electricity engineering of photovoltaic systems", Progensa,1994

REFERENCE BOOKS:

1. Frank S. Barnes & Jonah G. Levine, "Large Energy storage Systems Handbook", CRC Press, 2011.
2. McNeils, Frenkel, Desai, "Solar & Wind Energy Technologies", Wiley Eastern, 1990 5 S.P. Sukhatme, "Solar Energy", Tata McGraw Hill,1987.

WEB REFERENCES:

1. <https://www.loc.gov/rr/scitech/tracer-bullets/solar-updatetb.html>
2. <https://www.oxfordreference.com/view/10.1093/oi/authority.20110803100516798>
3. <https://link.springer.com/journal/11949>

E-TEXT BOOKS

1. https://courses.edx.org/c4x/DelftX/ET.3034TU/asset/solar_energy_v1.1.pdf
2. bookstore.teri.res.in/books/9788179935736

MOOC COURSE:

1. <https://nptel.ac.in/courses/112105051/>
2. <https://nptel.ac.in/courses/121106014/18>
3. <https://nptel.ac.in/courses/112105050/>
4. <https://nptel.ac.in/syllabus/112105051/>

INSTRUMENTATION AND CONTROL

VIII Semester: OPEN ELECTIVE -IV

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EE61	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES

1. To introduce the basic principles of all measuring instruments
2. To deal with the measurement of voltage, current measurements
3. To understand students how different types of meters work and their construction
4. To understand the different ways of system representations such as Transfer function representation and state space representations and to assess the system dynamic response
5. To assess the system performance using time domain analysis and methods for improving it

COURSE OUTCOMES:

The students should be able to

1. Identify the instruments suitable for typical measurements.
2. Apply the knowledge about transducers and instrument transformers to use them effectively.
3. Improve the system performance by selecting a suitable controller for a specific application
4. Apply various time domain techniques to assess the system performance

UNIT-I INTRODUCTION TO MEASURING INSTRUMENTS

Classification —deflecting, control and damping torques — Ammeters and Voltmeters — PMMC, moving iron type instruments — expression for the deflecting torque and control torque — Errors and compensations, extension of range using shunts and series resistance. Electrostatic Voltmeters-electrometer type and attracted disc type — Extension of range of E.S. Voltmeters.

UNIT-II POTENTIOMETERS & INSTRUMENT TRANSFORMERS

Principle and operation of D.C. Crompton's potentiometer—standardisation — Measurement of unknown resistance, current, voltage. A.C. Potentiometers: polar and coordinate types standardisation — applications. CT and PT — Ratio and phase angle errors.

UNIT-III TRANSDUCERS & OSCILLOSCOPES

Definition of transducers, Classification of transducers, Advantages of Electrical transducers, Characteristics and choice of transducers; Principle operation of LVDT and capacitor transducers; LVDT Applications, Strain gauge and its principle of operation, gauge factor, Thermistors, Thermo couples, Piezo electric transducers, photovoltaic, photo conductive cells, photo diodes.

CRO:Cathode ray oscilloscope-Cathode ray tube-time base generator- horizontal and vertical amplifiers-CRO probes-applications of CRO- Measurement of phase and frequency-lissajous patterns.

UNIT-IV CLASSIFICATION OF CONTROL SYSTEMS

Concepts of Control Systems- Open Loop and closed loop control systems and their differences-Different examples of control systems- Classification of control systems
Feed-Back Characteristics, Effects of feedback. Mathematical models – Differential equations – Impulse Response and transfer functions. Block diagram algebra – Representation by Signal flow graph – Reduction using mason's gain formula.

UNIT-V TIME RESPONSE ANALYSIS

Standard test signals – Time response of first order systems –Characteristic Equation of Feedback control systems, Transient response of second order systems – Time domain specifications .Effects of proportional derivative, proportional integral systems.

TEXT BOOKS:

1. Electrical and Electronic Measurements and Instrumentation, R. K. Rajput, S. Chand & Company Ltd.
2. Electrical Measuring Instruments and Measurements, S. C. Bhargava, BS Publications.
3. I. J. Nagrath and M. Gopal, "Control Systems Engineering", New Age International (P) Limited, Publishers, 5th edition, 2009
4. "B. C. Kuo", "Automatic Control Systems", John Wiley and sons, 8th edition, 2003.

REFERENCE BOOKS:

1. Electrical & Electronic Measurement & Instruments, A.K.Sawhney Dhanpat Rai & Co. Publications.
2. Electrical and Electronic Measurements, G. K. Banerjee, PHI Learning Pvt. Ltd.
3. Electrical Measurements and Measuring Instruments, Golding and Widdis, Reem Publications.
4. N. K. Sinha, "Control Systems", New Age International (P) Limited Publishers, 3rd Edition, 1998.
5. "NISE", "Control Systems Engineering", John Wiley, 6th Edition, 2011.
6. "Katsuhiko Ogata", "Modern Control Engineering", Prentice Hall of India Pvt. Ltd., 3rd edition, 1998.

WEB REFERENCES:

1. home.mit.bme.hu/~virosztek/docs/mt.../Principles_of_electrical_measurement.pdf
2. https://www.mccdag.com/handbook/chapt_4.aspx

E-TEXT BOOKS:

1. www.vssut.ac.in/lecture_notes/lecture1423813026.pdf
2. www.vssut.ac.in/lecture_notes/lecture1423904331.pdf
3. www.ent.mrt.ac.lk/~rohan/teaching/EN5001/Reading/DORFCH1.pdf

MOOC COURSE:

- <https://nptel.ac.in/syllabus/108106070/>
2. <https://nptel.ac.in/courses/108105064/>
 - <https://nptel.ac.in/courses/108101037/>
 3. <https://nptel.ac.in/courses/107106081/>

ENERGY AUDIT AND MANAGEMENT SYSTEMS

VIII Semester: OPEN ELECTIVE -IV

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EE62	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES

1. To understand energy efficiency, scope, conservation and technologies.
2. To design energy efficient lighting systems.
3. To estimate/calculate power factor of systems and propose suitable compensation techniques.
4. To understand energy conservation in HVAC systems.
5. To calculate life cycle costing analysis and return on investment on energy efficient technologies.

COURSE OUTCOMES:

The students should be able to

1. Explain energy efficiency, conservation and various technologies.
2. Design energy efficient lighting systems.
3. Calculate power factor of systems and propose suitable compensation techniques.
4. Explain energy conservation in HVAC systems.
5. Calculate life cycle costing analysis and return on investment on energy efficient technologies.

UNIT-I	ENERGY AUDIT AND MANAGEMENT ENERGY AUDIT	
Basic Principles of Energy Audit and management Energy audit – Definitions – Concept – Types of audit – Energy index – Cost index – Pie charts – Sankey diagrams – Load profiles – Energy conservation schemes and energy saving potential – Numerical problems – Principles of energy management – Initiating, planning, controlling, promoting, monitoring, reporting – Energy manager – Qualities and functions – Language – Questionnaire – Check list for top management.		
UNIT-II	LIGHTING	
Modification of existing systems – Replacement of existing systems – Priorities: Definition of terms and units – Luminous efficiency – Polar curve – Calculation of illumination level – Illumination of inclined surface to beam – Luminance or brightness – Types of lamps – Types of lighting – Electric lighting fittings (luminaries) – Flood lighting – White light LED and conducting Polymers – Energy conservation measures.		
UNIT-III	POWER FACTOR AND ENERGY INSTRUMENTS	
Power factor – Methods of improvement – Location of capacitors – Power factor with non linear loads – Effect of harmonics on Power factor – Numerical problems. Energy Instruments – Watt-hour meter – Data loggers – Thermocouples – Pyrometers – Lux meters – Tong testers – Power analyzer.		
UNIT-IV	SPACE HEATING AND VENTILATION	
Ventilation – Air-Conditioning (HVAC) and Water Heating: Introduction – Heating of buildings – Transfer of Heat-Space heating methods – Ventilation and air-conditioning – Insulation-Cooling load – Electric water heating systems – Energy conservation methods.		
UNIT-V	ECONOMIC ASPECTS AND ANALYSIS	
Economics Analysis – Depreciation Methods – Time value of money – Rate of return – Present worth method – Replacement analysis – Life cycle costing analysis – Energy efficient motors (basic concepts) Computation of Economic Aspects : Calculation of simple payback method – Net present worth method – Power factor correction – Lighting – Applications of life cycle costing analysis – Return on investment.		

TEXT BOOKS:

1. Energy management by W.R. Murphy & G. McKay Butterworth, Elsevier publications. 2012
2. Energy efficient electric motors by John .C. Andreas, Marcel Dekker Inc Ltd–2nd edition, 1995

REFERENCE BOOKS:

1. Electric Energy Utilization and Conservation by S C Tripathy, Tata McGraw hill publishing company Ltd. New Delhi.
2. Energy management by Paul o' Callaghan, Mc–Graw Hill Book company–1st edition, 1998.
3. Energy management hand book by W.C.Turner, John wiley and sons.
4. Energy management and conservation –k v Sharma and pvenkata seshiah-I K International Publishing House pvt.ltd,2011.

WEB REFERENCES:

1. <https://beeindia.gov.in/sites/default/files/1Ch3.pdf>
2. https://www.researchgate.net/publication/309463130_Energy_Audit_Management

E-TEXT BOOKS:

1. [.lab.fs.uni-lj.si/kes/erasmus/Energy%20Management%20Handbook.pdf](http://lab.fs.uni-lj.si/kes/erasmus/Energy%20Management%20Handbook.pdf)
2. <https://fenix.tecnico.ulisboa.pt/downloadFile/.../Energy%20Audit%202013.pdf>
3. enersupply.euke.sk/wp-content/uploads/Introduction-to-EM-and-EA_P1_Fotini.pdf

MOOC COURSE

1. <https://nptel.ac.in/courses/108106022>
2. <https://nptel.ac.in/courses/105107156/12>
3. shodhganga.inflibnet.ac.in/bitstream/10603/46067/11/11_chapter%201.pdf

ENERGY STORAGE SYSTEMS

VIII Semester: OPEN ELECTIVE -IV

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EE63	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES

- To enable the student to understand the need for energy storage, devices and technologies available and their applications

COURSE OUTCOMES:

The students should be able to

- Analyze the characteristics of energy from various sources and need for storage
- Classify various types of energy storage and various devices used for the purpose
- Identify various real time applications.

UNIT-I ELECTRICAL ENERGY STORAGE TECHNOLOGIES

Characteristics of electricity, Electricity and the roles of EES, High generation cost during peak-demand periods, Need for continuous and flexible supply, Long distance between generation and consumption, Congestion in power grids, Transmission by cable.

UNIT-II NEEDS FOR ELECTRICAL ENERGY STORAGE

Emerging needs for EES, More renewable energy, less fossil fuel, Smart Grid uses, The roles of electrical energy storage technologies, The roles from the viewpoint of a utility, The roles from the viewpoint of consumers, The roles from the viewpoint of generators of renewable energy.

UNIT-III FEATURES OF ENERGY STORAGE SYSTEMS

Classification of EES systems, Mechanical storage systems, Pumped hydro storage (PHS), Compressed air energy storage (CAES), Flywheel energy storage (FES), Electrochemical storage systems, Secondary batteries, Flow batteries, Chemical energy storage, Hydrogen (H₂), Synthetic natural gas (SNG).

UNIT-IV TYPES OF ELECTRICAL ENERGY STORAGE SYSTEMS

Electrical storage systems, Double-layer capacitors (DLC), Superconducting magnetic energy storage (SMES), Thermal storage systems, Standards for EES, Technical comparison of EES technologies.

UNIT-V APPLICATIONS

Present status of applications, Utility use (conventional power generation, grid operation & service), Consumer use (uninterruptable power supply for large consumers), New trends in applications, Renewable energy generation, Smart Grid, Smart Micro grid, Smart House, Electric vehicles, Management and control hierarchy of storage systems, Internal configuration of battery storage systems, External connection of EES systems, Aggregating EES systems and distributed generation (Virtual Power Plant), Battery SCADA– aggregation of many dispersed batteries.

TEXT BOOKS:

- “James M. Eyer, Joseph J. Iannucci and Garth P. Corey”, “Energy Storage Benefits and Market Analysis”, Sandia National Laboratories, 2004.
- The Electrical Energy Storage by IEC Market Strategy Board.

REFERENCE BOOKS:

1. “Jim Eyer, Garth Corey”, Energy Storage for the Electricity Grid: Benefits and Market Potential Assessment Guide, Report, Sandia National Laboratories, Feb 2010.

WEB REFERENCES:

1. <https://onlinelibrary.wiley.com/doi/abs/10.1002/9781118991978.hces212>

E-TEXT BOOKS:

1. https://www.pewtrusts.org/~media/.../energy_storage-backs_up_power_supply.pdf
2. <https://energy.mit.edu/wp-content/uploads/2018/04/Energy-Storage-for-the-Grid.pdf>
3. <https://www.adb.org/sites/default/files/.../handbook-battery-energy-storage-system.pdf>

MOOC COURSE:

1. nptel.ac.in/courses/112105221/56
2. nptel.ac.in/courses/108108036/9
3. <https://nptel.ac.in/courses/108102047/7>
4. <https://nptel.ac.in/courses/108105058/37>

OPEN ELECTIVES OFFERED BY IT DEPARTMENT

FUNDAMENTALS OF DATA STRUCTURES

V SEMESTER: OPEN ELECTIVE - I

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4IT11	OEC	L	T	P	C	CIA	SEE	Total
		3	0	0	3	30	70	100

COURSE OBJECTIVES

To learn

1. Impart the basic concepts of data structures and algorithms.
2. Understand concepts linked lists and their applications.
3. Understand basic concepts about stacks, queues and their applications.
4. Understand basic concepts of trees, graphs and their applications.
5. Enable them to write algorithms for sorting and searching and hashing.
6. Use advanced data structures like B-Trees, AVL-trees etc., for efficient problem solving.

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Evaluate algorithms in terms of time and memory complexity.
2. Formulate new solutions for problems or improve existing code using data structures and algorithms.
3. Implement basic data structures such as arrays, linked lists, stacks and queues.
4. Solve problem involving graphs, trees and heaps
5. Apply Algorithms for solving problems like sorting, searching, and hashing.
6. Implement advanced data structures such as B-Trees, Red-Black, and AVL-Trees

UNIT-I	INTRODUCTION TO DATA STRUCTURES	
Basic concepts- Algorithm Specification-Introduction, Recursive algorithms, Data Abstraction, Performance analysis- time complexity and space complexity, Asymptotic Notation-Big O, Omega and Theta notations Introduction to Linear and Non Linear data structures-Singly Linked Lists- Operations-Insertion, Deletion, Concatenating singly linked lists, Circularly linked lists-Operations for Circularly linked lists, Doubly Linked Lists- Operations- Insertion, Deletion.		
UNIT-II	STACKS AND QUEUES	
Stacks-Stack ADT, definition, operations, array and linked implementations in C, applications-infix to postfix conversion, Postfix expression evaluation. Queues-Queue ADT, definition and operations, array and linked Implementations in C, Circular queues-Insertion and deletion operations, Dequeue (Double ended queue)ADT, array and linked implementations in C.		
UNIT-III	TREESANDGRAPHS	
Trees – Terminology, Representation of Trees, binary tree ADT, Properties of Binary Trees, Binary Tree Representations-array and linked representations, Binary Tree traversals. Max Priority Queue-ADT- implementation-Max Heap-Definition, Insertion into a Max Heap, Deletion from a Max Heap. Graphs , Introduction, Definition, Terminology, Graph ADT, Graph Representations-Adjacency matrix, Adjacency lists, Graph traversals- DFS and BFS.		
UNIT-IV	SEARCHINGAND SORTING	
Searching- Linear Search, Binary Search, Comparison of search techniques. Sorting- Insertion Sort, Selection Sort, Radix Sort, Quick sort, Merge Sort, Heap Sort, Comparison of Sorting methods.		
UNIT-V	BINARY SEARCH TREES	

Search Trees-Binary Search Trees, Definition, Operations- Searching, Insertion and Deletion, AVL Trees- Definition and Examples, Insertion into an AVL Tree, B-Trees, Definition, B-Tree of order m, operations-Insertion and Searching, Comparison of Search Trees.

Pattern matching algorithm- The Knuth-Morris-Pratt algorithm, Tries (examples only)

TEXT BOOKS:

1. Fundamentals of Data Structures II, Illustrated Edition by Ellis Horowitz, Sartaj Sahni, Computer Science Press.
2. Fundamentals of Data structures in C, 2nd Edition, E.Horowitz, S.Sahni and Susan Anderson-Freed, Universities Press

REFERENCE BOOKS:

1. Algorithms, Data Structures, and Problem Solving with C++, Illustrated Edition by Mark Allen Weiss, Addison- Wesley Publishing Company
2. HowtoSolveitbyComputerII, 2nd Impression by R.G.Dromey, Pearson Education

SOFTWARE ENGINEERING PRINCIPLES

V Semester: OPEN ELECTIVE - I

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4IT12	OEC	L	T	P	C	CIA	SEE	Total
		3	0	0	3	30	70	100

COURSE OBJECTIVES

To learn

1. Be familiar with basic Software engineering methods and practices, and its applications.
2. Apply the implementation of software engineering layered technology and Process framework.
3. Understand the concepts of software measurement and software risks.
4. Be familiar with software requirements and the SRS documents.
5. Be familiar with role of project management including planning, scheduling, risk management.

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Apply basic knowledge and understanding of the analysis, synthesis and design of complex systems.
2. Apply software engineering principles and techniques.
3. Design and evaluate large-scale software systems.
4. Explain the principles, tools and practices of IT project management.
5. Apply the fundamental knowledge of science in emerging technologies.

UNIT-I INTRODUCTION TO SOFTWARE ENGINEERING

The evolving role of software, Changing Nature of Software, Software myths.

A Generic view of process: Software engineering- A layered technology, a process framework, The Capability Maturity Model Integration (CMMI), Process patterns, process assessment, personal and team process models.

UNIT-II PROCESS MODELS

The waterfall model, Incremental process models, Evolutionary process models, The Unified process.

Software Requirements: Functional and non-functional requirements, User requirements, System requirements, Interface specification, the software requirements document.

Requirements engineering process: Feasibility studies, Requirements elicitation and analysis, Requirements validation, Requirements management.

UNIT-III SYSTEM MODELS

Context Models, Behavioral models, Data models, Object models, structured methods.

Design Engineering: Design process and Design quality, Design concepts, the design model.

Creating an architectural design: Software architecture, Data design, Architectural styles and patterns, Architectural Design.

UNIT-IV OBJECT-ORIENTED DESIGN

Objects and object classes, An Object-Oriented design process, Design evolution.

Performing User interface design: Golden rules, User interface analysis and design, interface analysis, interface design steps, Design evaluation.

Testing Strategies: A strategic approach to software testing, test strategies for conventional software, Black-Box and White-Box testing, Validation testing, System testing, the art of Debugging.

UNIT-V RISK MANAGEMENT

Reactive vs. Proactive Risk strategies, software risks, Risk identification, Risk projection, Risk refinement, RMMM, RMMM Plan.

Quality Management: Quality concepts, Software quality assurance, Software Reviews, Formal technical reviews, Statistical Software quality Assurance, Software reliability, The ISO 9000 quality

standards.

TEXT BOOKS:

1. Software Engineering a practitioner's Approach, Roger S Pressman, 6th edition. Mc. Graw Hill International Edition.
2. Software Engineering, Ian Somerville, 7th edition, Pearson education.

REFERENCE BOOKS:

1. Software Engineering, A Precise Approach, Pankaj Jalote, Wiley India, 2010.
2. Software Engineering: A Primer, Waman S Jawadekar, Tata McGraw-Hill, 2008
3. Fundamentals of Software Engineering, Rajib Mall, PHI, 2005

COMPUTER ORGANIZATION AND OPERATING SYSTEMS

V Semester: OPEN ELECTIVE - I

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4IT13	OEC	L	T	P	C	CIA	SEE	Total
		3	0	0	3	30	70	100

COURSE OBJECTIVES

To learn

1. To understand the basic structure and operation of a digital computer.
2. To discuss in detail the operation of the arithmetic unit including the algorithms & implementation of fixed-point and floating-point addition, subtraction, multiplication & division.
3. To study different ways of communicating with I/O devices and standard I/O interfaces.
4. To study hierarchical memory system including cache memories and virtual memory.
5. To demonstrate the knowledge of functions of operating system memory management scheduling, file system and interface, distributed systems, security and dead locks.
6. To implement a significant portion of an Operating System

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Explain different types of Addressing modes
2. Explain different synchronous and asynchronous data transfer techniques?
3. Explain different I/O data transfer techniques with performance comparison?
4. Outline various concepts and features of Operating systems.

UNIT-I	BASIC STRUCTURE OF COMPUTERS	
Computer Types, Functional UNIT, Basic Operational Concepts, Bus Structures, Software, Performance, Multiprocessors and Multi Computers, Data Representation, Fixed Point Representation, Floating - Point Representation. Register Transfer Language and Micro Operations: Register Transfer Language, Register Transfer Bus and Memory Transfers, Arithmetic Micro Operations, Logic Micro Operations, Shift Micro Operations, Arithmetic Logic Shift Unit, Instruction Codes, Computer Registers Computer Instructions - Instruction Cycle. Memory - Reference Instructions, Input - Output and Interrupt, STACK Organization, Instruction Formats, Addressing Modes, Data Transfer and Manipulation.		
UNIT-II	MICRO PROGRAMMED CONTROL	
Control Memory, Address Sequencing, Micro Program Examples, Design of Control Unit, Hard Wired Control, Micro Programmed Control. Input-Output Organization: Peripheral Devices, Input-Output Interface, Asynchronous Data Transfer Modes, Priority Interrupt, Direct Memory Access, Input-Output Processor (IOP).		
UNIT-III	OPERATING SYSTEMS OVERVIEW	
Overview of Computer Operating Systems Functions, Protection and Security, Distributed Systems, Special Purpose Systems, Operating Systems Structures-Operating System Services and Systems Calls, System Programs, Operating System Generation. Principles of Deadlock: System Model, Deadlock Characterization, Deadlock Prevention, Detection and Avoidance, Recovery from Deadlock.		
UNIT-IV	MEMORY MANAGEMENT	
Swapping, Contiguous Memory Allocation, Paging, Structure of the Page Table, Segmentation, Virtual Memory, Demand Paging, Page-Replacement Algorithms, Allocation of Frames, Thrashing Case Studies -UNIX, Linux, Windows.		
UNIT-V	PRINCIPLES OF DEADLOCK	

System Model, Deadlock Characterization, Deadlock Prevention, Detection and Avoidance, Recovery from Deadlock.

File System Interface: The Concept of a File, Access Methods, Directory Structure, File System Mounting, File Sharing, Protection. Allocation Methods, Free-Space Management.

TEXT BOOKS:

1. Computer Organization - Carl Hamacher, Zvonks Vranesic, SafeaZaky, 5th Edition, McGraw Hill.
2. Computer System Architecture - M. moris mano, 3rd edition, Pearson
3. Operating System Concepts - Abreham Silberchatz, Peter B. Galvin, Greg Gagne, 8th Edition, John Wiley.

REFERENCE BOOKS:

1. Computer Organization and Architecture - William Stallings 6th Edition, Pearson
2. Structured Computer Organization - Andrew S. Tanenbaum, 4th Edition, PHI
3. Fundamentals of Computer Organization and Design - Sivaraama Dandamudi, Springer Int. Edition
4. Operating Systems - Internals and Design Principles, Stallings, 6th Edition - 2009, Pearson Education.
5. Modern Operating Systems, Andrew S Tanenbaum 2nd Edition, PHI
6. Principles of Operating System, B. L. Stuart, Cengage Learning, India Edition

BASICS OF PYTHON PROGRAMMING

VI Semester: OPEN ELECTIVE - II

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4IT14	OEC	L	T	P	C	CIA	SEE	Total
		3	0	0	3	30	70	100

COURSE OBJECTIVES

To learn

1. To Describe the basic elements of the Python language and the Python interpreter and discuss the differences between Python and other modern languages.
2. To Describe Python dictionaries and demonstrate the use of dictionary methods.
3. Define, analyze and code the basic Python conditional and iterative control structures and explain how they can be nested and how exceptions can be used.
4. To Explain and demonstrate methods of error handling and Python exceptions.
5. To demonstrate the understanding of —magic methodsll through use of these in the context of a Python application.

COURSE OUTCOMES:

Up on successful completion of the course, the student is able to

1. Write and debug Python programs which make use of the fundamental control structures and method-building techniques common to all programming languages. Specifically, the student will use data types, input, output, iterative, conditional, and functional components of the language in his or her programs.
2. Use object-oriented programming techniques to design and implement a clear, well-structured Python program. Specifically, the student will use and design classes and objects in his or her programs.
3. Outline the specific features of Python which made it more powerful programming language.

UNIT-I INTRODUCTION TO PYTHON

Overview, Basic. Python Installation, Comments in Python, Concept of Indentation in python.

UNIT-II DATA TYPES

Tuples, Lists More advanced data types (dictionary, string), Python operators, control flows, Loops, Functions.

UNIT-III MODULES AND PACKAGES

File handling in python, Module and Packages, Object oriented programming.

UNIT-IV ADVANCED FUNCTIONS

Exceptions, sorting, advanced function: map, filter, and reduce.

UNIT-V INTRODUCTION TO STANDARD LIBRARIES

Multi-Processing And Multi-Threading, Introduction To Standard Libraries (pandas,Turtle, numpy, os).

TEXT BOOKS:

1. Learning Python, by Shroff Pub& Dist., O'relly publications, Publication Year: 2013.

REFERENCE BOOKS:

1. Python Programming for Beginners: Python Programming Languageby Joseph Joyner

HUMAN COMPUTER INTERACTION

VI Semester: OPEN ELECTIVE - II

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4IT15	OEC	L	T	P	C	CIA	SEE	Total
		3	0	0	3	30	70	100

COURSE OBJECTIVES:

To learn

1. The human components functions.
2. The Computer components functions.
3. The Interaction between the human and computer components.
4. Paradigms
5. Interaction design basics
6. HCI in the software process
7. Design rules
8. Implementation supports
9. Evaluation techniques

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Explain the human components functions regarding interaction with computer
2. Explain Computer components functions regarding interaction with human
3. Demonstrate Understanding of Interaction between the human and computer components.
4. Use Paradigms
5. Implement Interaction design basics
6. Use HCI in the software process

UNIT-I	INTRODUCTION	
Importance of user Interface – definition, importance of good design, Benefits of good design. A brief history of Screen design, The graphical user interface – popularity of graphics, the concept of direct manipulation, graphical system, Characteristics, Web user – Interface popularity, characteristics-Principles of user interface.		
UNIT-II	DESIGN PROCESS	
Human interaction with computers, importance of human characteristics human consideration, Human interaction speeds, understanding business junctions. Screen Designing : Design goals – Screen planning and purpose, organizing screen elements, ordering of screen data and content – screen navigation and flow – Visually pleasing composition – amount of information – focus and emphasis – presentation information simply and meaningfully – information retrieval on web – statistical graphics – Technological consideration in interface design.		
UNIT-III	WINDOWS	
New and Navigation schemes selection of window, selection of devices based and screen based controls. Components – text and messages, Icons and increases – Multimedia, colors, uses problems, choosing colors.		
UNIT-IV	SOFTWARE TOOLS	
Specification methods, interface – Building Tools.		
UNIT-V	INTERACTION DEVICES	

Keyboard and function keys – pointing devices – speech recognition digitization and generation – image and video displays – drivers.

TEXT BOOKS:

1. The essential guide to user interface design, Wilbert O Galitz, Wiley Dreamtech.
2. Designing the user interface. 3rd Edition Ben Shneidermann , Pearson Education Asia.

REFERENCE BOOKS:

1. Human – Computer Interaction. ALAN DIX, JANET FINCAY, GRE GORYD, ABOWD, RUSSELL BEALG, PEARSON.
2. Interaction Design PRECE, ROGERS, SHARPS. Wiley Dreamtech,
3. User Interface Design, Soren Lauesen , Pearson Education.

SOFTWARE TESTING FUNDAMENTALS

VI Semester: OPEN ELECTIVE - II

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4IT16	OEC	L	T	P	C	CIA	SEE	Total
		3	0	0	3	30	70	100

COURSE OBJECTIVES:

To learn

Software Testing has different **goals** and objectives.

1. Finding defects which may get created by the programmer while developing the software.
2. Gaining **confidence** in and providing information about the level of quality.
3. To prevent defects.

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Understand software testing methods and apply various testing techniques
2. Design and conduct a software test process for a software testing project
3. Designing solutions for various software testing problems by selecting appropriate software test model
4. Implement various practice oriented software testing projects

UNIT-I	INTRODUCTION	
Basics of software testing, Testing objectives, Principles of testing, Test Life Cycle, Types of testing, Software defect tracking.		
UNIT-II	TESTING METHODOLOGIES	
White Box And Black Box Testing, Static Testing, Static Analysis Tools, Structural Testing, Unit/Code functional, testing, Code coverage testing, Code complexity testing, Black Box testing, Requirements based testing.		
UNIT-III	INTEGRATION TESTING	
Integration, System, and Acceptance Testing Top down and Bottom up integration, Functional versus Non-functional testing, Deployment testing, Beta testing, Scalability testing, Reliability testing, Stress testing, Acceptance testing		
UNIT-IV	TEST SELECTION & MINIMIZATION FOR REGRESSION TESTING	
Test Selection & Minimization for Regression Testing Regression testing, Regression test process, Initial Smoke or Sanity test, Selection of regression tests, Execution Trace, Dynamic Slicing, Test Minimization, Tools for regression testing, Ad hoc Testing: Pair testing, Exploratory testing, Iterative testing, Defect seeding.		
UNIT-V	TEST MANAGEMENT AND AUTOMATION TEST PLANNING	
Test Management and Automation Test Planning, Management, Execution and Reporting, Software Test Automation: Scope of automation, Design & Architecture for automation, Generic requirements for test tool framework, Test tool selection.		
TEXT BOOKS:		
1. S. Desikan and G. Ramesh, "Software Testing: Principles and Practices", Pearson Education. 2. Aditya P. Mathur, "Fundamentals of Software Testing", Pearson Education.		
REFERENCE BOOKS:		
1. Naik and Tripathy, "Software Testing and Quality Assurance", Wiley 2. K. K. Aggarwal and Yogesh Singh, "Software Engineering", New Age International Publication.		

E - COMMERCE

VII Semester: OPEN ELECTIVE - III

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4IT17	OEC	L	T	P	C	CIA	SEE	Total
		3	0	0	3	30	70	100

COURSE OBJECTIVES

To learn

1. To become familiar with state of the art Electronic Model, Payment Mechanisms.
2. To understand the basic principal of E-Government, Securities, Supply Chain Mechanisms.
3. To Evaluate and observe various online businesses Management.

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Understand how the E-Commerce is done in the IT environment when the data is huge and in a variety of formats (e.g. multi media) like on the Internet and in digital libraries.
2. To enable them to use the search and retrieval techniques taught in any of the IT applications that they may develop in future.
3. Understand the concept of electronic data interchange.
4. To learn different types of payment modes.

UNIT-I	ELECTRONIC COMMERCE	
Electronic Commerce - Frame work, anatomy of E-Commerce applications, E-Commerce Consumer applications, and E-Commerce organization applications, Consumer Oriented Electronic commerce - Mercantile Process models.		
UNIT-II	ELECTRONIC PAYMENT SYSTEMS	
Electronic payment systems – Digital Token-Based, Smart Cards, Credit Cards, Risks in Electronic Payment systems. Inter-Organizational Commerce – EDI, EDI Implementation, Value added networks.		
UNIT-III	INTRA ORGANIZATIONAL COMMERCE	
Intra Organizational Commerce – work Flow, Automation Customization and internal Commerce, Supply chain Management.		
UNIT-IV	CORPORATE DIGITAL LIBRARY	
Corporate Digital Library - Document Library, digital Document types, corporate Data Warehouses. Advertising and Marketing - Information based marketing, Advertising on Internet, on-line marketing process, market research.		
UNIT-V	CONSUMER SEARCH AND RESOURCE DISCOVERY & MULTIMEDIA	
Consumer Search And Resource Discovery: Information search and Retrieval, Commerce Catalogues, Information Filtering Multimedia - key multimedia concepts, Digital Video and electronic Commerce, Desktop video processing, Desktop video conferencing.		
TEXT BOOKS:		
1. Frontiers of electronic commerce – Kalakata, Whinston, Pearson. 2. E-Commerce fundamentals and applications Hendry Chan, Raymond Lee, Tharam Dillon, Ellizabeth Chang, John Wiley.		
REFERENCE BOOKS:		

1. E-Commerce, S.Jaiswal – Galgotia.
2. E-Commerce, Efrain Turbon, Jae Lee, David King, H.Michael Chang.
3. Electronic Commerce – Gary P.Schneider – Thomson.
4. E-Commerce – Business, Technology, Society, Kenneth C.Taudon, Carol

CYBER FORENSICS

VII Semester: OPEN ELECTIVE - III

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4IT18	OEC	L	T	P	C	CIA	SEE	Total
		3	0	0	3	30	70	100

COURSE OBJECTIVES

To learn

1. A brief explanation of the objective is to provide digital evidences which are obtained from digital media.
2. In order to understand the objectives of computer forensics, first of all, people have to recognize the different roles computer plays in a certain crime.
3. According to a snippet from the United States Security Service, the functions computer has in different kinds of crimes.

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Students will understand the usage of computers in forensic, and how to use various forensic tools for a wide variety of investigations.
2. It gives an opportunity to students to continue their zeal in research in computer forensics.

UNIT-I	COMPUTER FORENSICS FUNDAMENTALS	
What is Computer Forensics?, Use of Computer Forensics in Law Enforcement, Computer Forensics Assistance to Human Resources/Employment Proceedings, Computer Forensics Services, Benefits of Professional Forensics Methodology, Steps taken by Computer Forensics Specialists Types of Computer Forensics Technology: Types of Military Computer Forensic Technology, Types of Law Enforcement — Computer Forensic Technology — Types of Business Computer Forensic Technology Computer Forensics Evidence and Capture: Data Recovery Defined — Data Back-up and Recovery — The Role of Back-up in Data Recovery — The Data-Recovery Solution.		
UNIT-II	EVIDENCE COLLECTION AND DATA SEIZURE	
Why Collect Evidence? Collection Options — Obstacles — Types of Evidence — The Rules of Evidence — Volatile Evidence — General Procedure — Collection and Archiving — Methods of Collection — Artifacts — Collection Steps — Controlling Contamination: The Chain of Custody Duplication and Preservation of Digital Evidence: Preserving the Digital Crime Scene — Computer Evidence Processing Steps — Legal Aspects of Collecting and Preserving Computer Forensic Evidence Computer Image Verification and Authentication: Special Needs of Evidential Authentication — Practical Consideration — Practical Implementation.		
UNIT-III	COMPUTER FORENSICS ANALYSIS AND VALIDATION	
Determining what data to collect and analyze, validating forensic data, addressing data-hiding techniques, performing remote acquisitions Network Forensics: Network forensics overview, performing live acquisitions, developing standard procedures for network forensics, using network tools, examining the honeynet project. Processing Crime and Incident Scenes: Identifying digital evidence, collecting evidence in private-sector incident scenes, processing law enforcement crime scenes, preparing for a search, securing a computer incident or crime scene, seizing digital evidence at the scene, storing digital evidence, obtaining a digital hash, reviewing a case.		
UNIT-IV	CURRENT COMPUTER FORENSIC TOOLS	
Evaluating computer forensic tool needs, computer forensics software tools, computer forensics hardware tools, validating and testing forensics software E-Mail Investigations: Exploring the role of e-mail in investigation, exploring the roles of the client and server in e-mail, investigating e-mail crimes and violations, understanding e-mail servers, using specialized e-mail forensic tools. Cell phone and mobile device forensics: Understanding mobile device forensics, understanding acquisition procedures for cell		

phones and mobile devices.

UNIT-V**WORKING WITH WINDOWS AND DOS SYSTEMS**

Understanding file systems, exploring Microsoft File Structures, Examining NTFS disks, Understanding whole disk encryption, windows registry, Microsoft startup tasks, MS-DOS startup tasks, virtual machines.

TEXT BOOKS:

1. Computer Forensics, Computer Crime Investigation by John R. Vacca, Firewall Media, New Delhi.
2. Computer Forensics and Investigations by Nelson, Phillips Enfinger, Steuart, CENGAGE Learning

REFERENCE BOOKS:

1. Real Digital Forensics by Keith J. Jones, Richard Bejtlich, Curtis W. Rose, Addison Wesley Pearson Education
2. Forensic Compiling, A Tractitioneris Guide by Tony Sammes and Brian Jenkinson, Springer International edition.
3. Computer Evidence Collection & Presentation by Christopher L.T. Brown, Firewall Media.
4. Homeland Security, Techniques & Technologies by Jesus Mena, Firewall Media.
5. Software Forensics Collecting Evidence from the Scene of a Digital Crime by Robert M. Slade, TMH 2005
6. Windows Forensics by Chad Steel, Wiley India Edition

DISCRETE STRUCTURES

VII Semester: OPEN ELECTIVE - III

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4IT19	OEC	L	T	P	C	CIA	SEE	Total
		3	0	0	3	30	70	100

COURSE OBJECTIVES

To learn

1. Reason mathematically about basic data types and structures (such as numbers, sets, graphs, and trees) used in computer algorithms and systems;
2. Distinguish rigorous definitions and conclusions from merely plausible ones; synthesize elementary proofs, especially proofs by induction.
3. Model and analyze computational processes using analytic and combinatorial methods.
4. Apply principles of discrete probability to calculate probabilities and expectations of simple random processes.
5. Work in small teams to accomplish all the objectives above.

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Use logical notation to define and reason about fundamental mathematical concepts such as sets, relations, functions, and integers.
2. Synthesize induction hypotheses and simple induction proofs.
3. Prove elementary properties of modular arithmetic and explain their applications in Computer Science, for example, in cryptography and hashing algorithms.
4. Apply graph theory models of data structures and state machines to solve problems of connectivity and constraint satisfaction, for example, scheduling.
5. Calculate numbers of possible outcomes of elementary combinatorial processes such as permutations and combinations.

UNIT-I	MATHEMATICAL LOGIC	
Statements and Notations, Connectives, Statement Formulas and Truth Tables, well formed formulas, Tautologies, Equivalence of Formulas, Normal Forms: Disjunctive Normal Forms (DNF), Conjunctive Normal Forms (CNF), Principle Disjunctive Normal Forms (PDNF), Principle Conjunctive Normal Forms (PCNF).		
UNIT-II	PREDICATES	
The Predicate calculus, Free and Bound Variables, Rules of Inference, Consistency Of Premises and Indirect Method of Proof, Automatic Theorem Proving.		
UNIT-III	RELATIONS AND ORDERING	
Relations, Properties of Binary Relations in a Set, Equivalence Relations, Compatibility Relations, and Partial Ordering, Partial Ordered Set – Representation and Associated Terminology. FUNCTIONS: Definition and Introduction, Composition of Functions, Inverse Functions, Recursive Functions.		
UNIT-IV	TREES	
Trees, Properties of trees, Representation of trees, Binary trees, Tree traversals, spanning trees, DFS, BFS, Minimal spanning trees, Prim's algorithm and Kruskal's algorithm.		
UNIT-V	GRAPHS	

Representation of Graphs, Planar Graphs, Graph theory and applications. Basic concepts of Isomorphism and sub graphs, Multi graph and Euler circuits, Hamiltonian Graphs, chromatic number.

TEXT BOOKS:

1. J. P. Trembly, R. Manohar (1997), Discrete Mathematical Structures with Applications to Computer Science, Tata McGraw Hill, India.
2. Joe L. Mott, Abraham Kandel, Theodore P. Baker (2011), Discrete Mathematics for Computer Scientists and Mathematicians, 2nd edition, Prentice Hall of India Learning Private Limited, New Delhi, India.

REFERENCE BOOKS:

1. Kenneth H. Rosen (2007), Discrete Mathematics and its Applications, 6th edition, Tata McGraw Hill, India.
2. C.L. Liu, D.P. Mohapatra (2008), Elements of Discrete Mathematics, 3rd edition, McGraw Hill, India.
3. Ralph P. Grimaldi, B.V. Ramana (2006), Discrete and Combinatorial Mathematics - An Applied Introduction, 5th Edition, Pearson Education, India.

INTRODUCTION TO MOBILE APPLICATION DEVELOPMENT

VIII Semester: OPEN ELECTIVE - IV

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4IT20	OEC	L	T	P	C	CIA	SEE	Total
		3	0	0	3	30	70	100

COURSE OBJECTIVES

To learn

1. Produce apps for Android platform devices
2. Gain a basic understanding of computer architecture and object-oriented programming
3. Develop a working knowledge of Apple's Xcode app development tool
4. Understand mobile design principles
5. Identify need and opportunity in app markets

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Describe those aspects of mobile programming that make it unique from programming for other platforms,
2. Critique mobile applications on their design pros and cons,
3. Utilize rapid prototyping techniques to design and develop sophisticated mobile interfaces,
4. Program mobile applications for the Android operating system that use basic and advanced phone features, and
5. Deploy applications to the Android marketplace for distribution.

UNIT-I INTRODUCTION TO ANDROID

The Android Platform, Android SDK, Android Installation, Building your First Android application, Understanding Anatomy of Android Application, Android Manifest file.

UNIT-II ANDROID APPLICATION DESIGN ESSENTIALS

Android terminologies, Application Context, Activities.

UNIT-III ANDROID USER INTERFACE DESIGN ESSENTIALS

User Interface Screen elements, Designing User Interfaces with Layouts.

UNIT-IV TESTING ANDROID APPLICATIONS

Testing Android applications, Publishing Android application, Using Android preferences, managing Application resources in a hierarchy, working with different types of resources.

UNIT-V USING COMMON ANDROID APIS

Using Android Data and Storage APIs, Managing data using SQLite, Sharing Data between Applications with Content Providers.

TEXT BOOKS:

1. Lauren Darcey and Shane Conder, —Android Wireless Application DevelopmentII, Pearson Education, 2nd ed. (2011)

REFERENCE BOOKS:

- | |
|--|
| <ol style="list-style-type: none">1. R1. Reto Meier, —Professional Android 2 Application DevelopmentII, Wiley India Pvt Ltd2. R2. Mark L Murphy, —Beginning AndroidII, Wiley India Pvt Ltd.3. R3. Android Application Development All in one for Dummies by Barry Burd, Edition. |
|--|

DISTRIBUTED DATABASES

VIII Semester: OPEN ELECTIVE - IV

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4IT21	OEC	L	T	P	C	CIA	SEE	Total
		3	0	0	3	30	70	100

COURSE OBJECTIVES

To learn

1. Introducing Distributed Database Management System and its Design issues
2. Exploring several algorithms for processing queries and be able to use them
3. Describe the methods to translate complex conceptual data models into logical and Physical database designs
4. Demonstrating query optimization and its algorithms
5. Enumerating the concepts behind distributed transaction processing

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Identify the introductory distributed database concepts and its structures.
2. Describe terms related to distributed object database design and management.
3. Produce the transaction management and query processing techniques in DDBMS.
4. Relate the importance and application of emerging database technology.

Pre-requisites: Basic concept of DBMS and RDBMS.

UNIT-I	INTRODUCTORY CONCEPTS AND DESIGN OF (DDBMS)	
Data Fragmentation; Replication, and allocation techniques for DDBMS; Methods for designing and implementing DDBMS, designing a distributed relational database; Architectures for DDBMS: cluster federated, parallel databases and client server architecture.		
UNIT-II	QUERY PROCESSING & TRANSACTION MANAGEMENT	
Overview Of Query Processing: Query processing problem; Objectives of Query Processing; Complexity of Relational Algebra operations; characterization of Query processors; Layers of Query Processing Introduction To Transaction Management: Definition of Transaction, Properties of Transaction, types of transaction ; Distributed Concurrency Control: Serializability theory; Taxonomy of concurrency control mechanisms; locking bases concurrency control algorithms.		
UNIT-III	DISTRIBUTED OBJECT DATABASE MANAGEMENT SYSTEMS	
Fundamental Object concepts and Object models, Object distribution design, Architectural issues, Object management, Distributed object storage, object query processing.		
UNIT-IV	CURRENT TRENDS & DEVELOPMENTS RELATED TO DISTRIBUTED DATABASE APPLICATIONS TECHNOLOGIES	
Distributed Object/component-based DBMS; Database Interoperability including CORBA; DCOM and Java RMI; Distributed document-based systems; XML and Workflow management.		
UNIT-V	EMERGING RELATED DATABASE TECHNOLOGIES	
Parallel Database Mobile database; Multimedia Database; Spatial Database and Web Databases.		
TEXT BOOKS:		
1. 1 Distributed Databases - Principles and Systems; Stefano Ceri; Guiseppe Pelagatti; Tata McGraw Hill; 1985. 2. Fundamental of Database Systems; Elmasri & Navathe; Pearson Education; Asia. 3. Database System Concepts; Korth & Sudarshan; TMH. 4. Principles of Distributed Database Systems; M. Tamer Özsu; and Patrick Valduriez Prentice Hall.		

REFERENCE BOOKS:
1. Data Base Management System; Leon & Leon; Vikas Publications 2. Introduction to Database Systems; Bipin C Desai; Galgotia.

BIG DATA								
VIII Semester: OPEN ELECTIVE - IV								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4IT22	OEC	L	T	P	C	CIA	SEE	Total
		3	0	0	3	30	70	100
COURSE OBJECTIVES								
To learn								
1. To introduce the terminology, technology and its applications								
2. To introduce the concept of Analytics and Visualization								
3. To demonstrate the usage of various Big Data tools and Data Visualization tools								
COURSE OUTCOMES:								
Upon successful completion of the course, the student is able to								
1. Compare various file systems and use an appropriate file system for storing different types of data.								
2. Demonstrate the concepts of Hadoop ecosystem for storing and processing of unstructured data.								
3. Apply the knowledge of programming to process the stored data using Hadoop tools and generate reports.								
4. Connect to web data sources for data gathering, Integrate data sources with hadoop components to process streaming data.								
5. Tabulate and examine the results generated using hadoop components.								
UNIT-I	INTRODUCTION TO BIG DATA							
Data and its importance, Big Data- Definition, V's of Big Data, Hadoop Ecosystem								
HADOOP ARCHITECTURE								
Hadoop Storage : HDFS, Hadoop								
Processing : Map Reduce Framework								
Hadoop Server Roles : Name Node, Secondary Name Node and Data Node, Job Tracker, Task Tracker								
HDFS-HADOOP DISTRIBUTED FILE SYSTEM								
Design of HDFS, HDFS Concepts, HDFS Daemons, HDFS High Availability, Block Abstraction, FUSE- File System in User Space. HDFS Command Line Interface (CLI),Concept of File Reading and Writing in HDFS.								
UNIT-II	MAPREDUCE PROGRAMMING MODEL							
Introduction to Map Reduce Programming model to process BigData, key features of MapReduce, MapReduce Job skeleton, Introduction to MapReduce API, Hadoop Data Types, Develop MapReduce Job using Eclipse, build a MapReduce Job export it as a java archive(.jar file).								
MAPREDUCE JOB LIFE CYCLE: How Mapreduce Works? Understanding Mapper ,Combiner, Partitioner ,Shuffle & Sort and Reduce phases of MapReduce Application, Developing Map Reduce Jobs based on the requirement using given datasets like weather dataset.								
MAPREDUCE API: Understanding new MapReduce API from org.apache.hadoop.mapreduce and its sub packages to develop MapReduce applications ,key difference between old MapReduce API and the new MapReduce API.								
UNIT-III	INTRODUCTION TO PIG							
Understanding pig and pig Platform, introduction to Pig Latin Language and Execution engine, running pig in different modes, Pig Grunt Shell and its usage.								

PIG LATIN LANGUAGE–DATA TYPES IN PIG

Pig Latin Basics, Key words ,Pig Data types ,Understanding Pig relation, bag, tuple and writing pig relations or statements using Grunt Shell ,expressions, Data processing operators, using Built in functions.

WRITING PIG SCRIPTS USING PIG LATIN: Writing pig scripts and saving them in text editor, running pig scripts from command line.

UNIT-IV**INTRODUCTION TO HIVE**

Understanding Hive Shell, Running Hive, Understanding Schema on read and Schema on write.

HIVE QL DATA TYPES, SEMANTICS: Introduction to Hive QL (Query Language), Language semantics, Hive Data Types.

HIVE DDL, DML AND HIVE SCRIPTS: Hive Statements, Understanding and working with Hive Data Definition Languages and Manipulation Language statements, Creating Hive Scripts and running them from hive terminal and commands line.

UNIT-V**SQOOP, FLUME, OOZIE**

SQOOP: Introduction to Sqoop tool, commands to connect databases and list databases and tables, command to import data from RDBMS into HDFS, Command to export data from HDFS into required tables of RDBMS.

FLUME: Introduction to Flume agent, understanding Flume components Source, Channel and Sink, Writing flume configuration file, running flume configuration file to ingest the data into HDFS.

OOZIE: Introduction to Oozie, Understanding work flow and how to create Work flow using Work Flow definition language in XML, running a basic Oozie workflow to run a Map Reduce job.

TEXT BOOKS:

1. Hadoop: The Definitive Guide, 4th Edition - O'Reilly Media
2. Big Data and Hadoop- Learn by Example, Mayank Bhushan

REFERENCE BOOKS:

1. Michael Berthold, David J. Hand, "Intelligent Data Analysis", Springer, 2007.
2. Paul Zikopoulos ,Dirk DeRoos , Krishnan Parasuraman , Thomas Deutsch , James Giles , David Corigan , "Harness the Power of Big Data The IBM Big Data Platform ", Tata McGraw Hill Publications, 2012.

**OPEN ELECTIVES
OFFERED BY
MECH DEPARTMENT**

ELEMENTS OF MECHANICAL ENGINEERING

V Semester: OPEN ELECTIVE-I

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME60	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OVERVIEW:

This course deals with fundamental concepts of design, manufacturing and thermal engineering.

COURSE OBJECTIVES:

The course aims to enable the students learn basic concepts of mechanics, materials, thermodynamics, manufacturing and CAD/CAM.

COURSE OUTCOMES:

After completing this course the student must demonstrate the knowledge and ability:

1. To demonstrate basic concepts of mechanics of rigid and deformable bodies.
2. To explain fundamentals of metallurgy and material science.
3. To discuss basic concepts and principles of thermo dynamics.
4. To demonstrate the elements of fabrication processes and equipment.
5. To explain basic concepts of CAD/CAM.

UNIT-I	MECHANICS	
MECHANICS OF RIGID BODIES: Force, Force system, Classification of force systems, triangle law, parallelogram law-derivation and problems, polygon law, resolution of forces in coplanar-concurrent force system-problems; Definition of friction, laws of friction, coefficient of friction, types of friction, angle of friction, angle of repose, definition of centroid and centre of gravity, pappus and guldinus theorems, problems on centroid of plane areas; Definition of area moment of inertia, parallel axis theorem, perpendicular axis theorem, simple problems on area moment of inertia. MECHANICS OF DEFORMABLE BODIES: Stress, Strain, Types of stresses and strain, Poisson's ratio, stress-strain curve of mild steel, Hooke's Law, factor of safety, mechanical properties of materials, bulk modulus, shear modulus, bars of uniform and varying sections-simple problems.		
UNIT-II	METALLURGY & MATERIAL SCIENCE	
Bonds in Solids, crystallization of metals, Classification of steels, , Necessity of alloying, types of solid solutions, lever rule, phase rule, cooling curve of pure iron, Iron-Iron carbide diagram, Heat treatment- Annealing, quenching, normalizing, Hardening, tempering; Ceramic materials-classification, properties and applications.		
UNIT-III	THERMO DYNAMICS	
System, Control Volume, Surrounding, Boundaries, Universe, Types of Systems, Macroscopic and Microscopic viewpoints, Concept of Continuum, Thermodynamic Equilibrium, State, Property, Process, Zeroth Law of Thermodynamics – Concept of Temperature – Principles of Thermometry, First law of Thermodynamics – Corollaries, First law applied to a Closed System, applied to a flow system, Steady Flow Energy Equation and Limitations of the First Law, Second Law of Thermodynamics, Kelvin-Planck and Clausius Statements, Carnot cycle and its specialties.		
UNIT-IV	MANUFACTURING	
Casting-Types of casting, advantage of casting and its applications, defects in castings; Patterns - Pattern making, Types, Materials used for patterns, pattern allowances; Welding-classification, Heat affected zone in welding, welding defects-causes and remedies; Basic extrusion process and its characteristics, Forging operations and principles; Basic machining operations-drilling, milling, turning, step turning, threading, grinding and shaping; Limits, fits and tolerances- Unilateral and bilateral tolerance system, hole and shaft basis system.		

UNIT-V	CAD/CAM	
Fundamentals of CAD/CAM- Benefits of CAD, Computer configuration for CAD applications, Computer peripherals for CAD; Wire frame modeling: Definition, advantages, dis-advantages, wire frame entities-analytic entities and synthetic entities; Surface modeling: Definition, advantages, dis-advantages, surface entities-analytic entities and synthetic entities; Solid Modeling: Definition, constructive solid geometry, advantages, modeling entities.		
TEXT BOOKS:		
1. Singer's Engineering Mechanics by K.Vijay Kumar Reddy and J.Suresh Kumar, B.S Publications 2. Strength of Materials By Dr.Sadhu Singh, Khanna Publishers. 3. Introduction to physical metallurgy by Avner, TATA McGrawHill Edition. 4. Thermodynamics by P.K Nag, McGrawHill Education. 5. Production Technology by R.K Jain, Khanna Publishers. 6. CAD/CAM Theory & Practice by Ibrahim Zeid & R.Sivasubramanian, McGrawHill Education.		
REFERENCE BOOKS:		
1. Engineering Mechanics By R.K.Bansal, Laxmi Publications 2. Mechanics of Solids By GVV Satyanarayana & Pakirappa, Durga Publishing House 3. Material science and metallurgy by OP Khanna, Dhanpat Ray Publications 4. Engineering Thermodynamics By Pakirappa, V.Naveen Kumar, V.Naresh, Durga Publishing House 5. Production Technology by Dr. P.C Sharma, S.CHAND 6. CAD/CAM By P.Radha Krishnan, S.Subramanyan and V.Raju, New Age International Publishers.		
WEB REFERENCES:		
1. https://www.iitg.ac.in/rkbc/me101/Presentation/L01-03.pdf 2. https://nptel.ac.in/courses/113106032/ 3. https://nptel.ac.in/downloads/112108148/ 4. https://lecturenotes.in/subject/195/cad-and-cam-cac		
E-TEXT BOOKS:		
1. https://books.google.co.in/books?id=8kA8BAAAQBAJ&printsec=frontcover&source=gbg_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=XYorDAAAQBAJ&printsec=frontcover&source=gbg_ge_summary_r&cad=0#v=onepage&q&f=false		
MOOC COURSE:		
1. https://ocw.mit.edu/courses/mechanical-engineering/		

FUNDAMENTALS OF ENGINEERING MATERIALS

V Semester: OPEN ELECTIVE-I

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME61	PCC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	30	70	100

COURSE OBJECTIVES:

This course will enable students to understand basic structure and crystal arrangement of materials, the phase diagrams, advantages of heat treatment, various heat treatment processes, the need and application of composite materials.

COURSE OUTCOMES:

At the end of course students are able to

1. Explain basic concepts of crystal structure such as unit cells, crystal systems of metals etc.
2. Demonstrate the concept of alloying and formation of different types of phases in alloys.
3. Differentiate ferrous and non ferrous alloys.
4. Explain various heat treatment processes.
5. Classify and explain polymers, ceramics and composites.

UNIT-I	CRYSTAL STRUCTURE	
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Unit cells, crystal systems of metals, Imperfection in solids: Point, line, interfacial and volume defects; dislocation strengthening mechanisms and slip systems, determination of grain size, effect of grain size on the properties of alloys.

UNIT-II	ALLOYS & PHASE DIAGRAMS	
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Alloys- substitutional and interstitial solid solutions.
Phase diagrams: Interpretation of binary phase diagrams and microstructure development; eutectic, peritectic, peritectoid and monotectic reactions. Iron Iron-carbide phase diagram and microstructure of ledeburite, austenite, ferrite and cementite.

UNIT-III	FERROUS AND NON FERROUS ALLOYS	
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Alloying of steel, properties of stainless steel and tool steels, maraging steels; cast irons-grey, white, malleable and spheroidal cast irons; copper and copper alloys- brass, bronze and cupro-nickel; Aluminium and Aluminium alloys.

UNIT-IV	HEAT TREATMENT OF STEEL	
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Annealing, tempering, normalizing and spheroidising, austempering, martempering, case hardening, carburizing, nitriding, cyaniding, carbo-nitriding, flame and induction hardening, vacuum and plasma hardening.

UNIT-V	POLYMERS, CERAMICS AND COMPOSITES	
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Classification, properties and applications of polymers, ceramics, composites and nano materials.

TEXT BOOKS:

1. V. Raghavan, "Material Science and Engineering", Prentice Hall of India Private Limited, 1999.
2. U. C. Jindal, "Engineering Materials and Metallurgy", Pearson, 2011.
3. Sidney H. Avener (2007,) *Introduction to Physical Metallurgy*, 2nd edition, Tata McGraw hill education (P) Ltd, New Delhi, India.

REFERENCE BOOKS:

1. W. D. Callister, 2006, "Materials Science and Engineering-An Introduction", 6th Edition, Wiley India.
2. Kenneth G. Budinski and Michael K. Budinski, "Engineering Materials", Prentice Hall of India Private Limited, 4th Indian Reprint, 2002.

3. V. D. Kodgire (2006), Material Science and Metallurgy for engineers, 1st Edition, Everest, Pune, India.
WEB REFERENCES:
1. https://nptel.ac.in/courses/113106032/ 2. http://web.pdx.edu/~pmoeck/phy381/lecture1.pdf
E-TEXT BOOKS:
1. https://books.google.co.in/books?id=XXE8BAAAQBAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=kBM8BAAAQBAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false
MOOC COURSE:
1. https://ocw.mit.edu/courses/materials-science-and-engineering/ 2. https://www.mooc-list.com/tags/materials-science

FUNDAMENTALS OF THEORY OF MACHINES

V Semester: OPEN ELECTIVE-I

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME62	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OVERVIEW:

Theory of Machines may be defined as that branch of Engineering-science, which deals with the study of relative motion between the various parts of a machine, and forces which act on them. The knowledge of this subject is very essential for an engineer in designing the various parts of a machine.

COURSE OBJECTIVES:

1. This course aims to enable students to learn basic concepts of kinematics and dynamics of machinery.

COURSE OUTCOMES:

At the end of the course the students are able to:

1. Demonstrate various kinematic links, pairs, chains, mechanisms and their inversions.
2. Illustrate and analyze exact and approximate straight line motion mechanisms.
3. Explain concepts of higher pairs.
4. Demonstrate and analyze the effect of gyroscopic couples on stability of motor cycle, motor car and aeroplane.
5. Illustrate and analyze various governors.

UNIT-I	MECHANISMS	
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Elements or Links – Classification – Rigid Link, flexible and fluid link – Types of kinematic pairs – Types of constrained motion – kinematic chain – Mechanism-Machine-Structure – Kleins criteria, Degrees of freedom, Grubler's Criterion, Grashof's law – inversion of mechanism – inversions of four bar chain, single and double slider crank chains.

UNIT-II	STRAIGHT LINE MOTION MECHANISMS	
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Exact, approximately copied and generated types – Peaucellier - Hart and Scott Russel – Grasshopper – Watt – Tchebichef's and Robert Mechanisms - Pantograph.

UNIT-III	GEARS & GEAR TRAINS	
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Friction wheels and toothed gears – types – law of gearing, condition for constant velocity ratio for transmission of motion, Form of teeth-cycloid and involute profiles, phenomena of interference.

Gear trains: Introduction – Types – Simple, compound and reverted gear trains-simple problems.

UNIT-IV	GYROSCOPES	
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Gyroscopes – effect of precession – motion on the stability of moving vehicles such as motorcycle motorcar and aeroplane.

UNIT-V	GOVERNORS	
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Governors: Types of governors - Watt, Porter and Proell governors. Spring loaded governors – Hartnell and Hartung with auxiliary springs. (No derivations)

TEXT BOOKS:

1. Theory of Machines by S.S.Rattan.
2. Theory of Machines by R.S.Khurmi

REFERENCE BOOKS:

1. Theory of Machines by Shigley, Mc Graw Hill Publishers
2. Theory of Machines by Thomas Bevan, CBS Publishers
3. Theory of Machines by R.K.Bansal (Lakshmi publications)
4. Mechanism and Machine Theory by JS Rao and RV Duggipati, New Age International

WEB REFERENCES:

1. <http://nptel.ac.in/courses/112104121/>
2. <https://freevideolectures.com/course/2359/kinematics-of-machines>
3. <https://nptel.ac.in/courses/112104114/>
4. <https://freevideolectures.com/course/2364/dynamics-of-machines>

E-TEXT BOOKS:

1. https://books.google.co.in/books?id=T_WlquM4EI4C&printsec=frontcover&source=gbg_ge_summary_r&cad=0#v=onepage&q&f=false
2. https://books.google.co.in/books?id=YI7vAwAAQBAJ&printsec=frontcover&source=gbg_ge_summary_r&cad=0#v=onepage&q&f=false

MOOC COURSE:

1. <https://cosmolearning.org/courses/kinematics-of-machines>
2. <https://nptel.ac.in/courses/112101096/>
3. <https://www.mooc-list.com/course/mechanics-kinematics-and-dynamics-edx>

FUNDAMENTALS OF MECHATRONICS

VI Semester: OPEN ELECTIVE -II

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME63	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSEOVERVIEW:

The aim is to introduce students to the fundamental concepts and principles mechatronics. It builds upon the awareness and necessity of interdisciplinary dependency of 21st century. It aim is also to engage students to understand the introduction to mechatronics with emphasis on analog electronics, digital electronics, sensors and transducers, actuators, and microprocessors. Lectures are intended to provide the student with foundational concepts in mechatronics and practical familiarity with common elements making up mechatronic systems.

COURSEOBJECTIVES:

1. To understand the basic principles of mechatronics, Differentiating between conventional and mechatronics design process and product.
2. Differentiate various sensors, actuators, electromechanical systems.
3. To understand and apply interfacing of electromechanical products to microcontrollers and microprocessor.
4. To understand the working of various programmable logic controllers (PLCs), Architecture of PLCs.
5. Evaluatedifferent mechatronic systems.

COURSEOUTCOMES:

At the end of the course students are able to

1. Demonstrate various elements underlying mechatronic systems, electronics, control systems and able differentiate their purpose in the system.
2. Analyze and select sensors, actuators, electromechanical components needed for an application.
3. To evaluate microprocessor and microcontroller interfacing to mechanical application.
4. To choose PLC to a mechanical application.
5. To design mechatron systems for mechanical applications.

UNIT-I	INTRODUCTION	
Introduction to Mechatronics --Mechatronics systems - Mechatronics design process - Mechatronics in Manufacturing --Adoptive and distributed control systems – Modelling and simulation of mechatronics systems.		
UNIT-II	SENSORS AND ACTUATORS	
Sensors and actuators: Overview of sensors and transducers – Microsensors - Signal conditioning – Operational amplifiers – Protection – Filtering - Analog and Digital converters. Electro – pneumatics and Electro – hydraulics - Solenoids – Direct Current motors – Servomotors – Stepper motors - Micro actuators; Drives selection and application.		
UNIT-III	INTERFACING	
Interfacing: Microprocessor based Controllers Architecture of microprocessor and microcontroller – System interfacing for a sensor, keyboard, display and motors - Application cases for temperature control, warning and process control systems		
UNIT-IV	PLCs	
PLCs: Programmable Logic Controllers Architecture of Programmable Logic Controllers – Input/Outputmodules – programming methods – Timers and counters – Master control – Branching – Data handling – Analog input/output – Selection of PLC and troubleshooting.		

UNIT-V	ARTIFICIAL INTELLIGENCE	
AI: Intelligent Mechatronics and Case Studies Fuzzy logic control and Artificial Neural Networks in mechatronics – Algorithms – Computer – based instrumentation - Real-time Data Acquisition and Control – Software integration – Man Machine interface -Vision system – Mechatronics system case studies.		
TEXT BOOKS:		
1. Introduction to Mechatronics and Measurement Systems, Tata McGraw Hill		
REFERENCE BOOKS:		
1. Designing Intelligent Machines, Michel B. Histan and David G. Alciatore, Open University London		
2. Control Sensors and Actuators, ICW. Desiha, Prentice Hall		
WEB REFERENCES:		
1. https://lecturenotes.in/subject/137/mechatronics-mech		
2. https://sites.google.com/site/profnarendralakal/Home/presentations/mtrx		
E-TEXT BOOKS:		
1. https://books.google.co.in/books?id=NCEeONKWzX4C&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false		
2. https://books.google.co.in/books?id=z8yEUou06cEC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false		
MOOC COURSE:		
1. https://ocw.mit.edu/courses/mechanical-engineering/2-737-mechatronics-fall-2014/		
2. https://nptel.ac.in/courses/112103174/		
3. https://www.igi-global.com/chapter/massive-online-open-course-assisted-mechatronics-learning/137327		

BASICS OF THERMODYNAMICS

VI Semester: OPEN ELECTIVE -II

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME64	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OVERVIEW:

Thermodynamics is the field of physics that deals with the relationship between heat and work in a substance during a thermodynamic process. This course covers fundamental concepts of thermodynamics such as laws of thermodynamics, pure substances, perfect gas laws, mixture of perfect gases, mollier charts and Psychrometry.

COURSE OBJECTIVES:

To understand the treatment of classical Thermodynamics and to apply the First and Second laws of Thermodynamics for the analysis of thermal equipment

COURSE OUTCOMES:

At the end of the course, the student should be able to

1. Demonstrate basic concepts of thermodynamics.
2. Explain laws of thermodynamics.
3. Explain pure substances and power cycles.
4. Explain perfect gas laws and concepts of mixture of perfect gases.
5. Discuss fundamental concepts of psychrometry.

UNIT-I	INTRODUCTION	
Introduction: System, Control Volume, Surrounding, Boundaries, Universe, Types of Systems, Macroscopic and Microscopic viewpoints, Concept of Continuum, Thermodynamic Equilibrium, State, Property, Process, Exact & Inexact Differentials, Cycle – Reversibility – Quasi – static Process, Irreversible Process, Causes of Irreversibility – Energy in State and in Transition, Zeroth Law of Thermodynamics – Concept of Temperature – Principles of Thermometry		
UNIT-II	LAWS OF THERMO DYNAMICS	
First law of Thermodynamics – Corollaries – First law applied to a Closed System – applied to a flow system – Steady Flow Energy Equation. Limitations of the First Law Second Law of Thermodynamics, Kelvin-Planck and Clausius Statements and their Equivalence / Corollaries, Entropy, Principle of Entropy Increase – Energy Equation, Availability and Irreversibility – Elementary Treatment of the Third Law of Thermodynamics		
UNIT-III	PURE SUBSTANCES & POWER CYCLES	
Pure Substances, p-V-T- surfaces, T-S and h-s diagrams, Mollier Charts, Phase Transformations – Triple point at critical state properties during change of phase, Dryness Fraction – Clausius – Clapeyron Equation Property tables. Power Cycles: Otto, Diesel, Dual Combustion cycles, Sterling Cycle – Description and representation on P-V and T-S diagram, Thermal Efficiency, Mean Effective Pressures on Air standard basis – comparison of Cycles.		
UNIT-IV	PERFECT GAS LAWS & MIXTURE OF PERFECT GASES	
Perfect Gas Laws – Equation of State, specific and Universal Gas constants – various Non-flow processes, properties, end states Mixtures of perfect Gases – Mole Fraction, Mass fraction Gravimetric and volumetric Analysis – Dalton's Law of partial pressure, Avogadro's Laws of additive volumes – Mole fraction, Volume fraction and partial pressure.		

UNIT-V	PSYCHROMETRY	
Psychrometric Properties – Dry bulb Temperature, Wet Bulb Temperature, Dew point Temperature, Thermodynamic Wet Bulb Temperature, Specific Humidity, Relative Humidity, saturated Air, Vapour pressure, Degree of saturation – Adiabatic Saturation, Carrier's Equation – Psychrometric chart.		
TEXT BOOKS:		
1. Engineering Thermodynamics, P.K. Nag, 6th Edition, Mc Graw Hill Education. 2. Thermodynamics an engineering approach, Yunus A. Cengel & Michael A. Boles, 8th Edition, Mc Graw Hill Companies. 3. Fundamentals of Thermodynamics, Richard E. Sonntag Claus Borgnakke, 7th Edition, Wiley.		
REFERENCE BOOKS:		
1. Fundamentals of engineering thermodynamics, Rathakrishnan. E, 2nd Edition, Prentice hall of India Pvt Ltd., 2006. 2. Thermodynamics, Arora.C.P, Tata Mc Graw Hill, New Delhi. 3. Applied Thermodynamics, Onkar Singh, 3rd Edition, New Age, India		
WEB REFERENCES:		
1. https://www.askiitians.com/revision-notes/physics/thermodynamics/ 2. http://www.ncert.nic.in/ncerts//kech106.pdf		
E-TEXT BOOKS:		
1. https://books.google.co.in/books?id=91MZ2ZO3n8C&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=w8GhW3J8RHIC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false		
MOOC COURSE		
1. https://www.coursera.org/learn/thermodynamics-intro 2. https://www.edx.org/course/thermodynamics-2		

FUNDAMENTALS OF SMART MATERIALS

VI Semester: OPEN ELECTIVE -II

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME65	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OVERVIEW:

Smart materials are materials that sense and react to environmental conditions or stimuli (e.g., mechanical, chemical, electrical or magnetic signals). This course is designed to impart knowledge on fundamentals of various smart materials.

COURSE OBJECTIVES:

The objective of this course is to enable the students to learn principles of smart materials such as electronic materials, ionic polymer matrix composites etc.

COURSE OUTCOMES:

At the end of the course students are able to

1. Demonstrate various smart materials and their principles.
2. Explain different types of electro-active materials.
3. Explain principles of piezoelectricity.
4. Discuss principles of Magnetostriction.
5. Explain about various shape memory alloys.

UNIT-I	INTRODUCTION	
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Overview of Smart Materials: Introduction and definition of Smart Materials, principles of smart materials, input – output decision ability, devices based on conductivity changes, devices based on changes in optical response, biological systems smart materials. Devices based on magnetization, artificial structures, surfaces, hetero structures, polycrystalline, amorphous, liquid crystalline materials

UNIT-II	ELECTRO ACTIVE MATERIALS & PIEZO ELECTRICITY	
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Electro-active Materials: Electronic Materials, Electro-active Polymers, Ionic Polymer Matrix Composites (IPMC)

UNIT-III	PIEZOELECTRICITY	
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Piezoelectricity: Principles of Piezoelectricity, Perovskite Piezoceramic Materials, Single Crystals vs Polycrystalline Systems, Piezoelectric Polymers

UNIT-IV	MAGNETOSTRICTION	
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Principles of Magnetostriction, Rare earth Magnetostrictive materials, Giant Magnetostriction and Magnetoresistance Effect, magnetorheological fluids, electrorheological materials and their applications

UNIT-V	SHAPE MEMORY ALLOYS	
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Shape Memory Alloys: Shape Memory Effect, Shape Memory Alloys, Shape Memory Polymers, Self-Sensing Piezoelectric Transducers, Energy Harvesting Materials, Self-healing materials.

TEXT BOOKS:

1. B. Culshaw, Smart Structures and Materials, Artech House, 2000
2. P. Gauenzi, Smart Structures: Physical Behaviour, Mathematical Modelling and Applications, Wiley Publishers, 2009
3. W. G. Cady, Piezoelectricity, Dover Publication, 1964
4. M. Addington, Schodek, L. Daniel.: Smart materials and new technologies, Architectural Press, 2005
5. W.D Callister, Materials Science and Engineering an Introduction, Willey 1999

REFERENCE BOOKS:

1. M. Schwartz, Encyclopedia of Smart Materials, Volumes 1-2 ,Willey, 2002
2. Smith, C.: Smart material systems, Ralph, SIAM, 2005
3. K Vijay, K.Varadan, J. Vinoy, S.Gopalakrisham, Smart Material Systems and MEMS: Design and Development Methodologies , Willey 2006

WEB REFERENCES:

1. <https://www.sciencedirect.com/topics/chemistry/smart-material>
2. <https://www.mide.com/collections/smart-materials>

E-TEXT BOOKS:

1. https://books.google.co.in/books?id=KdBr38J-bxoC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false
2. https://books.google.co.in/books?id=CQxwDQAAQBAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false

MOOC COURSE

1. https://nptel.ac.in/noc/individual_course.php?id=noc18-me60
2. <https://nptel.ac.in/courses/112104251/>

FUNDAMENTALS OF OPERATIONS RESEARCH

VII Semester: OPEN ELECTIVE-III

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME66	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE DESCRIPTION:

Operations research (OR) is an analytical method of problem-solving and decision-making that is useful in the management of organizations. In operations research, problems are broken down into basic components and then solved in defined steps by mathematical analysis. This course gives insight of Linear Programming, Transportation, assignment problems, sequencing etc.

COURSE OBJECTIVES:

The objectives of this course are to learn quantitative methods and techniques for effective decisions-making; model formulation and applications that are used in solving business decision problems.

COURSE OUTCOMES:

At the end of course students will be able to

1. Describe types of models and solve linear programming problem.
2. Solve transportation and assignment problems.
3. Analyze sequencing and replacement models and apply them for optimization.
4. Apply gaming theory for optimal decision making.
5. Analyze inventory models to optimize the cost

UNIT-I	INTRODUCTION & ALLOCATION	
Development – Definition– Characteristics and Phases – Types of models – Operations Research models – applications. ALLOCATION: Linear Programming Problem - Formulation – Graphical solution – Simplex method – Artificial variables techniques: Two-phase method, Big-M method; Duality Principle.		
UNIT-II	TRANSPORTATION & ASSIGNMENT PROBLEMS	
TRANSPORTATION PROBLEM: Formulation – Optimal solution, unbalanced transportation problem – Degeneracy. ASSIGNMENT PROBLEM: Formulation – Optimal solution - Variants of Assignment Problem; Traveling Salesman problem.		
UNIT-III	SEQUENCING & REPLACEMENT	
SEQUENCING: Introduction – Flow –Shop sequencing – n jobs through two machines – n jobs through three machines REPLACEMENT: Introduction – Replacement of items that deteriorate with time – when money value is not counted and counted – Replacement of items that fail completely.		
UNIT-IV	THEORY OF GAMES	
THEORY OF GAMES: Introduction –Terminology– Solution of games with saddle points and without saddle points- 2 x 2 games –m x 2 & 2 x n games - graphical method – m x n games - dominance principle.		
UNIT-V	INVENTORY	
INVENTORY: Introduction – Single item, Deterministic models – Types - Purchase inventory models with one price break and multiple price breaks –Stochastic models – demand discrete variable or continuous variable – Single Period model with no setup cost.		

TEXT BOOKS:

1. Operation Research by J.K.Sharma, MacMilan.
2. Operations Research by ACS Kumar, Yesdee

REFERENCE BOOKS:

1. Operations Research: Methods and Problems by Maurice Saseini, Arthur Yaspan and Lawrence Friedman
2. Operations Research by A.M.Natarajan, P.Balasubramaniam, A. Tamilarasi, Pearson Education.
3. Operations Research by Wagner, PHI Publications.
4. Introduction to O.R by Hillier & Libermann, TMH.

WEB REFERENCES:

1. <http://www.cs.toronto.edu/~stacho/public/IEOR4004-notes1.pdf>
2. http://www.mathcity.org/msc/notes/operation_research
3. http://biobharati.com/admin/img/gall/1467763196_BCA-504.pdf

E-TEXT BOOKS:

1. https://books.google.co.in/books?id=6khDDAAQBAJ&printsec=frontcover&source=gbg_ge_summary_r&cad=0#v=onepage&q&f=false
2. https://books.google.co.in/books?id=cwoWJrgCrP8C&printsec=frontcover&source=gbg_ge_summary_r&cad=0#v=onepage&q&f=false
3. https://books.google.co.in/books?id=rj6bBMVzfPsC&printsec=frontcover&source=gbg_ge_summary_r&cad=0#v=onepage&q&f=false

MOOC COURSE:

1. <http://www.orcomplete.com/internet/enesbilgin/open-courses-on-operations-research>
2. <https://www.coursera.org/learn/wharton-operations>

FUNDAMENTALS OF ENERGY SOURCES AND APPLICATIONS

VII Semester: OPEN ELECTIVE-III

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME67	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OVERVIEW:

This course focuses on various renewable energy sources such as solar energy, wind energy, geothermal energy etc.

COURSE OBJECTIVES:

The objectives of this course are to study classifications and comparison of various renewable energy sources, flat plate and concentrating collectors, wind energy, principles of bio mass, geothermal and ocean energy sources and principles of direct energy conversion.

COURSE OUTCOMES:

At the end of the course, the student shall be able to

1. Classify renewable energy sources and explain environmental impact of solar energy, solar radiation, their measurement and instruments.
2. Classify concentrating collectors and explain in detail about various solar gadgets.
3. Explain about wind mill, types of wind mills, its construction, wind energy conversion systems and principles of bio mass.
4. Explain in detail geothermal and ocean energies.
5. Demonstrate the concepts of direct energy conversion.

UNIT-I	INTRODUCTION	
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Introduction – Renewable energy sources – Limitations – Classifications – Comparison – Global and Indian energy scenario. Solar Energy: Introduction – Environmental impact of solar power – extraterrestrial and terrestrial solar radiation – solar radiation on tilted surface – Solar Radiation Measurement and Instruments

UNIT-II	SOLAR ENERGY	
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Flat plate and concentrating collectors – classification of concentrating collectors – orientation and thermal analysis – advanced collectors. Solar Gadgets: Solar furnace, Solar air heater, Solar water heater, Solar cooker, solar still, solar pond, solar pump, solar dryer, solar cell.

UNIT-III	WIND ENERGY & BIO MASS	
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Wind energy utilization: Introduction – Origin and nature of winds – Types of wind mills and its construction – Wind energy conversion systems – performance characteristics.

Bio-Mass: Principles of Bio-Conversion, Anaerobic/aerobic digestion, types of Bio-gas digesters, combustion characteristics of bio-gas, applications and economic aspects.

UNIT-IV	GEOTHERMAL ENERGY & OCEAN ENERGY	
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Geothermal Energy: Resources, types of wells, methods of harnessing the energy, potential in India.

Ocean Energy: OTEC, Principles utilization, setting of OTEC plants. Tidal and wave energy: Potential and conversion techniques, mini-hydel power plants, and their economics.

UNIT-V	DIRECT ENERGY CONVERSION	
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Direct Energy Conversion: Need, limitations, principles of DEC. Thermo-electric generators, seebeck, peltier and Joule-Thomson effects – applications, MHD generator principle, Fuel cells: Classification of fuel cells; Operating principles. Bio-fuels, Urban waste to energy conversion

TEXT BOOKS:

1. Rai G.D, *Non-Conventional Energy Sources*, Khanna Publishers, New Delhi, 2011.
2. Twidell, J.W. & Weir, A., *Renewable Energy Sources*, EFN Spon Ltd., UK, 2006

REFERENCE BOOKS:

1. Ashok V Desai, *Non-Conventional Energy*, Wiley Eastern Ltd, New Delhi, 2003.
2. Mittal K M, *Non-Conventional Energy Systems*, Wheeler Publishing Co. Ltd, New Delhi, 2003.
3. Khan B H, *Non-Conventional Energy Resources*, Tata McGraw Hill, 2nd Edn. New Delhi, 2009.
4. Ramesh R & Kumar K U, *Renewable Energy Technologies*, Narosa Publishing House, New Delhi, 2004
2. Tiwari. G.N., *Solar Energy – Fundamentals Design, Modelling & Applications*, Narosa Publishing House, New Delhi, 2002.
- 3.

WEB REFERENCES:

1. http://www.afdc.energy.gov/fuels/natural_gas.html
2. <https://cleancities.energy.gov/>
3. <http://www.firmgreen.com/>
4. <https://www.renewableresourcescoalition.org/alternative-energy-sources/>
5. <https://www.edx.org/learn/renewable-energy>

E-TEXT BOOKS:

1. <https://www.sciencedirect.com/book/9780126561531/renewable-energy>
2. <https://www.pdfdrive.com/renewable-energy-resources-e14705840.html>
3. <https://www.pdfdrive.com/search?q=renewable&pagecount=&pubyear=&searchin=>

MOOC COURSE:

1. <https://www.coursera.org/learn/renewable-energy-entrepreneurship>
2. <https://www.coursera.org/learn/wind-for-renewable-energies>
3. <https://online-learning.tudelft.nl/courses/sustainable-energy-design-a-renewable-future/>

BASICS OF ROBOTICS

VII Semester: OPEN ELECTIVE-III

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME68	OEC	L	T	P	C	CIE	SEE	Total
		3		-	3	30	70	100

COURSE OVERVIEW:

Today robot finds applications in industries, medical and other fields. For example, in eye surgery (replacement of retina), where a cylindrical portion needs to be replaced, the operation is best done by robots. Mobile robots like walking machines, hopping machines are examples of robots, Nuclear and power plants use fish like robots which move inside pipes for purpose of inspection. This course focuses on various types of industrial robots, their kinematic and kinetic aspects, different types of grippers, mechanics of grippers, trajectory planning etc.

COURSE OBJECTIVES:

The objectives of this course are to

1. Study about basic components, architecture, anatomy and specifications of robot.
2. Study about various end effectors and actuators.
3. Study homogeneous transformation, D-H notation and direct kinematics
4. Study Inverse kinematics and Trajectory planning techniques
5. Study feedback components and applications of robots.

COURSE OUTCOMES:

At the end of the course, the student will be able to

1. Demonstrate different types of robots, specifications of robots and different end effectors used in robots.
2. Explain various types of end effectors
3. Evaluate rotation matrices, forward kinematics of RR, RP and 3R Manipulators.
4. Explain inverse kinematics of RR manipulator, RP manipulator and trajectory planning techniques.
5. Explain feedback components used in robots and industrial applications.

UNIT-I	INTRODUCTION	
Introduction: Automation and Robotics, Asimov's laws, Robot Architecture, Components, Anatomy of robot, Factors to be considered in the selection of robot, present and future applications, Specifications-Degree of freedom, Payload, Parts per hour, Accuracy, Repeatability, Speed, Work space, Work volume, Work envelope, classification of robots based on configuration and control systems		
UNIT-II	END EFFECTORS & ACTUATORS	
End effectors: Mechanical and Non-mechanical grippers, requirements for the design of grippers, considerations for the selection of grippers, Types of actuation mechanisms. Actuators: Pneumatic, Hydraulic actuators, electric & stepper motors, comparison of Actuators		
UNIT-III	MOTION ANALYSIS & DIRECT KINEMATICS	
Motion Analysis: Basic Rotation Matrices, Composite Rotation Matrices. Homogeneous transformations as applicable to rotation and translation – problems. Manipulator Kinematics: D-H notation, D-H method of Assignment of frames, D-H Transformation Matrix, joint coordinates and world coordinates, Forward kinematics of 2R, RP and 3R manipulators		
UNIT-IV	INVERSE KINEMATICS & TRAJECTORY PLANNING	
Inverse kinematics : Inverse kinematics of 2R and RP manipulators. Trajectory Planning: Definition of Trajectory planning, Path, Trajectory, Knot points, Steps involved in trajectory planning, Trajectory planning techniques-Joint space and Cartesian space techniques, Cubic		

polynomial trajectory		
UNIT-V	FEEDBACK COMPONENTS & APPLICATIONS	
Feedback Components: Position sensors – potentiometers, resolvers, optical encoders, Velocity sensor, Contact Sensors-Touch sensors, Tactile and Range sensors, Force and Torque sensors, Proximity sensor, Inductive sensor. Robot Application in Manufacturing: Material Transfer - Material handling, loading and unloading- Processing - spot and continuous arc welding & spray painting - Assembly and Inspection.		
TEXT BOOKS:		
1. Industrial Robotics by Groover M P, Pearson Edu. 2. Robotics by Fu K S, McGraw Hill. 2. 3. Theory of Applied Robotics (kinematics,Dynamics and Control-Jazar, Springer.		
REFERENCE BOOKS:		
1. Robotics and Control by Mittal R K & Nagrath I J, TMH. 2. Robot Dynamics and Controls by Spony and Vidyasagar, John Wiley 3. Robot Analysis and control by Asada and Slotine, Wiley Inter-Science 4. Introduction to Robotics by John J Craig, Pearson Education		
WEB REFERENCES:		
1. https://www.sciencedaily.com/terms/industrial_robot.htm 2. https://www.robotics.org/robotics/industrial-robot-industry-and-all-it-entails		
E-TEXT BOOKS:		
1. https://books.google.co.in/books?id=dr9IAI7wucUC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=rESVUHwMcvYC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false		
MOOC COURSE:		
1. https://www.classcentral.com/tag/robotics 2. https://www.mooc-list.com/tags/robotics		

INTRODUCTION TO MATERIAL HANDLING

VIII Semester: OPEN ELECTIVE-IV

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME69	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OVERVIEW:

Material handling is the movement, protection, storage and control of materials and products throughout manufacturing, warehousing, distribution, consumption and disposal. As a process, material handling incorporates a wide range of manual, semi-automated and automated equipments and systems that support logistics and make the supply chain work. This course will give an insight material handling equipment such as transport systems and storage systems, various inventory models and forecasting techniques.

COURSE OBJECTIVES:

The aim of this course is to enable the students to learn basic concepts of material handling, consideration in material handling system, materials transport and storage system, various inventory models and forecasting techniques.

COURSE OUTCOMES:

At the end of the course students are able to

1. Demonstrate basic concepts and principles of material handling.
2. Discuss about various material transport systems.
3. Explain conventional and automated storage systems.
4. Explain various inventory models.
5. Demonstrate different forecasting techniques.

UNIT-I	INTRODUCTION	
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Introduction: Overview of material handling equipment, considerations in material handling system design, The 10 principles of material handling.

UNIT-II	MATERIAL TRANSPORT SYSTEMS	
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Material Transport Systems: Industrial trucks, Automated guided vehicle systems, Monorails and other rail guided vehicles, conveyor systems, cranes and hoists, Analysis of material transport systems.

UNIT-III	STORAGE SYSTEMS	
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Storage Systems: Storage system performance, storage location strategies, conventional storage methods and equipment, automated storage systems and analysis of automated storage systems.

UNIT-IV	INVENTORY	
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Inventory: Introduction, Inventory management, Static Inventory models, ABC Analysis, VED analysis, Just in Time, Aggregate inventory analysis, Inventory models with quantity discounts.

UNIT-V	FORECASTING	
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Forecasting: Forecasting techniques, Make or Buy decision, Acceptance sampling, Materials requirement planning, objectives of master schedule.

TEXT BOOKS:

1. Mikell P. Groover, Automation Production Systems and Computer Integrated Manufacturing, Pearson International Edition.
2. A.K Singh, Materials Management, Laxmi Publications (P) Ltd.

REFERENCE BOOKS:

1. Dr.K.C Arora, Vikas V.Shinde, Aspects of Material Handling, Laxmi Publications (P) Ltd.
2. Charles Reese, Material Handling Systems, Taylor & Francis Network

WEB REFERENCES:

1. <https://www.scribd.com/doc/222647028/Material-Handling-Full-Notes>
2. [http://users.encs.concordia.ca/~andrea/indu421/Presentation%207%20\(MH\).pdf](http://users.encs.concordia.ca/~andrea/indu421/Presentation%207%20(MH).pdf)

E-TEXT BOOKS:

1. https://books.google.co.in/books?id=AN8dzzzp1wC&printsec=frontcover&source=gb_s_ge_summary_r&cad=0#v=onepage&q&f=false
2. https://books.google.co.in/books?id=d0mhPIDEm0gC&printsec=frontcover&source=gb_s_ge_summary_r&cad=0#v=onepage&q&f=false

MOOC COURSE

1. <https://www.mooc-list.com/course/materials-and-materials-processing-saylororg>
2. <https://www.mooc-list.com/tags/material-processing>

NON CONVENTIONAL ENERGY SOURCES

VIII Semester: OPEN ELECTIVE-IV

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME70	OEC	L	T	P	C	CIE	SEE	Total
		3	0	0				

COURSE OVERVIEW:

The energy has become an important and one of the basic infrastructures for the economic development of the country. It is imperative for the sustained growth of the economy. This course envisages the new and renewable source of energy, available in nature and to expose the students on sources of energy crisis and the alternates available, also stress up on the application of non-conventional energy technologies.

COURSE OBJECTIVES:

To impart the knowledge of basics of different non conventional types of power generation & power plants in detail so that it helps them in understanding the need and role of Non-Conventional Energy sources particularly when the conventional sources are scarce in nature.

COURSE OUTCOMES:

At the end of the course students are able to

1. Demonstrate various energy sources and their availability.
2. Explain concepts and applications of solar radiation
3. Discuss elaboratively the principles of wind energy conversion
4. Explain Biomass conversion techniques and Geothermal Sources and resources
5. Explain about tidal energy and Ocean Thermal Energy.

UNIT- I

ENERGY SOURCES & THEIR AVAILABILITY

Energy Sources & their Availability - Importance of Non Conventional Energy Resources - Classification of NCES - Solar, Wind, Geothermal, Bio-mass, Nuclear Energy & Fuel Cells, Ocean Energy Sources, Comparison of these Energy Sources, Prospects of Renewable Energy Sources - Criteria for Assessing the Potential of NCES, Statistics on Conventional Energy Sources and supply in Developing Countries

UNIT- II

INTRODUCTION TO SOLAR RADIATION & APPLICATIONS OF SOLAR ENERGY

Introduction to Solar radiation - beam and diffuse radiation, solar constant, earth sun angles, attenuation and measurement of solar radiation, local solar time, derived solar angles, sunrise, sunset and day length. Solar Energy Storage – Collectors: Flat plate and Concentrating collectors, collector efficiency, Focusing and non-focusing type

Applications of Solar Energy - solar water heater- Solar Cooker-Box type- Solar dryer - solar greenhouse— Summer and winter greenhouse-solar electric power generation- Solar Photo- Voltaic, Solar Cell Principle, Conversion efficiency and power output, Basic Photo Voltaic System for Power Generation.

UNIT- III

PRINCIPLE OF WIND ENERGY CONVERSION

Principle of Wind Energy Conversion - Basic components of Wind Energy Conversion Systems : Wind data and energy estimation, Site Selection Considerations - Wind mill Components, Various Types and their Constructional Features - Effect of Density, Frequency Variances, Angle of attack, and Wind Speed - Design Considerations of Horizontal and Vertical Axis Wind Machines - Analysis of Aerodynamic Forces Acting on Wind Mill Blades and Estimation of Power Output.

UNIT- IV

BIOMASS CONVERSION TECHNIQUES , GEOTHERMAL SOURCES AND RESOURCES

Biomass conversion techniques - Biogas Generation - Factors affecting biogas Generation-Types of biogas plants - Advantages and disadvantages of biogas plants - site selection, digester design consideration, filling a digester for starting, maintaining biogas production, Fuel properties of bio gas, utilization of biogas - urban waste to energy conversion.

Geothermal Sources and resources like hydrothermal, geo-pressured hot dry rock, magma. advantages, disadvantages and application of geothermal energy, prospects of geothermal energy in India - exhaust types of conventional steam turbines.

UNIT- V	TIDAL ENERGY & OCEAN THERMAL ENERGY CONVERSION	
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Tidal Energy - Principle of working, performance and limitations. Wave Energy - Principle of working, performance and limitations.

Ocean Thermal Energy Conversion - Availability, theory and working principle, performance and limitations. OTEC power plants, Operational of small cycle experimental facility, Economics of OTEC, Environmental impacts of OTEC.

TEXT BOOKS:

1. G. D. Rai, "Non-Conventional Energy Sources", 4th Edition, Khanna Publishers, 2000
2. B H Khan, " Non-Conventional Energy Resources", 2nd Edition, Tata Mc Graw Hill Education Pvt Ltd, 2011

REFERENCE BOOKS:

1. S.Hasan Saeed and D.K.Sharma , "Non-Conventional Energy Resources", 3rd Edition, S.K.Kataria & Sons, 2012
2. Renewable energy sources and conversion technology by N.K. Bansal, M. Kleemann, M. Heliss, Tata McGraw Hill 1990.
3. Ashok V Desai, Non-Conventional Energy, Wiley Eastern Ltd, New Delhi, 2003
4. K M, Non-Conventional Energy Systems, Wheeler Publishing Co. Ltd, New Delhi, 2003
5. S. P. Sukhatme, "Solar Energy Principles and Application", TMH, 2009

WEB REFERENCES:

1. <https://lecturenotes.in/subject/57/non-conventional-energy-systems-nces>
2. <https://www.pdfdrive.com/non-conventional-energy-sources-e10086374.html>
3. http://www.vssut.ac.in/lecture_notes/lecture1428910296.pdf

E-TEXT BOOKS:

1. <http://nptel.ac.in/courses/121106014>
2. <http://nptel.ac.in/courses/121106014/13>
3. <http://nptel.ac.in/courses/12110601427>

MOOC COURSE:

1. <https://nptel.ac.in/courses/121106014/>
2. <https://nptel.ac.in/syllabus/121106014/>
3. <https://nptel.ac.in/downloads/121106014/>

ASPECTS OF HEAT TRANSFER

VIII Semester: OPEN ELECTIVE-IV

Course Code	Category	Hours / Week			Credit s	Maximum Marks		
A4ME22	OEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	30	70	100

COURSE OBJECTIVES:

1. To understand the mechanisms of heat transfer under steady and transient conditions.
2. To understand the concepts of heat transfer through free and forced convection.
3. To learn the thermal analysis and sizing of heat exchangers.
4. To understand the principles of radiation heat transfer.

COURSE OUTCOMES:

1. Demonstrate basic modes of heat transfer and compute temperature distribution in steady state and unsteady state heat conduction.
2. Analyze the heat transfer through variable thermal conductivity, system with heat generation and extended surfaces.
3. Interpret and analyze the free and forced convection heat transfer.
4. Comprehend the concepts of free convection and heat exchangers.
5. Explain the principles of heat transfer with phase change.

UNIT-I INTRODUCTION

Introduction: Modes and mechanisms of heat transfer, Basic laws of heat transfer, Applications of heat transfer. Fourier rate equation, general heat conduction equations in Cartesian, Cylindrical and Spherical coordinates. Different forms of the field equation, steady, unsteady and periodic heat transfer – Initial and boundary conditions.

UNIT-II ONE DIMENSIONAL STEADY STATE HEAT CONDUCTION

One dimensional steady state heat conduction: Homogeneous slabs, hollow cylinders and spheres, Overall heat transfer coefficient, Electrical analogy, Critical radius of insulation. Variable thermal conductivity and Systems with internal heat generation. Extended surfaces (Fins), Long, Short and insulated tips.

UNIT-III CONVECTIVE HEAT TRANSFER

Convective Heat Transfer: Classification of systems based on causation of flow, condition of flow, configuration of flow and medium of flow – Dimensional analysis as a tool for experimental investigation – Buckingham Theorem and method, application for developing semi – empirical non- dimensional correlation for convection heat transfer – Significance of non-dimensional numbers – Concepts of Continuity, Momentum and Energy Equations.

UNIT-IV FREE CONVECTION & HEAT EXCHANGERS

Free Convection: Development of Hydrodynamic and thermal boundary layer along a vertical plate - Use of empirical relations for Vertical plates and pipes.
Heat Exchangers: Classification of heat exchangers – overall heat transfer Coefficient and fouling factor .

UNIT-V RADIATION HEAT TRANSFER

Radiation Heat Transfer : Emission characteristics and laws of black-body radiation – Irradiation – total and monochromatic quantities – laws of Planck, Wien, Kirchoff, Lambert, Stefan and Boltzmann– heat exchange between two black bodies – concepts of shape factor – Emissivity – heat exchange between grey bodies – radiation shields – electrical analogy for radiation networks.

TEXT BOOKS:

1. Yunus A. Cengel (2012), Heat Transfer a Practical Approach, 4th edition, Tata McGraw hill education (P) Ltd, New Delhi, India.
2. R. C. Sachdeva (2012), Fundamentals of Engineering, Heat and Man Transfer, 3rd edition, New Age, New Delhi, India.

REFERENCE BOOKS:

1. Holman (2012), Heat Transfer (SI Units), 10th edition, Tata McGraw hill education (P) Ltd, New Delhi, India.
2. P. S. Ghoshdastidar (2012), Heat Transfer, 2nd edition, Oxford University Press, New Delhi, India.
3. Incropera, Dewitt (2012), Fundamentals of Heat Transfer, 6th edition, John Wiley, UK.

WEB REFERENCES:

1. <http://nptel.ac.in/courses/112104113/>
2. <http://nptel.ac.in/courses/112105123/>
3. [3. https://www.udemy.com/fundamentals-of-engineering-heat transfer](https://www.udemy.com/fundamentals-of-engineering-heat transfer)

E-TEXT BOOKS:

1. <https://www.pdfdrive.net/heat-transfer-e18770201.html>
2. <https://www.pdfdrive.net/fundamentals-of-heat-transfer-e18762062.html>
3. [3. https://www.pdfdrive.net/heat-transfer-e15057935.html](https://www.pdfdrive.net/heat-transfer-e15057935.html)

MOOC COURSE

1. <https://onlinecourses.nptel.ac.in/>
2. <https://www.coursera.org/learn/heat-transfer-intro>
3. [3. https://swayam.gov.in/courses/4642-july-2018-conduction_and_convection](https://swayam.gov.in/courses/4642-july-2018-conduction_and_convection)

**OPEN ELECTIVES
OFFERED BY
HSM DEPARTMENT**

BUSINESS ECONOMICS AND FINANCIAL ANALYSIS								
OPEN ELECTIVE-I								
Course Code	Category	Hours / Week			Credit s	Maximum Marks		
A4HS08	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100
COURSE OBJECTIVES: To enable the student to understand and appreciate, with a particular insight, the importance of certain basic issues governing the business operations namely; demand and supply, production function, cost analysis, markets, forms of business organizations, capital budgeting and financial accounting and financial analysis.								
COURSE OUTCOMES: At the end of the course, the student will								
1. Understand the market dynamics namely, demand and supply, demand forecasting, elasticity of demand and supply, pricing methods and pricing in different market structures. 2. Gain an insight into how production function is carried out to achieve least cost combination of inputs and cost analysis. 3. Develop an understanding of 4. Analyse how capital budgeting decisions are carried out. 5. Understanding the framework for both manual and computerised accounting process 6. Know how to analyse and interpret the financial statements through ratio analysis.								
UNIT-I								
Introduction & Demand Analysis: Definition, Nature and Scope of Managerial Economics. Demand Analysis: Demand Determinants, Law of Demand and its exceptions. Elasticity of Demand: Definition, Types, Measurement and Significance of Elasticity of Demand. Demand Forecasting, Factors governing demand forecasting, methods of demand forecasting.								
UNIT-II								
Production & Cost Analysis: Production Function - Isoquants and Isocosts, MRTS, Least Cost Combination of Inputs, Cobb-Douglas Production function, Laws of Returns, Internal and External Economies of Scale. Cost Analysis: Cost concepts. Break-even Analysis (BEA)-Determination of Break-Even Point (simple problems) - Managerial Significance.								
UNIT-III								
Markets & New Economic Environment: Types of competition and Markets, Features of Perfect competition, Monopoly and Monopolistic Competition. Price-Output Determination in case of Perfect Competition and Monopoly. Pricing: Objectives and Policies of Pricing. Methods of Pricing. Business: Features and evaluation of different forms of Business Organisation: Sole Proprietorship, Partnership, Joint Stock Company, Public Enterprises and their types, New Economic Environment: Changing Business Environment in Post-liberalization scenario.								
UNIT-IV								
Capital Budgeting: Capital and its significance, Types of Capital, Estimation of Fixed and Working capital requirements, Methods and sources of raising capital - Trading Forecast, Capital Budget, Cash Budget. Capital Budgeting: features of capital budgeting proposals, Methods of Capital Budgeting: Payback Method, Accounting Rate of return (ARR) and Net Present Value Method (simple problems).								
UNIT-V								
Introduction to Financial Accounting & Financial Analysis: Accounting concepts and Conventions - Introduction IFRS - Double - Entry Book Keeping, Journal, Ledger, Trial Balance - Final Accounts (Trasing Account, Profit and Loss Account and Balance Sheet with simple adjustments). Financial Analysis: Analysis and Interpretation of Liquidity Ratios, Activity Ratios, and Capital structure Ratios and Profitability ratios. Du								

Pont Chart.

TEXT BOOKS:

1. Varshney & Maheswari: Managerial Economics, Sultan Chand, 2009.
2. S.A. Siddiqui & A.S. Siddiqui, Managerial Economics and Financial Analysis, New Age international Publishers, Hyderabad 2013.
3. M. Kasi Reddy & Saraswathi, Managerial Economics and Financial Analysis, PHI New Delhi, 2012.

REFERENCE BOOKS:

1. Ambrish Gupta, Financial Accounting for Management, Pearson Education, New Delhi, 2012.
2. H. Craig Peterson & W. Cris Lewis, Managerial Economics, Pearson, 2012.
3. Lipsey & Chrystel, Economics, Oxford University Press, 2012.
4. Domnick Salvatore: Managerial Economics In a Global Economy, Thomson, 2012.
5. Narayanaswamy: Financial Accounting - A Managerial Perspective, Pearson, 2012.
6. S.N. Maheswari & S.K. Maheswari, Financial Accounting, Vikas, 2012.
7. Truet and Truet: Managerial Economics: Analysis, Problems and Cases, Wiley, 2012.
8. Dwivedi: Managerial Economics, Vikas, 2012.
9. Shailaja & Usha: MEFA, University Press, 2012.
10. Aryasri: Managerial Economics and Financial Analysis, TMH, 2012.
11. Vijay Kumar & Appa Rao, Managerial Economics & Financial Analysis, Cengage 2011.
12. J.V. Prabhakar Rao & P.V. Rao, Managerial Economics & Financial Analysis, Maruthi Publishers, 2011.

BASIC OF ENTREPRENEURSHIP OPEN ELECTIVE-I

V Semester: B.Tech/MBA

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A2HS51	Foundation	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

OBJECTIVES:

The course should enable the students to:

The curriculum helps students

1. Understand and **discover** entrepreneurship
2. Build a strong foundation for students to **start, build, and grow** a viable and sustainable venture
3. Develop an entrepreneurial outlook and **mindset, critical skills and knowledge**
4. Mitigate three types of **risks**: Customer, Business Model, and Product/Technical

UNIT-I	DISCOVER YOURSELF AND IDENTIFY PROBLEMS WORTH SOLVING	
Discover Yourself: Find your flow, Effectuation, Case Study: Tristan Walker: The extroverted introvert, Identify your entrepreneurial style, Identify Problems Worth Solving: What is a business opportunity and how to identify it, Find problems around you that are worth solving, Methods for finding and understanding problems - (Observation, Questioning, DT, Jobs to be done (JTBD), How to run problem interviews to understand the customer's worldview, Introduction to Design Thinking – Process and Examples, Generate ideas that are potential solutions to the problem identified – DISRUPT, Class Presentation: Present the problem you "love"		
UNIT-II	CUSTOMER, BUSINESS MODEL, VALIDATION	
CUSTOMER :Identify Your Customer Segments and Early Adopters - The difference between a consumer and a customer (decision maker), Market Types, Segmentation and Targeting, Defining the personas; Understanding Early Adopters and Customer Adoption Patterns, Identify the innovators and early adopters for your startup; Craft Your Value Proposition - Come up with creative solutions for the identified problems, Deep dive into Gains, Pains and “Jobs-To- Be-Done” (using Value Proposition Canvas, or VPC), Identify the UVP of your solution using the Value Proposition section of the VPC, Outcome-Driven Innovation, Class Presentation: Communicating the Value Proposition- 1 min Customer Pitch . BUSINESS MODEL: Get Started with Lean Canvas. VALIDATION: Develop the Solution, Sizing the Opportunity, Building an MVP.		
UNIT-III	MONEY AND TEAM	
MONEY: Revenue Streams - Basics of how companies make money, Understand income, costs, gross and net margins, Identify primary and secondary revenue, streams ; Pricing and Costs - Value, price, and costs; Different pricing, Understand product costs and operations costs; Basics of unit costing strategies; Financing Your New Venture - How to finance business ideas, Various sources of funds available to an entrepreneur and pros and cons of each, What investors expect from you, Practice Pitching to Investors and Corporates. TEAM: Team Building - Shared Leadership, Role of a good team in a venture's success; What to look for in a team; How do you ensure there is a good fit? Defining clear roles and responsibilities, How to pitch to candidates to join your startup, Explore collaboration tools and techniques - Brainstorming, Mind mapping, Kanban Board, Slack.		
UNIT-IV	MARKETING & SALES	
MARKETING & SALES: Positioning - Understand the difference between product and brand and the linkbetween them,•Define the positioning statement for your product/service and how it should translate into what your customers should see about that brand in the market place. Channels & Strategy : • Building Digital Presenceand leveraging Social media,Creating your company		

profile page, Measuring the effectiveness of selected channels, Budgeting and planning.		
UNIT-V	PLANNING & TRACKING	
Sales Planning : Understanding why customers buy and how buying decisions are made; Listening, Sales planning, setting targets, •Unique Sales Proposition (USP); Art of the sales pitch (focus on customers needs, not on product features, Follow-up and closing a sale; Asking for the sale Planning & Tracking : Importance of project management to launch and track progress, Understanding time management, workflow, and delegation of tasks. Business Regulation : Basics of business regulations of starting and operating a business; Importance of being compliant and keeping proper documentation, How to find help to get started.		
TEXT BOOKS:		
1. S. R. Bhowmik & M. Bhowmik, Entrepreneurship, New Age International, 2007. 2. Steven Fisher, Ja-nae' Duane, The Startup Equation -A Visual Guidebook for Building Your Startup, Indian Edition, Mc Graw Hill Education India Pvt. Ltd, 2016. 3. D F Kuratko and T V Rao —Entrepreneurship-A South-Asian Perspective —Cengage Learning, 2012. (For PPT, Case Solutions Faculty may visit : login.cengage.com)		
REFERENCE BOOKS:		
1. Vasant Desai —Small Scale industries and entrepreneurship Himalaya publishing 2012. 2. Rajeev Roy —Entrepreneurship 2e, Oxford, 2012. 3. B. Janakiram and M. Rizwan Entrepreneurship Development: Text & Cases, Excel Books, 2011. 4. Stuart Read, Effectual Entrepreneurship, Routledge, 2013. 5. Nandan H, Fundamentals of Entrepreneurship, PHI, 2013		
E-RESOURCES		
1. http://freevideolectures.com/Course/3641/Entrepreneurship-Through-the-Lens-of-Venture-Capital 2. http://www.onlinevideolecture.com/?course=mba-programs&subject=entrepreneurship 3. http://nptel.ac.in/courses/122106032/Pdf/7_4.pdf 4. https://www.scribd.com/doc/21516826/Entrepreneurship-Notes 5. http://freevideolectures.com/Course/3514/Economics-/-Management-/-Entrepreneurship/50 6. Journal of Entrepreneurship & Organization Management, Vikalpa, IIMA, IIMB Review, Decision, IIMC, Vision, HBR.		

HUMAN VALUES AND PROFESSIONAL ETHICS OPEN ELECTIVE-I

Course Code	Category	Hours / Week			Credit s	Maximum Marks		
A4HS52	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

1. To help the students appreciate the essential complementarity between 'VALUES' and 'SKILLS' to ensure sustained happiness and prosperity which are the core aspirations of all human beings.
2. To facilitate the development of a Holistic perspective among students towards life, profession and happiness, based on a correct understanding of the Human reality and the rest of Existence. Such a holistic perspective forms the basis of Value based living in a natural way.
3. To highlight plausible implications of such a Holistic understanding in terms of ethical human conduct, trustful and mutually satisfying human behaviour and mutually enriching interaction with Nature.

COURSE OUTCOMES:

1. It ensures students sustained happiness through identifying the essentials of human values and skills.
2. It facilitates a correct understanding between profession and happiness
3. It helps students understand practically the importance of trust, mutually satisfying human behavior and enriching interaction with nature.
4. Ability to develop appropriate technologies and management patterns to create harmony in professional and personal life.

UNIT-I

Course Introduction - Need, basic Guidelines, Content and Process for Value Education: Understanding the need, basic guidelines, content and process for Value Education. Self Exploration - what is it? - its content and process; 'Natural Acceptance' and Experiential Validation - as the mechanism for self exploration. Continuous Happiness and Prosperity - A look at basic Human Aspirations. Right understanding, Relationship and Physical Facilities - the basic requirements for fulfillment of aspirations of every human being with their correct priority. Understanding Happiness and Prosperity correctly - A critical appraisal of the current scenario. Method to fulfill the above human aspirations: understanding and living in harmony at various levels.

UNIT-II

Understanding Harmony in the Human Being - Harmony in Myself! : Understanding human being as a co-existence of the sentient 'I' and the material 'Body'. Understanding the needs of Self ('I') and 'Body' - Sukh and Suvidha. Understanding the Body as an instrument of 'I' (I being the doer, seer and enjoyer). Understanding the harmony of I with the Body: Sanyam and Swasthya; correct appraisal of Physical needs, meaning of Prosperity in detail. Programs to ensure Sanyam and Swasthya.

UNIT-III

Understanding Harmony in the Family and Society - Harmony in Human - Human Relationship: Understanding harmony in the Family the basic unit of human interaction. Understanding values in human - human relationship; meaning of Nyaya and program for its fulfillment to ensure Ubhay-tripti; **Trust (Vishwas) and Respect (Samman) as the foundational values of relationship.** Understanding the meaning of Vishwas; Difference between intention and competence. Understanding the meaning of Samman, Difference between respect and differentiation; the other salient values in relationship. Understanding the harmony in the society (society being an extension of family): Samadhan, Samridhi, Abhay, Sah-astiva as comprehensive Human Goals. Visualizing a universal harmonious order in society - Undivided Society (Akhand Samaj), Universal Order (Sarvabhaum Vyawastha) - from family to world family!

UNIT-IV

Understanding Harmony in the nature and Existence - Whole existence as Co-existence: Understanding the harmony in the Nature. Interconnectedness and mutual fulfillment among the four orders of nature - recyclability and self-regulation in nature. Understanding Existence as Co-existence (Sah-astiva) of mutually interacting units in all-pervasive space. Holistic perception of harmony at all levels of existence.

UNIT-V

Implications of the above Holistic Understanding of Harmony on Professional Ethics: Natural acceptance of human values, Definitiveness of Ethical Human Conduct, Basic for Humanistic Education, Humanistic Constitution and Humanistic Universal Order. Competence in professional ethics:
 Ability to utilize the professional competence for augmenting universal human order,
 Ability to identify the scope and characteristics of people-friendly and eco-friendly production systems,
 Ability to identify and develop appropriate technologies and management patterns for above production systems.
 Case studies of typical holistic technologies, management models and production systems. Strategy for transition from the present state to Universal Human Order.
 At the level of individual: as socially and ecologically responsible engineers, technologists and managers
 At the level of society: as mutually enriching institutions and organizations.

TEXT BOOKS:

1. R. R. Gaur, R Sangal, G P Bagaria, 2009, A Foundation Course in Human Values and Professional Ethics.
2. Prof. K. V. Subba Raju, 2013, Success Secrets for Engineering Students, Smart Student Publications, 3rd Edition.

REFERENCE BOOKS:

1. Ivan Illich, 1974, Energy & Equity, The Trinity Press, Worcester, and HarperCollins, USA
2. E. F. Schumacher, 1973, Small is Beautiful: a study of economics as if people mattered. Blond & Briggs, Britain.
3. A Nagraj, 1998 Jeevan Vidya ek Parichay, Divya Path Sansthan, Amarkantak.
4. Sussan George, 1976, How the Other Half Dies, Penguin Press, Reprinted 1986, 1991.
5. P. L. Dhar, R. R. Gaur, 1990, Science and Humanism, Commonwealth Publishers.
6. A. N. Tripathy, 2003, Human Values, New Age International Publishers.
7. Subhas Palekar, 2000, How to practice Natural Farming, Pracheen(Vaidik) Krishi Tantra Shodh, Amravati.
8. Donella H. Meadows, Dennis L. Meadows, Jorgen Randers, William W. Behrens III, 1972, Limits to Growth - Club of Rome's report, Universe Books.
9. E G Seebauer & Robert L. Berry, 2000, Fundamentals of Ethics for Scientists & Engineers, Oxford University Press.
10. M Govindrajan, S Natrajan & V. S Senthil kumar, Engineering Ethics (including Humna Values), Eastern Economy Edition, Prentice Hall of India Ltd.

RELEVANT CDS, MOVIES, DOCUMENTARIES & OTHER LITERATURE:

1. value Education website, <http://www.uptu.ac.in>
2. Story of Stuff, <http://www.storyofstuff.com>
3. Al Gore, An Inconvenient Truth, Paramount Classics, USA
4. Charle Chaplin, Modern Times, United Artists, USA
5. IIT Delhi, Modern Technology - the Untold Story

ADVANCED ENTREPRENEURSHIP OPEN ELECTIVE-2

VI Semester: B.Tech/MBA

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A2HS53	Foundation	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

OBJECTIVES:

The course should enable the students to:

The curriculum helps students

1. Understand and **discover** entrepreneurship
2. Build a strong foundation for students to **start, build, and grow** a viable and sustainable venture
3. Develop an entrepreneurial outlook and **mindset**, critical **skills** and **knowledge**
4. Mitigate three types of **risks**: Customer, Business Model, and Product/Technical

UNIT-I ORIENTATION TO GROWTH

ORIENTATION TO GROWTH : Getting Ready for Growth-Why growth stage is different compared to startup phase, Why Product-Market fit is not enough, Case study, To assess readiness for growth, To chart a growth path.

UNIT-II CUSTOMER, BUSINESS MODEL, VALIDATION

CUSTOMERS : Expanding Customer Base-Revisit your business model and develop few variants (more business model types), Identify additional customer segments that your solution can address, Evaluate business models for the new customer segments, Relook at the Problem Statement (can you expand the scope and scalability of your business by repositioning your problem statement?), Explore additional ways to monetize.

BUSINESS MODEL: Get Started with Lean Canvas.

VALIDATION: Develop the Solution, Sizing the Opportunity, Building an MVP.

UNIT-III TRACTION AND TEAM

TRACTION : Scaling - How to gain traction beyond early customers, Defining traction (in quantifiable terms) and identifying the most important metrics to measure traction, Calculate cost of new customer acquisition, Estimate your customer lifetime value (LTV), Identifying waste in your operations and focusing your team on what is important for traction.

Channels and Strategy - The Bullseye framework, Identify Channels using Bulls Eye Framework, Measuring the effectiveness of selected channels, Budgeting and planning.

UNIT-IV MONEY & SALES

MONEY : Growing Revenues - Stabilizing key revenue streams, Developing additional revenue streams (licensing, franchising), Exploring new channels and partnerships, Sales Planning - Understanding why customers buy and how buying decisions are made; Listening skills, Sales planning, setting targets, Unique Sales Proposition (USP); Art of the sales pitch (focus on customer's needs, not on product features, Follow-up and closing a sale; Asking for the sale.

Strengthening Sales - Follow-up and closing a sale; Asking for the sale, Building a professional sales team, Sales compensation and incentives, Sales planning, setting targets.

Improving Margins - Testing price elasticity, Optimizing costs and operational expenses,

Advanced concepts of unit costing. Financial Modeling - Financial modeling of your venture's growth, Analyzing competitor and peer's financial models.

UNIT-V SUPPORT

SUPPORT: Legal - Overview of legal issues and their impact on entrepreneurs, Importance of getting professional help (legal and accounting), Importance of being compliant and keeping proper documentation, Patents and Intellectual property, Trademarks.

Mentors, Advisors, and Experts - The importance of a Mentor and how to find one, Role of business

advisors and experts for specific targets in your growth plan

TEXT BOOKS:

1. Bygrave, W., & Zacharakis, A. (2017) Entrepreneurship, 4th Edition (3rd Edition is ok too) Wiley.
2. [Carree, M. A., Thurik, A. R. "The impact of entrepreneurship on economic growth" In: Audretsch, D. B., Acs, Z. J. \(eds\). Handbook of Entrepreneurship Research. Berlin: Springer Verlag, 2010.](#)

REFERENCE BOOKS:

1. Eric, Reis (2017) The Startup Way: How Entrepreneurial Management Transforms Culture and Drives Growth
2. Thurik, A. R., Audretsch, D. B., Stam, E. "The rise of the entrepreneurial economy and the future of dynamic capitalism" *Technovation*
3. [Audretsch, D. B., Grilo, I., Thurik, A. R. \(eds\). The Handbook of Research on Entrepreneurship Policy. Cheltenham, UK: Edward Elgar, 2007.](#)

E RESOURCES:

1. <https://www.ediindia.org/EDILibrary.aspx>
2. <https://www.entrepreneurshipsecret.com/sources-of-information-for-entrepreneurship-development-contd>
Key reference: [3]

INDIAN ETHOS AND BUSINESS ETHICS

OPEN ELECTIVE-3

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A2HS54	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

OBJECTIVES:

1. To understand the importance of ethics in business
2. To acquire knowledge and capability to develop ethical practices for effective management
3. To understand the Business Ethics and to provide best practices of business ethics.
4. To learn the values and implement in their careers to become a good managers.
5. To develop various ethical Responsibilities and practice in their professional life
6. To imbibe the ethical issues and to adhere to the ethical codes.

COURSE OUTCOMES:

1. Understand the dynamics of morality
2. Identify the constant in morality
3. Recognize the variable values in morality
4. Students will be able to understand the business ethics.
5. The student will be able to analyze various ethical codes in corporate governance
6. Student will be able to analyze the Employees conditions and Business Ethics

UNIT-I

Introduction to Indian Ethos: History & Relevance, Principles Practiced by Indian Companies, Role of Indian Ethos in Managerial Practices, Management Lessons from Vedas, Mahabharata, Bible and Quran.

UNIT-II

Understanding Values in Business: Kautilya's Arthashastra, Indian Heritage in Business, Management-Production and Consumption. Ethics v/s Ethos , Indian v/s Western Management, Work Ethos and Values for Indian Managers- Relevance of Value Based Management in Global Change- Impact of Values on Stakeholders, Trans-Cultural Human Values, Secular v/s Spiritual Values , Value System in Work Culture, Stress Management-Meditation for mental health, Yoga.

UNIT-III

Contemporary Approaches to Indian Ethos: Contemporary Approaches to Leadership- Joint Hindu Family Business-Leadership Qualities of Karta, Indian Systems of Learning-Gurukul System of Learning, Advantages- Disadvantages of Karma, importance of Karma to Managers-Nishkama Karma-Laws of Karma, Law of Creation- Law of Humility- Law of Growth- Law of Responsibility- Law of Connection- Corporate Karma Leadership.

UNIT-IV

Understanding the Business Ethics : Understanding the need for ethics, Ethical values, myths and ambiguity, ethical codes, Ethical Principles in Business; Theories of Ethics, Absolutism verses Relativism, Teleological approach, the Deontological approach, Kohlberg's six stages of moral development (CMD).

UNIT-V

Ethical Culture in Organization: Ethical Culture in Organization, Developing codes of Ethics and conduct, Ethical and value based leadership. Role of scriptures in understanding ethics, Indian wisdom & Indian approaches towards business ethics.

TEXT BOOKS

1. M.G. Velasquez, Business Ethics, Prentice Hall India Limited, New Delhi,
2. R.C. Sekhar, Ethical Choices in Business, Response Books, New Delhi, 2007

REFERENCE BOOKS:

1. Chakraborty S.K., —Management Transformation by Values, New Delhi, Sage Publication, 1990.
2. Chakraborty, S.K., Ethics in Management-Vedantic Approach, New Delhi, Oxford India Ltd. 1995.
3. Fernando A.C., Business Ethics: An Indian Perspective, Pearson, 2009.
4. Kautilya's Arthashastra, King, Governance, and Law in Ancient India, Oxford University Press, 2016.
5. Murthy, C.S.R. Business Ethics, Himalaya Publishing House, Mumbai, 2009.
6. Narayana G., —The Responsible Leader: A Journey through Gitall, Ahmedabad, AMA 2000.

E RESOURCES

1. <http://sib.co.uk/lms/wp-content/uploads/2015/02/Indian-Ethos-and-Management.pdf>
2. www.vikaspublishing.com/books/business-economics/management/ethics

MANAGEMENT SCIENCE OPEN ELECTIVE-4

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A2HS55	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

Course Objectives:

1. Familiarize & obtain Knowledge with the process of management and to provide basic insights into management practices.
2. Understand the structure & Designing of an Organization.
3. Knowledge on the aspects of Production.
4. Analyze the market and the strategies involved in Marketing.
5. Knowledge on concepts related to Human Resources.
6. Understand the techniques used in Project Management.
7. Familiarize with strategies used for analysis of an Organization.
8. Understand the Contemporary Management Issues.
9. Familiarize with the management skills which can be applied in the Organizational context to achieve Organizational goals.

Course Outcomes:

1. Knowledge on management theories and practices.
2. Understanding designing organizational structure.
3. Understanding on the methods & charts used in operations management.
4. Ability to understand the market and its environment.
5. Understand the processes, functions etc in Human Resources Management.
6. Ability to solve problems in managing the Project.
7. Knowledge on Strategic alternatives.
8. Familiar with the practices implemented in management.
9. Understand the social responsibilities of Management.
10. Understand the basic concepts of Management.

UNIT-I

INTRODUCTION TO MANAGEMENT AND ORGANIZATION: Concepts of Management and organization- nature, importance and Functions of Management, Systems Approach to Management- Taylor's Scientific Management Theory- Fayal's Principles of Management- Maslow's theory of Hierarchy of Human Needs- Douglas McGregor's Theory X and Theory Y - Herzberg Two Factor Theory of Motivation - Leadership Styles, Social responsibilities of Management, Designing Organizational Structures: Basic concepts related to Organization - Departmentation and Decentralization, Types and Evaluation of mechanistic and organic structures of organization and suitability.

UNIT-II

OPERATIONS AND MARKETING MANAGEMENT: Principles and Types of Plant Layout-Methods of Production(Job, batch and Mass Production), Work Study - Basic procedure involved in Method Study and Work Measurement - Business Process Reengineering(BPR) - Statistical Quality Control: control charts for Variables and Attributes (simple Problems) and Acceptance Sampling, TQM, Six Sigma, Deming's contribution to quality, Objectives of Inventory control, EOQ, ABC Analysis, Purchase Procedure, Stores Management and Store Records - JIT System, Supply Chain Management, Functions of Marketing, Marketing Mix, and Marketing Strategies based on Product Life Cycle, Channels of distribution.

UNIT-III

HUMAN RESOURCES MANAGEMENT(HRM): Concepts of HRM, HRD and Personnel Management and Industrial Relations (PMIR), HRM vs PMIR, Basic functions of HR Manager: Manpower planning, Recruitment, Selection, Training and Development, Placement, Wage and Salary Administration, Promotion, Transfer, Separation, Performance Appraisal, Grievance Handling and Welfare Administration, Job Evaluation and Merit Rating - Capability Maturity Model (CMM) Levels - Performance Management System.		
UNIT-IV		
PROJECT MANAGEMENT (PERT/ CPM): Network Analysis, Programme Evaluation and Review Technique (PERT), Critical Path Method (CPM), Identifying critical path, Probability of Completing the project within given time, Project Cost Analysis, Project Crashing (simple problems).		
UNIT-V		
STRATEGIC MANAGEMENT AND CONTEMPORARY STRATEGIC ISSUES: Mission, Goals, Objectives, Policy, Strategy, Programmes, Elements of Corporate Planning Process, Environmental Scanning, Value Chain Analysis, SWOT Analysis, Steps in Strategy Formulation and Implementation, Generic Strategy alternatives. Bench Marking and Balanced Score Card as Contemporary Business Strategies.		
TEXT BOOKS		
1. A.R.Aryasri : Management Science, TMH, (Latest edition) 2. Stoner, Freeman, Gilbert, Management, 6th Ed, Pearson Education, New Delhi, 2004.		
REFERENCE BOOKS:		
1. Kotler Philip and Keller Kevin Lane: Marketing Management, Pearson, 2012. 2. Koontz and Weihrich: Essentials of Management, McGraw Hill, 2012. 3. Thomas N. Duening and John M. Ivancevich Management - Principles and Guidelines, Biztantra, 2012. 4. Kanishka Bedi, Production and Operations Management, Oxford University Press, 2012. 5. Samuel C. Certo: Modern Management, 2012. 6. Schermerhorn, Capling, Poole and Wiesner: Management, Wiley, 2012. 7. Parnell: Strategic Management, Cengage, 2012. 8. Lawrence R Jauch, R. Gupta and William F. Glueck: Business Policy and Strategic Management Science, McGraw Hill, 2012.		
E RESOURCES		
1. https://www.smartworld.com/notes/management-science-pdf-notes-ms-notes-pdf 2. https://www.alljntuworld.in/download/management-science-notes/		

INTELLECTUAL PROPERTY RIGHTS OPEN ELECTIVE-4

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A2HS56	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

1. To introduce fundamental aspects of Intellectual property Rights to students who are going to play a major role in development and management of innovative projects in industries.
2. To disseminate knowledge on patents, patent regime in India and abroad and registration aspects
3. To disseminate knowledge on copyrights and its related rights and registration aspects
4. To aware about current trends in IPR and Govt. steps in fostering IPR
5. To familiarize them with the kind of rights, remedies and licensing regime associated with each kind of intellectual property so that students can have a basic understanding of Intellectual Property laws.

COURSE OUTCOMES:

1. Skill to understand the concept of intellectual property rights.
2. Develops procedural knowledge to Legal System and solving the problem relating to intellectual property rights.
3. Skill to pursue the professional programs in Company Secretaryship, Law, Business(MBA), International Affairs, Public Administration and Other fields.
4. Employability as the Compliance Officer, Public Relation Officer and Liaison Officer.
5. Establishment of Legal Consultancy and service provider.

UNIT-I

Introduction to Intellectual property: Introduction, types of intellectual property, international organizations, agencies and treaties, importance of intellectual property rights.

UNIT-II

Trade Marks: Purpose and function of trademarks, acquisition of trade mark rights, protectable matter, selecting and evaluating trade mark, trade mark registration processes.

UNIT-III

Law of copy rights: Fundamental of copy right law, originality of material, rights of reproduction, rights to perform the work publicly, copy right ownership issues, copy right registration, notice of copy right, international copy right law.

Law of patents: Foundation of patent law, patent searching process, ownership rights and transfer.

UNIT-IV

Trade Secrets: Trade secrete law, determination of trade secrete status, liability for misappropriations of trade secrets, protection for submission, trade secrete litigation.

Unfair competition: Misappropriation right of publicity, False advertising.

UNIT-V

New development of intellectual property: new developments in trade mark law; copy right law, patent law, intellectual property audits.

International overview on intellectual property, international - trade mark law, copy right law, international patent law, international development in trade secrets law.

TEXT BOOKS

1. Intellectual property right, Deborah, E. Bouchoux, cengage learning.
2. Intellectual property right - Unleashing the knowledge economy, prabuddha ganguli, Tata McGraw Hill Publishing Company Ltd.

REFERENCE BOOKS:

1. Deborah. E. Bouchoux (2009), Intellectual property, Cengage learning, India.
2. Deborah. E. Bouchoux (2001), Protecting your companies intellectual property, AMACOM, USA.
3. Prabuddaganguli (2003), Intellectual property right, Tata McGraw Hill Publishing company Ltd., India.
4. Robert Hisrich, Michael P. PETER, Dean A. Shepherd (201), Entrepreneurship, Tata McGraw Hill., India

E RESOURCES

1. Subramanian, N., & Sundararaman, M. (2018). Intellectual Property Rights – An Overview. Retrieved from <http://www.bdu.ac.in/cells/ipr/docs/ipr-eng-ebook.pdf>
2. World Intellectual Property Organisation. (2004). WIPO Intellectual property Handbook. Retrieved from https://www.wipo.int/edocs/pubdocs/en/intproperty/489/wipo_pub_489.pdf
3. Reference Journal: 1. Journal of Intellectual Property Rights (JIPR): NISCAIR

NUMBER THEORY

IV B. Tech II Semester: OPEN ELECTIVE - IV

Course Code:	Category	Hours / Week			Credits	Maximum Marks		
A4HS57	HSMC	L	T	P	C	CIA	SEE	Total
		3	0	-	3	30	70	100
Contact Classes: 42	Tutorial Classes: 00	Practical Classes:--				Total Classes: 42		
COURSE OBJECTIVES: To learn 1. Divisibility test and Euclidian Algorithms. 2. The Arithmetic function and Dirichlet multiplication. 3. The properties of congruences. 4. application of Lagrange's theorem. 5. Quadratic Residues and Reciprocity Law.								
COURSE OUTCOMES: Upon successful completion of the course, the student is able to 1. Apply the Fundamental theorem of Arithmetic.. 2. Identify the Liouville's function and divisor function. 3. Apply the Euler Fermat theorem in polynomial congruence modulo P 4. Apply Chinese remainder theorem. 5. Find Quadratic residues.								
UNIT-I								
Divisibility, GCD, Prime Numbers, Fundamental theorem of Arithmetic, the series of reciprocal of the Primes, The Euclidean Algorithm.								
UNIT-II								
A Arithmetic function and Dirichlet Multiplication, The functions $\varphi(n)$, $\mu(n)$ and a relation connecting them, Product formulae for $\varphi(n)$, Dirichlet Product, Dirichlet inverse and Mobius inversion formula and Mangoldt function $\Lambda(n)$, multiplication function, multiplication function and Dirichlet multiplication, Inverse of a completely multiplication function, Liouville's function $\lambda(n)$, the divisor function is $\sigma_{\alpha}(n)$								
UNIT-III								
Congruences, Properties of congruences, Residue Classes and complete residue system, linear congruences conversion, reduced residue system and Euler Fermat theorem, polynomial congruence modulo P								
UNIT-IV								
Lagrange's theorem, Application of Lagrange's theorem, Chinese remainder theorem and its application, polynomial congruences with prime power module.								
UNIT-V								
Quadratic residue and quadratic reciprocity law, Quadratic residues, Legendre's symbol and its properties, evaluation of $(-1/p)$ and $(2/p)$, Gauss Lemma, the quadratic reciprocity law and its applications								
TEXT BOOKS:								

-
- | |
|--|
| 1. “ Introduction to analytic Number Theory “by Tom M. Apostol. Chapters 1, 2, 5, 9. |
|--|

REFERENCE BOOKS:

- | |
|---|
| <ol style="list-style-type: none">1. Number Theory by Joseph H.Silverman.2. Theory of Numbers byK.Ramchandra.3. Elementary Number Theory by James KStrayer.4. Elementary Number Theory by James Tattusall. |
|---|

PHYSICS AND TECHNOLOGY OF THIN FILMS								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4HS58	HSMC	L	T	P	C	CIA	SEE	Total
		3	0	0	3	30	70	100
Contact Classes: 42	Tutorial Classes: NIL	Practical Classes: NIL			Total Classes: 42			
OBJECTIVES: The course should enable the students to: 1. Understand the bonding, nucleation process in materials. 2. Gain the knowledge on fabrication of thin films. 3. Learn the kinetics of different thin films. 4. Understand the properties of thin films by using characterization methods. 5. Learn the applications of thin films in different fields.								
OUTCOMES: The student will able to: 1. Analyze the defects in solids and kinetics. 2. Evaluate the epitaxy of thin films and growth process. 3. Justify how the growth kinetics is more efficient than nucleation kinetics in thin film technology. 4. Recommend appropriate synthesis method and explain the characterization techniques. 5. Analyze which thin films are used in different applications.								
UNIT-I	A Review of Materials Science							
Introduction, Structure Defects in Solids Bonding of Materials, Thermodynamics of Materials, Kinetics, Nucleation and Conclusion.								
UNIT-II	Preparation of Thin-films:							
Classifications of vacuum ranges – Vacuum pumps - Rotary, Diffusion, Turbo molecular and Ion Pumps –Thin film (epitaxy) – definition & advantages – Types of epitaxy. Different Growth Techniques: Liquid Phase Epitaxy, Molecular Beam Epitaxy, Metal Organic Vapour Phase Epitaxy, Sputtering (RF & DC), Pulsed Laser Deposition. Thickness Measurement: Microbalance technique, Photometry, Interferometry (MBI, FECO)								
UNIT-III	Kinetics of Thin films							
Nucleation Kinetics: types of nucleation – kinetic theory of nucleation – energy formation of a nucleus – critical nucleation parameters; spherical and non-spherical nucleus (cap, disc and cubic shaped)on the substrates. Growth Kinetics: Kinetics of binary (GaAs, InP, etc.), ternary (Al _{1-x} Ga _x As, Ga _{1-x} In _x P, etc.) and quaternary (Ga _{1-x} In _x As _{1-y} P _y , etc.) semiconductors – derivation of growth rate and composition expressions.								
UNIT-IV	Properties & Characterization of Thin Films							
Properties of Thin Films: Dielectric properties – Important parameters, Measurement of dielectric properties- Effect of annealing and film thickness. Optical properties – Optical constants, determination of optical constants by Brewster angle method, Normal incidence method and graphical method. Mechanical properties – Concept and origin of stress and strain, Lattice misfit, Thermal misfit, Hardness test and Bulge test. Characterization X-ray diffraction –Photoluminescence –UV-Vis-IR spectrophotometer – Atomic Force Microscope – Scanning Electron Microscope – Hall Effect – Vibrational Sample Magnetometer – Secondary Ion MassSpectrometry.								
UNIT-V	Applications of Thin Films							
Optoelectronic devices: LED, LASER and Solar cell – Micro Electromechanical Systems (MEMS) –								

Fabrication of thin film capacitor – application of ferromagnetic thin films; Data storage, Giant Magneto resistance (GMR).

TEXT BOOKS:

1. A. Goswami, Thin Film Fundamentals, New Age international (P) Ltd. Publishers, New Delhi(1996).

REFERENCE BOOKS:

1. K. L. Chopra, Thin Film Phenomena, McGraw- Hill book company New York,(1969).
2. L. Eckertova, Physics of Thin Films, Plenum press, New York(1977).
3. Milton Ohring, Material Science of Thin films, 2nd Edition, Academic Press(2002).

E-TEXT BOOKS:

1. <https://www.safaribooksonline.com/library/view/materials-science-of/9780125249751/?ar&orpg>
2. <http://doi-ds.org/doilink/12.2018-75375189/>
3. <https://ieeexplore.ieee.org/document/5488896>

MOOC COURSE:

1. <https://nptel.ac.in/courses/113106034/44>
2. www.facweb.iitkgp.ac.in/~jdas/mfm/Teaching.html
2. https://swayam.gov.in/nd1_noc19_mm13

Polymer Chemistry Open Elective Course								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4HS59	OEC	L	T	P	C	CIA	SEE	Total
		3	0	0	3	30	70	100
Contact Classes:	Tutorial Classes: NIL	Practical Classes: NIL			Total Classes: 48			
COURSE OBJECTIVES: The course should enable the students to: 1. Understand the structure of the polymers, identify and explain types of polymerization. 2. Gain the knowledge to use techniques of polymerization. 3. Understand the properties, processing and applications of natural rubber. 4. Gain the knowledge to design polymers for industrial applications. 5. Learn the structure properties of polymers ,inorganic and its characterization techniques								
COURSE OUTCOMES: The student will able to: 1. Describe the structures of the polymers, identify and explain types of polymerization and account for the concept of the molecular weight. 2. Explain the various properties of polymers and use techniques of polymerization. 3. Account the properties, processing and applications of natural rubber. 4. Design polymers for industrial application using suitable additives. 5. Correlate structure-properties of polymer composite, inorganic, conducting polymers and its characterization techniques.								
UNIT-I	Basics of Polymers							
Introduction: Polymer Definition, Classifications of Polymers: Natural & Synthetic, Biopolymers, Thermoplastic, Thermosets, Elastomers, Fibers etc. Types of Polymerization: Addition & Condensation Polymerization. Molecular Weight : No. Average and Weight Average Molecular Weight of Polymers, Distribution & Poly dispersity Index, Structure of Polymers: Amorphous, Semi crystalline and Crystalline states								
UNIT-II	Properties & Polymerization Techniques							
Polymerization Techniques: Mass Polymerization, Bulk Polymerization, Solution Polymerization, Emulsion Polymerization, Suspension Polymerization, Mechanisms with explanation. Structure Property Relationship in Polymers: glass transition, melting and crystallization temperature. Effect of structure on the chemical, mechanical, electrical and optical properties of polymers.								
UNIT-III	Natural Rubber : Properties, Processing and Applications							
Natural Rubber (NR): Origin – Natural Rubber Latex, Tapping, Processing- Vulcanization, Properties and Applications – Conversion of Latex into Dry Rubber – Properties of Dry Rubber. Synthetic Polyisoprene (IR) Rubbers: Preparation, Properties and Applications, Thermoplastic Elastomers Based on Plastics.								
UNIT-IV	Non Rubber Additives							
Non Rubber Additives: Part A: Vulcanizing ingredients & other additives: Vulcanizing ingredients & their sequence of mixing: Activators and Accelerators: mechanisms of action Part B: Fillers Carbon black-Its preparation, structure, properties and their effect on rubber properties Silica fillers & coupling agents, Other fillers: Clay, Calcium carbonate, titania etc. Nano-fillers: Reinforcement by filler: Reinforcement, Part C: Processing aids & other additives: Processing aids, plasticizers, process additives, release agents, Prevention of Ozone Attack with the use of waxes & saturated polymer for Ozone Protection.								
UNIT-V	Inorganic, Composite and Conducting Polymer, Characterization Techniques of Polymers							

Inorganic Polymers: Synthesis, Structure- property Relationship and Applications of Polyphosphates, Polysiloxanes, Polymer Composite- Properties and its Applications. Conducting Polymers- Poly Aniline (PaNi)- Mechanism of Conduction and Applications.

Characterization Techniques of Polymers: Thermo gravimetric Analysis (TGA), Differential Scanning Calorimetry (DSC).

REFERENCE BOOKS:

1. F. W. Billmeyer Jr., Text Book of Polymer Science, Ed. Wiley-Interscience, 1984.
2. V. T. Gowariker, N. V. Viswanathan, and J. Sreedar, Polymer Science, 1988.
3. M. Morton, Rubber Technology, Chapman Hall, 1995.
4. Robert J. Young and Peter A. Lovell, Introduction to Polymers, 3rd Edition, 2011.

PROGRAM OUTCOMES (POs):

Engineering Graduates will be able to:

PO1. Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2. Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3. Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4. Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5. Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6. The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7. Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8. Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9. Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10. Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11. Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12. Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs):

PSO1: Understand the structure, evolutionary changes and development methodologies of software systems to address modern computing challenges.

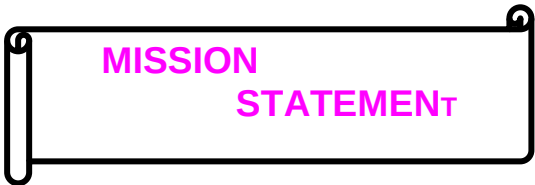
PSO2: Develop intelligent and autonomous systems to cater societal needs especially in the fields of health care, ecommerce, banking, agriculture, cyber security, and insurance.



VISION STATEMENT

VISION OF THE INSTITUTION

Promote academic excellence, research, innovation, and entrepreneurial skills to produce graduates with human values and leadership qualities to serve the nation.



MISSION STATEMENT

MISSION OF THE INSTITUTION

Provide student-centric education and training on cutting-edge technologies to make the students globally competitive and socially responsible citizens. Create an environment to strengthen the research, innovation and entrepreneurship to solve societal problems.